

Pure Geometry with Directed Angles

Version 2.0

11-4-2023

Contents

Introduction	3
1 Angles and Lengths	5
1.1 Directed Angles	5
1.2 Directed Lengths	12
2 Angle Chasing, Similar Triangles and Concyclicity	14
3 Transformations	25
3.1 Rigid Motions	25
3.2 Reflection	28
3.3 Homothety	30
4 Triangle Geometry	37
4.1 Centroid and Midpoints	37
4.2 Circumcircle and Circumcentre	41
4.3 Orthocentre and Nine-point Circle	45
4.4 Incentre and Excentres	60
5 Power and Radical Axes	69
5.1 Power	69
5.2 Radical Axes and Radical Centre	79
6 Special Quadrilaterals	90
6.1 Parallelograms	90
6.2 Isosceles Trapezoids	94
6.3 Circumscribed Quadrilaterals	97
7 Spiral Similarity and Miquel Point	103
7.1 Spiral Similarity	103
7.2 Miquel Point	113
8 Isogonal Conjugate	121
8.1 Isogonal Conjugate in Triangles	121
8.2 Isogonal Conjugate in Polygons	136
9 More on Triangle Geometry	143
9.1 Symmedians	143
9.2 Humpty Point and Dumpty Point	148
9.3 Sharky-Devil Point	158
10 Reconstruction of Figures	163
Links	178

Introduction

In everywhere of this note and MO, geometry mainly refers to plane geometry. Plane geometry is among the least important topics in advanced mathematics, but the case is totally different in Olympiad. Students who are weak in geometry could hardly do well. It should be remarked that there is no algebraic technique which can completely replace pure geometry. In any way of doing geometry problems, we need to use our knowledge in geometry to deduce useful facts. Experiences and observations are both essential.

For experiences, we first have to learn a lot of facts and properties. People who are good at geometry can always recall various useful facts immediately which are related to the configuration. The things that seem to be nontrivial observations to us might be ‘well-known’ facts to them. Since this is supposed to be an intermediate note in pure geometry, we are not going to introduce all the basic definitions and results, but we will recall the more important things that we should know.

After gaining sufficient knowledge, we can have more practices, which is the only way to perform well in MO. Good observations can also be trained in this way. One standard thing that we need to do is to draw an accurate figure. This helps us observe important facts more quickly. In particular, observing groups of concyclic points is a necessary skill. Another suggestion is that we should always draw some standard figures (such as triangles, cyclic quadrilaterals, complete quadrilaterals, etc.) in more or less the same way when we first try on a new problem. This allows us to connect the figures with those in our old practices. Sometimes we may be able to guess the meaning of a special point in the figure because of this connection.

In this note, we will sometimes use directed angles for convenience. This is usually not necessary if we only aim at finding a solution to the problem, but it is helpful for both our consideration and the presentation of the solution in many cases. For this reason, we first give a simple introduction to this topic in section 1.

In section 2, we use the basic tools, i.e. angle chasing, similar triangles and concyclic points, to solve some elementary geometry problems. We do not need much knowledge to solve those simple problems, but all these basic tools are needed in solving all geometry problems, and so one should be skillful in using these tools.

Next, section 3 is another preliminary section, where the basic transformations are introduced. Among these, reflection and homothety are quite useful.

Section 4 is about triangle geometry, which is one major area in plane geometry. We will revise the important properties of the few basic triangle centres, including the centroid, the circumcentre, the orthocentre, the nine-point centre, the incentre and the excentres. It is a must for us to be familiar with this topic in order to solve problems in triangle geometry.

Another useful technique which is commonly used is power and radical axes. These will be included in section 5. They can be used to solve some problems with one or more circles, which are usually considered as some harder problems.

In section 6, we shall investigate several special quadrilaterals, including parallelograms, isosceles trapezoids and circumscribed quadrilaterals. The first two kinds of quadrilaterals are basic, but they have some useful symmetry that can be used to solve a number of problems. Even if the problems do not involve these special figures, we may construct parallelograms and isosceles trapezoids on our own by defining some new points. This is one possible idea to construct new points in the figure. On the other hand, circumscribed quadrilaterals are not frequently used, but they have some special features which are worthwhile to discuss.

Spiral similarity is another important tool because it is closely related to similar triangles. We shall investigate more on this transformation in section 7. We shall also introduce the Miquel point of quadrilaterals and its relation with spiral similarities.

Then in sections 8 and 9, we discuss some more advanced tools and facts. They are not as useful as the previous sections, but provide some more insight to us which could be useful in solving a particular problem. The topics include isogonal conjugates, symmedians, the Lemoine point and the Sharky-Devil point.

Finally, the last section 10 is not about geometric knowledge. Rather than that, we will discuss an extremely useful technique which can be used in every geometry problem. Although we will only provide a few examples in this note, one should find the power of this technique if one understands the principle.

1 Angles and Lengths

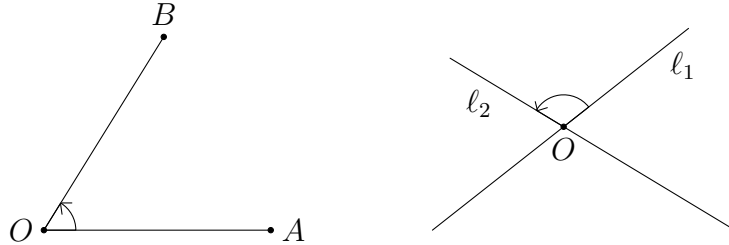
The most fundamental elements that build up the plane geometry are points and lines. To express the relations between points and lines more precisely, we need to have various quantities to measure, among which angles and lengths are the most important ones.

1.1 Directed Angles

In this note, we will sometimes use the usual angles. This means $\angle ABC$ always refer to the angle at B lying between 0° and 180° . There is no difference between $\angle ABC$ and $\angle CBA$. However, sometimes it is more convenient to use directed angles to handle various cases together. Therefore, we shall introduce the notations we use and the simple manipulations of directed angles in the following.

Definition 1.1. The *directed angle* (有向角) $\sphericalangle AOB$ refers to the angle spread by the line OA when it is rotated anticlockwise about O until it overlaps the line OB .

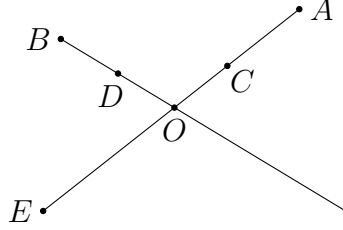
More generally, the directed angle spread by a line ℓ_1 when it is rotated anticlockwise about O until it overlaps a line ℓ_2 is denoted by $\sphericalangle(\ell_1, \ell_2)$ (or $\sphericalangle(\ell_1, \ell_2)$).



In this note, all directed angles (in radian) we use are taken modulo π . This means two angles x and y are equal (or congruent) if and only if (iff) their difference is an integral multiple of π (e.g. $0, \pm\pi, \pm2\pi$). This is useful in many situations since we need not distinguish between x and $x + \pi$. For example, in the definition, we need not specify whether ℓ_1 overlaps ℓ_2 for the first time or not. Indeed, it overlaps ℓ_2 again after rotating by another π rad.

With the abuse of notation, we still write $x = y$ instead of $x \equiv y \pmod{\pi}$. Therefore, we have relations like $0 = \pi$ and $\frac{\pi}{2} = -\frac{\pi}{2}$. However, it should be noted that such equalities are indeed congruences. In particular, we cannot divide a nonzero constant from both sides. For example, $0 = \pi$ does not imply $0 = \frac{\pi}{2}$. Also, since the angles used are taken modulo π , they cannot be used together with the trigonometric functions $\sin$ and $\cos$, except when only the absolute values are considered.

By definition, we have $\sphericalangle AOB = \sphericalangle(OA, OB)$. Since we are only interested in the relative positions of the two lines, it does not matter which points on the lines we choose to represent the lines. For example, if A, C, E, O are collinear, B, D, O are collinear, then all of $\sphericalangle AOB$, $\sphericalangle COB$, $\sphericalangle AOD$, $\sphericalangle COD$ or even $\sphericalangle EOB$ are referring to the same angle. Also, there is no difference between $\sphericalangle(OA, OB)$, $\sphericalangle(AO, OB)$, $\sphericalangle(AE, BD)$ (but not $\sphericalangle(OB, OA)$), etc.



Here gives the basic properties of directed angles.

Proposition 1.1. Let O, A, B, C, D be distinct points and let ℓ_1, ℓ_2, ℓ_3 be any lines on the plane. Then the following hold.

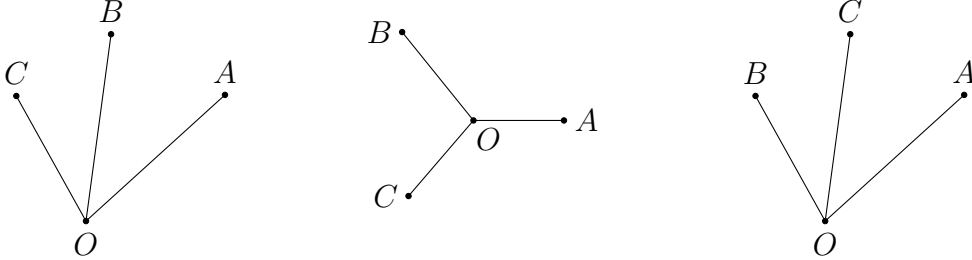
- (a) $\sphericalangle AOB = -\sphericalangle BOA$, $\sphericalangle(\ell_1, \ell_2) = -\sphericalangle(\ell_2, \ell_1)$
- (b) $\sphericalangle AOB + \sphericalangle BOC = \sphericalangle AOC$, $\sphericalangle(\ell_1, \ell_2) + \sphericalangle(\ell_2, \ell_3) = \sphericalangle(\ell_1, \ell_3)$
- (c) A, B, C are collinear iff $\sphericalangle ABC = 0$
- (d) ℓ_1 and ℓ_2 are parallel iff $\sphericalangle(\ell_1, \ell_2) = 0$
- (e) ℓ_1 and ℓ_2 are perpendicular iff $\sphericalangle(\ell_1, \ell_2) = \frac{\pi}{2}$ iff $\sphericalangle(\ell_2, \ell_1) = \frac{\pi}{2}$
- (f) $\sphericalangle ABC + \sphericalangle BCA + \sphericalangle CAB = 0$
- (g) $AB = AC$ iff $\sphericalangle ABC = -\sphericalangle ACB$
- (h) A, B, C, D are concyclic iff $\sphericalangle ABC = \sphericalangle ADC$

In view of (a), it is crucial to distinguish between $\sphericalangle AOB$ and $\sphericalangle BOA$, unlike the case for the usual angle notation. Therefore, we must be careful when writing the order of the labels. This suggests the name ‘directed’ angles.

Part (b) can be easily understood in view of the definition, since $\sphericalangle(\ell_1, \ell_2) + \sphericalangle(\ell_2, \ell_3)$ measures the angle spread by ℓ_1 when it is rotated until it overlaps ℓ_2 , and then rotated until it overlaps ℓ_3 . Note that the formula $\sphericalangle AOB + \sphericalangle BOC = \sphericalangle AOC$ is not always true if the usual angle notation is used. This is because we would obtain different (yet similar) formulae in different configurations. For example, the following figures illustrate some other possible cases, where we may have

$$\begin{aligned}\sphericalangle AOB + \sphericalangle BOC + \sphericalangle COA &= 360^\circ, \\ \sphericalangle AOB &= \sphericalangle BOC + \sphericalangle COA,\end{aligned}$$

etc. All these become the same when directed angles are used. This suggests the main reason why we would like to use directed angles whenever needed, i.e. to consider all configurations together.



Similarly, (c) is true regardless of the relative positions of A, B, C on the line. Both $\angle ABC = 0^\circ$ and $\angle ABC = 180^\circ$ are the same when taken modulo π .

Next, (f) is the angle sum of triangle. It is not hard to remember the order of the labels, since the labels only shift in a cyclic way. We may also write

$$\angle ACB + \angle CBA + \angle BAC = 0.$$

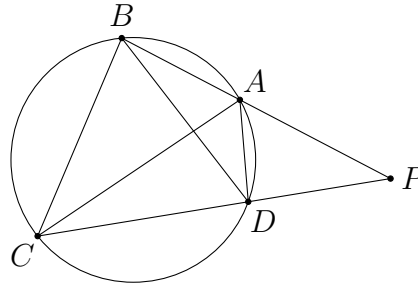
Note that the formula for the exterior angle of triangle is essentially the same as the angle sum of triangle. In general, we have the angle sum of polygon

$$\angle A_1A_2A_3 + \angle A_2A_3A_4 + \cdots + \angle A_nA_1A_2 = 0$$

for the polygon $A_1A_2\cdots A_n$, as the left-hand side is the same as

$$\angle (A_1A_2, A_2A_3) + \angle (A_2A_3, A_3A_4) + \cdots + \angle (A_nA_1, A_1A_2).$$

Part (h) is the most useful fact for our use, since we always need to consider concyclic points. In the usual notation, we have various facts about concyclic points. For example, in the figure below, we have $\angle CBD = \angle CAD$, $\angle BAD + \angle DCB = 180^\circ$ and $\angle CBA = \angle PDA$, etc. Conversely, each of these implies A, B, C, D are concyclic. In directed angles, all these are essentially the same as the single formula given in (h) up to symmetry. The formula is easy to remember since only the middle letter is changed. Also, note that A, B, C, D being concyclic includes the case when A, B, C, D are collinear. Usually, this case can be easily rejected.



In the limiting case where $C = D$, line CD refers to the tangent ℓ at C . This gives $\angle(AB, BC) = \angle(AC, \ell)$. Another way is to write $\angle ABC = \angle ACP$ where P is any point on ℓ . In the usual sense, this is just the angle in alternate segment.

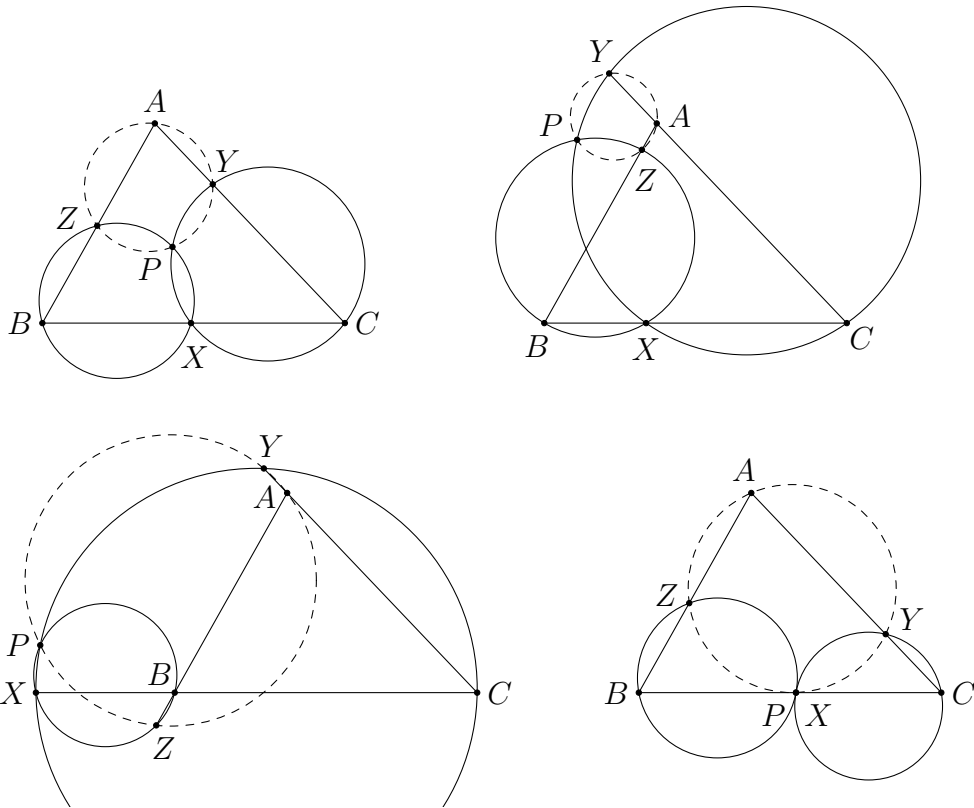
Problem 1.1. Let X, Y, Z be points on the lines BC, CA, AB of $\triangle ABC$ respectively which are different from A, B, C . Prove that $(AYZ), (BZX), (CXY)$ pass through a common point.

Analysis. The notation $(P_1P_2\cdots P_n)$ means the circle passing through the concyclic points $P_1, P_2, \dots, P_n$.

When new points are added, we need to be careful whether it can lie on the whole line, on the (line) segment, or on an open ray. Sometimes there might be more than one case for us to consider. For an easy problem like this, we must prove the assertion for all possible cases. Usually, writing something like ‘the other cases are similar’ is not enough.

Here, we would use directed angles to combine all cases into one. To do so, we can first construct a proof for one of the cases (for example, when X, Y, Z lie on the segments BC, CA, AB) and then change the angles to directed ones.

Solution.



Here, we have given a few possible configurations of this problem. Suppose we deal with the case when X, Y, Z lie on the sides (meaning segments) BC, CA, AB as the first figure. Let P be the second intersection of (BZX) and (CXY) . Then it is not hard to deduce A, Y, P, Z are concyclic. Indeed, this is a direct consequence by noting that $\angle AYP = \angle CXP = \angle BZP$ from the concyclic points. However, such an argument fails in most of the other configurations.

For instance, in the top right figure, $\angle AYP$ and $\angle CXP$ are not equal to each other, but they sum up to 180° . However, as we also have $\angle CXP + \angle BZP = 180^\circ$, the final result is still true. For the bottom left figure, we have $\angle AYP + \angle CXP = 180^\circ$ and $\angle CXP = \angle BZP$. These yield $\angle AYP + \angle BZP = 180^\circ$. This time it is the relation we want in order to prove A, Y, P, Z are concyclic.

Noting the trouble arises from various configurations, it is better for us to use directed angles. To do so, we just need to alter the above proof slightly. Indeed, we have

$$\angle AYP = \angle CYP = \angle CXP = \angle BXP = \angle BZP = \angle AZP.$$

This implies A, Y, Z, P are concyclic. The only trick of using directed angles in this proof is to note two things: represent the same lines using different labels (like $\angle AYP = \angle CYP$ since AY and CY are the same line), and deduce results from the concyclic points (like $\angle CYP = \angle CXP$ as C, X, Y, P are concyclic). Once we have noted the groups of collinear points and concyclic points, we can write down the proof without drawing any figures. $\square$

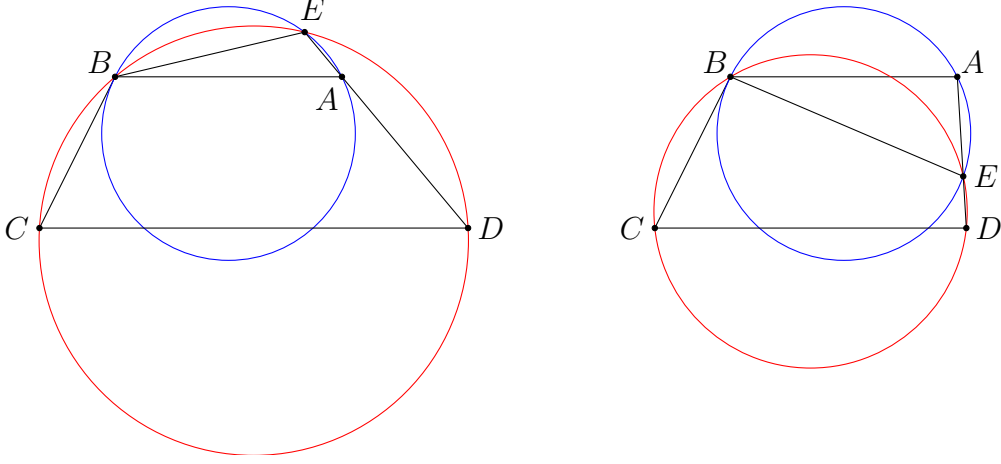
Remarks. Strictly speaking, the proof we have given is still incomplete! The bottom right figure on the previous page provides a case for which our proof fails, i.e. when X and P coincide (which means (BZX) and (CXY) are tangent at X). In that case, $\angle CXP$ and $\angle BXP$ are undefined. The same is true even if we use the other notation $\angle(XC, XP)$ for $\angle CXP$. However, we can treat this as the limiting case and allow XP to mean the tangent at X to the two circles (in case they are tangent), so that $\angle(XC, XP)$ is still meaningful. Another way to see is to give some explanations for these limiting cases, for example, we may say that the same holds in these cases by continuity.

The intersection point P of the three circles is called the *Miquel point*(密克點) of $\triangle XYZ$ with respect to(w.r.t.) $\triangle ABC$.

Problem 1.2. Suppose $ABCD$ is a trapezoid with $AB \parallel CD$. The circumcircle of triangle BCD meets the line AD again at E . Prove that the circumcircle of triangle ABE is tangent to BC .

Analysis. This is a simple exercise in angle chasing. One should be able to give a proof based on a certain configuration. However, there are 3 possible configurations in total (2 are given below). We demonstrate the use of directed angles to handle the 3 cases together. We would also illustrate how to use the parallel condition and prove the tangency.

Solution.



Our goal is to show $\angle CBE = \angle BAE$. Firstly, since B, C, D, E are concyclic, we have $\angle CBE = \angle CDE$. Note that this is the angle $\angle(CD, DE)$. As $CD \parallel AB$, we can replace CD by AB . This gives

$$\angle CBE = \angle(AB, DE) = \angle(AB, AE) = \angle BAE.$$

From this, we know that BC is tangent to (ABE) . □

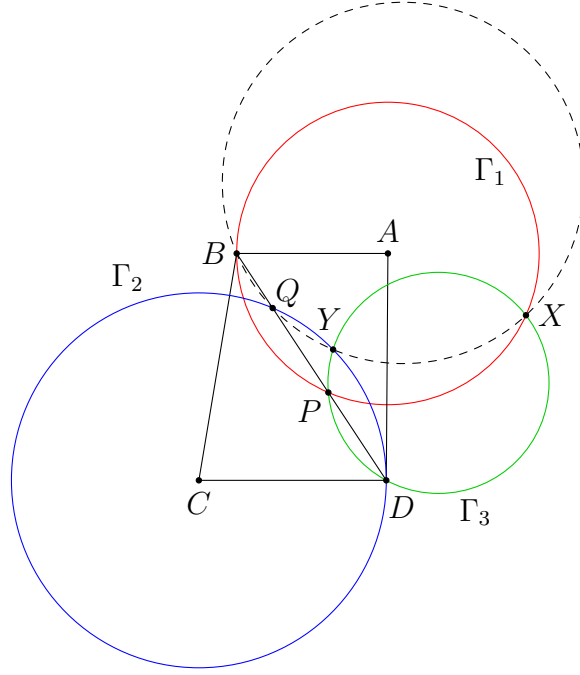
Remarks. As can be seen, we can replace a line by another parallel line when using the notation $\angle(\ell_1, \ell_2)$. This is how we use parallel conditions. All angle relations we have learnt for parallel lines (alternate angles, corresponding angles, etc.) are the same as this replacement.

Problem 1.3. Suppose $ABCD$ is a trapezoid with $AB \parallel CD$. Let Γ_1 be the circle with centre A passing through B , and Γ_2 be the circle with centre C passing through D . Suppose the line BD meets Γ_1 and Γ_2 again at P and Q respectively. Another circle Γ_3 passes through P and D . Let X be the second intersection of Γ_1 and Γ_3 , and Y be the second intersection of Γ_2 and Γ_3 . Prove that B, Q, X, Y are concyclic.

Analysis. Without the help of software, it is not an easy task to draw a nice figure, especially for those who are weak at drawing figures. One advantage of using directed angles is that it is not necessary to have a figure to find out a proof using angle chasing. Indeed, we only need to know which points are collinear and which points are concyclic.

Solution. Since Q and X are not so related, it is hard to chase an angle having these as vertices. Thus, a better way is to show $\angle BXY = \angle BQY$ to deduce the concyclicity. Since B, X, Y are related to Γ_1 and Γ_3 , another intersection point P of the two circles would be useful. Thus, the first step is to rewrite

$$\angle BXY = \angle BXP + \angle PXY.$$



For $\angle BXP$, we only know that B, X, P lie on a circle with centre A . Thus, we can transform this angle to some angle in the given trapezoid $ABCD$, which should be much better. (If every angle can be expressed in terms of the angles of $ABCD$, then the problem is trivial.) Indeed, as $\angle BXP$ is the angle at circumference while $\angle BAP$ is the angle at centre, we have

$$\angle BXP = \frac{\pi}{2} - \angle PBA = \frac{\pi}{2} - \angle DBA.$$

(The first equality may sound obvious for the usual angles. For a more detailed discussion for directed angles, we shall postpone to subsection 4.2 where we talk about circumcentre.)

For $\angle PXY$, as P, X, Y, D are concyclic, it is natural to write $\angle PXY = \angle PDY$. This is equal to $\angle BDY$, which is the simplest form that we can get since Y arises from Γ_3 , which is a circle that can vary. Hence, we stop here and conclude

$$\begin{aligned} \angle BXY &= \frac{\pi}{2} - \angle DBA + \angle BDY = \frac{\pi}{2} + \angle (AB, BD) + \angle (BD, DY) \\ &= \frac{\pi}{2} + \angle (AB, DY). \end{aligned}$$

Next, we consider $\angle BQY = \angle DQY$. As D, Q, Y lie on a circle with centre C , we get

$$\angle DQY = \frac{\pi}{2} - \angle YDC.$$

But then it is obvious that $-\angle YDC = \angle (CD, DY) = \angle (AB, DY)$ since $AB \parallel CD$. It follows that $\angle BXY = \angle BQY$. This proves B, Q, X, Y are concyclic. $\square$

Remarks. Apart from the use of directed angles, one should also learn the idea of angle chasing from this solution. Sometimes we need to break an angle into two. The usual way is that the points involved in one angle should be somehow related (by concyclicity or similar triangles, etc.) and can be transformed to those ‘fundamental angles’, meaning the angles which arise at the very beginning of the construction of the figure. Such angles are easy to compare.

1.2 Directed Lengths

Similar to the case of angles, it is sometimes convenient to express lengths of segments with directions. For each line, we assign one of its two directions to be the positive direction (and the other one as the negative direction). Therefore, the segments AB and BA have the same magnitude but are considered as opposite to each other in the sense that they have different directions. These are called *directed lengths* (有向長度). We have the following basic facts.

Proposition 1.2. Let A, B, C be points on a line in any order. Then the following hold.

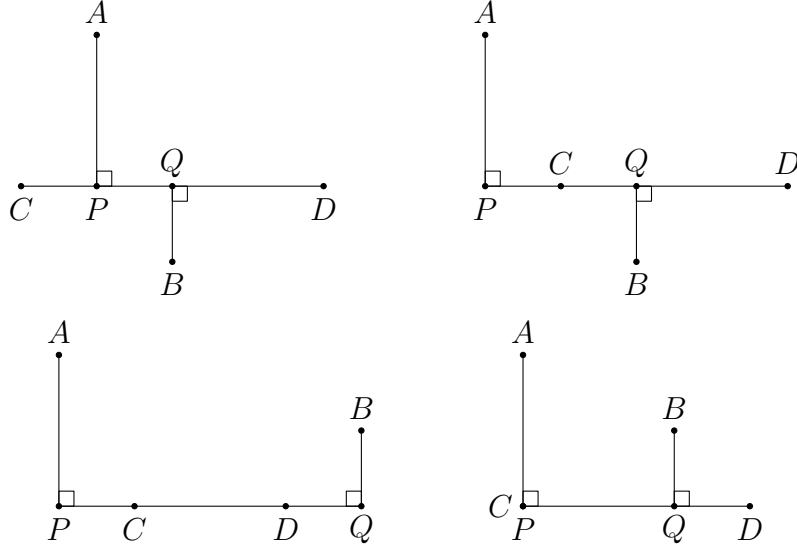
- (a) $AA = 0$
- (b) $AB = -BA$
- (c) $AB + BC = AC$
- (d) $AB - AC = CB$

Since there is no standard notation for directed lengths, we need to mention when we are using directed lengths. Note that two segments on two different lines cannot be compared as the two directions can be chosen randomly. Therefore, directed lengths are meaningful only when we are working on a single line. Usually, we use directed lengths to deduce relations among ratios of side lengths (on the same line). So choosing which direction to be positive does not matter.

Problem 1.4. Let A, B, C, D be four distinct points on the plane. Then $AB \perp CD$ iff $AC^2 - AD^2 = BC^2 - BD^2$.

Analysis. This problem can be viewed as a generalization of Pythagoras’ theorem. Since all terms in the equation are squares of side lengths, we probably need to apply Pythagoras’ theorem. To construct more right-angled triangles, one way is to draw feet of perpendiculars from the points to a line. Note that the lengths in the equation are not necessarily directed as they may belong to different lines. But we can restrict them to directed lengths after projections since XY^2 and $YX^2 = (-XY)^2$ are indeed the same.

Solution.



Let P and Q be the projections from A and B to the line CD respectively. By Pythagoras' theorem, we have

$$AC^2 - AD^2 = (AP^2 + CP^2) - (AP^2 + DP^2) = CP^2 - DP^2.$$

Similarly, $BC^2 - BD^2 = CQ^2 - DQ^2$. Therefore, $AC^2 - AD^2 = BC^2 - BD^2$ is the same as $CP^2 - DP^2 = CQ^2 - DQ^2$. Now, all segments belong to the same line. We may assume they are written with directions since all terms are squares. Our goal is to prove that this is equivalent to $P = Q$, which means exactly the same as $AB \perp CD$. Indeed,

$$\begin{aligned}
 & CP^2 - DP^2 = CQ^2 - DQ^2 \\
 \Leftrightarrow & (CP - DP)(CP + DP) = (CQ - DQ)(CQ + DQ) \\
 \Leftrightarrow & CD(CP + DP) = CD(CQ + DQ) \\
 \Leftrightarrow & CP + DP = CQ + DQ \\
 \Leftrightarrow & CD + 2DP = CD + 2DQ \\
 \Leftrightarrow & 2DP = 2DQ \\
 \Leftrightarrow & DP = DQ \\
 \Leftrightarrow & P = Q.
 \end{aligned}$$

□

Remarks. In the steps $CP - DP = CD$ and $CP = CD + DP$, the deductions are not always true if the usual lengths are used. But our proof is fine since directed lengths are used. Also $DP = DQ$ implies $P = Q$ is only true when DP and DQ have the same direction. Otherwise, it could happen that D is the midpoint of P and Q .

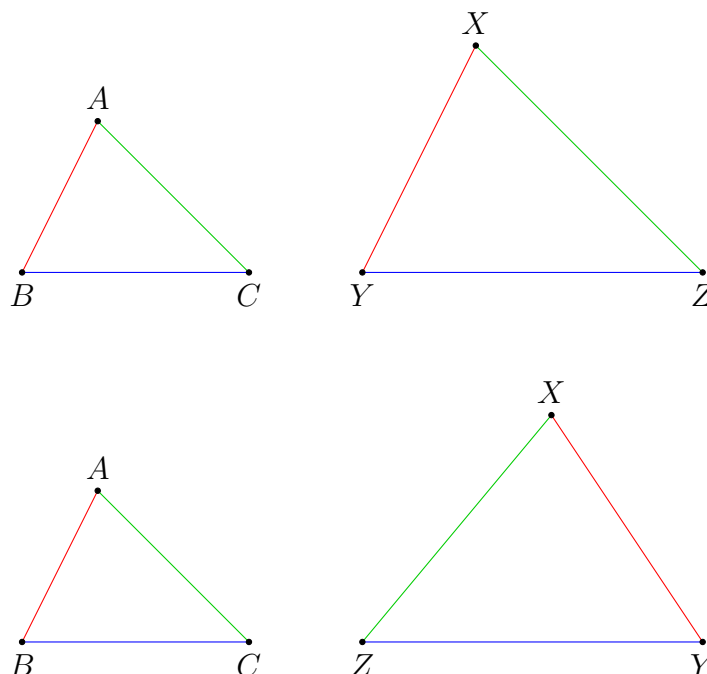
The assertion in this problem is useful in proving two lines are perpendicular to each other. But since it is related to side length calculation, solutions using this fact may not be pure enough. Yet it will be used when we talk about radical axis later.

2 Angle Chasing, Similar Triangles and Concyclicity

In the sense of Olympiad, doing pure geometry usually means carrying out angle chasing, deducing similar triangles and concyclic points, etc. Almost all geometry problems are solved with the use of these basic tools. They provide a lot of relations between lengths and angles, and they occur naturally in every geometric setting. To solve a geometry problem, the first step is to find out pairs of similar triangles and groups of concyclic points. Some may be obvious, some seem to be true but hard to prove, while some are implicit. As mentioned at the beginning, having a good observation is important to deduce these. Even if we cannot prove some assertions, the problem can be simplified as it has been divided into several parts.

Before working on some basic problems, we first discuss the relations between similar triangles and directed angles. There is a slight difference if the usual angle notation is not used.

When we say that $\triangle ABC$ and $\triangle XYZ$ are similar (or congruent), it could happen in two cases as shown below. The first case is when both triangles are oriented in the same direction. This means the corresponding vertices A, B, C and X, Y, Z lie in the anticlockwise (or clockwise) direction simultaneously. The second case is when their orientations are different.



In the usual notation, the two cases mentioned do not differ too much since the corresponding angles are still equal and the corresponding sides are in proportion. However, if directed angles are used, we do not have $\angle ABC = \angle XYZ$ in the second

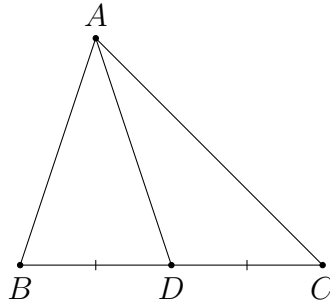
case, but $\angle ABC = -\angle XYZ$ instead. Therefore, we need to distinguish between these cases carefully when using directed angles. This yields the following definitions.

Definition 2.1. $\triangle ABC$ and $\triangle XYZ$ are said to be *directly similar* (順向相似) if they are similar and have the same orientation.

Definition 2.2. $\triangle ABC$ and $\triangle XYZ$ are said to be *oppositely similar* (逆向相似) if they are similar and have different orientations.

There are no standard notations for directly similarity and oppositely similarity. Therefore, we need to express these properties by words. To show that $\triangle ABC$ and $\triangle XYZ$ are directly similar, we can only use ‘equiangular’ (angles should be written in directed angles). This means $\triangle ABC \sim \triangle XYZ$ if $\angle ABC = \angle XYZ$ and $\angle BCA = \angle YZX$. In the same way, we can prove that two triangles are oppositely similar by adding negative signs to both sides of each angle equation.

Note that ‘3 sides proportional’ can only be used to deduce the similarity but cannot be used to determine the orientations of the triangles. (Directed lengths are not useful since all segments could probably lie on different lines.) Moreover, ‘ratio of 2 sides, included angles’ is insufficient to show the similarity if directed angles are used. For example, we have $\frac{AD}{BD} = \frac{AD}{CD}$ and $\angle ADB = \angle ADC$ in the figure below, but $\triangle ADB$ and $\triangle ADC$ may not be similar.



This shows directed angles sometimes do not work well with similar triangles. Therefore, when working on problems with similar triangles, it should be better to use the usual angles, unless one is familiar with all these things.

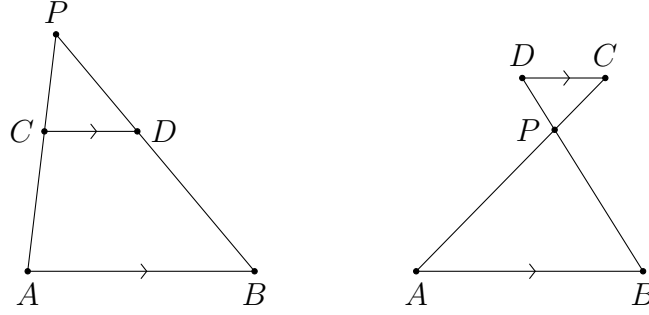
Next, we provide some common occurrences of similar triangles.

Proposition 2.1. Let AB and CD be parallel lines and let P be the intersection point of the lines AC and BD . Then $\triangle PAB$ and $\triangle PCD$ are directly similar.

Proof. This comes directly from

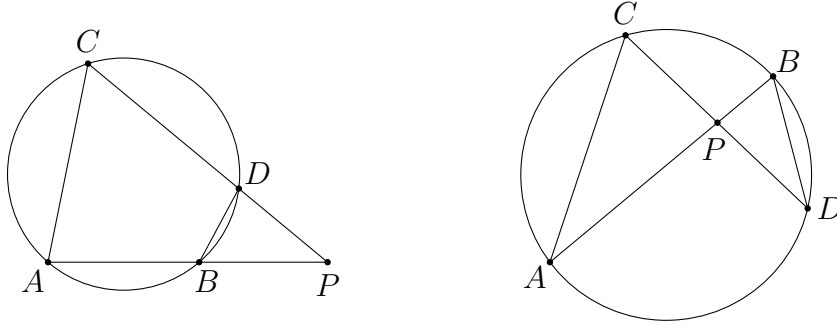
$$\angle PAB = \angle(PA, AB) = \angle(PC, CD) = \angle PCD$$

and $\angle PBA = \angle PDC$ due to symmetry. □



Proposition 2.1 implies $\frac{PA}{PC} = \frac{PB}{PD}$, and also $\frac{PC}{CA} = \frac{PD}{DB}$ (the **intercept theorem**(截線定理), which is useful for the first figure). Conversely, these ratios can be used to prove the parallelism. In the case when C and D are the midpoints of PA and PB respectively, we have the **midpoint theorem**(中點定理), which states that $CD \parallel AB$ and $CD = \frac{1}{2}AB$.

Proposition 2.2. Let A, B, C, D be concyclic points and let P be the intersection point of the lines AB and CD . Then $\triangle PAC$ and $\triangle PDB$ are oppositely similar.



Proof. This comes directly from

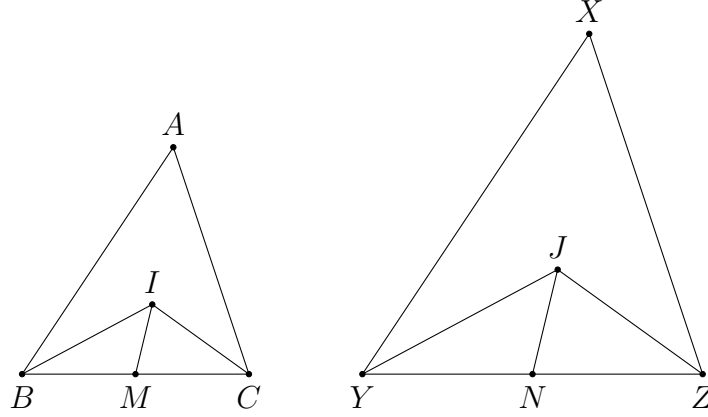
$$\angle PAC = \angle BAC = \angle BDC = \angle BDP = -\angle PDB$$

and $\angle PCA = -\angle PBD$ due to symmetry. $\square$

Proposition 2.2 implies $\frac{PA}{PC} = \frac{PD}{PB}$. The case is rather similar to the previous case with parallel lines. Therefore, AC and BD are called **antiparallel**(逆平行) w.r.t. the lines AB and CD since this configuration shares a lot of similar properties with parallelism.

Lastly, we emphasize another useful fact concerning similar triangles. Suppose $\triangle ABC \sim \triangle XYZ$. Whenever corresponding points are constructed on the pair of

similar triangles, we obtain a lot more pairs of similar triangles under the same similarity.

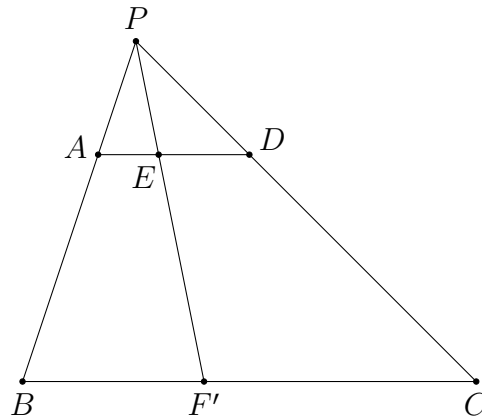


For example, let M be the midpoint of BC and N be the midpoint of YZ . Then we get $\triangle ABM \sim \triangle XYN$. Let I and J be the incentres of $\triangle ABC$ and $\triangle XYZ$ respectively. Then $\triangle IBC \sim \triangle JYZ$ and $\triangle IMC \sim \triangle JNZ$, etc. Also, the orientations are preserved when we deduce new pairs of similar triangles from old ones. This often enables us to find more similar triangles quickly.

Problem 2.1. ([Intercept theorem](#)(截線定理)) Suppose $ABCD$ is a trapezoid with $AD \parallel BC$. Let E and F be points on the sides AD and BC respectively such that $AE : ED = BF : FC$. Prove that AB, CD, EF are concurrent.

Analysis. This is an elementary problem and the result is also well-known. However, for a beginner, it is not obvious how we can prove that some lines are concurrent because there is no elementary result which enables us to do so. A common technique used in solving geometry problem is the method of false position. More will be discussed in section 10.

Solution.



Let AB meet CD at P , and let PE meet BC at F' . It suffices to show $F' = F$. Since the given condition involves ratio of lengths, a simple way would be proving $\frac{BF'}{F'C} = \frac{BF}{FC}$. This suggests using similar triangles to relate the ratios.

By [proposition 2.1](#), We have $\triangle PAE \sim \triangle PBF'$. This implies $\frac{AE}{BF'} = \frac{PE}{PF'}$. Similarly, from $\triangle PDE \sim \triangle PCF'$, we obtain $\frac{DE}{CF'} = \frac{PE}{PF'}$. Combining these results, we get $\frac{AE}{BF'} = \frac{DE}{CF'}$, and hence

$$\frac{BF'}{CF'} = \frac{AE}{DE} = \frac{BF}{CF}.$$

As both F and F' lie on the side BC , and they cut the segment in the same ratio, we must have $F' = F$. This means AB, CD, EF are concurrent at P . $\square$

Remarks. The term ‘concurrent’ here includes the limiting case where AB, CD, EF are parallel. In that case, our proof fails because AB and CD have no intersection point. It is a good habit to consider such limiting cases as well, but usually (not always!) it does not cause much trouble if one ignores them.

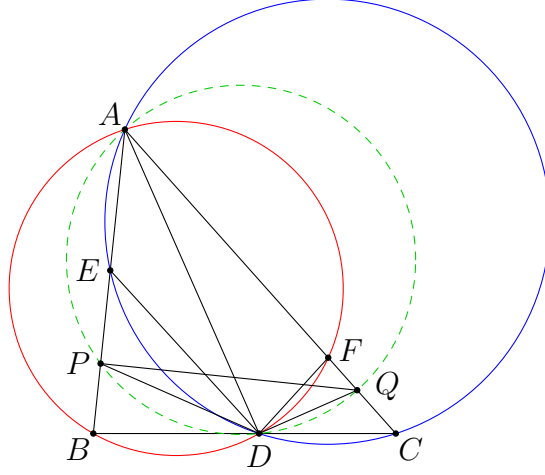
Problem 2.2. In $\triangle ABC$, $\angle B > \angle C$. Let D be the point on the side BC such that $\angle DAC = \frac{B - C}{2}$. The circumcircle of $\triangle ACD$ meets the side AB again at E . The circumcircle of $\triangle ABD$ meets the side AC again at F . The internal angle bisector of $\angle BDE$ meets the side AB at P . The internal angle bisector of $\angle CDF$ meets the side AC at Q . Prove that PQ and AB are perpendicular.

Analysis. Since points P and Q are defined separately, it is hard to express $\angle APQ$ in terms of A, B, C directly. A natural way is to find another right-angled triangle which is similar to $\triangle APQ$ or to find concyclic points involving P and Q . In practice, since the use of protractor is sometimes prohibited, it may not be easy to draw an accurate figure in the general case. However, we can first study some special cases, like $\angle B = 90^\circ$ and $\angle C = 30^\circ$. Such a special case is easy to draw and one can easily observe some groups of concyclic points other than those given. Although some are not true in the general case, we only need to test for a few possibilities. Considering the limiting case when $\angle B = \angle C$ may also be useful, though one should be careful about the definition of $\odot ACD$ when C and D coincide.

Other than considering special cases, we may also mark the useful angles in the figure to see which angles are equal. It is not hard to find that A, P, D, Q are concyclic.

Solution. From the concyclic points, we have $\angle BDE = A = \angle CDF$. This gives $\angle BDP = \angle CDQ = \frac{A}{2}$. We find that

$$\angle PDQ = 180^\circ - \angle BDP - \angle CDQ = 180^\circ - A.$$



Therefore, A, P, D, Q are concyclic. It follows that

$$\angle APQ = \angle ADQ = 180^\circ - \angle DAC - \angle ACD - \angle CDQ = 180^\circ - \frac{B+C}{2} - C - \frac{A}{2} = 90^\circ.$$

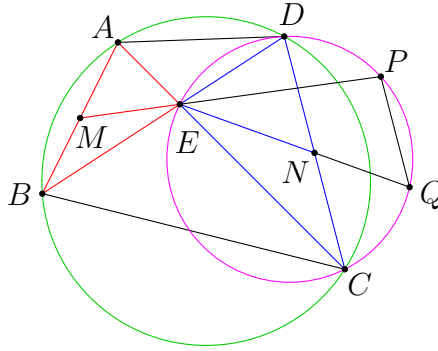
□

Remarks. After finding the concyclic points, the remaining steps are simply angle chasing. This means expressing angles in terms of other angles, usually using concyclicity and similar triangles. To do so, one only needs to have in mind which angles can be expressed in terms of A, B, C easily. However, note that in Olympiad problems, it is always impossible that the problem can be solved by finding all angles alone (using angle sum of triangles, etc.). It is necessary to make nontrivial observations like A, P, D, Q being concyclic. This is the reason why finding groups of concyclic points is important.

Problem 2.3. Let $ABCD$ be a cyclic quadrilateral. Its diagonals AC and BD intersect at point E . Suppose M and N are the midpoints of the sides AB and CD respectively. Extend ME and EN to meet the circumcircle of $\triangle CDE$ at points P and Q respectively. Prove that CD and PQ are parallel.

Analysis. As suggested by [proposition 2.2](#), triangles EAB and EDC are similar. As M and N are the midpoints of the corresponding sides AB and CD , we can easily find more pairs of similar triangles. This in turn gives us more relations between the angles, and enables us to deduce the parallelism.

Solution. Note that $\triangle EAB \sim \triangle EDC$. As M and N are the midpoints of the corresponding sides AB and DC respectively, we have $\triangle EAM \sim \triangle EDN$. This implies $\angle AEM = \angle DEN$, so that $\angle CEP = \angle DEQ$. The arcs $\widehat{CP}$ and $\widehat{DQ}$ opposite to these angles thus have the same length. Therefore, CD and PQ are parallel. □

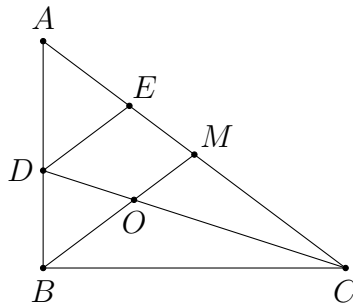


Remarks. The characterization of a pair of parallel lines on a circle in terms of the arc length is an useful result. In any case, it is not hard to deduce the parallel lines using angles once we have obtained $\angle CEP = \angle DEQ$.

Problem 2.4. (IMO HK TST 2003 Test 1 Problem 2) Let $\triangle ABC$ be a triangle. M is the midpoint of AC , D is a point on AB . BM and CD meet at O , with $AB = CO$. E is a point on AC such that $DE \parallel BM$. Prove that $AB \perp BC$ if and only if $ADOM$ is a cyclic quadrilateral.

Analysis. In view of [proposition 2.1](#), we immediately obtain pairs of similar triangles from $DE \parallel BM$, namely, $\triangle ADE \sim \triangle ABM$ and $\triangle CMO \sim \triangle CED$. These similar triangles are proved via angles. Therefore, the best way to use them is to consider the relations of side lengths. We can use the give facts $AB = CO$ and $AM = CM$ to simplify the expressions. We shall find that this allows us to obtain an isosceles triangle, and the remaining angle chasing is straightforward.

Solution.



From $\triangle ADE \sim \triangle ABM$, we have $\frac{AB}{BD} = \frac{AM}{ME}$. Also, $\triangle CMO \sim \triangle CED$ implies $\frac{CM}{ME} = \frac{CO}{OD}$. Since $AM = CM$, these ratios are all equal. In particular, $\frac{AB}{BD} = \frac{CO}{OD}$. Now, using the given condition $AB = CO$, we have $BD = OD$.

Now, the quadrilateral $ADOM$ is cyclic iff $\angle DAM = \angle DOM$. Note that

$$\begin{aligned}
& \angle DAM = \angle DOM \\
\Leftrightarrow & \angle BAM = \angle DOB \\
\Leftrightarrow & \angle BAM = -\angle DBO \\
\Leftrightarrow & \angle BAM = -\angle ABM \\
\Leftrightarrow & MB = MA (= MC) \\
\Leftrightarrow & AB \perp BC.
\end{aligned}$$

Hence, we are done. $\square$

Remarks. The last equivalence comes from an important characterization of right-angled triangles, namely, the median from the right angle is half the length of the hypotenuse.

We have used directed angles in case there are other possible configurations. For example, the problem does not emphasize whether D lies on the side AB or on its extension. In most cases, AB should refer only to the segment AB . However, it is always better to consider all possible cases, or ask for clarification if this is a problem in tests/competitions.

Problem 2.5. (CGMO 2003 Problem 3) Quadrilateral $ABCD$ is inscribed in a circle with AC as its diameter, $BD \perp AC$, and E the intersection of AC and BD . Extend line segments DA and BA through A to F and G respectively, such that $DG \parallel BF$. Extend GF to H such that $CH \perp GH$. Prove that points B, E, F and H lie on one circle.

Analysis. The first step is to decide what is the simplest way to prove the assertion. As E and H are not related, it should be better to show $\angle BHF + \angle BEF = 180^\circ$. The main condition related to point H is that $\angle CHG = 90^\circ$. As we also have $\angle CBG = 90^\circ$, we note that B, C, G, H are concyclic. Therefore, it is easy to express $\angle BHF$ in terms of $\angle BGC$. The assertion is then equivalent to $\angle AEF = \angle AGC$. This means $\triangle AFE$ and $\triangle ACG$ should be similar, and the problem is solved once we have shown this. The question is now unrelated to point H .

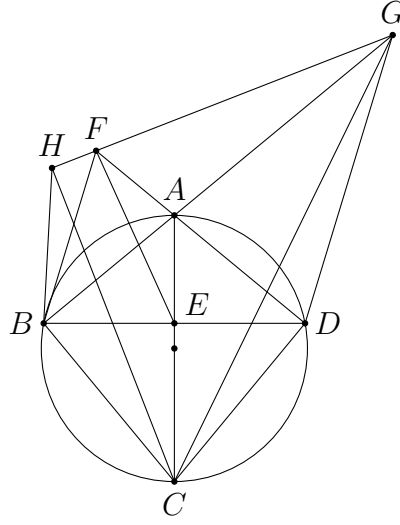
As the conclusion is a pair of equal angles, we should use ratios of lengths to prove the similarity. These ratios can be obtained from other obvious pairs of similar triangles.

Solution. From $BF \parallel DG$, we have $\triangle AFB \sim \triangle ADG$. This gives

$$\frac{AF}{AB} = \frac{AD}{AG}. \quad (1)$$

Next, we have to find a relationship between AE and AC . One way is to note that $\triangle AEB \sim \triangle ABC$, as both are right-angled triangles sharing a common angle. This gives

$$\frac{AE}{AB} = \frac{AB}{AC}. \quad (2)$$



Combining (1) and (2), we have

$$AF \times AG = AB \times AD = AB^2 = AE \times AC,$$

so that $\frac{AF}{AE} = \frac{AC}{AG}$. Together with

$$\angle FAE = \angle FAB + \angle BAE = \angle DAG + \angle CAD = \angle CAG,$$

we find that $\triangle FAE \sim \triangle CAG$. Hence, $\angle AEF = \angle AGC$.

Next, note that B, C, G, H are concyclic since $\angle CHG = \angle CBG = 90^\circ$. Therefore,

$$\angle BHF = 90^\circ + \angle BHC = 90^\circ + \angle BGC = 90^\circ + \angle AEF = 180^\circ - \angle BEF.$$

This implies B, E, F, H are concyclic. □

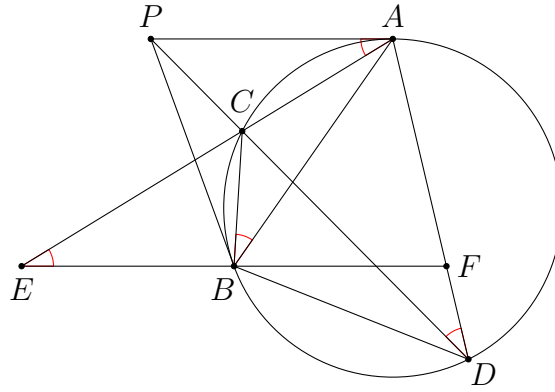
Remarks. The ideas we use to solve this problem are very useful in solving some basic geometry problems. First of all, we can start from the last point defined in the question (the point H), look for its properties, and rephrase the question in a way so that this point can be removed. By working backwards step by step, eventually we can obtain an equivalent question which is simple enough since the number of points is reduced.

Secondly, the way that we prove a pair of similar triangles or a group of concyclic points suggests how we should use this information, and vice versa. For example, if we prove a pair of similar triangles by ‘ratio of 2 sides, included angles’, then we can use this to deduce some pairs of equal angles that cannot be obtained in a trivial way. Very often more observations can be made in this way, and eventually a complete solution can be built up.

Problem 2.6. (CWMO 2005 Problem 2) Given three points P, A, B and a circle such that the lines PA and PB are tangent to the circle at the points A and B , respectively. A line through the point P intersects that circle at two points C and D . Through the point B , draw a line parallel to PA ; let this line intersect the lines AC and AD at the points E and F , respectively. Prove that $BE = BF$.

Analysis. The construction is symmetric in E and F . Therefore, we can first focus on how to find the length BE . To do so, we may first see if there exists a pair of similar triangles involving the length BE . In any case, from the construction of point E , the angle $\angle AEB$ should be important. Therefore, we can use the parallel lines and tangent lines to mark all the angles equal to this in the figure. It is then not hard to observe some pairs of similar triangles. The proof follows in a natural way.

Solution.



From the figure, we observe that $\triangle ABE \sim \triangle ACB$. Indeed, this follows from

$$\angle AEB = \angle EAP = \angle CBA$$

and the common angle $\angle BAE = \angle BAC$. This implies

$$\frac{BE}{CB} = \frac{AB}{AC} = \frac{AE}{AB}.$$

The first equality is more useful since we can now express BE in terms of other lengths not involving E . Indeed, we have $BE = \frac{AB \times CB}{AC}$. Similarly, since directed angles are used, we have the analogous result for BF , namely, $BF = \frac{AB \times DB}{AD}$. To show that $BE = BF$, it remains to show $\frac{CB}{AC} = \frac{DB}{AD}$.

Now, points E and F do not play any roles. The figure is simplified and we simply need to use the standard pairs of similar triangles found in figures involving tangents and secants (limiting case of [proposition 2.2](#)). Indeed, from $\triangle PCB \sim \triangle PBD$, we have

$$\frac{PC}{PB} = \frac{CB}{BD}.$$

Similarly, from $\triangle PCA \sim \triangle PAD$, we have

$$\frac{PC}{PA} = \frac{CA}{AD}.$$

Combining these and $PA = PB$, we obtain $\frac{CB}{BD} = \frac{CA}{AD}$, and hence $\frac{CB}{AC} = \frac{DB}{AD}$. This is the desired relation. The proof is complete. $\square$

Remarks. There are usually two simple ways to spot similar triangles (and concyclic points) in a given figure. The first way is to recall some basic facts like [propositions 2.1](#) and [2.2](#), or any other configurations that one has come across before. The second way is to start from the conclusion to see which pairs of triangles, if similar, are useful.

Let us recall the techniques we have seen in this section.

- finding similar triangles ([problems 2.1, 2.3, 2.4, 2.5, 2.6](#))
- finding concyclic points ([problems 2.2, 2.5](#))
- angle chasing using similar triangles and concyclic points ([problems 2.2, 2.3, 2.4, 2.5](#))
- length chasing using similar triangles ([problems 2.1, 2.4, 2.5, 2.6](#))
- finding corresponding points of similar triangles ([problem 2.3](#))

3 Transformations

Transformations(變換) (in plane geometry) refer to mapping points to new points under certain rules. Sometimes it is easier to deduce results from the new figure than the original one. This in turn provides information of the original figure by using the invariance of such transformations. Basic transformations include rigid motions, reflection and homothety.

3.1 Rigid Motions

In general, a *rigid motion*(剛體運動) (*rigid transformation* or *isometry*(等距同構)) refers to any transformation which preserves distances between every pair of points. Strictly speaking, the orientation should also be preserved. In particular, translation and rotation are two basic kinds of rigid motion.

For *translation*(平移), we need to specify a segment PP' (where $P \neq P'$) so that each point A is mapped to a new point A' for which AA' has the same length and direction as PP' . Clearly, lengths and directions are all preserved under translation. In particular, each line is mapped to a line parallel to it.

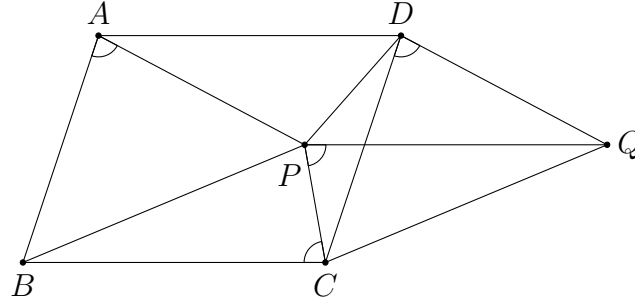
For *rotation*(旋轉), we need to specify a point O as centre and a fixed angle θ . Each point A apart from O is then mapped to A' for which $OA' = OA$ and $\angle AOA' = \theta$. Here the angle is not taken modulo π , or otherwise there are two such points A' , but the angle is still measured in the anticlockwise direction. As a rigid motion, lengths are preserved. But each line is rotated by θ .

Translation and rotation may not be widely used because the configuration does not vary too much after the transformation. Here, we will only provide some standard problems involving these transformations.

Problem 3.1. Let $ABCD$ be a parallelogram and let P be a point inside the parallelogram such that $\angle PAB = \angle PCB$. Show that $\angle PDC = \angle PBC$.

Analysis. Note that as the usual angles are used, $\angle PAB = \angle PCB$ is not the same as A, B, C, P being concyclic. Indeed, from the configuration, $\angle PAB$ and $\angle PCB$ must be opposite angles. However, by applying a translation to point P , we can ‘move’ the angles to new positions. This allows us to show that some points are concyclic. As we shall see, this is one of the simplest way to apply transformation to get more information.

Solution. Consider the translation which maps A to D (and hence B to C). Let Q be the image of P . Then $\triangle ABP \cong \triangle DCQ$ and $BCQP$ is a parallelogram.



Therefore,

$$\angle CDQ = \angle BAP = \angle PCB = \angle CPQ,$$

which implies P, C, Q, D are concyclic. From this, we have

$$\angle PDC = \angle PQC = \angle PBC$$

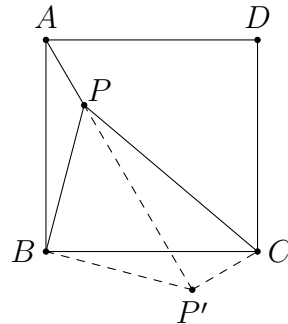
as desired. $\square$

Remarks. Instead of defining Q using a translation, we may also define Q as the point on the different side of the line CD as P such that $\triangle ABP \cong \triangle DCQ$. In other words, the main idea in this solution is to construct a pair of congruent triangles. Sometimes it is also useful to construct a pair of similar triangles.

Problem 3.2. Let P be a point inside a square $ABCD$. Suppose $PA = 1$, $PB = 2$ and $PC = 3$. Find $\angle APB$.

Analysis. This is a classical problem in geometry. The elegant solution involves applying a rotation with centre B . It is hard to think of such a transformation if one has never seen it. But everyone in this area should learn all these classical tricks. Once again, the construction is equivalent to constructing a pair of congruent triangles to ‘move’ the segments.

Solution.



Consider the rotation about B by 90° clockwise. It maps A to C and P to a point P' (note that A, P, P' need not be collinear, though it is the case indeed). This implies $\triangle ABP \cong \triangle CBP'$. In particular, we have $BP' = BP = 2$ and $CP' = AP = 1$.

Note that $\triangle PBP'$ is a right-angled isosceles triangle. we have $PP' = 2\sqrt{2}$ by Pythagoras' theorem. Consider $\triangle PP'C$. As

$$PP'^2 + P'C^2 = (2\sqrt{2})^2 + 1^2 = 9 = PC^2,$$

we find that $\triangle PP'C$ is a right-angled triangle with $\angle PP'C = 90^\circ$. Then

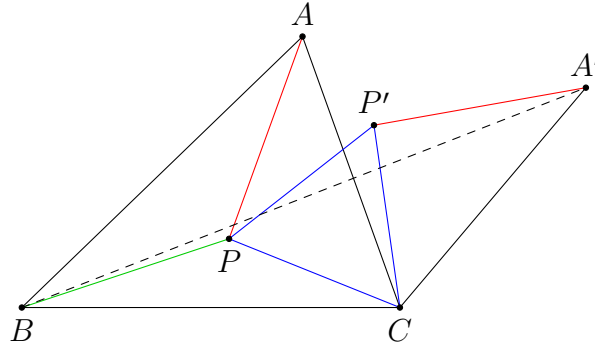
$$\angle APB = \angle CP'B = \angle CP'P + \angle PP'B = 90^\circ + 45^\circ = 135^\circ.$$

□

Problem 3.3. Let $\triangle ABC$ be acute. Find the point P which minimizes the sum of lengths $PA + PB + PC$.

Analysis. Since the point P is a random point, it is difficult to compare the lengths PA, PB, PC directly. A standard trick is to put the segments in such a way that the endpoint of one coincides with the initial point of the other, so that the lengths can be added up easily. One way to 'move' the segments is to construct equilateral triangles, or equivalently, to rotate by 60° .

Solution.

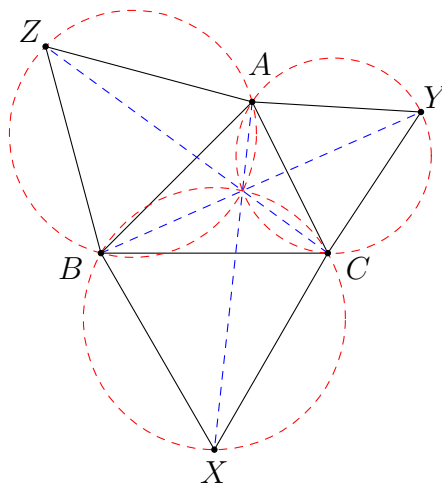


Rotate points A and P by 60° clockwise about C to points A' and P' respectively. Firstly, $\triangle CPP'$ is an equilateral triangle, so that $CP = PP'$. Secondly, $\triangle ACP$ and $\triangle A'CP'$ are congruent, and hence $AP = A'P'$. Note that A' is a fixed point independent of P . Therefore, by the triangle inequality, we have

$$PA + PB + PC = P'A' + PB + PP' \geq A'B.$$

This means the distance is minimized when B, P, P', A' are collinear. This is the case when $\angle BPC = 180^\circ - 60^\circ = 120^\circ$ and $\angle CPA = \angle CP'A' = 180^\circ - 60^\circ = 120^\circ$, i.e. when $\angle APB = \angle BPC = \angle CPA = 120^\circ$. □

Remarks.



Since $\triangle ABC$ is acute, the point P specified in the solution does lie inside the triangle. Indeed, the same proof works for the case when all interior angles of $\triangle ABC$ are at most 120° . If, for example, $\angle BAC > 120^\circ$, then one can show that when B, P, P', A' are collinear, P and P' cannot lie inside segment BA' simultaneously, and so the proof fails. In fact, it can be proved that $PA + PB + PC$ is minimized when $P = A$ in that case.

To locate point P geometrically, we can construct equilateral $\triangle BCX$, $\triangle CAY$, $\triangle ABZ$ outside $\triangle ABC$. Then P is the intersection point of the circumcircles of these equilateral triangles. This is because, for example, the angle on $\widehat{AB}$ not containing Z is $180^\circ - 60^\circ = 120^\circ$. In addition, it can be shown that P is the concurrent point of AX, BY, CZ . This point is called the *1st Fermat point* (第一費馬點) of $\triangle ABC$.

3.2 Reflection

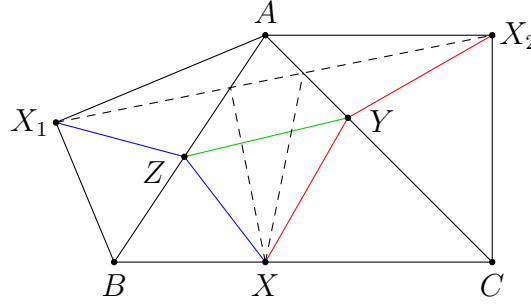
For *reflection* (反射), we need to specify a line ℓ so that each point A is mapped to a new point A' for which ℓ is the perpendicular bisector of AA' . Again, lengths are preserved under reflection. However, the orientation is changed so that it is normally not considered as a rigid motion.

As the orientation is changed, if directed angles are used, we need to add a negative sign after reflection. For example, let A', B', C' be the images of A, B, C under a reflection, then $\angle ABC = -\angle A'B'C'$. Triangles after reflection are oppositely congruent to the original triangles.

Problem 3.4. (Fagnano's problem(法格乃諾問題)) Let $\triangle ABC$ be acute and let X, Y, Z be points on the sides BC, CA, AB respectively. Find the points X, Y, Z which minimize the perimeter of $\triangle XYZ$.

Analysis. This problem is similar to the previous one. This time we find that reflection is more suitable than rotation to align the segments.

Solution.



Reflect X in AB to X_1 , and reflect X in AC to X_2 . Then

$$XY + YZ + ZX = X_2Y + YZ + ZX_1 \geq X_1X_2.$$

When X is fixed, then Y and Z should be the points such that X_1, Z, Y, X_2 are collinear, and the perimeter of $\triangle XYZ$ is X_1X_2 . As $AX_1 = AX = AX_2$ and

$$\angle X_1AX_2 = \angle X_1AX + \angle XAX_2 = 2\angle BAX + 2\angle CAX = 2\angle BAC,$$

we get $X_1X_2 = 2AX \sin \angle BAC$. As $\angle BAC$ is fixed, this is minimized when AX is the altitude of $\triangle ABC$. In that case, from $\angle AXB = \angle AX_1B = 90^\circ$, A, X, B, X_1 are concyclic. Also, since A is the centre of (XX_1X_2) , we have

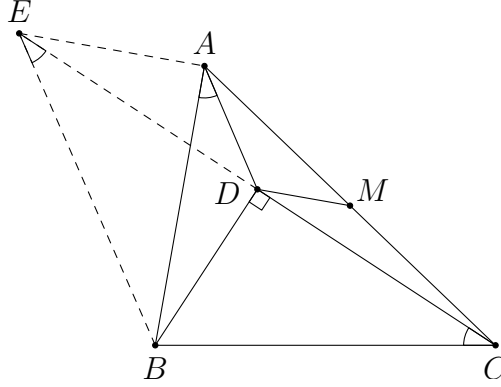
$$\angle XX_1Y = \angle XX_1X_2 = \frac{1}{2}\angle XAX_2 = \angle XAY,$$

implying that X, X_1, A, Y are concyclic. This means A, Y, X, B, X_1 lie on a single circle with diameter AB . Then $\angle AYB = 90^\circ$, and similarly $\angle CZA = 90^\circ$. In other words, X, Y, Z are the feet of altitudes of $\triangle ABC$, i.e. $\triangle XYZ$ is the orthic triangle. $\square$

Problem 3.5. (IMO Prelim 2012 Problem 16) Let D be a point inside $\triangle ABC$ such that $\angle BAD = \angle BCD$ and $\angle BDC = 90^\circ$. If $AB = 5$, $BC = 6$ and M is the midpoint of AC , find the length of DM .

Analysis. Again, the key idea is to use the condition $\angle BAD = \angle BCD$. Usually, we try to rearrange the angles so that we can obtain some concyclic points using the condition. This time we try to reflect C in the line BD .

Solution.



Let E be the reflection of C in the line BD . Since $\angle BDC = 90^\circ$, it is the same as taking the symmetric point of C w.r.t. point D . In other words, D is the midpoint of CE . By the midpoint theorem, we have $EA = 2DM$. It remains to find the length of EA .

Based on our idea, we have $\angle BED = \angle BCD = \angle BAD$, so that A, E, B, D are concyclic. Since $\angle EDB = 90^\circ$, BE is a diameter of the circle. In particular, $\triangle AEB$ is a right-angled triangle with $\angle EAB = 90^\circ$. We also have $AB = 5$ and $BE = BC = 6$. Therefore,

$$EA = \sqrt{6^2 - 5^2} = \sqrt{11},$$

so that $DM = \frac{\sqrt{11}}{2}$. □

More problems using reflection will be discussed later when we go through triangle geometry and isogonal conjugate.

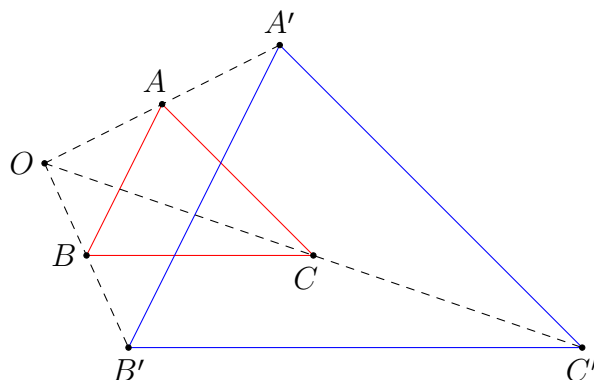
3.3 Homothety

Next, we consider a transformation which does not preserve lengths, but is much more useful in Olympiad. To define a *homothety* (位似) (or *dilation*), we need to specify a homothetic centre O and a nonzero ratio r so that each point A distinct from O is mapped to a new point A' for which O, A, A' are collinear and $\frac{OA'}{OA} = r$ in directed lengths. The image of O is defined to be O .

If the ratio $r > 1$, the homothety can be viewed as a process to enlarge the figure, while for $0 < r < 1$, the figure is shrunk. In both cases, A and its image A' lie on the same side of the centre O on the line OA . When $r < 0$, O lies between A and A' , and the figure is enlarged or shrunk depending on whether $|r| > 1$ or $0 < |r| < 1$. In

the special case $r = -1$, the transformation is the same as reflecting in the point O , or rotating about O by 180° .

After a homothety, all side lengths are multiplied by r , and all (directed) angles are preserved. In particular, if A' and B' are the images of A and B respectively, then we have $\frac{A'B'}{AB} = r$ and $A'B' \parallel AB$. Indeed, the resultant figure is similar to the one before we apply the homothety. For example, if C' is the image of another point C , then $\triangle ABC$ and $\triangle A'B'C'$ are directly similar. This suggests the usefulness of homothety, since it is closely related to similar triangles.



Conversely, if we are given points A, B, C, A', B', C' such that $AB \parallel A'B'$, $BC \parallel B'C'$ and $CA \parallel C'A'$ (hence $\triangle ABC \sim \triangle A'B'C'$), then we can find a homothety mapping A, B, C to A', B', C' respectively (except a special case where homothety is replaced by translation). The centre is the concurrent point of AA', BB', CC' since it must lie on the three lines by definition. For a proof of this useful fact, simply consider a homothety (with centre $AA' \cap BB'$) mapping A, B to A', B' respectively, and show that it also maps C to C' .

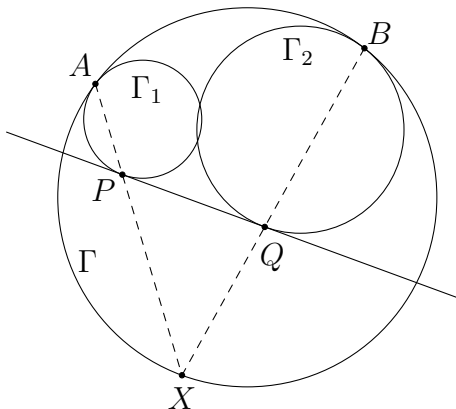
Problem 3.6. Let Γ_1 and Γ_2 be two circles internally tangent to a larger circle Γ at points A and B respectively. A common external tangent to Γ_1 and Γ_2 touches them at points P and Q respectively. Prove that AP and BQ meet on Γ .

Analysis. In case there are two circles tangential to each other, we may consider a homothety with centre being the tangential point that takes one circle to the other. Such a homothety exists because all circles are similar to each other.

Solution. Consider the homothety which sends Γ_1 to Γ (with centre A and ratio being the proportion of the radii of the two circles). Let X be the image of P . By the similarity, the tangent at P to Γ_1 is mapped to the tangent at X to Γ . Also, these tangents are parallel as homothety preserves angles.

Similarly, by considering the homothety which sends Γ_2 to Γ , point Q is mapped to a point Y such that the tangent at Y to Γ is parallel to the tangent at Q to Γ_2 , i.e. the line PQ . Hence, the tangents at X and Y to Γ are parallel to each other. As

PQ is a common external tangent to Γ_1 and Γ_2 , the configuration suggests X and Y should lie on the same side of the line PQ . Therefore, the tangents at X and Y coincide and $X = Y$. This means AP and BQ meet at X on Γ . $\square$



Remarks. In the solution, we have proved that the tangent at the intersection point X is parallel to the common external tangent PQ . If PQ meets Γ at S and T , then X is the midpoint of $\widehat{ST}$.

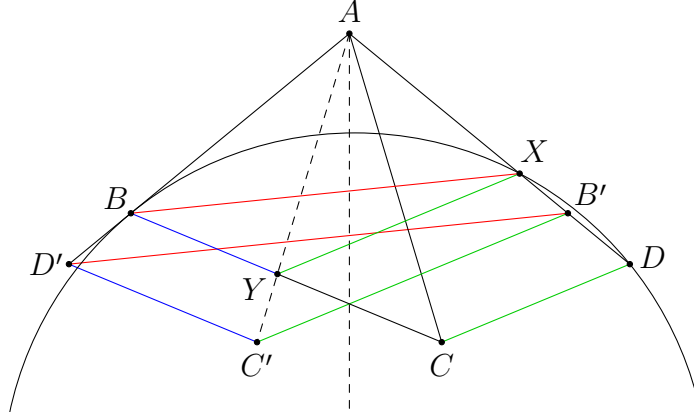
Problem 3.7. Let $ABCD$ be a convex quadrilateral such that $\angle ABC = \angle ADC$. A circle Γ passes through points B, D and is tangent to the line AB at B . Suppose that Γ intersects the line AD again at X . Let Y be a point on the line BC such that $XY \parallel DC$. Prove that $\angle BAY = \angle CAD$.

Analysis. The condition $\angle ABC = \angle ADC$ does not imply A, B, C, D are concyclic. Instead of reflecting D in AC so that the image lies on (ABC) , we shall introduce another useful trick. This is to reflect the quadrilateral in the angle bisector of $\angle BAD$ so that the image of CD is parallel to BC .

Next, we note that BX and BD are antiparallel w.r.t. AB and AD . (This is the limiting case when B and B coincide. The line BB is then the tangent at B , i.e. AB .) Therefore, the reflection of BD in the angle bisector of $\angle BAD$ is parallel to BX , as a property of antiparallelism. This is another reason of applying the reflection mentioned.

After constructing a lot of parallel lines, we may try to consider a homothety which maps some of the points to the others. Note that the conclusion $\angle BAY = \angle CAD$ is equivalent to the image of C under the reflection lies on the line AY . This also suggests the use of such a reflection.

Solution. Consider the reflection in the angle bisector of $\angle BAD$. Let B', C', D' be the images of points B, C, D under this reflection respectively. Clearly, B' lies on the line AD and D' lies on the line AB . We note that there are three groups of parallel lines.



Firstly, as AB is a tangent to (BDX) , we have

$$\begin{aligned}\angle(AX, BX) &= \angle(DX, BX) = \angle(DB, AB) = -\angle(D'B', AB') \\ &= \angle(AB', D'B') = \angle(AX, D'B').\end{aligned}$$

This shows $BX \parallel D'B'$. Secondly,

$$\angle YBA = \angle CBA = \angle ADC = -\angle AD'C' = \angle C'D'A,$$

which yields $BY \parallel D'C'$. Thirdly,

$$\angle ADC = \angle CBA = -\angle C'B'A = \angle AB'C',$$

which implies $B'C' \parallel DC$. Thus, $B'C' \parallel DC \parallel XY$.

Now, consider the homothety with centre A which sends B to D' . Since $BX \parallel D'B'$, the image of X under the homothety is B' . Also, the image of Y , called Y' , satisfies $BY \parallel D'Y'$ and $XY \parallel B'Y'$. The unique point satisfying these properties is exactly C' , so that $Y' = C'$. In particular, A, Y, C' are collinear. Then

$$\angle BAY = \angle BAC' = -\angle B'AC' = -\angle DAC' = \angle CAD.$$

As B, Y, C are collinear, this must imply $\angle BAY = \angle CAD$ in any case. $\square$

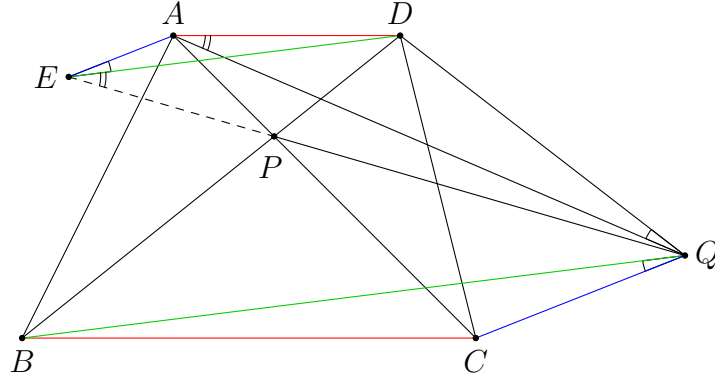
Remarks. The configuration is not unique. Therefore, we need to use directed angles. However, the last assertion $\angle BAY = \angle CAD$ implies $\angle BAY = \angle CAD$ must be dealt with the configuration.

Problem 3.8. (IMO Shortlist 2007 G3/Baltic Way 1998 Problem 13) The diagonals of a trapezoid $ABCD$ intersect at point P . Point Q lies between the parallel lines BC and AD such that $\angle AQD = \angle CQB$, and line CD separates points P and Q . Prove that $\angle BQP = \angle DAQ$.

Analysis. The equal angles given cannot be compared easily using the existing figure. Again, one simple trick is to move the angles to obtain concyclic points.

Therefore, we may consider to add a point E such that $\angle DEA = \angle BQC$. This implies D, A, E, Q are concyclic. There are many points E which satisfy this relation. The most natural one to consider is to take the point E for which $\triangle AED$ and $\triangle CQB$ are directly similar. By simple angle chasing, one can see that it remains to prove that E, P, Q are collinear.

Solution.



Let E be the point such that $\triangle AED$ and $\triangle CQB$ are directly similar. Since one pair of corresponding sides AD and CB are parallel, all corresponding sides of these triangles are parallel (as directly similarity preserves angles).

As mentioned, there exists a homothety which sends $\triangle AED$ to $\triangle CQB$. The centre of this homothety is the concurrent point of the lines joining the corresponding vertices, which means the intersection point P of AC and BD . It follows that E, P, Q are collinear.

Now, from $\angle DEA = \angle BQC = \angle DQA$, we know D, A, E, Q are concyclic. Then $\angle QAD = \angle QED = \angle PQB$ from the concyclic points and parallel lines. $\square$

Remarks. The point E can also be defined via a homothety with centre P . The direction we use in this solution should be easier to think of. The only trick is to put the things in the right places to obtain more results.

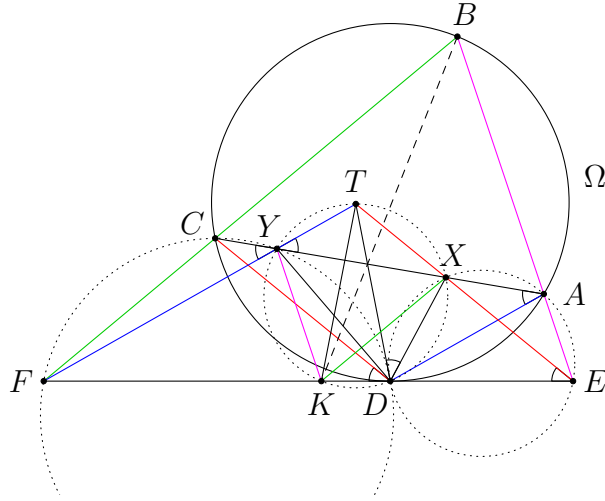
Problem 3.9. (IMO Shortlist 2021 G4) Let $ABCD$ be a quadrilateral inscribed in a circle Ω . Let the tangent to Ω at D intersect the rays BA and BC at points E and F , respectively. A point T is chosen inside the triangle ABC so that $TE \parallel CD$ and $TF \parallel AD$. Let $K \neq D$ be a point on the segment DF such that $TD = TK$. Prove that the lines AC, DT and BK intersect at one point.

Analysis. This is a harder problem, which requires more observations. The first idea is to mark some equal angles in the figure using the concyclic points, the tangent line and the parallel lines. By doing so, one should find it natural to define the intersect points $X = AC \cap ET$ and $Y = AC \cap FT$. The reason is that we can use the equal angles to show two groups of concyclic points, say X, D, E, A .

It is not hard to draw a nice figure. If one is good at recognizing groups of concyclic points (which is an essential technique in solving geometry problems), one should make a correct guess that T, Y, K, D, X are concyclic. Again, this can be proved solely by angle chasing.

Homothety is only used in the final step, which is used to conclude the concurrency by finding a homothety between two groups of points. This idea comes from the fact that there are many parallel lines in the figure (in addition to the two given pairs). In other words, it is often useful to consider homothety when there are some parallel lines.

Solution.



Let AC meet ET and FT at X and Y respectively. Since

$$\angle XAD = \angle CAD = \angle CDF = \angle XED,$$

the points X, D, E, A are concyclic. By symmetry, Y, D, F, C are concyclic. Next, as

$$\angle XDY = 180^\circ - \angle EDX - \angle YDF = 180^\circ - \angle BAC - \angle ACB = \angle CBA$$

and $\angle YTX = \angle ADC$ (since $TY \parallel DA$ and $TX \parallel DC$), we obtain

$$\angle XDY + \angle YTX = 180^\circ.$$

This implies D, X, T, Y are concyclic. To show that K also lies on the circle containing these points, we must use the condition $TD = TK$. There are many ways to chase the angles. For example, as

$$\angle XDT = \angle XYT = \angle CYF = \angle CDF = \angle XED,$$

we have

$$\angle DKT = \angle TDK = 180^\circ - \angle EDX - \angle XDT = 180^\circ - \angle EDX - \angle XED = \angle DXE.$$

This implies K, D, X, T are concyclic. Therefore, T, Y, K, D, X are concyclic. Using this, we can find two more pairs of parallel lines. Indeed, we have

$$\angle KYX = \angle EDX = \angle BAX,$$

which yields $KY \parallel BA$. Similarly, we have $KX \parallel BC$. Finally, note that there is a homothety mapping $\triangle DAC$ to $\triangle TYX$ since their corresponding sides are parallel. In addition, as $AB \parallel YK$ and $CB \parallel XK$, the homothety maps B to K . Therefore, DT, AY, CX, BK are concurrent at the centre of this homothety. Equivalently, AC, DT, BK are concurrent. $\square$

The following summarizes the ideas we have used along with some basic transformations in this section.

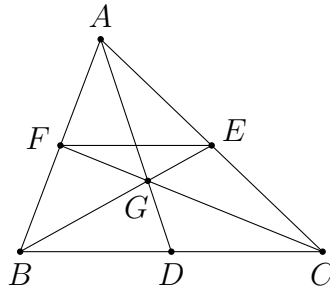
- constructing congruent triangles ([problems 3.1, 3.2, 3.3, 3.5](#))
- generating concyclic points ([problems 3.1, 3.5, 3.8](#))
- moving the segments to sum up the lengths ([problems 3.3, 3.4](#))
- using parallel lines and homothety ([problems 3.6, 3.7, 3.8, 3.9](#))

4 Triangle Geometry

Triangle geometry is one of the most popular topics in plane geometry. To solve problems in this area, we must be familiar with the properties of the triangle centres. The most fundamental triangle centres include the centroid, the circumcentre, the orthocentre, the incentre, excentres and the nine-point centre. We are assumed to have learned these things before. Therefore, we will only revise the more important facts in this section, and add some ingredients involving directed angles.

4.1 Centroid and Midpoints

In $\triangle ABC$, let D, E, F be the midpoints of the sides BC, CA, AB respectively. The *centroid* (重心) G of $\triangle ABC$ is defined to be the concurrent point of the three medians AD, BE, CF . It always lies inside $\triangle ABC$. The only important fact about the centroid is that it cuts a median in the ratio $2 : 1$, i.e. $AG : GD = 2 : 1$. Hence, the homothety with centre G and ratio -2 takes $\triangle DEF$ to $\triangle ABC$. Most properties of the medial $\triangle DEF$ come from this homothety. For example, G is also the centroid of $\triangle DEF$.



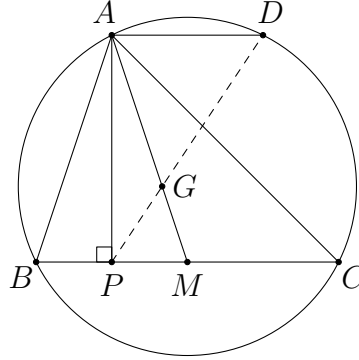
Since the centroid is closely related to midpoints of sides, we usually add the midpoints of the sides in the figure when working on problems involving the centroid, even if they are not defined in the question. Then we can make use of the properties of the midpoints and the above mentioned ratio to solve the problems.

Problem 4.1. Let G be the centroid of $\triangle ABC$. Let P be the foot of altitude from A . The line through A parallel to BC meets the circumcircle of $\triangle ABC$ again at D . Prove that P, G, D are collinear.

Analysis. At first glance, it looks like the three points are unrelated. As suggested, the first idea is to draw some midpoints and medians of $\triangle ABC$. Since the construction is symmetric in B and C , we may define the midpoint M of BC . This helps us relate the point G with other points in the figure. In particular, as $AD \parallel BC$ and P, G, D should be collinear, $\triangle GAD$ and $\triangle GMP$ should be similar to each other. As

before, finding pairs of similar triangles is always a strategy, which means we should be on the right track. To prove that they are similar, we have to make use of the important property $AG : GM = 2 : 1$.

Solution.



As $ABCD$ is cyclic and $AD \parallel BC$, we know that $ABCD$ is an isosceles trapezoid. Thus, the midpoint M of BC lies on the perpendicular bisector of AD . This implies $AD = 2MP$. Hence,

$$\frac{AD}{MP} = 2 = \frac{AG}{MG}.$$

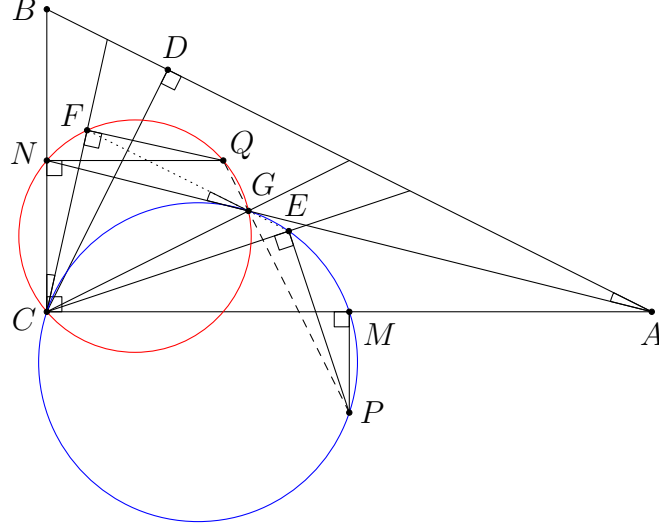
Together with $\angle GAD = \angle GMP$, we deduce $\triangle GAD \sim \triangle GMP$. It follows that $\angle DGA = \angle PGM$, and hence P, G, D are collinear. $\square$

Problem 4.2. (Dutch TST 2021 Test 3 Problem 2) Let ABC be a right triangle with $\angle C = 90^\circ$ and let D be the foot of the altitude from C . Let E be the centroid of triangle ACD and let F be the centroid of triangle BCD . The point P satisfies $\angle CEP = 90^\circ$ and $CP = AP$, while point Q satisfies $\angle CFQ = 90^\circ$ and $CQ = BQ$. Prove that PQ passes through the centroid of triangle ABC .

Analysis. Again, the first step is to define some midpoints and draw the corresponding medians. The midpoints M and N of AC and BC should be useful. For example, as NQ is the perpendicular bisector of BC , we deduce $\angle CNQ = 90^\circ$. The strange condition $\angle CFQ = 90^\circ$ can then be used to deduce a group of concyclic points. The three medians from C of the three triangles should also be considered as they share the same vertex.

There are two more important observations in this figure. The first one is due to the standard configuration involving a right-angled $\triangle ABC$ and the altitude CD . All three triangles in this configuration are similar to each other. The second one is that the centroid of $\triangle ABC$ should indeed lie on both (CFQ) and (CEP) . A good drawing should provide a good hint to this. If this can be proved, the result would be obvious.

Solution.



Let G be the centroid of $\triangle ABC$. Let M and N be the midpoints of AC and BC respectively. Since $\angle CFQ = \angle CNQ = 90^\circ$, the points C, Q, F, N are concyclic. It is well-known (and obvious) that $\triangle BCD \sim \triangle BAC$. As F and G are the centroids of these triangles, we have $\triangle BCF \sim \triangle BAG$. In particular, we have

$$\angle FCB = \angle GAB.$$

Also, consider the three medians from C of $\triangle ACD$, $\triangle BCD$ and $\triangle ABC$. As the centroids of these triangles cut these medians in the same ratio $2:1$, E, F, G must be collinear and $EF \parallel AB$ (for example, consider the homothety with centre C and ratio $\frac{2}{3}$). This implies

$$\angle FGN = \angle BAN = \angle FCN.$$

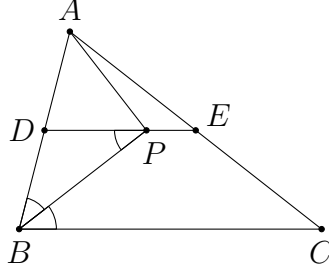
It follows that G lies on $(CQFN)$, and hence $\angle QGC = 90^\circ$. Similarly, we have $\angle PGC = 90^\circ$. This clearly implies P, G, Q are collinear. $\square$

As midpoints are closely related to the centroid. Let us go through some problems about midpoints.

Problem 4.3. ([Iran lemma](#)(伊朗引理)) Let D and E be the midpoints of the sides AB and AC of $\triangle ABC$ respectively. Suppose the internal angle bisector of $\angle ABC$ meets the line DE at P . Prove that $AP \perp BP$.

Analysis. The only crucial fact is that $DE \parallel BC$ by the midpoint theorem. By a bit of angle chasing, it is not hard to deduce $\angle APB = 90^\circ$. Note that P could possibly lie outside $\triangle ABC$.

Solution.



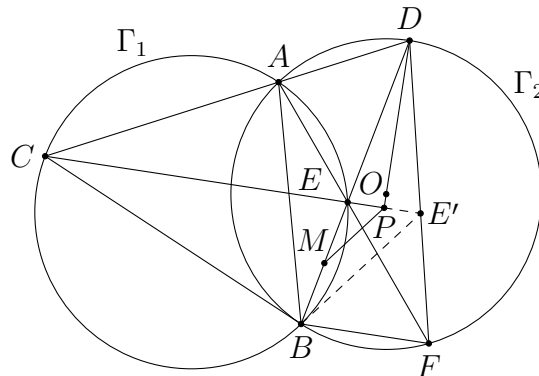
By the midpoint theorem, $DE \parallel BC$. Hence, $\angle DPB = \angle CBP = \angle PBA$, so that $\triangle DBP$ is isosceles with $DB = DP$. As D is the midpoint of AB , we also have $DA = DB = DP$. Therefore, $\triangle APB$ is right-angled with $AP \perp BP$. $\square$

Remarks. We will use this fact to solve a number of problems afterwards. There is another part of the Iran lemma given in [problem 4.17](#).

Problem 4.4. Circles Γ_1 and Γ_2 intersect at two distinct points A and B . The tangent to Γ_2 at B meets Γ_1 again at C . Extend CA and meet Γ_2 again at D . Let E be the intersection of the segment BD and Γ_1 , and let F be the second intersection of the line AE and Γ_2 . Denote by O the centre of Γ_2 . Let P be the intersection of DO and CE , and let M be the midpoint of BE . Prove that $MP = \frac{1}{2}EF$.

Analysis. The assertion suggests the use of the midpoint theorem. Since M is the midpoint of BE , we may try to reflect E in P for consideration. With some angle chasing which relates the tangent line and the concyclic points, it is not hard to see that $\triangle DBF$ is an isosceles triangle. Therefore, one way to prove the result is to use the reflection symmetry in this isosceles triangle.

Solution.



By the properties of tangent lines and concyclic points, we have

$$\angle DFB = \angle DBC = \angle FAD = \angle FBD.$$

So $\triangle DBF$ is isosceles with $DB = DF$. In addition, we have

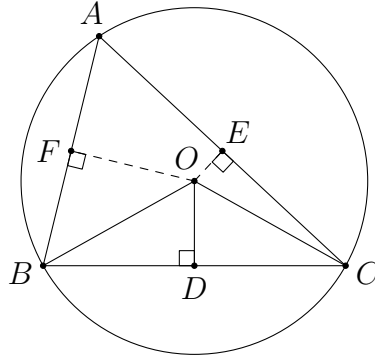
$$\angle FBD = \angle DFB = \angle CAB = \angle CEB.$$

This implies $CE \parallel BF$. As $\triangle DBF$ is isosceles and $OB = OF$, DO is the perpendicular bisector of BF . Therefore, we have $DO \perp CE$. Thus, reflecting E in P is the same as reflecting E in the line DO . The reflection image E' must lie on DF . By the midpoint theorem and by symmetry, we have $MP = \frac{1}{2}BE' = \frac{1}{2}EF$. $\square$

Apart from the midpoint theorem, problems involving midpoints can often be solved using parallelograms. This trick will be discussed in subsection 6.1.

4.2 Circumcircle and Circumcentre

The *circumcircle*(外接圓) of $\triangle ABC$ is the unique circle that passes through the three vertices A, B, C . Its centre, the *circumcentre*(外心), is usually denoted by O , which is the concurrent point of the perpendicular bisectors of the three sides. The *circumradius*(外接圓半徑) $OA = OB = OC$ is denoted by R . Point O lies inside $\triangle ABC$ iff $\triangle ABC$ is acute, lies outside iff $\triangle ABC$ is obtuse, and lies on the side BC iff $\angle BAC = 90^\circ$. In the last case, O must be the midpoint of BC .



Most properties of the circumcircle and the circumcentre come directly from the properties of circles. In particular, we have the *central angle theorem*(圓心角定理) $\angle BOC = 2\angle BAC$, which is true in any case (even for obtuse $\triangle ABC$) as directed angles are used. Note that we cannot write $\frac{1}{2}\angle BOC = \angle BAC$. One way is to let D be the midpoint of BC and write $\angle BAC = \angle BOD = \angle DOC$. Alternatively, we can rewrite the central angle theorem as

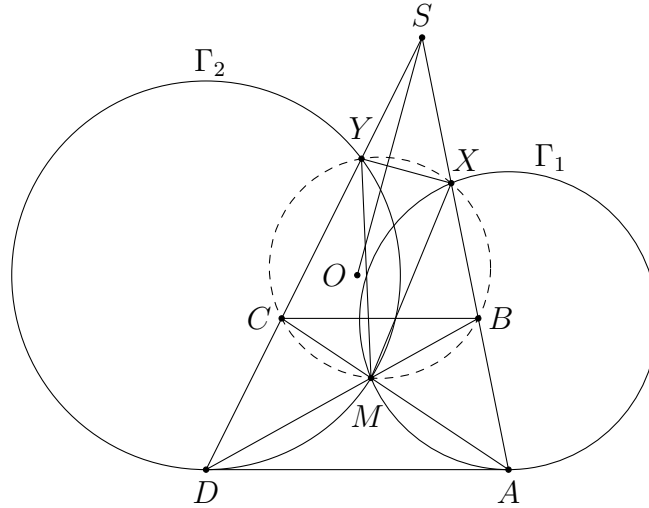
$$\angle OCB + \angle BAC = \angle CBO + \angle BAC = \frac{\pi}{2}.$$

The advantage of this relation is that we do not need to define the midpoint D .

Problem 4.5. (Argentina TST 2011 Problem 3) Let $ABCD$ be a trapezoid with bases $BC \parallel AD$, where $AD > BC$, and non-parallel legs AB and CD . Let M be the intersection of AC and BD . Let Γ_1 be a circle that passes through M and is tangent to AD at point A ; let Γ_2 be a circle that passes through M and is tangent to AD at point D . Let S be the intersection of the lines AB and CD , X the intersection of Γ_1 with the line AS , Y the intersection of Γ_2 with the line DS , and O the circumcentre of triangle ASD . Show that $SO \perp XY$.

Analysis. With some angle chasing, it is equivalent to showing B, C, Y, X are concyclic. The rest is again just simple angle chasing. We use this simple problem to illustrate the use of the directed angle version of the central angle theorem.

Solution.



Firstly, from the tangent and parallel lines, we have

$$\angle MXB = \angle MXA = \angle MAD = \angle(AC, AD) = \angle(AC, BC) = \angle MCB.$$

This shows C, M, B, X are concyclic. By symmetry, C, M, B, Y are concyclic. Thus, C, M, B, X, Y are concyclic.

By the central angle theorem, since O is the circumcentre of $\triangle ASD$, we have

$$\angle DSO + \angle SAD = \frac{\pi}{2}.$$

It follows that

$$\angle(XY, SO) = \angle XYS + \angle DSO = \angle XBC + \left(\frac{\pi}{2} - \angle SAD\right) = \frac{\pi}{2}.$$

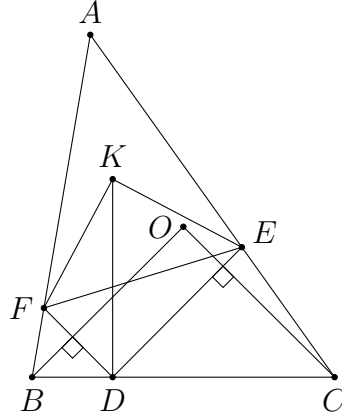
This means $SO \perp XY$. □

Problem 4.6. (EGMO 2012 Problem 1) Let ABC be a triangle with circumcentre O . The points D, E and F lie in the interiors of the sides BC, CA and AB respectively, such that DE is perpendicular to CO and DF is perpendicular to BO . Let K be the circumcentre of triangle AFE . Prove that the lines DK and BC are perpendicular.

Analysis. By assuming $DK \perp BC$, each of $\angle EDK, \angle KDF, \angle KEF, \angle EFK$ is equal to $\angle BAC$. So D, E, K, F should be concyclic. Therefore, proving this group of concyclic points should be the key idea. As D and K are not closely related, the best is to prove $\angle FKE = \angle FDE$.

In this solution, we use directed angles. In that case, the condition that D, E, F lie in the ‘interiors’ of the sides BC, CA, AB is redundant.

Solution.



Firstly, we have

$$\angle CDE = \frac{\pi}{2} - \angle OCB = \angle BAC$$

by the central angle theorem. By symmetry, we also have $\angle FDB = \angle BAC$. Therefore, we have

$$\angle FDE = \angle FDB + \angle CDE = \angle BAC + \angle BAC = 2\angle FAE = \angle FKE.$$

This implies F, D, E, K are concyclic. Finally, we check that

$$\angle CDK = \angle CDE + \angle EDK = \angle BAC + \angle EFK = \angle FAE + \angle EFK = \frac{\pi}{2}.$$

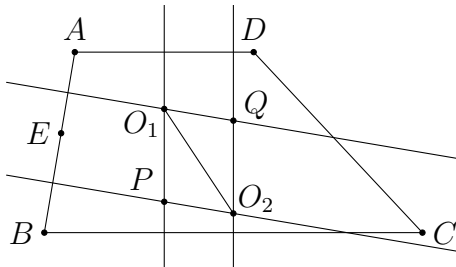
The last equality again follows from the central angle theorem. So we are done. $\square$

Problem 4.7. (CWMO 2002 Problem 5) Suppose a trapezoid $ABCD$ with $AD \parallel BC$ is given. Let E be a movable point on the side AB , and let O_1, O_2 be the circumcentres of $\triangle AED$ and $\triangle BEC$ respectively. Prove that the length of O_1O_2 is fixed.

Analysis. To locate the circumcentres, we choose to draw the perpendicular bisectors of AD, AE and BC, BE since there is little information about the perpendicular

bisectors of DE, CE . We note that these perpendicular bisectors always lie in the same directions when E moves along AB . Then it suffices to prove that the projections of O_1O_2 onto AD and AB have fixed lengths, which is clearly true.

[Solution.](#)



Let the perpendicular bisectors of AD and BE meet at P , while the perpendicular bisectors of BC and AE meet at Q . Then O_1P and QO_2 are parallel as both are perpendicular to AD (and BC). Also, O_1Q and PO_2 are parallel as both are perpendicular to AB .

Now, the projection of O_1Q onto AD has fixed length d , since this is the distance between the perpendicular bisectors of AD and BC . Also, the angle between O_1Q and AD is fixed. Hence,

$$O_1Q = \frac{d}{|\cos \angle (O_1Q, AD)|}$$

is fixed. Similarly, the projection of O_1P onto AB always has length $\frac{1}{2}AB$. Then

$$O_1P = \frac{AB}{2|\cos \angle (O_1P, AB)|}$$

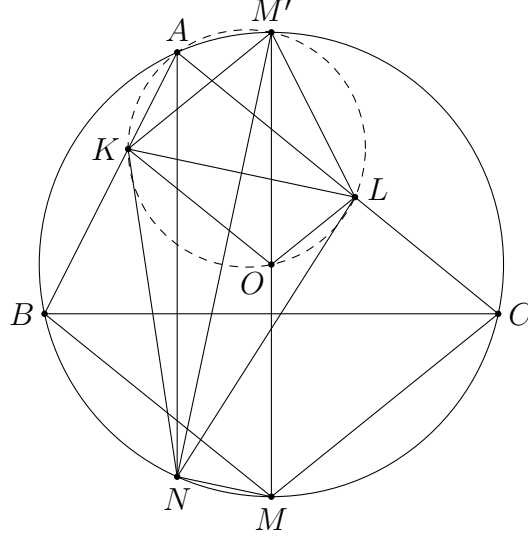
is also fixed. In other words, O_1PO_2Q is a parallelogram of fixed size. Its diagonal O_1O_2 has a fixed length. $\square$

Remarks. When using directed angles with the cosine function, we need to take its absolute value.

Problem 4.8. (Iran MO 2nd Round 2013 Problem 3) Let M be the midpoint of (the smaller) arc BC in circumcircle of triangle ABC . Suppose that the altitude drawn from A intersects the circle at N . Draw two lines through circumcentre O of $\triangle ABC$ parallel to MB and MC , which intersect AB and AC at K and L , respectively. Prove that $NK = NL$.

Analysis. Using the conditions, it is not hard to observe that A, K, O, L are concyclic and $KL \parallel NM$. In particular, in view of $KL \parallel NM$, the perpendicular bisector of KL should intersect (ABC) at N (according to the result we need) and another point M' for which MM' is a diameter. This additional point M' is important because it helps link up the points N and M in some ways. In addition, by drawing an accurate figure, one may notice that M' also lies on $(AKOL)$.

Solution.



Note that A, K, O, L are concyclic since

$$\angle AKO + \angle ALO = \angle ABM + \angle ACM = 180^\circ.$$

Next, we have

$$\begin{aligned} \angle(KL, NM) &= \angle LKA + \angle KAN + \angle ANM \\ &= \angle LOA + \left(\frac{\pi}{2} - \angle CBA\right) + \angle ACM \\ &= \angle(LO, OA) + \angle OAC + \angle ACM \\ &= \angle(LO, OA) + \angle(OA, AC) + \angle(AC, CM) \\ &= 0. \end{aligned}$$

This means $KL \parallel NM$. Let M' be the point such that MM' is a diameter of (ABC) . Then M, O, M' are collinear and $\angle MNM' = 90^\circ$. From

$$\angle M'OL = \angle M'MC = \angle M'AC,$$

we know that point M' also lies on $(AKOL)$. Also, we have

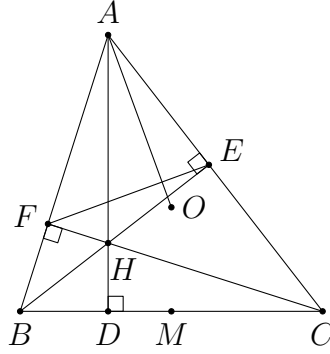
$$\angle M'KL = \angle M'OL = \angle M'MC = \angle M'MB = \angle M'OK = \angle M'LK.$$

This shows M' lies on the perpendicular bisector of KL . As $M'N \perp KL$, the same holds for N , and hence $NK = NL$. $\square$

4.3 Orthocentre and Nine-point Circle

Let D, E, F be the feet of altitudes from A, B, C of $\triangle ABC$ respectively. The *orthocentre* (垂心) H of $\triangle ABC$ is defined to be the concurrent point of the three altitudes

AD, BE, CF . Point H lies inside $\triangle ABC$ iff $\triangle ABC$ is acute, coincide with A iff $\angle BAC = 90^\circ$, and lies outside iff $\triangle ABC$ is obtuse. In the last case, if $\angle BAC$ is the obtuse angle, then H and A lie on the same side of the line BC . In any case, each of A, B, C, H is the orthocentre of the (non-degenerate) triangle formed by the other three.



Since there are many perpendicular lines, there are many groups of concyclic points in the figure. For example, B, C, E, F are concyclic, lying on a circle with centre M , the midpoint of the side BC . In particular, $MB = MF = ME = MC$. Apart from that, A, F, H, E are also concyclic. The centre of the corresponding circle is the midpoint of AH .

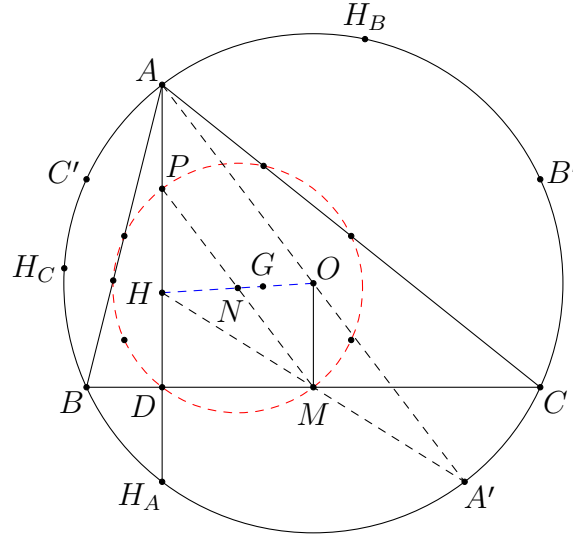
Let O be the circumcentre of $\triangle ABC$. Then $\angle BAH = \angle OAC$. In other words, AH and AO are symmetric in the internal angle bisector of $\angle BAC$. In particular, we have $AO \perp EF$ (see [proposition 8.1\(a\)](#)).

There are a few important properties relating the orthocentre and the circumcircle as given below.

- Firstly, one can show that $\angle BHC = -\angle BAC$. Therefore, if we let H_A be the reflection of H in BC , then $\angle BH_A C = \angle BAC$, so that H_A lies on the circumcircle of $\triangle ABC$.
- Secondly, the segments AH and OM are parallel (both are perpendicular to BC) and $AH : OM = 2 : 1$. This yields two other important facts as follows.
 - The centroid G lies on the segment HO with ratio $HG : GO = 2 : 1$ (indeed $\triangle AGH$ and $\triangle MGO$ are directly similar).
 - The symmetric point of H about M coincides with the symmetric point of A about O . We shall call this point A' . Note that A' lies on the circumcircle of $\triangle ABC$ as $OA = OA'$.

Now, we can define H_B, H_C, B', C' analogously. Then $A, B, C, H_A, H_B, H_C, A', B', C'$ all lie on the circumcircle of $\triangle ABC$. If we consider the homothety with centre H and ratio $\frac{1}{2}$, points H_A, H_B, H_C and A', B', C' are mapped to the feet of altitudes

and the midpoints of the sides respectively. Also, points A, B, C are mapped to the midpoints of AH, BH, CH respectively. These nine points are hence concyclic, lying on a circle with centre N , the midpoint of HO , and radius $\frac{R}{2}$. This is called the *nine-point circle* (九點圓/費爾巴哈圓) of $\triangle ABC$ and N is called the *nine-point centre* (九點圓圓心).



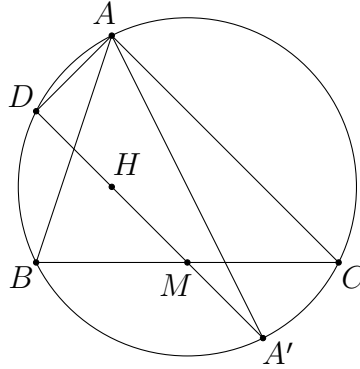
Many facts are simple consequences of this. For example, H, N, G, O are collinear in that order with ratio $HN : NG : GO = 3 : 1 : 2$. The line containing these points is called the *Euler line* (歐拉線) of $\triangle ABC$ (for non-equilateral $\triangle ABC$). Let P be the midpoint of AH . Then $AP = PH = OM$, so that $APMO$ and $PHMO$ are parallelograms, and N is the midpoint of PM . It follows that PM is a diameter of the nine-point circle.

Next, we go through a few problems related to the orthocentre and the nine-point circle. Recalling the well-known facts of these things is always the key for us to spot a solution.

Problem 4.9. An acute $\triangle ABC$ with orthocentre H is given. Let M be the midpoint of the side BC . Suppose the ray MH meets the circumcircle of $\triangle ABC$ at D . Prove that $\angle ADH = 90^\circ$.

Analysis. Let MH meet (ABC) at D and A' . As mentioned in the introductory part, we know that M is the midpoint of HA' , and AA' is a diameter of the circumcircle. This gives us a simple proof to this problem.

Solution. Let A' be the second intersection of MH and (ABC) . It is well-known that AA' is a diameter of (ABC) . Therefore, we have $\angle ADA' = 90^\circ$ as desired. $\square$



Remarks. This is also a well-known fact about the orthocentre, but sometimes a short proof is needed (to be safe, one may also need to explain why AA' is a diameter). We shall use this fact to do some other problems later. Apart from the result given in this problem, we can also obtain some related results. For example, note that the circle with diameter AH intersects (ABC) at points A and D since $\angle ADH = 90^\circ$. In particular, the former circle is simply the circumcircle of $AH_C H H_B$, where H_C and H_B are the feet of altitudes from C and B respectively.

Problem 4.10. (CHKMO 2013 Problem 3) Given $\triangle ABC$ with $CA > BC > AB$, let O and H be the circumcentre and orthocentre of $\triangle ABC$ respectively. Denote by D and E the midpoints of arcs $\widehat{AB}$ and $\widehat{AC}$ of the circumcircle of $\triangle ABC$ not containing the opposite vertices. Let D' be the reflection of D in side AB and E' be the reflection of E in side AC . Prove that O, H, D', E' are concyclic if and only if A, D', E' are collinear.

Analysis. The two statements are rather irrelevant. Indeed, one can easily find that the condition A, D', E' being collinear is the same as $\angle A = 60^\circ$. Thus, it remains to show that this is equivalent to O, H, D', E' being concyclic. This is the main part of the problem.

The construction in this problem involves reflection, the circumcircle and the orthocentre. This reminds us about the property of the orthocentre as discussed. Indeed, this allows us to conclude A, B, H, D' are concyclic, lying on a circle with the same radius as (ABC) . The remaining task is just angle chasing.

We shall use directed angles, so that the condition $CA > BC > AB$ is not necessary.

Solution. In the following, we denote $\alpha = \angle(AB, \ell_a)$, $\beta = \angle(BC, \ell_b)$, $\gamma = \angle(CA, \ell_c)$, where ℓ_a, ℓ_b, ℓ_c are the internal angle bisectors of $\angle A, \angle B, \angle C$ respectively. (One may regard α as $\frac{1}{2} \angle BAC$, but the latter is not well-defined modulo π .) Note that $\alpha + \beta + \gamma = \frac{\pi}{2}$.

Thus, by symmetry,

$$\begin{aligned}
& O, H, D', E' \text{ are concyclic} \\
\Leftrightarrow & \angle OD'H = \angle OE'H \\
\Leftrightarrow & -\angle ACD - \angle CBA = -\angle ABE - \angle BCA \\
\Leftrightarrow & -\gamma - 2\beta = \beta + 2\gamma \\
\Leftrightarrow & 3(\beta + \gamma) = 0 \\
\Leftrightarrow & \frac{\pi}{2} - 3\alpha = 0 \\
\Leftrightarrow & \alpha = \frac{\pi}{6}.
\end{aligned}$$

Therefore, the two statements are equivalent. $\square$

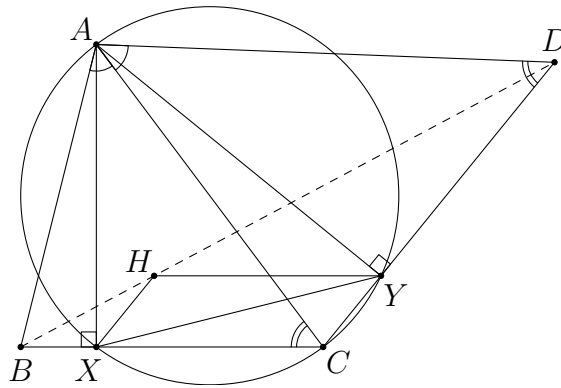
Remarks. Having the condition $CA > BC > AB$, one can stick to a unique configuration and use the ordinary angles. This should make things much simpler. The use of directed angles here is a bit tricky. Firstly, we have to be careful when handling half an angle. Writing something like $\frac{1}{2}\angle BAC$ should be avoided. Secondly, after we have deduced $\angle OD'H = -\gamma - 2\beta$, the corresponding formula for $\angle OE'H$ is not $-\beta - 2\gamma$ but $\beta + 2\gamma$! This is because the symmetry consists of a reflection of the two sides.

Problem 4.11. (Iran MO 2nd Round 2015 Problem 4) In quadrilateral $ABCD$, AC is the bisector of $\angle A$ and $\angle ADC = \angle ACB$. X and Y are the feet of perpendiculars from A to BC and CD , respectively. Prove that orthocentre of triangle AXY is on BD .

Analysis. The similarity $\triangle ABC \sim \triangle ACD$ can be seen immediately. But apart from this, it is important to notice that X and Y are corresponding points in this similarity.

Another important observation used in the solution is that $HXC Y$ is a parallelogram, where H is the orthocentre of $\triangle AXY$. This is also an immediate fact if one is familiar with the properties of orthocentre.

Solution.



Since $\angle AXC = \angle AYC = 90^\circ$, points A, X, C, Y are concyclic. In particular, AC is a diameter of the corresponding circle. Let H be the orthocentre of $\triangle AXY$. Then $HXC Y$ is a parallelogram (recall that the midpoint of XY is the midpoint of HC).

From the angle conditions, we have $\triangle ABC \sim \triangle ACD$. Points X and Y are the corresponding points in this pair of similar triangles. Therefore, we have $\frac{BX}{XC} = \frac{CY}{YD}$.

Since $CY = XH$ and $XC = HY$, we have $\frac{BX}{XH} = \frac{HY}{YD}$.

Together with $\angle BXH = \angle BCD = \angle HYD$, we obtain $\triangle BXH \sim \triangle HYD$. Hence,

$$\angle BHD = \angle BHX + \angle XHY + \angle YHD = \angle BHX + \angle HXB + \angle XBH = 180^\circ,$$

which means B, H, D are collinear. $\square$

Problem 4.12. (IMO Shortlist 1995 G2) Let A, B and C be non-collinear points. Prove that there is a unique point X in the plane of ABC such that

$$XA^2 + XB^2 + AB^2 = XB^2 + XC^2 + BC^2 = XC^2 + XA^2 + CA^2.$$

Analysis. We first consider an equality, say, $XA^2 + XB^2 + AB^2 = XB^2 + XC^2 + BC^2$. It can be simplified to $XA^2 + AB^2 = XC^2 + BC^2$, i.e. $XA^2 - XC^2 = BC^2 - BA^2$. One may recall the perpendicular condition given in [problem 1.4](#). However, there is a slight difference. We would like to have $XC^2 - XA^2 = BC^2 - BA^2$ instead. One way is to consider the reflection of X in the perpendicular bisector of AC . Then we can see that the locus of X satisfying this equation is a straight line.

Solution. The relation $XA^2 + XB^2 + AB^2 = XB^2 + XC^2 + BC^2$ is equivalent to $XA^2 - XC^2 = BC^2 - BA^2$. Let X_1 be the reflection of X in the perpendicular bisector of AC . Then this condition is the same as $X_1C^2 - X_1A^2 = BC^2 - BA^2$. By [problem 1.4](#), it means exactly $X_1B \perp AC$, i.e. X_1 lies on the altitude from B . Therefore, this holds iff X lies on ℓ_1 , the reflection of the altitude from B in the perpendicular bisector of AC .

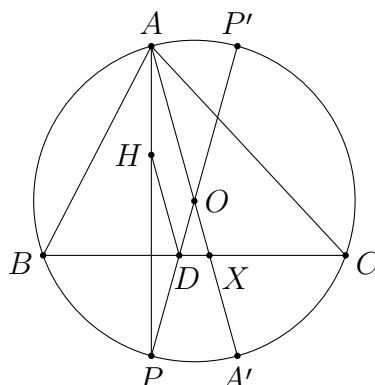
Similarly, $XB^2 + XC^2 + BC^2 = XC^2 + XA^2 + CA^2$ holds iff X lies on ℓ_2 , the reflection of the altitude from C in the perpendicular bisector of AB . Therefore, all the conditions are satisfied iff X lies on ℓ_1 and ℓ_2 . As ℓ_1 is perpendicular to AC and ℓ_2 is perpendicular to AB , and A, B, C are non-collinear, ℓ_1 and ℓ_2 cannot be parallel, and so there exists a unique point on both ℓ_1 and ℓ_2 , as desired. $\square$

Remarks. It is not hard to see that the point X satisfying the condition is the reflection of the orthocentre H in the circumcentre O .

Problem 4.13. (IMO Shortlist 2000 G3) Let O be the circumcentre and H the orthocentre of an acute triangle ABC . Show that there exist points D, E , and F on sides BC, CA , and AB respectively such that $OD + DH = OE + EH = OF + FH$ and the lines AD, BE , and CF are concurrent.

Analysis. It is difficult to compute the sum $OD + DH$ in general. The only point for which this is easy to evaluate is the point when $OD + DH$ is minimized. In that case, the sum is found by reflecting H in the line BC . This should be the case as the fact that this reflection point lies on the circumcircle of $\triangle ABC$ is the most useful property relating the orthocentre and the circumcircle. Using this fact, we can easily obtain the first relation. The only task is to prove that AD, BE, CF are concurrent.

Solution.



Let P be the reflection of H in BC . Then P lies on the circumcircle of $\triangle ABC$. Take D to be the point where PO meets BC . This gives $OD + DH = OD + DP = R$, the circumradius of $\triangle ABC$. Therefore, if we take the corresponding points E and F on the sides CA and AB , the relation $OD + DH = OE + EH = OF + FH$ is satisfied. Next, we shall prove that AD, BE, CF are concurrent as well.

Note that the lines AO and PO are symmetric in the line BC (as $AP \perp BC$). If we let X be the intersection of AO and BC , then D and X are symmetric in the midpoint of BC (one may also prove this by $\angle ODX = \angle OXD$). Define Y and Z to be the intersection of BO and CA , CO and AB respectively. For the same reason, Y and E are symmetric in the midpoint of CA , Z and F are symmetric in the midpoint of AB . Obviously, AX, BY, CZ are concurrent at O . By Ceva's theorem,

$$\frac{BX}{XC} \times \frac{CY}{YA} \times \frac{AZ}{ZB} = 1.$$

Due to the symmetry of the points, this yields

$$\frac{CD}{DB} \times \frac{AE}{EC} \times \frac{BF}{FA} = 1.$$

(For example, $BX = CD$ and $XC = DB$.) Again by Ceva's theorem, AD, BE, CF are concurrent. $\square$

Remarks. Given a point Q not lying on the sidelines BC, CA, AB , let X, Y, Z be the intersection points of AQ and BC , BQ and CA , CQ and AB respectively. Let X', Y', Z' be the symmetric points of X, Y, Z about the midpoints of BC, CA, AB respectively. Then the concurrent point of AX', BY', CZ' is called the *isotomic*

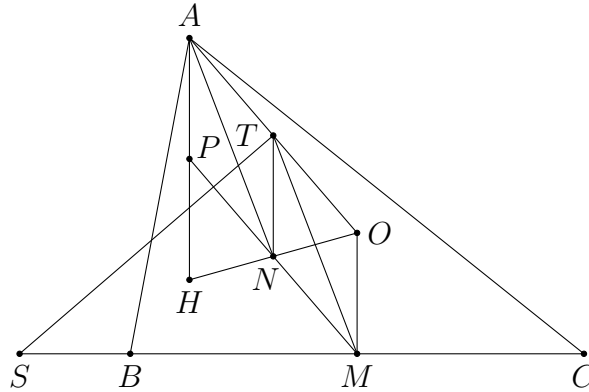
conjugate(等截共軛) of Q . Thus, the concurrent point in this problem is the isotomic conjugate of O .

Problem 4.14. (Belarus TST 2019 Test 2 Problem 2) Let O be the circumcentre and H be the orthocentre of an acute-angled triangle ABC . Point T is the midpoint of the segment AO . The perpendicular bisector AO intersects the line BC at point S . Prove that the circumcircle of the triangle AST bisects the segment OH .

Analysis. Note that we need to prove the nine-point centre N of $\triangle ABC$ lies on (AST) . Therefore, recalling the properties of N should be useful in solving this problem.

First of all, as there is a midpoint T of AO , adding the midpoint P of AH is helpful as this generates a lot of parallel lines and parallelograms. The midpoint M of BC should also be added because N is also the midpoint of PM . This further generates more parallelograms, including $ANMT$. This information is useful, because we are going to show $\angle ANS = 90^\circ$, which means we need to know the direction of AN .

Solution.



Let N be the nine-point centre of $\triangle ABC$. Let P and M be the midpoints of AH and BC respectively. Recall that $APMO$ is a parallelogram. Since T and N are the midpoints of the opposite sides AO and PM respectively, we have $TN \parallel AP$, and $ANMT$ is a parallelogram. It follows that $TN \perp SM$ and $MN \perp ST$ (because $AP \perp SM$ and $AT \perp ST$). Therefore, N is the orthocentre of $\triangle TSM$. This implies $SN \perp TM$, and hence $SN \perp AN$. Finally, this yields

$$\angle ANS = 90^\circ = \angle ATS,$$

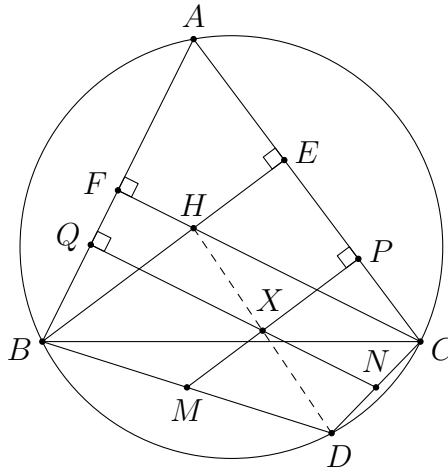
showing that A, S, N, T are concyclic. As N is the midpoint of OH , the result follows immediately. $\square$

Remarks. The diameter PM of the nine-point circle is useful when working on problems involving the nine-point centre.

Problem 4.15. Let D be a point on the circumcircle of $\triangle ABC$. Let M and N be the midpoints of BD and CD respectively. P is the foot of perpendicular from M to AC , and Q is the foot of perpendicular from N to AB . Prove that the intersection point of MP and NQ lies on the nine-point circle of $\triangle ABC$.

Analysis. Since the problem involves the circumcircle and the nine-point circle, we should recall the homothety between the two circles. Also, since there are some perpendicular lines to AB and AC , it may be useful to relate these lines with the altitudes from B and C . In both ideas, the orthocentre of $\triangle ABC$ naturally appears in the figure. The key observation is that the intersection point in the problem statement is the midpoint of the orthocentre and D . It is not hard to make this observation since this is the way that we can make use of the homothety.

Solution.



Let E and F be the feet of altitudes from B and C respectively, and let BE meet CF at the orthocentre H . Note that MP and BE are parallel as both are perpendicular to AC . Since M is the midpoint of BD , the homothety with centre D and ratio 2 maps M to B , hence it also maps the line MP to the line BE . Similarly, the same homothety maps the line NQ to the line CF . It follows that the intersection point X of MP and NQ is mapped to the intersection point of BE and CF , which is just H . This implies X is the midpoint of DH .

Finally, recall that the nine-point circle is the image of the circumcircle under the homothety with centre H and ratio $\frac{1}{2}$. It is clear that D is mapped to X under this homothety from above, which means X lies on the nine-point circle as desired. $\square$

Before proceeding to the next problem, we shall introduce the following proposition, which is a useful fact related to the orthocentre.

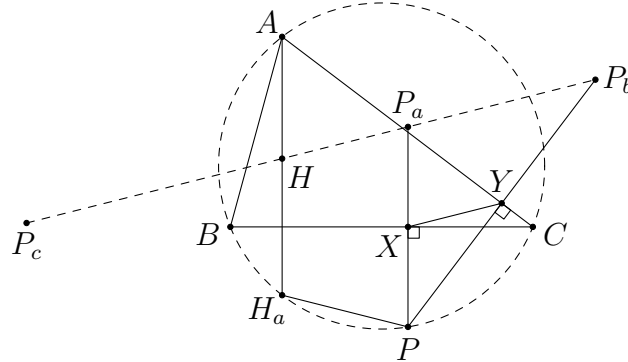
Proposition 4.1. Let H be the orthocentre of $\triangle ABC$.

- (a) Let P be a point on the plane. Then the reflection points of P in BC, CA, AB are collinear iff P lies on (ABC) . If this is the case, then the line on which these reflection points lie also passes through H . The line is called the *Steiner line* of P w.r.t. $\triangle ABC$.
- (b) Let ℓ be a line on the plane. Then the reflection lines of ℓ in BC, CA, AB are concurrent iff ℓ passes through H . If this is the case, then the point for which these reflection lines intersect lies on (ABC) . The concurrent point is called the *anti-Steiner point* of ℓ w.r.t. $\triangle ABC$.

Proof.

- (a) Let P_a, P_b, P_c be the reflections of P in BC, CA, AB respectively. Denote by X and Y the projections from P onto BC and CA respectively. Then X, Y are the midpoints of PP_a, PP_b respectively. By the midpoint theorem, $P_aP_b \parallel XY$. Also, C, P, X, Y are concyclic as $\angle CXP = \angle CYP = 90^\circ$. Hence,

$$\angle PP_aP_b = \angle PXY = \angle PCY = \angle PCA.$$



Due to symmetry, $\angle PP_aP_c = \angle PBA$. Then

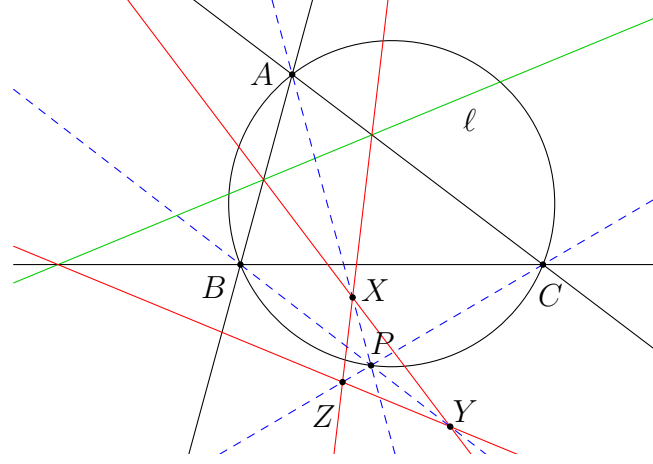
$$\begin{aligned} & P_a, P_b, P_c \text{ are collinear} \\ \Leftrightarrow & \angle PP_aP_b = \angle PP_aP_c \\ \Leftrightarrow & \angle PCA = \angle PBA \\ \Leftrightarrow & P, A, B, C \text{ are concyclic.} \end{aligned}$$

Next, suppose P does lie on (ABC) . Let H_a be the reflection of H in BC , which lies on (ABC) . By the reflection, PP_aHH_a is an isosceles trapezoid. In particular, its vertices are concyclic. Then

$$\angle PP_aH = \angle PH_aH = \angle PH_aA = \angle PCA = \angle PP_aP_b.$$

This shows H also lies on the Steiner line of P .

(b)



Let ℓ_a, ℓ_b, ℓ_c (the red lines) be the reflections of ℓ (the green line) in BC, CA, AB respectively. Define the intersection points $X = \ell_b \cap \ell_c$, $Y = \ell_c \cap \ell_a$ and $Z = \ell_a \cap \ell_b$. Denote by P the intersection of BY and CZ .

Firstly, we note that BA is an angle bisector of ℓ and ℓ_c , while BC is an angle bisector of ℓ and ℓ_a . Hence, B is the incentre or an excentre of the triangle formed by ℓ, ℓ_c, ℓ_a . Then BY is also an angle bisector of ℓ_a and ℓ_c . In the next subsection, we would introduce the following relation

$$\angle(BY, \ell_a) + \angle(BC, \ell) + \angle(BA, \ell_c) = \frac{\pi}{2}.$$

In the case when B is the incentre of this triangle, the above equation merely states the sum of half the angle at each vertex is $\frac{\pi}{2}$. In directed angles, this holds equally well for excentres. Now,

$$\begin{aligned} & \angle(BY, \ell_a) + \angle(BC, \ell) + \angle(BA, \ell_c) = \frac{\pi}{2} \\ \Rightarrow & \angle(BY, \ell) + \angle(BC, \ell_a) + \angle(BA, \ell_c) = \frac{\pi}{2} \\ \Rightarrow & \angle(BY, \ell) = \frac{\pi}{2} - \angle(BC, \ell_a) - \angle(BA, \ell_c) \\ \Rightarrow & \angle(BY, \ell) = \frac{\pi}{2} + \angle(BC, \ell) + \angle(BA, \ell). \end{aligned}$$

By symmetry, $\angle(CZ, \ell) = \frac{\pi}{2} + \angle(CB, \ell) + \angle(CA, \ell)$. This yields

$$\angle(BY, CZ) = \angle(BY, \ell) - \angle(CZ, \ell) = \angle(BA, CA).$$

In other words, $\angle BPC = \angle BAC$, implying P, A, B, C are concyclic. Therefore, P is the second intersection of the line BY and (ABC) . Similarly, AX also passes through P . That means AX, BY, CZ are concurrent at P .

Now, if ℓ_a, ℓ_b, ℓ_c are concurrent, then $X = Y = Z = P$ is the concurrent point, lying on (ABC) . Since P lies on ℓ_a, ℓ_b, ℓ_c , the reflections of P in BC, CA, AB lie on ℓ . Hence, ℓ is the Steiner line of P w.r.t. $\triangle ABC$. From (a), ℓ passes through H .

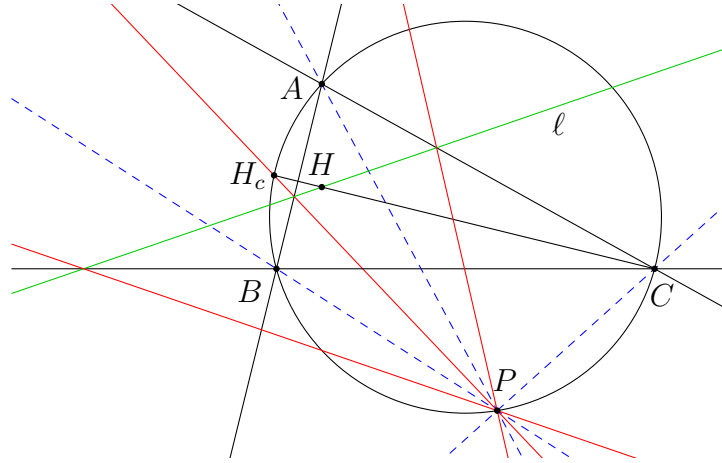
If H lies on ℓ , let H_c be the reflection of H in AB . Clearly, H_c lies on ℓ_c . We are going to show that Y lies on (ABC) . On the one hand,

$$\begin{aligned} \angle(\ell_c, BY) + \angle(\ell_a, BC) + \angle(\ell, BA) &= \frac{\pi}{2} \\ \Rightarrow \angle(\ell_c, BY) &= \frac{\pi}{2} - \angle(\ell_a, BC) - \angle(\ell, BA) \\ \Rightarrow \angle(\ell_c, BY) &= \frac{\pi}{2} + \angle(\ell, BC) - \angle(\ell, BA) \\ \Rightarrow \angle(\ell_c, BY) &= \frac{\pi}{2} + \angle(AB, BC). \end{aligned}$$

On the other hand,

$$\angle H_c CB = \angle(H_c C, AB) + \angle(AB, BC) = \frac{\pi}{2} + \angle(AB, BC).$$

This gives $\angle H_c Y B = \angle H_c C B$, and so Y, H_c, C, B are concyclic. As H_c lies on (ABC) , Y also lies on (ABC) . Therefore, $P = Y$, and similarly, $P = X = Y = Z$. This means ℓ_a, ℓ_b, ℓ_c are concurrent on (ABC) .



□

Remarks. In the proof of (b), we actually prove more than the results we want. For a general line ℓ , the corresponding lines AX, BY, CZ are concurrent on (ABC) (the fact that these lines are concurrent is a trivial consequence of Desargues' theorem). Also, the Steiner line of the concurrent point of ℓ_a, ℓ_b, ℓ_c is ℓ , suggesting the name 'anti-Steiner point'.

Problem 4.16. (IMO Shortlist 2011 G8/Problem 6) Let ABC be an acute triangle with circumcircle Γ . Let ℓ be a tangent line to Γ , and let ℓ_a, ℓ_b and ℓ_c be the lines obtained by reflecting ℓ in the lines BC, CA and AB , respectively. Show that the circumcircle of the triangle determined by the lines ℓ_a, ℓ_b and ℓ_c is tangent to the circle Γ .

Analysis. Since the problem involves reflection of a point in the sidelines, we may think of those properties related to the orthocentre. In particular, it is natural to consider [proposition 4.1](#). Let T be the tangential point of ℓ to Γ . Firstly, the reflections of T in BC, CA, AB are collinear with H . This fact might be helpful for us to draw the figure. Secondly, the reflections of TH in BC, CA, AB are concurrent at a point D on (ABC) .

Even with these facts, it is hard to solve this G8 problem. It would be nice if we could guess the tangential point of the two circles. Indeed, such a point must be symmetric to A, B, C and depend mainly on T . One point which satisfies these conditions is just the anti-Steiner point of HT , i.e. D . There are of course many more other candidates. But if the figure is sufficiently accurate, then it is a good evidence to support this guess.

After guessing the tangential point, the only mission is to deduce the concyclic points and do angle chasing. It is still troublesome (as directed angles must be used to handle the many configurations, though no partial marks would be deducted for this hard problem if the solution is in general correct), but not many tricks are needed. The only possibly difficulty is to observe the concyclic points. Again, drawing an accurate figure is important.

Solution. We first explain our figure and notations. The small purple circle is Γ . T is the tangential point of ℓ to Γ . The blue lines are ℓ_a, ℓ_b, ℓ_c as defined in the problem. They intersect pairwise at P_{ab}, P_{bc}, P_{ca} . The indices are used according to the symmetry of the corresponding sides. For example, P_{ab} is the intersection of ℓ_a and ℓ_b .

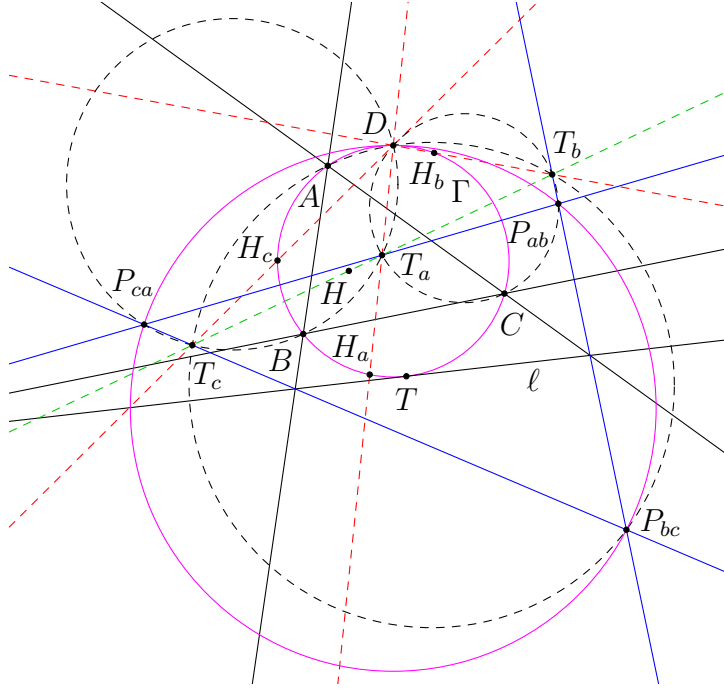
The reflections of T in BC, CA, AB are defined as T_a, T_b, T_c . Clearly, each of them lies on the corresponding line ℓ_a, ℓ_b, ℓ_c . Also, T_a, T_b, T_c, H are collinear by [proposition 4.1\(a\)](#) (the green line), where H is the orthocentre of $\triangle ABC$. Let H_a, H_b, H_c be the reflections of H in BC, CA, AB . Then H_a, H_b, H_c lie on Γ and $H_a T_a, H_b T_b, H_c T_c$ are concurrent at some point D on Γ by [proposition 4.1\(b\)](#). We shall prove that $(P_{bc} P_{ca} P_{ab})$ (the large purple circle) touches Γ at D .

The initial step is to prove C, D, T_a, T_b, P_{ab} are concyclic. Firstly, by the reflection in AC ,

$$\angle DT_b P_{ab} = \angle (H_b T_b, \ell_b) = -\angle (HT, \ell).$$

Due to symmetry, $\angle DT_a P_{ab} = -\angle (HT, \ell) = \angle DT_b P_{ab}$, so that D, T_a, T_b, P_{ab} are concyclic. For the point C , we find that

$$\angle DT_b C = \angle (H_b T_b, T_b C) = -\angle (HT, TC) = \angle CTH.$$



Again by symmetry, $\angle DT_aC = \angle CTH = \angle DT_bC$, so that D, T_a, T_b, C are concyclic. Hence, all C, D, T_a, T_b, P_{ab} are concyclic as claimed. Similarly, A, D, T_b, T_c, P_{bc} are concyclic, B, D, T_c, T_a, P_{ca} are concyclic (the dashed circles).

The next step is to show $D, P_{ab}, P_{bc}, P_{ca}$ are concyclic. In fact, this comes from the fact that D is the Miquel point of $\triangle T_a T_b T_c$ w.r.t. $\triangle P_{bc} P_{ca} P_{ab}$, and $\triangle T_a T_b T_c$ is a degenerate triangle (that means a straight line). The proof is simple. Indeed,

$$\begin{aligned} \angle P_{ca} D P_{ab} &= \angle P_{ca} D T_a + \angle T_a D P_{ab} = \angle P_{ca} T_c T_a + \angle T_a T_b P_{ab} \\ &= \angle(\ell_c, HT_a) + \angle(HT_a, \ell_b) = \angle(\ell_c, \ell_b) = \angle P_{ca} P_{bc} P_{ab}. \end{aligned}$$

This implies $D, P_{ab}, P_{bc}, P_{ca}$ are concyclic.

Lastly, it suffices to prove that the tangent t_1 at D to Γ coincides with the tangent t_2 at D to $(P_{bc} P_{ca} P_{ab})$. We are going to evaluate $\angle(t_1, t_2)$. This can be expressed as $\angle(t_1, AD) + \angle(AD, DP_{bc}) + \angle(DP_{bc}, t_2)$, where (using the concyclic points and reflection points)

$$\begin{aligned} \angle(t_1, AD) &= \angle(DH_a, AH_a) = \angle(T_a H_a, HH_a) = \angle T_a H_a H, \\ \angle(AD, DP_{bc}) &= \angle ADP_{bc} = \angle AT_b P_{bc} = \angle(AT_b, P_{bc} T_b) = -\angle(AT, \ell) \\ &= -\angle(AH_a, TH_a) = \angle(TH_a, HH_a), \\ \angle(DP_{bc}, t_2) &= \angle(\ell_b, DP_{ab}) = \angle(\ell_b, H_b T_b) + \angle T_b D P_{ab} = -\angle(\ell, HT) + \angle T_b T_a P_{ab} \\ &= -\angle(\ell, HT) + \angle(HT_a, \ell_a) = -\angle(\ell, HT) - \angle(H_a T, \ell) = \angle(HT, H_a T). \end{aligned}$$

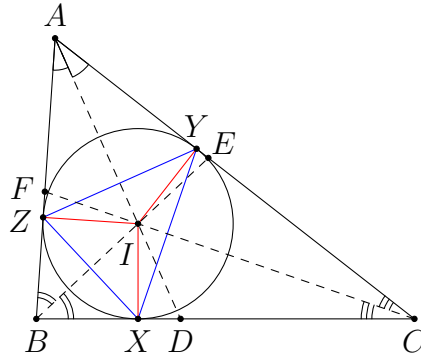
Adding these and noting that $HH_a T T_a$ is an isosceles trapezoid, we get

$$\angle(t_1, t_2) = \angle T_a H_a H + \angle T H H_a = 0.$$

Hence, Γ and $(P_{bc} P_{ca} P_{ab})$ are tangent at D . □

4.4 Incentre and Excentres

The *incircle*(内切圓) of $\triangle ABC$ is the unique circle inside $\triangle ABC$ tangential to the three sides BC, CA, AB at some points X, Y, Z respectively. Its centre, the *incentre*(内心), is usually denoted by I , which is the concurrent point of the internal angle bisectors AD, BE, CF of $\angle A, \angle B, \angle C$. The *inradius*(内切圓半徑) $IX = IY = IZ$ is denoted by r . Point I always lies inside $\triangle ABC$. Note that IX, IY, IZ are perpendicular to the three sides respectively. Also, we have $AZ = AY = s - a$, etc., where $s = \frac{a+b+c}{2}$ is the *semiperimeter*(半周長). So $AZIY$ is a kite and is cyclic.



Since AI is an angle bisector, we have $\angle BAI = -\angle CAI$. Again, it should be noted that we cannot write $\angle BAI = \frac{1}{2}\angle BAC$. One way to express this relation is to use the fact

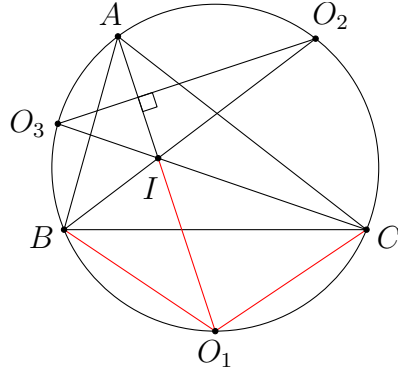
$$\angle BAI + \angle ACI + \angle CBI = \angle IAC + \angle ICB + \angle IBA = \frac{\pi}{2}.$$

We also have

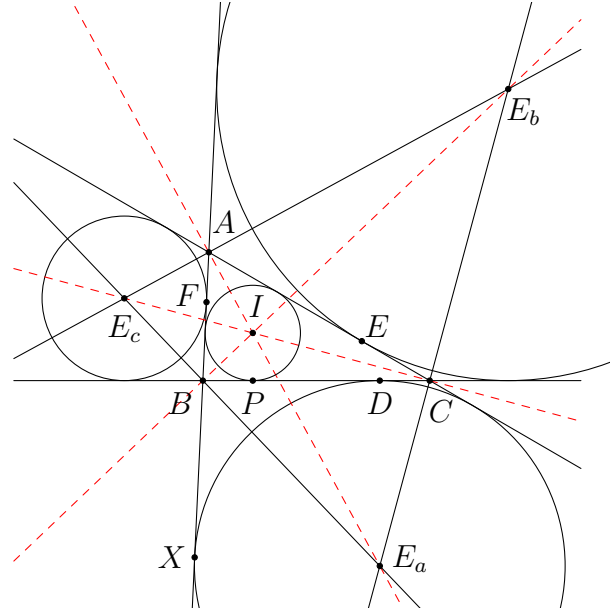
$$\angle BIC = \angle BCI + \angle IBC = -\angle ACI - \angle CBI = \frac{\pi}{2} + \angle BAI,$$

which is the same as the usual relation $\angle BIC = 90^\circ + \frac{1}{2}\angle BAC$. Indeed, as there is only one configuration, the use of directed angles is not necessary. However, we will soon note the usefulness of such notations when we go through excentres and excircles.

Next, the above relation implies $2\angle BIC = 2\angle BAI = \angle BAC$. Let O_1 be the circumcentre of $\triangle BIC$. Then $\angle BO_1C = 2\angle BIC = \angle BAC$, implying O_1 lies on (ABC) . In fact, the point O_1 is the intersection of the line AI and the perpendicular bisector of BC , which is also the midpoint of $\widehat{BC}$ not containing A . Therefore, we have $O_1B = O_1I = O_1C$.



If we define O_2 and O_3 to be the circumcentres of $\triangle CIA$ and $\triangle AIB$ respectively, then we have $O_2A = O_2I$ and $O_3A = O_3I$ from above. This shows O_2O_3 is the perpendicular bisector of AI . Thus, it is not hard to see that I is the orthocentre of $\triangle O_1O_2O_3$. Therefore, properties of the orthocentre are related to properties of the incentre. For example, the above result about the perpendicular bisectors is the same as the fact that the reflections of the orthocentre I in the sides O_2O_3, O_3O_1, O_1O_2 lie on the circumcircle (as they are just A, B, C respectively).



In the above, we only consider the internal angle bisectors of each interior angle. If we consider the external angle bisectors (which bisect the exterior angles), we also get many similar results. Firstly, the external angle bisectors of $\angle ABC$ and $\angle ACB$ intersect on the internal angle bisector of $\angle BAC$. Denote this point by E_a . Then there is a circle with centre E_a which touches the lines BC, CA, AB simultaneously. For this reason, E_a is called the *A-excentre* (旁心) while the corresponding circle is called the *A-excircle* (旁切圆).

By considering other intersections, we can define the *B-excentres* E_b, E_c and *B, C-excircles* analogously. The only circles which are tangent to all of the lines

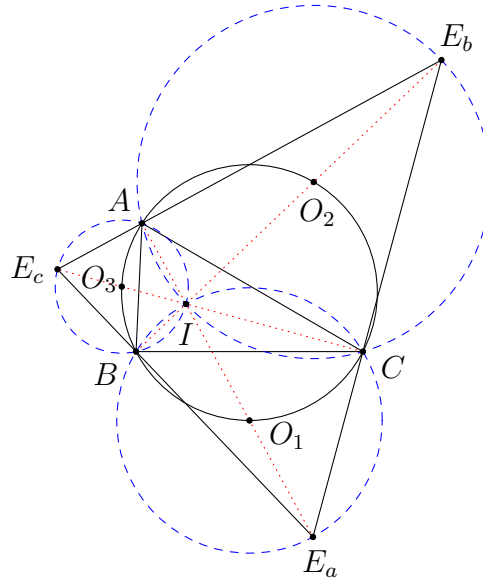
BC, CA, AB are the incircle and the three excircles. The excentres always lie outside $\triangle ABC$. Since the internal and external angle bisectors are perpendicular to each other, we have $AI \perp E_bE_c$. Hence, I is the orthocentre of $\triangle E_aE_bE_c$.

Let the A -excircle touch the side BC at D and the line AB at X . With some calculations, we can verify $BD = s - c$, $CD = s - b$ and $AX = s$. Let the incircle touch the side BC at P . One may note that D is the symmetric point of P in the midpoint of BC .

The excentres and the incentre share a lot of common properties. For example, the relation

$$\angle BAE_a + \angle ACE_a + \angle CBE_a = \angle E_aAC + \angle E_aCB + \angle E_aBA = \frac{\pi}{2}$$

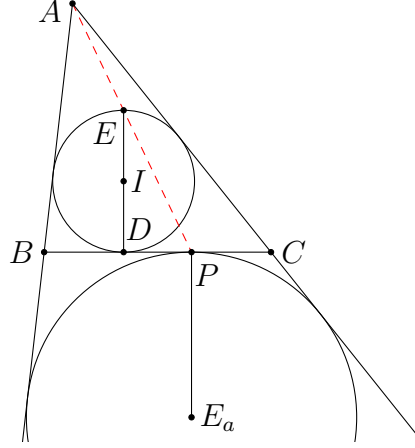
holds equally well by replacing I by E_a . Indeed, when two internal angle bisectors are replaced by external ones, the difference between the angles in these relations is the sum of two $\frac{\pi}{2}$, which is congruent to 0. Similarly, we have $\angle BE_aC = \frac{\pi}{2} + \angle BAE_a$.



Next, points B, I, C, E_a are concyclic. In particular, the circumcentre of $\triangle BCE_a$ lies on $\odot ABC$, which is the midpoint of IE_a . These can be deduced easily again from the fact that I is the orthocentre of $\triangle E_aE_bE_c$. We have mentioned that $\triangle O_1O_2O_3$ (defined as above) also has I as the orthocentre. Indeed, $\triangle O_1O_2O_3$ and $\triangle E_aE_bE_c$ are homothetic, with I being the centre and 2 being the ratio. Note that (ABC) is the nine-point circle of $\triangle E_aE_bE_c$. Hence, $\widehat{E_bE_c}$ meets $\odot ABC$ at A and the midpoint of E_bE_c , which is also the midpoint of $\widehat{BAC}$ (compared to the fact that the internal angle bisector meets the circumcircle at the midpoint of the opposite arc).

There is another important fact related to the incircle and the excircle.

Proposition 4.2. Let the incircle of $\triangle ABC$ touch the side BC at D , and the A -excircle touch the side BC at P . Suppose E is the point on the incircle such that DE is a diameter. Then A, E, P are collinear.

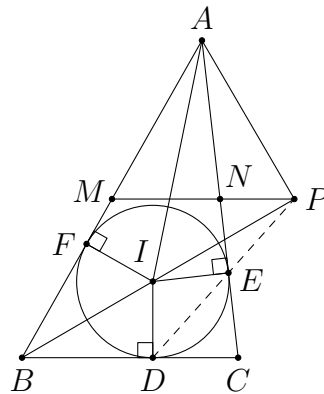


Proof. By considering the homothety with centre A mapping the incircle to the A -excircle, point E is mapped to P' on $\odot E_a$ whose tangent to $\odot E_a$ is parallel to the tangent at E to $\odot I$, which is parallel to BC . Hence, $P' = P$, so that A, E, P are collinear. $\square$

Problem 4.17. (Iran lemma(伊朗引理)) Let I be the incentre of $\triangle ABC$. The incircle of $\triangle ABC$ touches BC and AC at D and E respectively. Let M, N be the midpoints of the sides AB, AC respectively, and let P be the intersection point of the lines MN and BI . Prove that D, E, P are collinear.

Analysis. This is the second part of the Iran lemma. Recall from the first part of the Iran lemma that $\angle APB = 90^\circ$. This shows P lies on $(AFIE)$, where F is the contact point of the incircle and AB . The rest is some simple angle chasing related to the angle bisectors.

Solution.



By the first part of the Iran lemma, we have $\angle APB = 90^\circ$. Let F be the contact point of the incircle and AB . Then A, F, I, E, P are concyclic. It follows that

$$\angle FEP = \angle FAP = \angle BAP = \frac{\pi}{2} - \angle IBA.$$

Also, we have

$$\angle FED = \angle FEA + \angle CED = \left(\frac{\pi}{2} - \angle CAI\right) + \left(\frac{\pi}{2} - \angle ICA\right) = \angle IAC + \angle ACI.$$

It is easy to see that $\angle FEP = \angle FED$, implying D, E, P are collinear. $\square$

Remarks. This is also a useful fact about the incircle. One may derive a similar result for the case of excircles. In the above proof, the key step is to note that $\angle APB = 90^\circ$. This may not be too obvious if we do not know the Iran lemma. This shows how knowing a lot of geometric facts could help us solve some problems.

Problem 4.18. (CWMO 2004 Problem 2) $ABCD$ is a convex quadrilateral. I_1, I_2 are the incentres of $\triangle ABC, \triangle DBC$ respectively. The line through I_1, I_2 meets AB, DC at points E, F respectively. The extensions of AB and DC intersect at point P , where $PE = PF$. Prove that A, B, C, D are concyclic.

Analysis. There are two possible configurations in this problem. (If we interpret AB and DC as rays, then there is only one configuration.) We may take a look at the first case, where A and D lie on the sides of $\triangle PBC$. By simple angle chasing, A, B, C, D are concyclic iff B, I_1, I_2, C are concyclic. We would like to prove the latter assertion, since then points A, D do not play any more roles and we may ignore them to simplify the figure. This is a standard idea in solving geometry problems.

I_1 and I_2 are characterized by the fact that BI_1 bisects $\angle CBP$ and CI_2 bisects $\angle PCB$. If we let I be the incentre of $\triangle PBC$, then I_1 lies on BI and I_2 lies on CI . Also, $PE = PF$ implies $PI \perp EF$. These are sufficient to give the final result.

In the second case, I becomes the P -excentre instead of the incentre. So the calculation is exactly the same if we use directed angles.

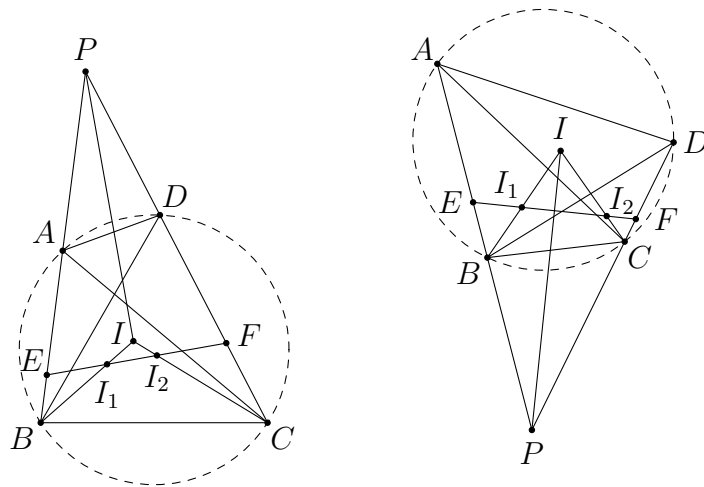
Solution. Let I be the intersection of BI_1 and CI_2 . Then I is either the incentre or P -excentre of $\triangle PBC$. The condition $PE = PF$ is the same as $PI \perp EF$. So

$$\angle I_2 I_1 B = \angle(EF, BI) = \angle(EF, PI) + \angle(PI, BI) = \frac{\pi}{2} + \angle PIB = \angle ICB = \angle I_2 CB.$$

This means I_2, I_1, B, C are concyclic. Next, we have

$$\angle BAC = 2\angle BI_1 C = 2\angle BI_2 C = \angle BDC.$$

Hence, A, B, C, D are concyclic. $\square$

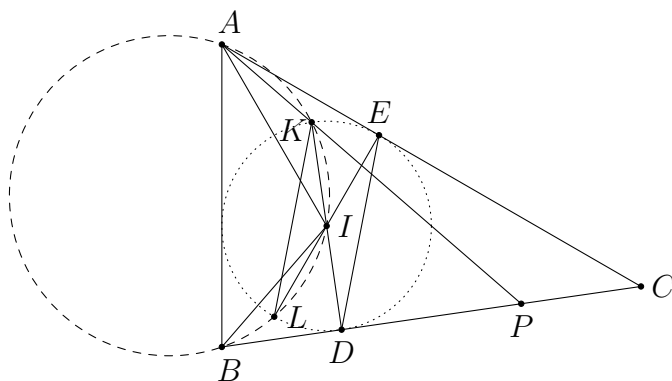


Problem 4.19. (IMO Shortlist 2005 G1) Given a triangle ABC satisfying $AC + BC = 3AB$, the incircle of triangle ABC has centre I and touches the sides BC and CA at the points D and E , respectively. Let K and L be the reflections of the points D and E with respect to I . Prove that the points A, B, K, L lie on one circle.

Analysis. If A, B, K, L are concyclic, then the centre O of this circle lies on the perpendicular bisectors of AB and KL . As KL is parallel to ED , and hence perpendicular to CI , O lies on the perpendicular bisector of AB and the line CI . It is well-known that O lies on (ABC) . Also, we know that $OI = OA = OB$, which means I should lie on the same circle. It would be simpler to prove A, K, I, L are concyclic (then A, B, K, I, L are concyclic by symmetry), since most of the angles inside $AKIL$ can be expressed as the angles of $\triangle ABC$ easily.

To use the condition $AC + BC = 3AB$ in a geometric way, we may rewrite it using the semiperimeter s , as there are some relations between s and the lengths of tangents to the incircle. This equation is the same as $s = 2c$, i.e. $s - c = c$. This suggests we can construct an isosceles triangle using the condition.

Solution.



Let a, b, c be the side lengths BC, CA, AB respectively and let $s = \frac{a+b+c}{2}$. The condition becomes $a+b=3c$. Therefore, $s=2c$, or $s-c=c$.

Suppose AK meets BC at P . By [proposition 4.2](#), since KD is a diameter of the incircle, P is the tangential point of the A -excircle to the side BC . Hence, $BP=s-c$. From above, $BP=BA$. As BI is the internal angle bisector of $\angle CBA$, BI is perpendicular to PA . From this, we deduce

$$\begin{aligned}\angle IAK &= \angle(AI, BI) + \angle(BI, AP) = \left(\frac{\pi}{2} + \angle ACI\right) + \frac{\pi}{2} = \angle ECI = \angle EDI \\ &= \angle EDK = \angle ELK = \angle ILK,\end{aligned}$$

so that A, L, I, K are concyclic. Due to symmetry, B, L, I, K are concyclic. In particular, A, B, K, L lie on the same circle. $\square$

Remarks. If the given condition is a relation between side lengths but we want to avoid too many computations, we usually try to rewrite the equation in the form so that it has some geometric meaning. For example, it can be used to deduce an isosceles triangle, a pair of similar triangles, or the midpoint of a segment, etc.

Problem 4.20. (CGMO 2016 Problem 7) $\triangle ABC$ is acute, $AB > AC$. I is the incentre of $\triangle ABC$, D is the perpendicular foot of I to the side BC . The line AH perpendicular to the side BC meets the lines BI, CI at points P, Q respectively. O is the circumcentre of $\triangle IPQ$, and the line AO meets BC at L . The circumcircle of $\triangle AIL$ intersects the line BC at points L and N ($N \neq L$). Prove that $\frac{BD}{CD} = \frac{BN}{CN}$.

Analysis. One difficult part is to think of how to use the circumcentre O of $\triangle IPQ$. If one tries to investigate $\triangle IPQ$, it is not hard to obtain its interior angles, which are $90^\circ - \frac{A}{2}$, $90^\circ - \frac{B}{2}$ and $90^\circ - \frac{C}{2}$ respectively. These are exactly the same as the interior angles of the intouch triangle, and hence it is natural to consider the similarity between $\triangle IPQ$ and the intouch triangle. We can find more corresponding points under this similarity. In particular, this provides a simple way to handle the most important point N .

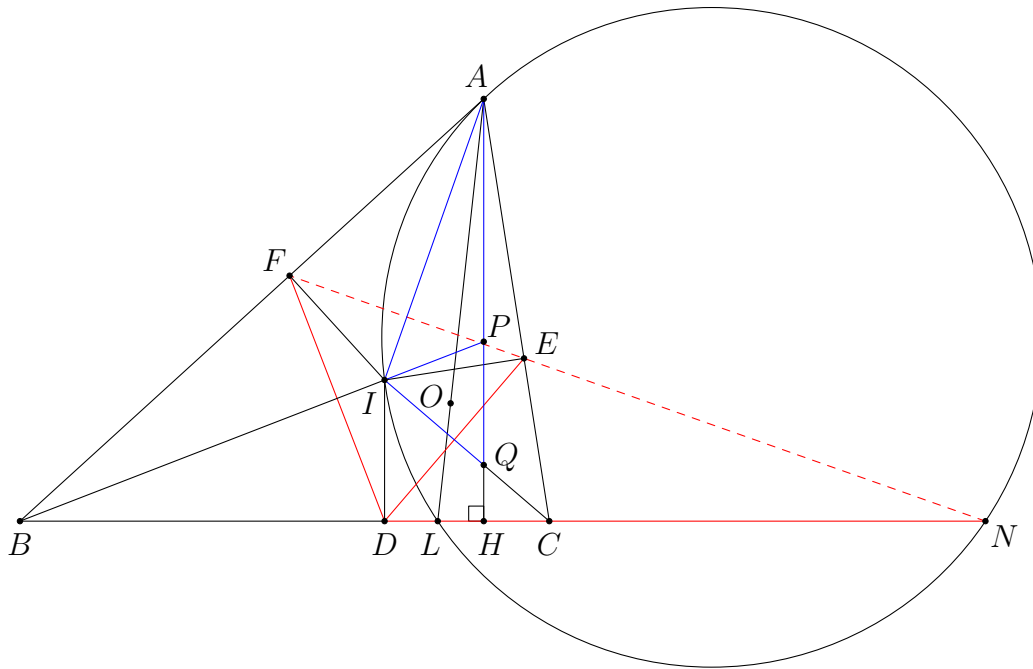
Solution. Let the incircle of $\triangle ABC$ touch the sides AC and AB at E and F respectively. We find that

$$\angle PQI = 90^\circ - \frac{C}{2} = \angle EFD$$

and

$$\angle IPQ = 90^\circ - \frac{B}{2} = \angle DEF.$$

These show $\triangle IPQ \sim \triangle DEF$. As O is the circumcentre of $\triangle IPQ$ and I is the circumcentre of $\triangle DEF$, points O and I are the corresponding points under this similarity.



Next, we check that

$$\angle PIA = 180^\circ - \angle AIB = 90^\circ - \frac{C}{2} = \angle PQI.$$

This shows AI is a tangent to the circumcircle of $\triangle IPQ$, and hence $\angle OIA = 90^\circ$. Correspondingly, ND is a tangent to the circumcircle of $\triangle DEF$, so that we have $\angle IDN = 90^\circ$. Together with $\angle IAO = \angle DNI$ from the concyclic points, the triangles OIA and IDN are similar. Therefore, A and N are the corresponding points under the similarity $\triangle IPQ \sim \triangle DEF$. Since A, P, Q are collinear, this shows N, E, F are collinear. By Menelaus' theorem,

$$\frac{BN}{NC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1.$$

Using $AF = AE$, $BD = BF$ and $CE = CD$, we obtain $\frac{BD}{CD} = \frac{BN}{CN}$. $\square$

Remarks. The main component in the above solution is to note the similarity between the two groups of points. This is one usage of similar figures.

Finally, let us revise the important properties of the basic triangle centres.

- centroid and midpoints
 - 2 : 1 ratio (problems 4.1, 4.2)
 - midpoint theorem (problems 4.3, 4.4)
 - Iran lemma (problems 4.3, 4.17)

- circumcentre
 - central angle theorem ([problems 4.5, 4.6](#))
- orthocentre
 - line joining orthocentre and midpoint ([problem 4.9](#))
 - reflection of orthocentre ([problems 4.10, 4.11, 4.13, 4.16](#))
- nine-point centre
 - diameter of nine-point circle ([problem 4.14](#))
 - homothety between nine-point circle and circumcircle ([problem 4.15](#))
- incentre and excentres
 - angles and lengths related to incentre and incircle ([problems 4.17, 4.18, 4.19, 4.20](#))
 - homothety between incircle and excircle ([problem 4.19](#))

5 Power and Radical Axes

The power of points and radical axes are vital knowledge in solving Olympiad problems in plane geometry. These have been used frequently in recent years. Although these concepts are simple consequences of similar triangles and Pythagoras' theorem, we include them as a single section to mark their importance.

5.1 Power

The concept of power is related to a useful property of concyclic points. We first introduce its definition.

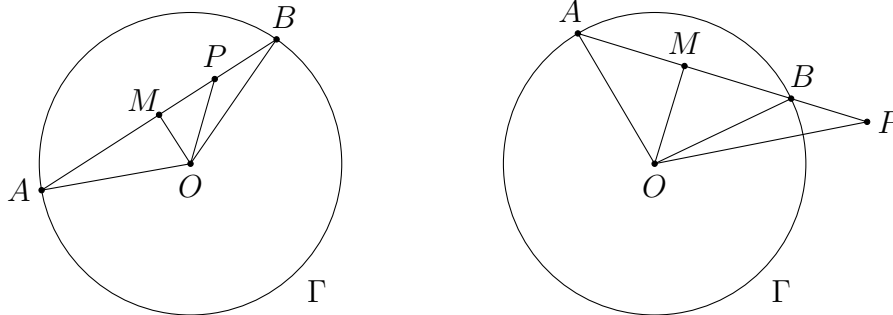
Definition 5.1. Let Γ be a circle with centre O and radius r . For any point P on the plane, the **power**(幂) of P w.r.t. Γ is defined to be the real number $OP^2 - r^2$.

After specifying the circle Γ , we may simply say the power of P for convenience. In view of the definition, we deduce the following basic facts.

- P lies outside Γ iff the power of P is positive
- P lies on Γ iff the power of P is 0
- P lies inside Γ iff the power of P is negative

Therefore, one may regard power as a measure of how far a point is to the circle. The locus of points having the same power is a circle whose centre coincides with that of Γ . The definition is a bit difficult to use in practice. In fact, we have the following important theorem about power.

Theorem 5.1. (**Power of a point theorem**(圓幂定理)) Let Γ be a circle and let P be any point on the plane. Suppose ℓ is any line passing through P which cuts Γ at points A and B . Then the power of P w.r.t. Γ is $PA \times PB$, where all lengths are measured in directed lengths.



Proof. Let M be the midpoint of AB . Then $OM \perp AB$. By Pythagoras' theorem, we have

$$OP^2 - r^2 = OP^2 - OA^2 = (PM^2 + OM^2) - (MA^2 + OM^2) = PM^2 - MA^2.$$

Here, we may assume directed lengths are used since only squares of side lengths are involved. Then

$$PM^2 - MA^2 = (PM - MA)(PM + MA) = (PM - BM)(PM + MA) = PB \times PA.$$

Hence, the power of P w.r.t. Γ is $PA \times PB$. $\square$

Since the line ℓ can be chosen randomly (as long as it passes through P and meets Γ at two points), the product $PA \times PB$ formed by different lines ℓ is a constant depending on P and Γ only. This is not a new fact, since if another line ℓ' through P meets Γ at points C, D , then $\triangle PAC \sim \triangle PDB$ (see [proposition 2.2](#)).

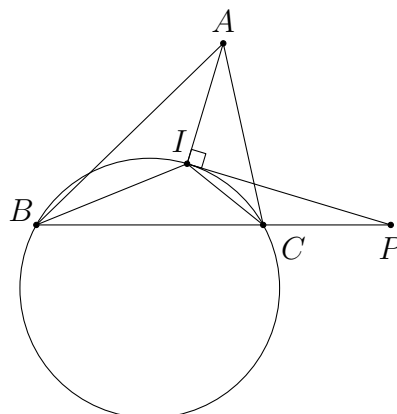
If P lies outside Γ , then ℓ can be chosen as a tangent through P to Γ , so that A and B coincide to give PA^2 as the power of P (where A is the tangent point). This limiting case is also useful in characterizing the power of points outside the circle.

The most important fact about power is that it is a constant for each point. Whenever we see something like $PA \times PB$, we may try to use power to handle this expression (this is the power of P w.r.t. any circle passing through A, B).

Problem 5.1. (IMO HK TST 2011 Test 1 Problem 1) Let I be the incentre of a triangle ABC with $AB > AC$. A line through I perpendicular to AI cuts BC produced at P . Show that $IP^2 = BP \cdot CP$.

Analysis. The assertion looks like the power of P . Indeed, it is two representations of the power of P w.r.t. (BCI) . In other words, we have to show that IP is a tangent to this circle.

Solution.



We have

$$\angle CIP = \angle CIA - \frac{\pi}{2} = \angle CBI.$$

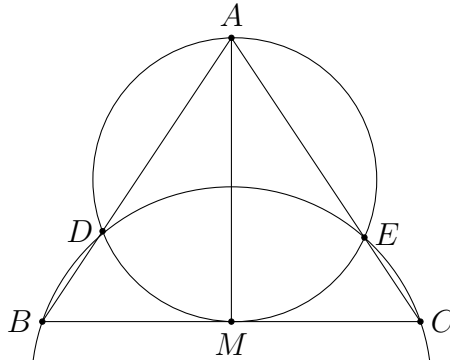
This shows IP is tangent to (BCI) . Hence, $IP^2 = BP \cdot CP$ as both sides are equal to the power of P w.r.t. (BCI) . $\square$

Remarks. For this simple problem, one can easily use similar triangles to deduce the same thing. The convenience of using power will appear after we have gone through more problems.

Problem 5.2. (IMO HK TST 2011 Test 2 Problem 1) Given a triangle ABC , let M be the midpoint of BC . The circle passing through A and tangent to BC at M cuts AB and AC at D and E respectively. Suppose B, C, E and D are concyclic. Show that $AB = AC$.

Analysis. Since the figure consists of two circles with some secants and tangents, we may try to consider the power of different points w.r.t. these circles.

Solution.



Firstly, consider the power of B w.r.t. $(ADME)$. This gives $BD \times BA = BM^2$. Similarly, we get $CE \times CA = CM^2$ by considering the power of C w.r.t. the same circle. As M is the midpoint of BC , the two expressions are equal. It follows that

$$BD \times BA = CE \times CA. \quad (1)$$

Next, since B, C, E, D are concyclic, we have

$$AD \times AB = AE \times AC \quad (2)$$

from the power of A . Adding (1) and (2), we obtain

$$AB^2 = AB \times BD + AB \times AD = AC \times CE + AC \times AE = AC^2.$$

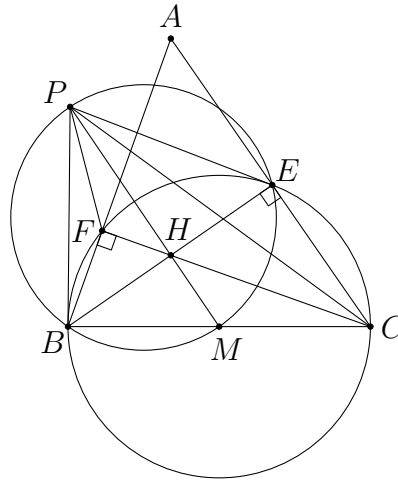
This gives $AB = AC$. $\square$

Remarks. In view of this solution, one simple use of power is to derive some relations of lengths.

Problem 5.3. In $\triangle ABC$, E and F are the feet of perpendiculars from B and C to the opposite sides respectively. H is the orthocentre of $\triangle ABC$. M is the midpoint of the side BC . The second intersection point of the line MH and the circumcircle of $\triangle BME$ is P . Prove that $\angle BPF = \angle CPE$.

Analysis. One way to use the concyclic condition is to deduce $HB \times HE = HM \times HP$ from the power of H . Note that we also have $HB \times HE = HC \times HF$ using another group of concyclic points. This immediately yields a nontrivial group of concyclic points P, F, M, C . In fact, this is another way to use power.

Solution.



Since P, B, M, E are concyclic, we have $HB \times HE = HM \times HP$. Next, from the condition $\angle BFC = \angle BEC = 90^\circ$, points B, F, E, C lie on the circle with diameter BC (which means M is the centre). From this we obtain $HB \times HE = HC \times HF$, so that $HC \times HF = HM \times HP$. This implies P, F, M, C are concyclic (by similar triangles or the converse of the power of a point theorem).

Now, as P, B, M, E are concyclic and $MB = ME$, we know that PM is the internal angle bisector of $\angle BPE$, and hence $\angle BPM = \angle EPM$. Similarly, as P, F, M, C are concyclic and $MF = MC$, we have $\angle FPM = \angle CPM$. It follows that

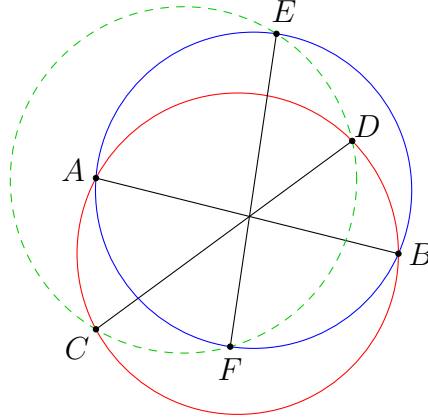
$$\angle BPF = \angle BPM - \angle FPM = \angle EPM - \angle CPM = \angle CPE$$

as desired. $\square$

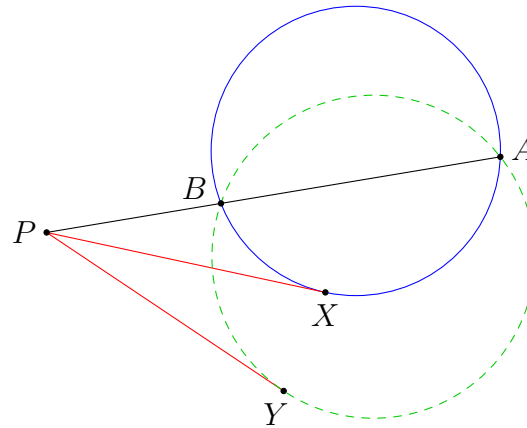
Remarks. Do you remember the line MH as discussed in [problem 4.9](#)? Indeed, the two problems are closely related since the point P in this problem coincides with the point D in that problem. Therefore, one may try to show the following facts: A, P, B, C are concyclic, $\triangle BPF \sim \triangle CPE$ and $\triangle BPC \sim \triangle FPE$.

The idea used in the above solution can be summarized as follows. Suppose the segments AB, CD, EF are concurrent at an interior point. If A, B, C, D are con-

cyclic and A, B, E, F are concyclic, then C, D, E, F are concyclic. The same holds if AB, CD, EF are concurrent at their extensions.



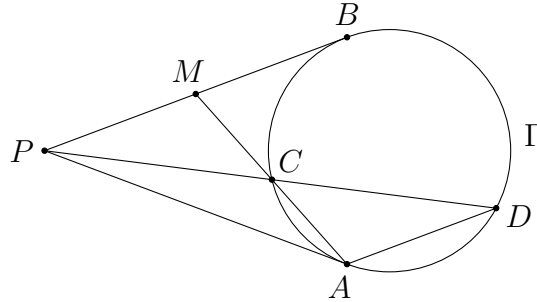
A similar idea works in some problems involving tangents. Suppose P is a point on the extension of AB such that PX is tangent to (XAB) for some point X . If $PX = PY$ for some point Y , then we have $PY^2 = PX^2 = PA \times PB$, which implies PY is tangent to (YAB) . This yields some pairs of equal angles (such as $\angle BYP = \angle BAY$) which are not easy to prove without the use of power.



Problem 5.4. Let P be a point lying outside a circle Γ . The contact points of the tangents from P to Γ are A and B respectively. Let M be the midpoint of PB . The line AM intersects Γ again at C , and the line PC intersects Γ again at D . Prove that $AD \parallel PB$.

Analysis. We use this simple problem to demonstrate how to use the above idea. Note that the isosceles triangle in the idea now becomes the condition of a midpoint, since we have $MB = MP$. We can use power to derive a tangent, and hence a pair of equal angles that can be used to prove the parallel lines.

Solution.



We have $MP^2 = MB^2 = MC \times MA$. This implies MP is tangent to (ACP) , and hence $\angle MPC = \angle PAC$. It follows that

$$\angle MPD = \angle MPC = \angle PAC = \angle ADC = \angle ADP.$$

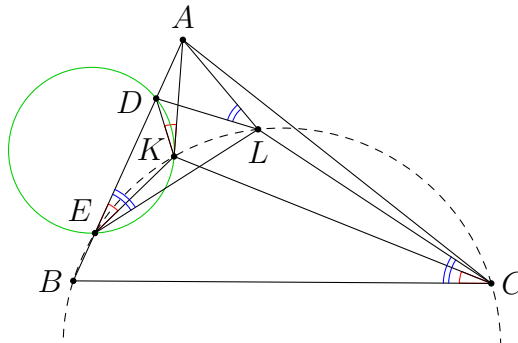
This shows $PM \parallel AD$ as desired. □

Problem 5.5. (CGMO 2022 Problem 5) Points K and L are in the interior of a triangle ABC . Point D lies on the side AB . It is known that points B, K, L, C are concyclic, and $\angle AKD = \angle BCK$, $\angle ALD = \angle BCL$. Prove that $AK = AL$.

Analysis. One way to solve a geometry problem is first to think of a plan based on some well-known facts. This may help us construct some extra points in the figure in a natural way. For example, in this problem, it seems not easy to relate the equal angles to the equal length. However, we may recall that power is a useful tool to relate the two things together.

The key idea is to look for a line XY through A such that AK is tangent to (KXY) and AL is tangent to (LXY) . If such points X and Y exist, we immediately finish the problem by using power (as $AK^2 = AX \times AY = AL^2$). We should probably use angles to prove the tangent lines. In view of the figure, a suitable choice of X is D as we know some information about $\angle AKD$ and $\angle ALD$. This leads to the natural construction of the point E in the following solution.

Solution.



Let E be the point on the line AB such that AK is tangent to (DEK) . Since we want to use the angle $\angle AKD$, we derive

$$\angle KED = \angle AKD = \angle KCB.$$

This implies E, B, C, K are concyclic, or equivalently, B, C, L, K, E are concyclic. In view of our final goal as discussed, we deduce

$$\angle ALD = \angle LCB = \angle LED.$$

It follows that AL is tangent to (DEL) , and hence

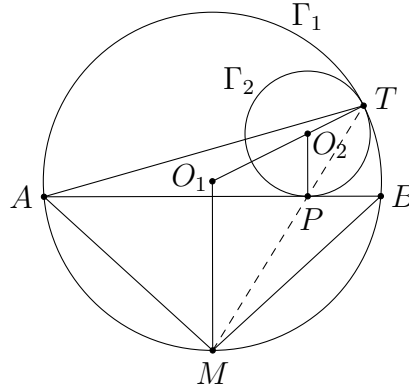
$$AL^2 = AD \times AE = AK^2.$$

This proves $AL = AK$. □

Problem 5.6. Let Γ_2 be a smaller circle internally tangent to Γ_1 at T . Let AB be a chord of Γ_1 tangent to Γ_2 at P . If M is the midpoint of $\widehat{AB}$ lying on different sides of AB to Γ_2 , prove that the tangent from M to Γ_2 has length equal to MA .

Analysis. Recall from [problem 3.6](#) that T, P, M should be collinear. The length of tangent from M to Γ_2 is equal to the square root of its power. From the figure, it may be easier to use $MP \times MT$ as the power of M . Hence, it suffices to show $\triangle MAP \sim \triangle MTA$, which is easy by angle chasing.

Solution.



The homothety with centre T mapping Γ_2 to Γ_1 takes the point P to a point M' on Γ_1 such that the tangent at M' to Γ_1 is parallel to AB (the tangent at P to Γ_2), and such that M' lies on different sides of AB to Γ_2 . This means exactly $M' = M$, and hence T, P, M are collinear.

Next, we have $\angle AMP = \angle TMA$ and $\angle MAP = \angle MBA = \angle MTA$ using $MA = MB$ and the fact that A, M, B, T are concyclic. These give $\triangle MAP \sim \triangle MTA$, so that $MA^2 = MP \times MT$. As $MP \times MT$ is the power of M w.r.t. Γ_2 , the length of tangent from M to Γ_2 is equal to MA as desired. □

Remarks. The assertion of this problem is a useful property for two circles internally tangent to each other. We will see one of its applications in the next problem.

Problem 5.7. Let D be a point on the side BC of $\triangle ABC$ and let I be the incentre of $\triangle ABC$. The unique circle $\odot O_1$ tangential to the sides AD, DC and (ABC) touches AD, DC at E, F respectively. The unique circle $\odot O_2$ tangential to the sides AD, DB and (ABC) touches AD, DB at P, Q respectively.

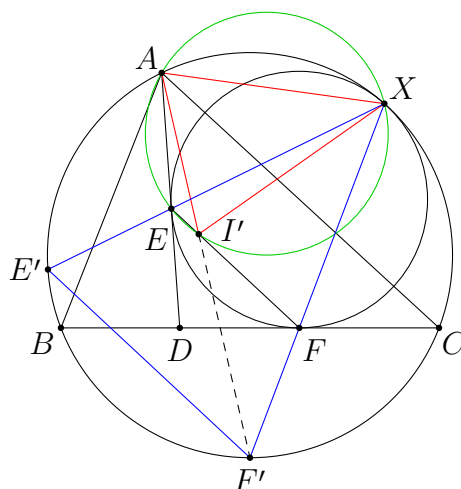
- (a) (**Y. Sawayama's lemma**(澤山定理)) Prove that E, F, I are collinear, and P, Q, I are collinear.
- (b) (**Thébault's theorem**(泰博定理)) Prove that O_1, O_2, I are collinear.

Analysis. Since the figure involves two tangential circles, we may use **problem 5.6** and the idea used in that solution. Let $\odot O_1$ touch (ABC) at X . Extend XF and meet (ABC) again at F' . Then we know that F' is the midpoint of $\widehat{BC}$, and so A, I, F' are collinear. Also, using the standard trick of power, we can show that $F'I$ is tangent to (IFX) .

Afterwards, if we assume the final result, it is not hard to see that A, E, I, X should be concyclic. However, it does not seem easy to prove the concyclic points directly. For this reason, we first construct a point I' on (AEX) , and show that $I' = I$. This works better since more properties can be deduced from the concyclic points.

Solution.

(a)



Let $\odot O_1$ touch (ABC) at X . Consider the homothety with centre X mapping $\odot O_1$ to (ABC) . Let E', F' be the images of E, F respectively. Suppose (AEX) meets EF again at I' .

$\triangle I'AX = \triangle I'EX = \triangle FEX = \triangle F'E'X = \triangle F'AX,$
 A, I', F' are collinear. Also,

which implies AI' is tangent to $(I'FX)$. Then

Here, the last equality comes from [problem 5.6](#). By the same problem, we know that F' is the midpoint of $\overline{BC}$. Hence, I also lies on the segment AF' and we have $F'I = F'B = F'I'$. This yields $I' = I$. In particular, E, F, I are collinear. Similarly, P, Q, I are collinear. $\square$

$$\Delta O_1FD, \Delta O_1NF, \Delta FND, \Delta DQO_2, \Delta DMQ, \Delta QMO_2$$
$$\frac{IN}{NO_1} = \frac{MD}{NO_1} = \frac{MQ}{NF} = \frac{MO_2}{ND} = \frac{MO_2}{IM},$$

and hence $\triangle INO_1 \sim \triangle O_2MI$. This gives

$$\angle O_2IO_1 = \angle O_2IM + 90^\circ + \angle NIO_1 = 180^\circ.$$

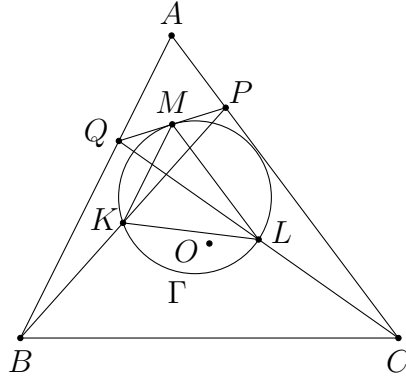
Thus, O_1, O_2, I are collinear. $\square$

Problem 5.8. (IMO Shortlist 2009 G2/Problem 2) Let ABC be a triangle with circumcentre O . The points P and Q are interior points of the sides CA and AB , respectively. Let K, L and M be the midpoints of the segments BP, CQ and PQ , respectively, and let Γ be the circle passing through K, L and M . Suppose that the line PQ is tangent to the circle Γ . Prove that $OP = OQ$.

Analysis. It is not simple to find the length OP . However, using the concept of power, it is the same as proving P and Q have the same power w.r.t. (ABC) . This means it suffices to show $PA \times PC = QA \times QB$.

On the other hand, it is rather easy to see how the conditions can be used. Indeed, we can only use the tangent condition to obtain some equal angles (other than that, nothing can be done if no further points are introduced). For the midpoints, one would probably think of the midpoint theorem.

Solution.



By the midpoint theorem, we have $MK \parallel QB$. As PQ is tangent to Γ , we have

$$\angle MLK = \angle QMK = \angle MQA.$$

By symmetry, we find that $\angle MKL = \angle MPA$. Therefore, $\triangle MKL \sim \triangle APQ$. Recall that we have to compare AP and AQ . Hence, we use the similar triangles to deduce

$$\frac{AP}{AQ} = \frac{MK}{ML} = \frac{\frac{QB}{2}}{\frac{PC}{2}} = \frac{QB}{PC},$$

again by the midpoint theorem. This can be rewritten as $PA \times PC = QA \times QB$. Since P and Q lie inside (ABC) , this means the power of P and Q w.r.t. (ABC) are equal. Therefore, $OP = OQ$. $\square$

5.2 Radical Axes and Radical Centre

When talking about the power of a point, the circle for which this power is defined should be fixed. Sometimes it may be useful if we consider the power w.r.t. more than one circle. For example, if we have two fixed circles Γ_1 and Γ_2 , we may ask the question: is there any point which has the same power w.r.t. both Γ_1 and Γ_2 ?

Indeed, let O_1, O_2 be the centres of Γ_1, Γ_2 and let r_1, r_2 be the radii of Γ_1, Γ_2 respectively. Any point A which has the same power w.r.t. both Γ_1 and Γ_2 is characterized by the relation

$$O_1A^2 - r_1^2 = O_2A^2 - r_2^2,$$

i.e.

$$O_1A^2 - O_2A^2 = r_1^2 - r_2^2,$$

which is a constant. If $O_1 = O_2$, then we may assume $r_1 \neq r_2$ (or otherwise, $\Gamma_1 = \Gamma_2$, which is not an interesting case). The left-hand side is 0 while the right-hand side is not. So there is no such point A . If $O_1 \neq O_2$, let P be the projection of A onto the line O_1O_2 . By [problem 1.4](#),

$$O_1P^2 - O_2P^2 = O_1A^2 - O_2A^2,$$

so it suffices to search for all points P on the line O_1O_2 satisfying

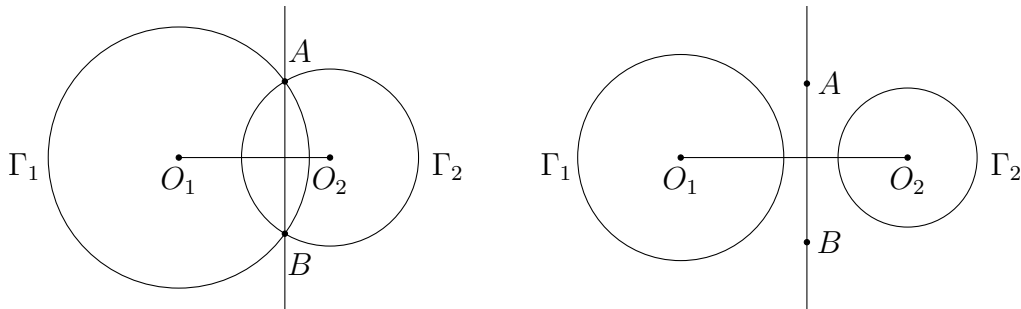
$$O_1P^2 - O_2P^2 = c$$

for some constant c . Using directed lengths,

$$\begin{aligned} & O_1P^2 - O_2P^2 = c \\ \Leftrightarrow & (O_1P - O_2P)(O_1P + O_2P) = c \\ \Leftrightarrow & O_1O_2(2O_2P + O_1O_2) = c. \end{aligned}$$

As $O_1O_2 \neq 0$, there is a unique solution of O_2P to this equation. In other words, there is exactly one point P with this property. From above, any point lying on the perpendicular from P to O_1O_2 also has equal power w.r.t. Γ_1 and Γ_2 . This is summarized in the following definition.

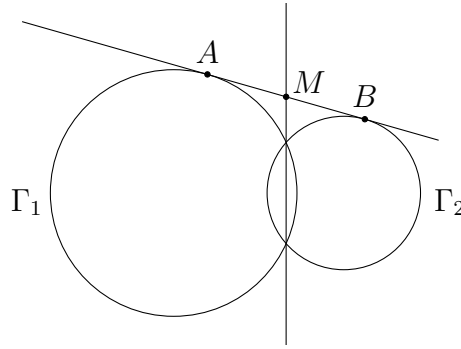
Definition 5.2. Let Γ_1 and Γ_2 be two circles having distinct centres O_1 and O_2 . The locus of points P having equal power w.r.t. Γ_1 and Γ_2 is a line perpendicular to O_1O_2 , called the *radical axis* (根軸) of Γ_1 and Γ_2 .



If Γ_1 and Γ_2 intersect at two points A and B , then the radical axis of Γ_1 and Γ_2 is simply the line AB . This is because the power of A (or B) w.r.t. both circles is 0 as it lies on these circles. In particular, the line joining the intersections of Γ_1 and Γ_2 is perpendicular to the line joining the centres. This also includes the limiting case when Γ_1 and Γ_2 touch at some point A (externally or internally). Their radical axis is then the tangent at A . If the two circles do not intersect, the radical axis will be a line not passing through either circle.

Next, we give a useful fact for circles having external common tangents.

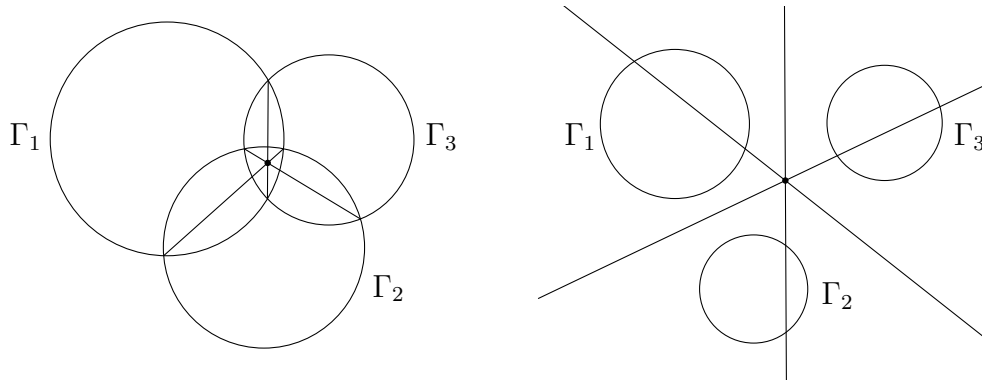
Proposition 5.1. Let Γ_1 and Γ_2 be two distinct circles with none of them lying entirely inside the other. Then the radical axis of Γ_1 and Γ_2 bisects the external common tangents to these circles.



Proof. Since none of the circles lies entirely inside the other, they have different centres and the external common tangents exist. As the radical axis of Γ_1 and Γ_2 is perpendicular to the line joining the centres, it cannot be parallel to the external common tangents (the angle between an external common tangent to the line joining the centres must be smaller than 90°). Therefore, we can let the radical axis meet an external common tangent at M , and let this tangent touch Γ_1 and Γ_2 at A and B respectively. Since the power of M w.r.t. the two circles are equal, we have $MA^2 = MB^2$, and hence $MA = MB$. This means M is the midpoint of AB . $\square$

When there are three circles $\Gamma_1, \Gamma_2, \Gamma_3$ with distinct centres, there are three radical axes by taking two circles each time. It is possible that all these radical axes are parallel, which occurs when the centres are collinear. Apart from this special case, suppose the radical axis of Γ_1, Γ_2 meets the radical axis of Γ_2, Γ_3 at a point P . Then the power of P w.r.t. Γ_1 is equal to that w.r.t. Γ_2 , and is in turn equal to that w.r.t. Γ_3 . Hence, P lies on the radical axis of Γ_1, Γ_3 . This gives the following definition.

Definition 5.3. Let $\Gamma_1, \Gamma_2, \Gamma_3$ be three circles having distinct centres. Denote by $\ell_{i,j}$ the radical axis of Γ_i and Γ_j for $1 \leq i < j \leq 3$. Then $\ell_{1,2}, \ell_{1,3}, \ell_{2,3}$ are parallel or concurrent. In the latter case, the concurrent point is called the *radical centre* (根心) of $\Gamma_1, \Gamma_2, \Gamma_3$.



The existence of the radical centre is an important fact. For example, if there are three circles for which every two intersect at two distinct points, then we immediately know that the three lines each joining the two corresponding intersection points are parallel (including coincide) or concurrent.

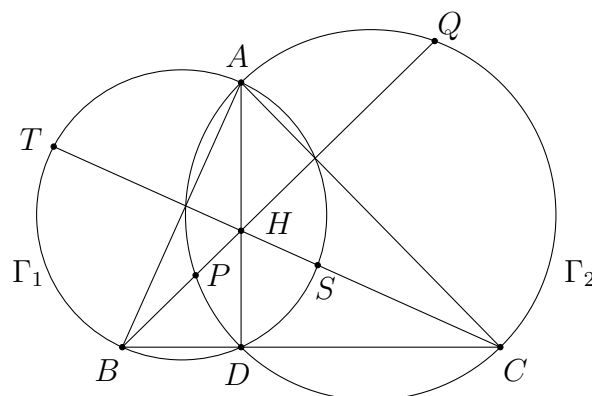
We shall go through some basic usages of the radical axis and the radical centre below.

Problem 5.9. An acute $\triangle ABC$ is given. Let the line through B perpendicular to AC meet the circle with diameter AC at points P and Q , and let the line through C perpendicular to AB meet the circle with diameter AB at points S and T . Prove that P, Q, S, T are concyclic.

Analysis. The two perpendicular lines are just the altitudes from B and C . They meet at the orthocentre H of $\triangle ABC$. In addition, the circles with diameter AB and AC pass through some of the feet of altitudes. Therefore, these points should be included on the circles when we draw the figure.

It is not at all easy to prove P, Q, S, T are concyclic by angle chasing, since the points do not have much relations. It seems to be better to use power. It suffices to prove $HP \times HQ = HS \times HT$, or the power of H w.r.t. the two circles are equal. In other words, we need to show that H lies on the radical axis of these circles.

Solution.



Note that BPQ is the altitude from B , while CST is the altitude from C . The two lines intersect at the orthocentre H of $\triangle ABC$. Let D be the foot of altitude from A . As $\angle ADB = \angle ADC = 90^\circ$, D lies on both the circle Γ_1 with diameter AB and the circle Γ_2 with diameter AC . Therefore, Γ_1 and Γ_2 intersect at the distinct points A and D . In particular, AD is the radical axis of Γ_1 and Γ_2 .

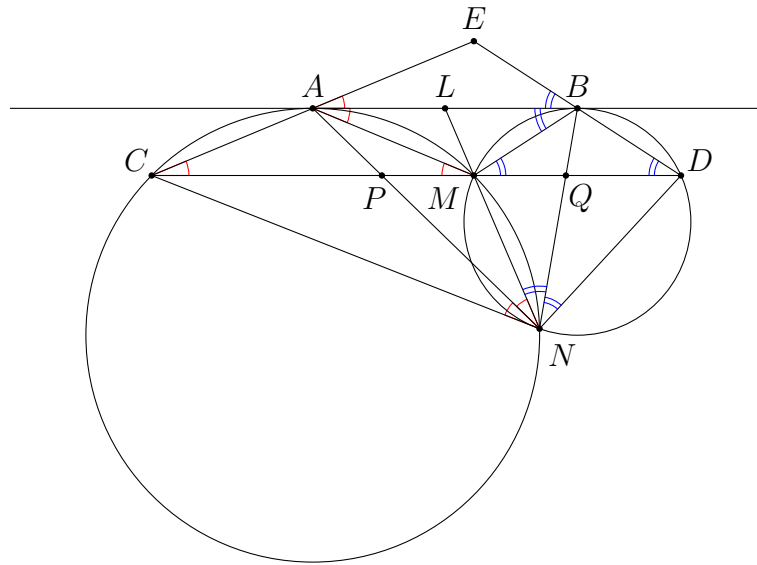
Next, since H lies on the radical axis AD , it has equal power w.r.t. Γ_1 and Γ_2 . Then $HP \times HQ = HS \times HT$. This implies P, Q, S, T are concyclic. $\square$

Remarks. The solution gives one common way of using power to prove concyclic points. To prove that W, X, Y, Z are concyclic where WX and YZ meet at a point E , we can prove that E lies on the radical axis of two circles, one of which passes through W, X and the other passes through Y, Z .

Problem 5.10. (IMO Shortlist 2000 G2/Problem 1) AB is tangent to the circles $CAMN$ and $NMBD$. M lies between C and D on the line CD , and CD is parallel to AB . The chords NA and CM meet at P ; the chords NB and MD meet at Q . The rays CA and DB meet at E . Prove that $PE = QE$.

Analysis. By [proposition 5.1](#), the radical axis MN bisects AB . Using homothety, this means M is the midpoint of PQ . Hence, the assertion $EP = EQ$ is the same as $EM \perp PQ$, which is not hard to see after labeling the equal angles.

Solution.



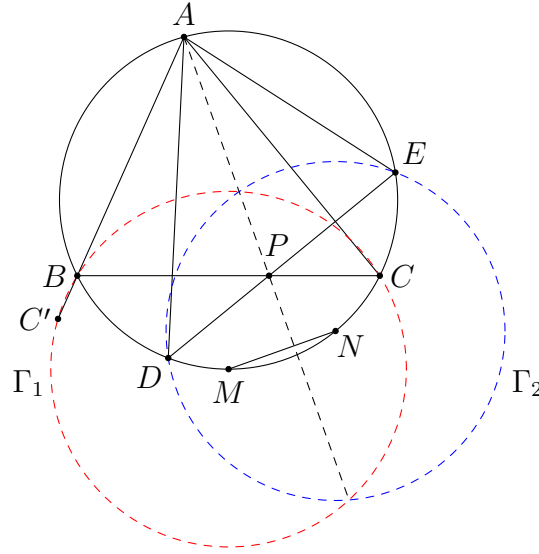
Let MN intersect AB at L . By [proposition 5.1](#), L is the midpoint of AB . Consider the homothety with centre N taking A to P . As $AB \parallel PQ$, B is mapped to Q while L is mapped to M . Then M is the midpoint of PQ . Hence, it suffices to prove $EM \perp PQ$.

Note that $\angle BAE = \angle MCA = \angle MAB$ and $\angle EBA = \angle BDM = \angle ABM$. This implies $EAMB$ is a kite. In particular, $EM \perp AB$. This means $EM \perp PQ$, and so $EP = EQ$. $\square$

Problem 5.11. Let A, B, D, C, E be points on a circle in that order. Let M and N be the midpoints of the arcs $\widehat{BC}$ and $\widehat{DE}$ not containing A respectively. The chords BC and DE meet at P . If AP is perpendicular to MN , prove that $AB \times AC = AD \times AE$.

Analysis. The equation in the assertion looks like an equality of power. However, it is not that straightforward since AB, AC, AD, AE are distinct lines. On the other hand, to use the tricky condition $AP \perp MN$, we may think of the property of radical axes. This should be a correct direction since power and radical axes are closely related. This means we should consider some circles with centres M, N or with centres A, P .

Solution.



Let Γ_1 be the circle with centre M passing through B and C , and let Γ_2 be the circle with centre N passing through D and E . As $PB \times PC = PD \times PE$, the point P lies on the radical axis of Γ_1 and Γ_2 . Since $AP \perp MN$, this shows AP is the radical axis of the two circles. In particular, the power of A w.r.t. the two circles are the same.

It is now not hard to guess $AB \times AC$ is the power of A w.r.t. Γ_1 . To verify this (which is indeed a well-known fact since Γ_1 is the circumcircle of $\triangle BIC$ where I is the incentre of $\triangle ABC$), we reflect C in the angle bisector of $\angle BAC$ to get C' . Then the points A, B, C' are collinear. As AM is the internal angle bisector of $\angle BAC$, we have $MC' = MC$, so that C' lies on Γ_1 . Hence, the power of A w.r.t. Γ_1 is

$$AB \times AC' = AB \times AC.$$

Similarly, the power of A w.r.t. Γ_2 is $AD \times AE$. This shows $AB \times AC = AD \times AE$ as desired. $\square$

Remarks. This problem suggests another usage of radical axes. We can use radical axes to handle perpendicular conditions or prove that two lines are perpendicular.

Problem 5.12. (CHKMO 2006 Problem 3) A convex quadrilateral $ABCD$ with $AC \neq BD$ is inscribed in a circle with centre O . Let E be the intersection of diagonals AC and BD . If P is a point inside $ABCD$ such that

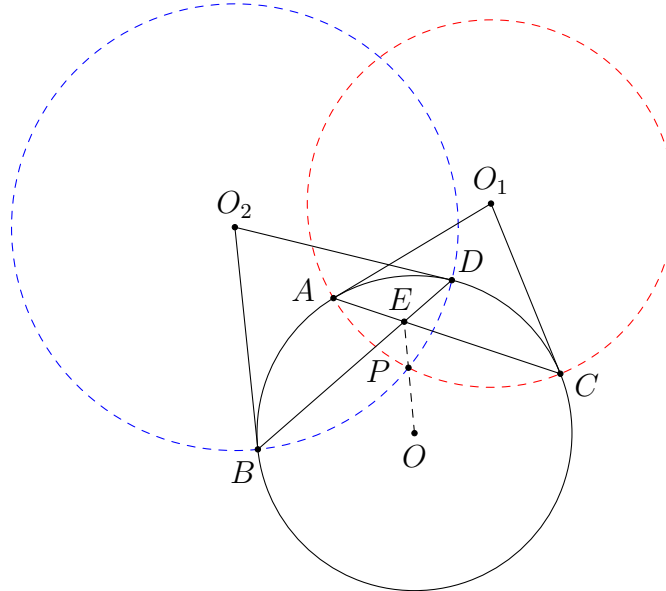
$$\angle PAB + \angle PCB = \angle PBC + \angle PDC = 90^\circ,$$

prove that O, P and E are collinear.

Analysis. Apart from using the perpendicular lines arisen from radical axes, we can prove that three points are collinear by proving all of them lie on the radical axis of two circles. This is an important application of radical axes, and is the strategy that we are going to apply.

As E is the intersection of AC and BD , we may consider two circles Γ_1 and Γ_2 such that Γ_1 passes through A, C while Γ_2 passes through B, D . From the fact that A, B, C, D are concyclic, we can easily deduce E lies on the radical axis of Γ_1 and Γ_2 . A simple way is to take $\Gamma_1 = (ACP)$ and $\Gamma_2 = (BDP)$. We can use the given condition on P to figure out what these circles are.

Solution.



We have

$$\angle APC = \angle(AP, AB) + \angle(AB, BC) + \angle(BC, CP) = \frac{\pi}{2} + \angle ABC.$$

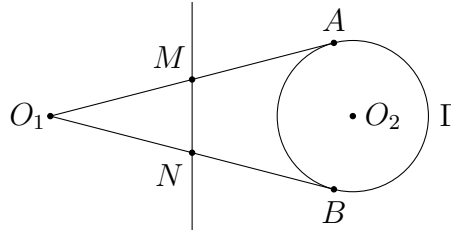
Let O_1 be the centre of (ACP) . Then

$$\angle AO_1C = 2\angle APC = 2\angle ABC = \angle AOC.$$

This shows A, C, O, O_1 are concyclic. As O, O_1 lie on the perpendicular bisector of AC , O_1 is just the intersection of the tangents at A and C to $(ABCD)$. Similarly, let O_2 be the intersection of the tangents at B and D to $(ABCD)$. Then O_2 is the centre of (BDP) .

Clearly, P lies on the radical axis of (ACP) and (BDP) . From $EA \times EC = EB \times ED$, E also lies on this radical axis. Lastly, the power of O w.r.t. the two circles are OA^2 and OB^2 respectively as OA and OB are tangents (from $OA \perp O_1A$ and $OB \perp O_2B$). As $OA = OB$, the point O has equal power w.r.t. the two circles, and hence it lies on the radical axis. Therefore, P, E, O are collinear. $\square$

The concept of radical axes also applies to point circles, which means circles with radii 0. For example, let O_1 be a point. We can regard this point as a circle with centre O_1 and radius 0. By definition, the power of a point P to this point circle is $O_1P^2 - 0^2 = O_1P^2$. Let Γ be a circle with centre $O_2 \neq O_1$. Then we can define the radical axis of the point circle O_1 and Γ as usual, which is a line perpendicular to O_1O_2 .

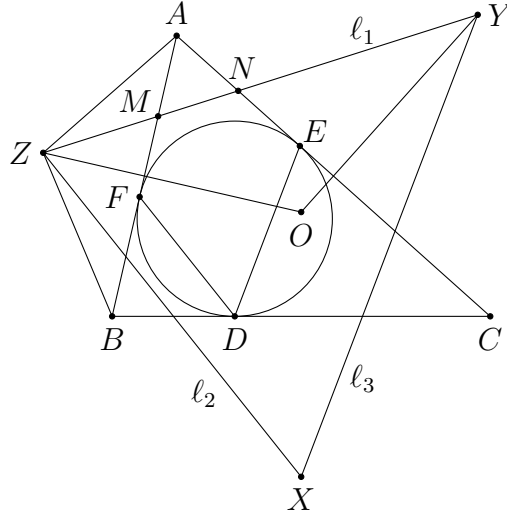


If O_1 lies outside Γ , let A, B be the contact points of the tangents from O_1 to Γ , and let M, N be the midpoints of O_1A, O_1B respectively. As a special case of [proposition 5.1](#), we know that M, N lie on the radical axis, and hence the radical axis is just MN .

Problem 5.13. The incircle of $\triangle ABC$ touches the sides BC, CA, AB at points D, E, F respectively. Let ℓ_1 be the line equidistant to point A and the line EF , let ℓ_2 be the line equidistant to point B and the line FD , and let ℓ_3 be the line equidistant to point C and the line DE . Prove that the circumcentre of the triangle bounded by ℓ_1, ℓ_2 and ℓ_3 coincides with the circumcentre of $\triangle ABC$.

Analysis. As ℓ_1 is equidistant to A and the line EF , it passes through the midpoints of AE and AF . The configuration looks like the one we have just discussed. In particular, we immediately see that ℓ_1 is the radical axis of the point circle A and the incircle. Using the radical axes, we are able to obtain some isosceles triangles, which are sufficient for our purpose.

[Solution.](#)



Let ℓ_1 meet ℓ_2 at Z , ℓ_2 meet ℓ_3 at X , ℓ_3 meet ℓ_1 at Y . Let M and N be the midpoints of AF and AE respectively. Note that M and N lie on ℓ_1 because of the given condition. As discussed, MN is the radical axis of the point circle A and the incircle. Thus, Z lies on this radical axis. Similarly, Z lies on the radical axis of B and the incircle. It follows that Z is the radical centre of A, B and the incircle. This implies $ZA = ZB$.

Let O be the circumcentre of $\triangle ABC$. Then $OA = OB$. Thus, $AZBO$ is a kite, and we have $AB \perp OZ$. Similarly, $AC \perp OY$. Therefore, we find that $\angle YOZ = 180^\circ - A$. Also, as $XY \parallel DE$ and $XZ \parallel DF$, we have

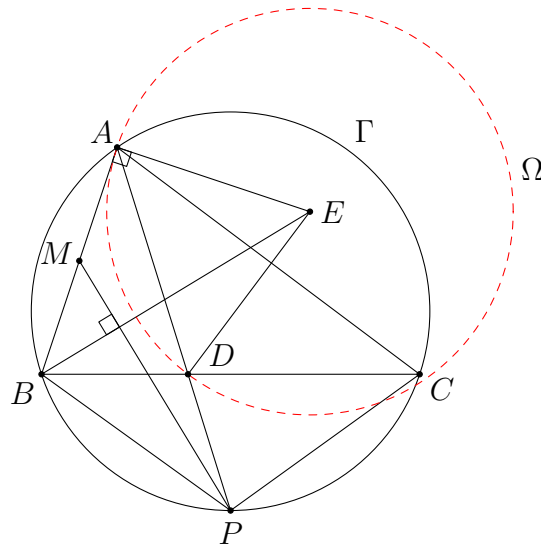
$$\angle YXZ = \angle EDF = 90^\circ - \frac{A}{2} = \frac{1}{2} \angle YOZ.$$

Similarly, we have $2\angle ZYX = \angle ZOY$. Therefore, O is the circumcentre of $\triangle XYZ$. $\square$

Problem 5.14. (IMO HK TST 2021 Test 2 Problem 3) Let $\triangle ABC$ be an acute triangle with circumcircle Γ , and let P be the midpoint of the minor arc BC of Γ . Let AP meet BC at D , and let M be the midpoint of AB . Also, let E be the point such that $AE \perp AB$ and $BE \perp MP$. Prove that $AE = DE$.

Analysis. The condition $BE \perp MP$ is hard to use because the few points are not closely related. One plan is to use radical axis to get rid of this. For example, we may consider two circles with centres B and E respectively. We would like to make a suitable choice of the circles, so that MP is the radical axis. In view of $MA = MB$ and the configuration we have seen, we may consider the point circle B and the circle with centre E and passing through A (with MA being a tangent). This should be

Solution.

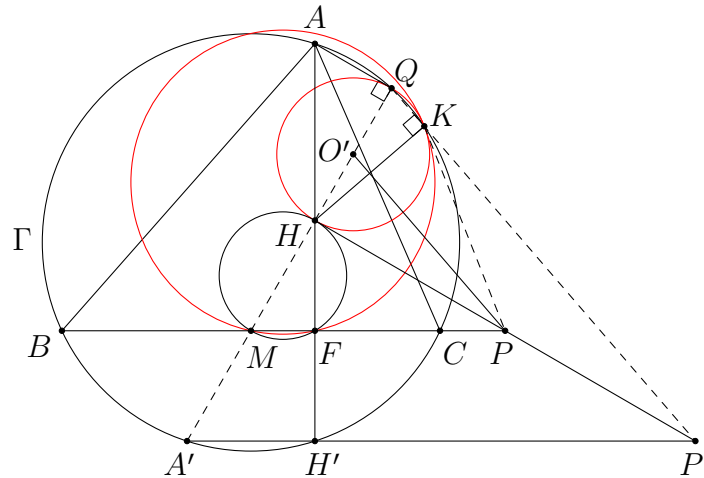


Note that ADP is the angle bisector of $\angle BAC$. Since $\angle PBD = \angle PAC = \angle PAB$, we have $\triangle PBD \sim \triangle PAB$, so $PB^2 = PD \times PA$. Therefore, the power of P w.r.t. Ω is $PD \times PA$. It follows that D lies on Ω , meaning that $AE = DE$. $\square$

Prove that the circumcircles of triangles KQH and FKM are tangent to each other.

There are three circles given in the problem. When there are a lot of circles, it is natural to consider the radical axes and radical centre of some of the circles. However, it is no use to consider the radical centre of Γ , (KQH) and (FKM) since all of them pass through point K . Therefore, we need to consider other circles in addition.

Solution.



By [problem 4.9](#), M, H, Q are collinear. Let P be the radical centre of (KQH) , (FKM) and (HMF) . Note that (KQH) and (HMF) have diameters HQ and HM respectively. Hence, they are tangent to each other and their radical axis is the line ℓ through H perpendicular to HM . Also, P clearly lies on FM , which is the line BC . As PK is the radical axis of (KQH) and (FKM) , if we can show that it is a tangent to (KQH) , then the two circles are tangent at K and we are done.

To find more properties of the point P , we may consider other radical centres. We would like ℓ to be a radical axis of some other circles so that the point P is involved. Also, we have not used Γ yet and probably this time we should consider it.

Let H' and A' be the reflections of H in F and M respectively. Then $\triangle HMF$ and $\triangle HA'H'$ are homothetic to each other. In particular, $\angle HH'A' = 90^\circ$. Let P' be the radical centre of (KQH) , $(HA'H')$ and Γ . Again, the radical axis of (KQH) and $(HA'H')$ is ℓ since the two circles are tangent at H . The other radical axes are clearly $A'H'$ and QK . In other words, P' is the intersection of ℓ , $A'H'$ and QK . As the homothety with centre H and ratio 2 maps the line MF to $A'H'$, we know that P is the midpoint of HP' .

Now, let O' be the centre of (KQH) , so that it is the midpoint of HQ . From $\angle HKP' = 90^\circ$ and $PH = PP'$, we have $PH = PP' = PK$. Together with $O'H = O'K$ and the common side $O'P$, we have $\triangle O'HP \cong \triangle O'KP$. Then $\angle O'KP = 90^\circ$ and hence PK is a tangent to (KQH) . This completes the proof. $\square$

The usages of power and radical axes are summarized below.

- power
 - handling equations involving product of lengths ([problems 5.1, 5.2](#))
 - proving concyclic points ([problem 5.3](#))

- relating tangents and equal lengths ([problems 5.4, 5.5, 5.6, 5.7](#))
 - computing distance from centre of circle ([problem 5.8](#))
- radical axes
 - proving concyclic points ([problem 5.9](#))
 - deducing equal power ([problems 5.9, 5.10, 5.11](#))
 - handling perpendicular conditions ([problems 5.11, 5.14](#))
 - proving collinear points ([problem 5.12](#))
 - using point circles ([problems 5.13, 5.14](#))
- radical centres ([problems 5.13, 5.15](#))

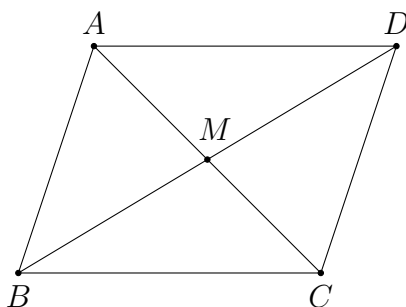
6 Special Quadrilaterals

In this section, we study three kinds of special quadrilaterals which often occurs in Olympiad. Their properties are simple, so the main purpose is to go through some related problems to see how to handle these quadrilaterals.

6.1 Parallelograms

A parallelogram $ABCD$ has all the following properties.

- $AB \parallel DC$, $AD \parallel BC$
- $AB = DC$, $AD = BC$
- $\angle A = \angle C$, $\angle B = \angle D$
- the diagonals AC and BD bisect each other

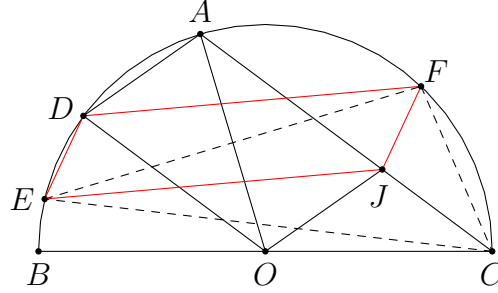


In particular, the last property is the most important fact that characterizes a parallelogram. In other words, if two segments AC and BD meet at M such that $MA = MC$ and $MB = MD$, then $ABCD$ is a parallelogram. Sometimes it is useful to construct some parallelograms to handle the midpoint condition in the question.

Problem 6.1. (IMO Shortlist 2002 G3/Problem 2) BC is a diameter of a circle with centre O . A is any point on the circle with $\angle AOC > 60^\circ$. EF is the chord which is the perpendicular bisector of AO . D is the midpoint of the minor arc AB . The line through O parallel to AD meets AC at J . Show that J is the incentre of triangle CEF .

Analysis. It is not hard to see that $AEOF$ is a rhombus with interior angles 60° and 120° . Also, it can be deduced that $ADOJ$ is a parallelogram. These show $DEJF$ is also a parallelogram as the diagonals of a parallelogram bisect each other. This is the crucial observation for this problem. The angles related to EJ and FJ can then be evaluated. We can prove the result based on the definition of the incentre.

Solution.



Without loss of generality(WLOG) we may assume B, E, F, C lie in this order on the circle (by renaming E and F if necessary). As EF is the perpendicular bisector of AO and $OE = OA = OF$, $AEOF$ is a rhombus where both $\triangle AEO$ and $\triangle AFO$ are equilateral. Next, from $\angle DOB = \frac{1}{2} \angle AOB = \angle ACB$, we have $DO \parallel AC$. Also, it is given that $AD \parallel JO$. So $ADOJ$ is a parallelogram. Then the midpoint of DJ coincides with the midpoint of AO , which is also the midpoint of EF as $AEOF$ is a rhombus (which is a special case of a parallelogram). This shows $DEJF$ is a parallelogram.

We now evaluate the angles as follows:

$$\begin{aligned}
 \angle(FE, FJ) + \angle(FC, FJ) &= \angle(FE, DE) + \angle(FC, DE) \\
 &= \angle FED + \angle CFD + \angle FDE \\
 &= \angle FAD + \angle CBD + \angle FAE \\
 &= \left(-\frac{\pi}{3} + \angle OAD\right) + \angle OBD - \frac{2\pi}{3} = 0, \\
 \angle(EC, EJ) + \angle(EF, EJ) &= \angle(EC, DF) + \angle(EF, DF) \\
 &= \angle CED + \angle EDF + \angle EFD \\
 &= \angle CBD + \angle EAF + \angle EAD \\
 &= \angle OBD + \frac{2\pi}{3} + \left(\angle OAD + \frac{\pi}{3}\right) = 0.
 \end{aligned}$$

These mean FJ is an angle bisector of $\angle EFC$ and EJ is an angle bisector of $\angle CEF$. From the configuration, these are the internal angle bisectors. So J is the incentre of $\triangle CEF$. $\square$

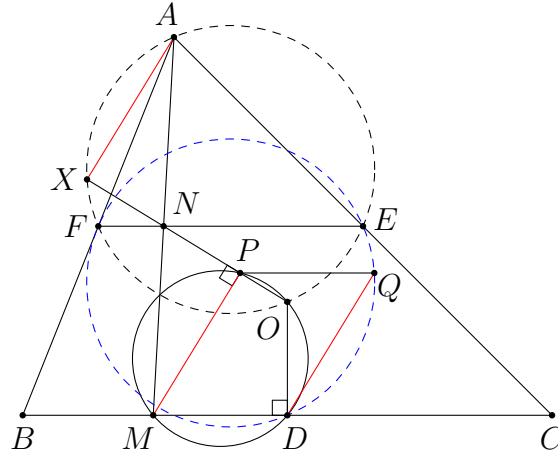
Problem 6.2. (Balkan MO Shortlist 2017 G4) The acute-angled triangle ABC with circumcentre O is given. The midpoints of the sides BC, CA and AB are D, E and F respectively. An arbitrary point M on the side BC , different of D , is chosen. The straight lines AM and EF intersects at the point N and the straight line ON cut again the circumscribed circle of the triangle ODM at the point P . Prove that the reflection of the point M with respect to the midpoint of the segment DP belongs on the nine-point circle of the triangle ABC .

Analysis. It is clear that $PMDQ$ is a parallelogram. Apart from this, there are many more parallelograms in the figure, such as $AFDE$, $BDEF$ and $CDFE$.

There are two given conditions which are hard to use. The first one is the collinearity of A, N, M . The simplest corollary of this is $AN = NM$, which is still unrelated to other conditions in this question. The second one is the concyclicity of M, D, O, P . We can deduce $\angle MPO = 90^\circ$ from this, which is again hard to be used directly.

The key idea in the following solution is to introduce the point X such that $AXMP$ is a parallelogram. The first reason of doing so is that the condition $AN = NM$ is closely related to this parallelogram, as it means N is the intersection of the diagonals. The second reason is that we can ‘move’ the right angle $\angle MPO$ to $\angle AXO$, which can be combined with some other right angles in the figure to yield some concyclic points.

Solution.



Let X be the point such that $AXMP$ is a parallelogram. Since FE is equidistant to A and BC , we have $AN = NM$. Therefore, N is the midpoint of XP by the property of parallelograms.

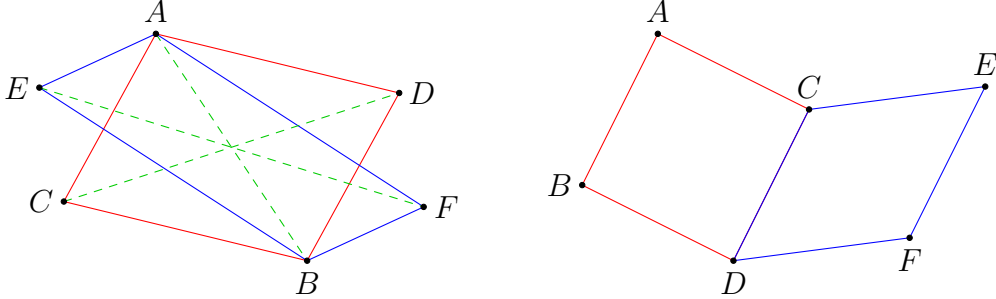
Note that $\angle MPO = 180^\circ - \angle MDO = 90^\circ$. As $AX \parallel PM$, we have $\angle AXO = 90^\circ$. Since $\angle AEO = \angle AFO = 90^\circ$, the points A, X, F, O, E are concyclic.

Next, observe that AX, PM, QD have the same length and direction since both $AXMP$ and $PMDQ$ are parallelograms. Therefore, $AXDQ$ is another parallelogram. In addition, $AFDE$ is a parallelogram since $AF \parallel ED$ and $AE \parallel FD$ by the midpoint theorem. This shows FE, AD, XQ share the same midpoint. Thus, by reflecting A, X, F, E in this common midpoint, we see that their images D, Q, E, F are concyclic, which means Q lies on the nine-point circle of $\triangle ABC$. $\square$

Remarks. As can be seen in the solutions of the previous two problems, the following gives some common ways of using parallelograms.

- If $ACBD$ and $AEBF$ are parallelograms, then AB, CD, EF share the same midpoint, and hence $CFDE$ is a parallelogram.

- If $ABDC$ and $CDFE$ are parallelograms, then AB, CD, EF have the same length and direction, and hence $ABFE$ is a parallelogram.



Problem 6.3. (IMO Shortlist 2010 G5) Let $ABCDE$ be a convex pentagon such that $BC \parallel AE$, $AB = BC + AE$, and $\angle ABC = \angle CDE$. Let M be the midpoint of CE , and let O be the circumcentre of triangle BCD . Given that $\angle DMO = 90^\circ$, prove that $2\angle BDA = \angle CDE$.

Analysis. The question provides many unrelated conditions. The main idea in the following solution is to construct some extra points in the figure to bring all conditions together. To use the midpoint condition $MC = ME$, we again think of the parallelogram trick.

Firstly, we can define the point F such that $FCDE$ is a parallelogram. Then M is also the midpoint of DF . The perpendicular condition $OM \perp DF$ can be used to show that F lies on (BCD) .

Secondly, we can define the point P such that $PEBC$ is a parallelogram. Then we have $BC = EP$. The condition $AB = BC + AE$ now becomes a much simpler condition $AB = AP$.

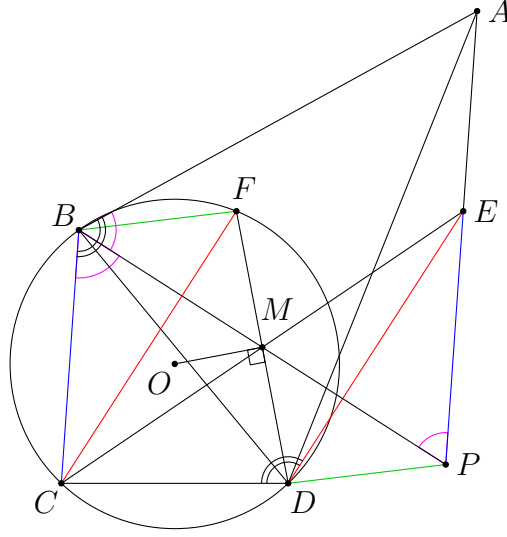
Using these, it is not hard to see that BP bisects $\angle ABC$. This gives us a way to represent $\frac{1}{2}\angle CDE$ as one of the angles in the figure, which is necessary to prove the final assertion.

Solution. Let F be the point such that $FCDE$ is a parallelogram. As M is the midpoint of CE , it is also the midpoint of DF . Since $OM \perp DF$, we have $OD = OF$, and hence F lies on (BCD) .

Let P be the point such that $PEBC$ is a parallelogram. Note that P lies on AE since $BC \parallel AE$. Then $AB = BC + AE = EP + AE = AP$. Using this, we obtain $\angle PBA = \angle APB = \angle CBP$.

Next, as the midpoints of DF and BP coincide, $BDPF$ is a parallelogram. This implies

$$\angle PDB = 180^\circ - \angle DBF = 180^\circ - \angle DCF = \angle EDC = \angle CBA = 180^\circ - \angle BAP.$$



Therefore, A, B, D, P are concyclic. It follows that

$$\angle ADB = \angle APB = \frac{1}{2} \angle CBA = \frac{1}{2} \angle EDC.$$

This completes the proof. $\square$

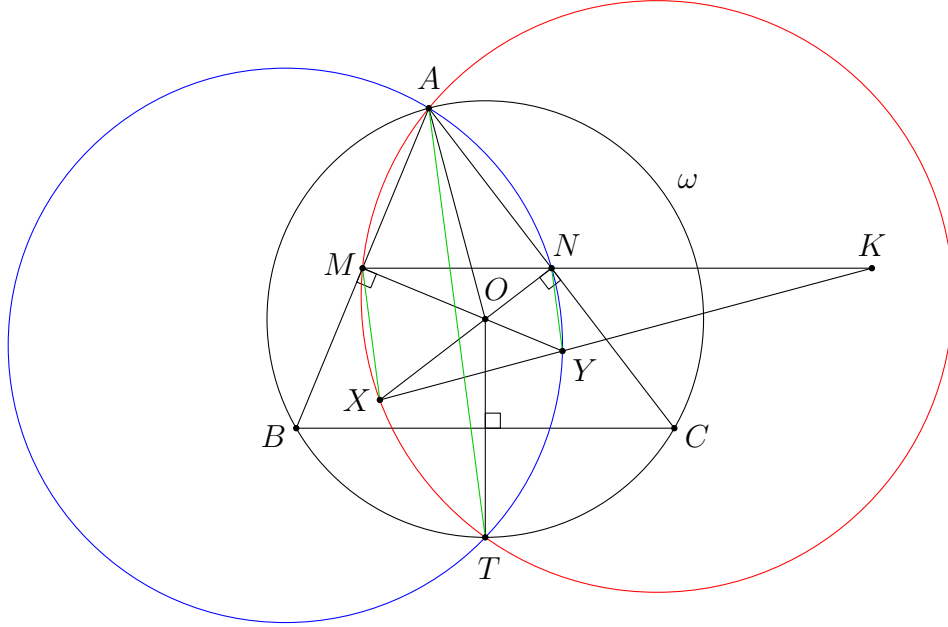
6.2 Isosceles Trapezoids

An isosceles trapezoid has the property that it is cyclic and has an axis of symmetry. This allows us to move the angles and lengths to different positions if an isosceles trapezoid exists. Thus, introducing a new point to construct an isosceles trapezoid is another possible approach to solve a geometry problem.

Problem 6.4. (IMO Shortlist 2013 G2) Let ω be the circumcircle of a triangle ABC . Denote by M and N the midpoints of the sides AB and AC , respectively, and denote by T the midpoint of the arc BC of ω not containing A . The circumcircles of the triangles AMT and ANT intersect the perpendicular bisectors of AC and AB at points X and Y , respectively; assume that X and Y lie inside the triangle ABC . The lines MN and XY intersect at K . Prove that $KA = KT$.

Analysis. Isosceles trapezoids sometimes appear in problems involving cyclic points and perpendicular bisectors. As the assertion $KA = KT$ means K lies on the perpendicular bisector of AT , we may wish to see if there are some isosceles trapezoids with AT as one of the bases. Sometimes we have to make such guesses when solving geometry problems. Very often our guesses are wrong, but if we can make one such correct guess, we are likely to solve the whole problem.

Solution.



Let O be the circumcentre of $\triangle ABC$. Note that O lies on both NX and MY , and $OT \perp BC$. We have

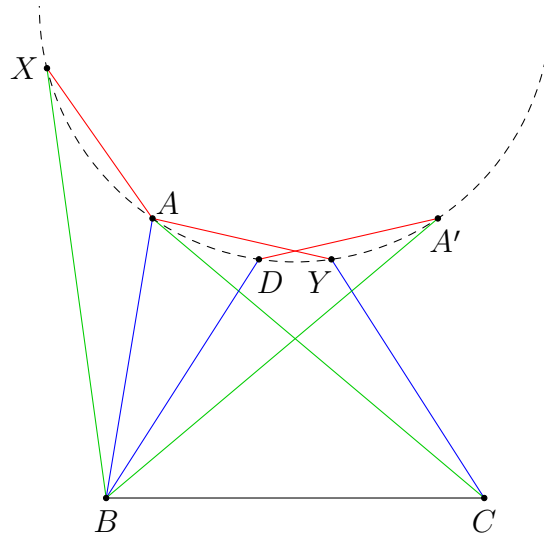
$$\angle MOT = 180^\circ - B = 180^\circ - \angle NOA = \angle AOX.$$

Since O lies on the perpendicular bisector ℓ of AT , the lines OA and OT are symmetric about ℓ . Thus, the lines OX and OM are symmetric about ℓ . This implies $MXTA$ is an isosceles trapezoid with ℓ being the axis of symmetry. Similarly, AT, MX, NY share the same perpendicular bisector ℓ . By symmetry, the intersection point K of MN and XY lies on ℓ . Therefore, $KA = KT$. $\square$

Problem 6.5. (Thailand Online MO 2021 Problem 7) Let ABC be an acute triangle. Construct a point X on the different side of C with respect to the line AB and construct a point Y on the different side of B with respect to the line AC such that $BX = AC$, $CY = AB$, and $AX = AY$. Let A' be the reflection of A across the perpendicular bisector of BC . Suppose that X and Y lie on different sides of the line AA' , prove that points A, A', X , and Y lie on a circle.

Analysis. Obviously, there is a pair of congruent triangles in the figure. However, the interior angles of these two triangles cannot be easily related. In view of the isosceles trapezoid $ABCA'$, we can reflect the segment AC to the position $A'B$, which is more related to XB . In addition, by introducing another isosceles trapezoid $BCYD$, we can move more segments to suitable positions to compare the lengths.

Solution.



Let D be the symmetric point of Y about the perpendicular bisector of BC . Then we have

$$\begin{aligned} AX &= AY = DA', \\ AB &= CY = DB, \\ BX &= AC = BA'. \end{aligned}$$

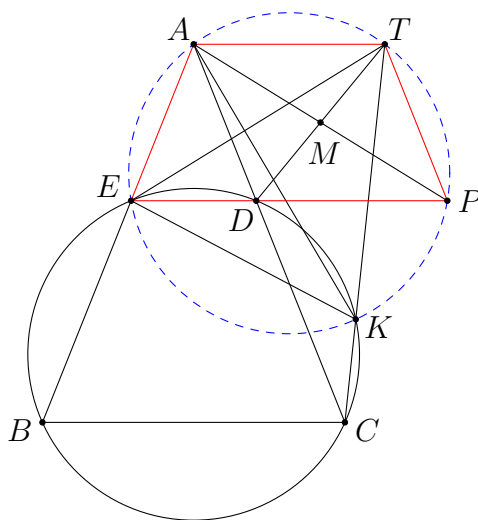
Therefore, $\triangle BAX \cong \triangle BDA'$, and the orientation is reversed. By symmetry, it is clear that $XADA'$ is an isosceles trapezoid. In particular, X, A, D, A' are concyclic. In addition, $ADY A'$ is an isosceles trapezoid, and so A, D, Y, A' are concyclic. Hence, X, A, D, Y, A' are concyclic, and we are done. $\square$

Problem 6.6. (All-Russian MO 2019 Grade 11 Problem 6) In the segment AC of an isosceles triangle $\triangle ABC$ with base BC is chosen a point D . On the smaller arc CD of the circumcircle of $\triangle BCD$ is chosen a point K . Line CK intersects the line through A parallel to BC at T . M is the midpoint of segment DT . Prove that $\angle AKT = \angle CAM$.

Analysis. To use the midpoint M , the first idea that comes to our mind is the parallelogram trick. By constructing a parallelogram $ADPT$, we are able to deduce $\angle CAM = \angle TPA$. The assertion is the same as showing A, K, P, T are concyclic. It is still hard to prove this directly, since K and P are not at all related.

This time we shall construct an isosceles trapezoid to add more points on the same circle. Then we can change our goal to prove some other groups of concyclic points. Sometimes the problem can be made simpler by moving the angles to different positions.

Solution.



Let P be the point such that $ADPT$ is a parallelogram. Then M is the midpoint of AP . Let E be the point such that $AEPT$ is an isosceles trapezoid with $AT \parallel EP$. Since $AE = TP = AD$ and $ED \parallel AT \parallel BC$, the point E lies on AB , and $BCDE$ is an isosceles trapezoid. In particular, E lies on $(BCKD)$.

Since B, C, K, E are concyclic, and $BC \parallel AT$, the points A, E, K, T are concyclic by Reim's theorem (see remark). Note that A, E, P, T are concyclic as $AEPT$ is an isosceles trapezoid. It follows that A, E, K, P, T are concyclic. Finally, we have

$$\angle AKT = \angle APT = \angle DAP = \angle CAM.$$

☐

Remarks. (Reim's theorem(雷姆定理)) Let A, E, B be collinear points, and let T, K, C be collinear points. If B, C, K, E are concyclic, and E, K, T, A are concyclic, then $AT \parallel BC$. Conversely, if B, C, K, E are concyclic, and $AT \parallel BC$, then E, K, T, A are concyclic.

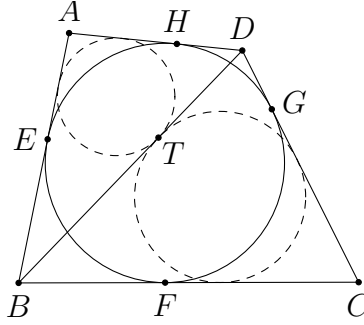
It is very easy to prove this theorem by simple angle chasing.

6.3 Circumscribed Quadrilaterals

Lastly, we introduce *circumscribed quadrilaterals*(圓外接四邊形). It means a quadrilateral which has an incircle. Problems involving circumscribed quadrilaterals are essentially problems involving a circle and many tangents. Apart from knowledge in circles and tangents, we introduce some more useful facts below.

Proposition 6.1. Let $ABCD$ be a circumscribed quadrilateral. Then the following hold.

- (a) (Pitot theorem(皮托定理)) $AB + CD = AD + CB$
- (b) the incircles of $\triangle ABD$ and $\triangle CDB$ touch each other on BD



Proof.

- (a) Let E, F, G, H be the contact points of the incircle of $ABCD$ and the sides AB, BC, CD, DA respectively. Using the tangents, we have

$$\begin{aligned} AB + CD &= (AE + EB) + (CG + GD) = AH + FB + CF + HD \\ &= (AH + HD) + (CF + FB) = AD + CB. \end{aligned}$$

- (b) Let the incircle of $\triangle ABD$ touch BD at T_1 and that of $\triangle CDB$ touch BD at T_2 . Then

$$BT_1 = \frac{BA + BD - AD}{2} = \frac{BC + BD - CD}{2} = BT_2$$

by Pitot's theorem. Hence, $T_1 = T_2$ as desired.

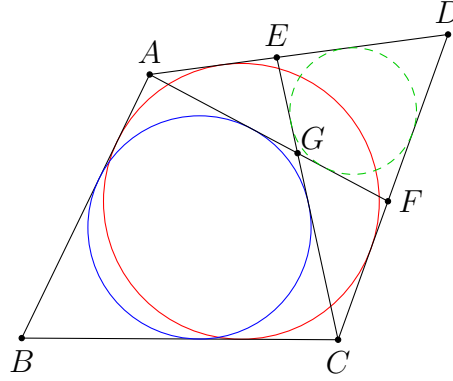
□

By false position (see section 10), one can show that each of (a) and (b) is a sufficient condition for $ABCD$ to have an incircle. This gives a practical way to show that $ABCD$ is circumscribed. Another straightforward way is to prove that the angle bisectors of $\angle A, \angle B, \angle C, \angle D$ are concurrent.

Problem 6.7. Let $ABCD$ be a circumscribed quadrilateral. Let E and F be points on the sides AD and CD respectively. Let G be the intersection point of AF and CE . If $ABCG$ is a circumscribed quadrilateral, prove that $DEGF$ is a circumscribed quadrilateral.

Analysis. It is not at all easy to prove the assertion by using the definition of circumscribed quadrilaterals only. We shall provide the usage of the Pitot theorem below. Note that there are some variations of the theorem. In the following, we apply the converse of the theorem to a concave quadrilateral.

Solution.



Since $ABCD$ and $ABCG$ are circumscribed, by Pitot's theorem, we have

$$AB + CD = BC + AD, \quad (1)$$

$$AB + CG = BC + AG. \quad (2)$$

Taking the difference of (1) and (2), we obtain

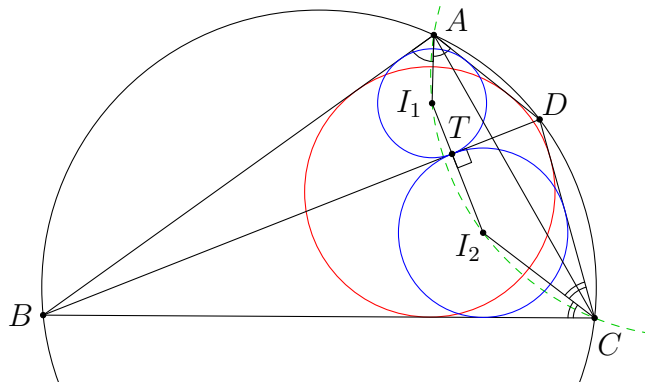
$$CD + AG = AD + CG.$$

This implies the concave quadrilateral $AGCD$ has an incircle, which means $DEGF$ is a circumscribed quadrilateral. $\square$

Problem 6.8. Let $ABCD$ be a bicentric quadrilateral. Let I_1 and I_2 be the incentres of $\triangle ABD$ and $\triangle BCD$ respectively. Prove that C, A, I_1, I_2 are concyclic.

Analysis. A *bicentric quadrilateral* (雙心四邊形) is a quadrilateral which is both cyclic and circumscribed. It is quite clear that this problem is related to [proposition 6.1\(b\)](#). After noticing $I_1 I_2 \perp BD$, the rest of the solution is just angle chasing.

Solution.



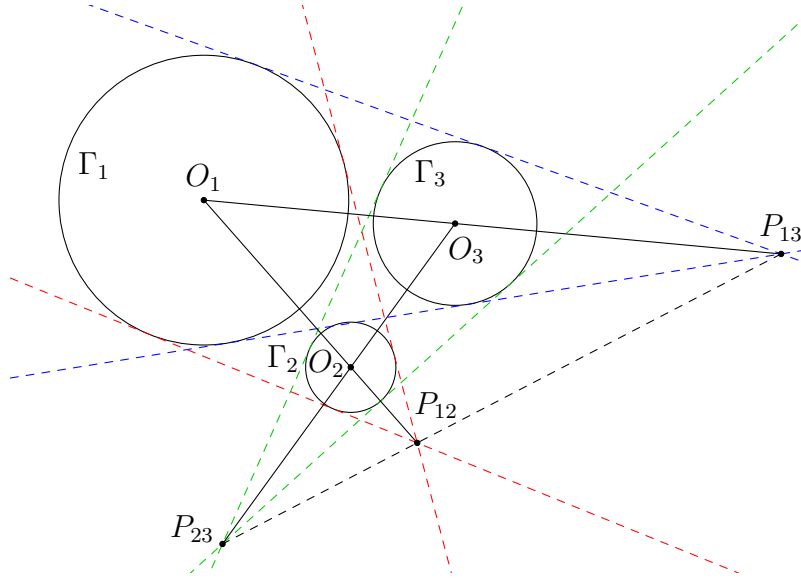
By [proposition 6.1\(b\)](#), we have $I_1I_2 \perp BD$. Let T be the intersection point of I_1I_2 and BD . Then we have

$$\begin{aligned}\angle I_1AC + \angle I_1I_2C &= \left(\frac{A}{2} - \angle CAD\right) + \left(360^\circ - 90^\circ - \angle BDC - \frac{C}{2}\right) \\ &= \frac{180 - C}{2} - \angle CBD + 270^\circ - \angle BDC - \frac{C}{2} \\ &= 360^\circ - C - \angle CBD - \angle BDC \\ &= 180^\circ.\end{aligned}$$

This proves C, A, I_1, I_2 are concyclic. $\square$

Before we move on to the next problem, we introduce one more result about circles which will be used.

Theorem 6.1. ([Monge's theorem](#)(蒙日定理)) Let $\Gamma_1, \Gamma_2, \Gamma_3$ be three circles on the plane such that no two of them have the same centre or the same radius. Denote by P_{ij} the external homothetic centre of Γ_i and Γ_j for $1 \leq i < j \leq 3$. Then P_{12}, P_{23}, P_{13} are collinear.



Proof. An external homothetic centre between two circles means the centre of homothety mapping one circle to the other with positive ratio.

Let O_1, O_2, O_3 be the centres of $\Gamma_1, \Gamma_2, \Gamma_3$ and let r_1, r_2, r_3 be the radii of $\Gamma_1, \Gamma_2, \Gamma_3$ respectively. Using directed lengths, we have $\frac{O_i P_{ij}}{O_j P_{ij}} = \frac{r_i}{r_j}$. Hence,

$$\frac{O_1 P_{12}}{P_{12} O_2} \times \frac{O_2 P_{23}}{P_{23} O_3} \times \frac{O_3 P_{13}}{P_{13} O_1} = \left(-\frac{r_1}{r_2}\right) \left(-\frac{r_2}{r_3}\right) \left(-\frac{r_3}{r_1}\right) = -1.$$

By Menelaus' theorem, P_{12}, P_{23}, P_{13} are collinear. $\square$

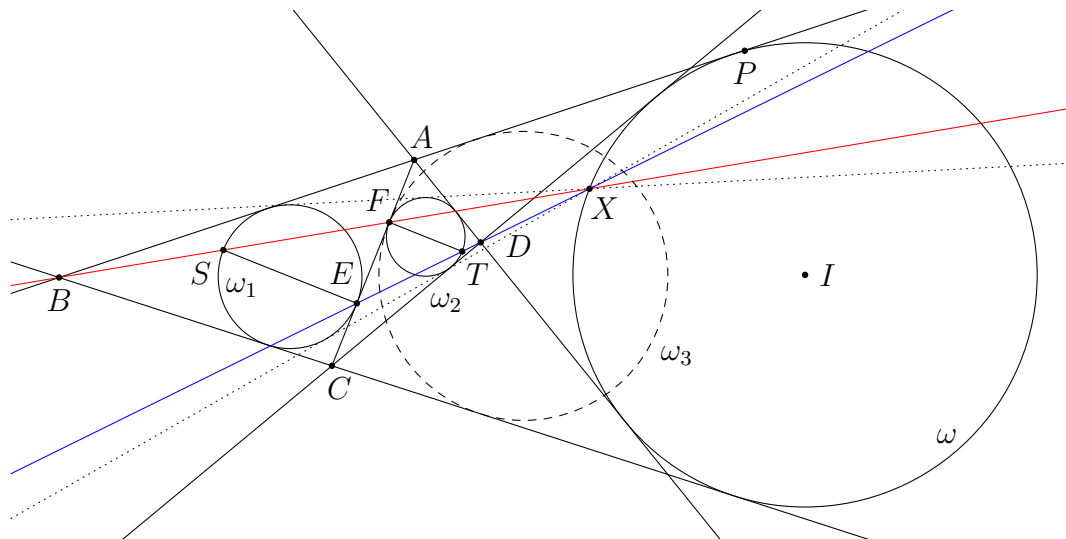
The external homothetic centre between two circles is the intersection of the two external common tangents, if they exist. Similarly, one can define the internal homothetic centre. There are at most two homothetic centres mapping one circle to the other. Monge's theorem also applies if two of the external homothetic centres are replaced by internal homothetic centres.

Problem 6.9. (IMO Shortlist 2008 G7/Problem 6) Let $ABCD$ be a convex quadrilateral with $|BA| \neq |BC|$. Denote the incircles of triangles ABC and ADC by ω_1 and ω_2 respectively. Suppose that there exists a circle ω tangent to the ray BA beyond A and to the ray BC beyond C , which is also tangent to the lines AD and CD . Prove that the common external tangents of ω_1 and ω_2 intersect on ω .

Analysis. Although $ABCD$ is not a circumscribed quadrilateral, it has an excircle which touches all its sidelines. Therefore, it should share some similar properties with circumscribed quadrilaterals. In particular, we can deduce similar results to [proposition 6.1](#).

Firstly, we can show that the contact points of ω_1 and ω_2 to AC are symmetric about the midpoint of AC . This reminds us the relation between the incircle and an excircle of a triangle. For example, we know that the B -excircle of $\triangle BCA$ is tangent to ω_2 . Since there are many circles and we have to consider the external homothetic centre X of ω_1 and ω_2 , we can use Monge's theorem. This gives some information about X , and we can use properties of homothety to complete the proof.

Solution.



Similar to the proof of the Pitot theorem, we get $BC + CD = BA + AD$. Let ω_1 touch AC at E and let ω_2 touch AC at F . Again, following the proof of [proposition 6.1\(b\)](#), points E and F are symmetric about the midpoint of AC . Hence, the B -excircle ω_3 of $\triangle BCA$ is internally tangent to ω_2 at F .

Let X be the intersection of the external common tangents of ω_1 and ω_2 . Apply Monge's theorem to $\omega_1, \omega_2, \omega_3$. Then B, F, X are collinear. Note that the line BF also passes through point S , where SE is a diameter of ω_1 , by [proposition 4.2](#). Due to symmetry, if we define point T such that TF is a diameter of ω_2 , then E, T, D, X are collinear. In other words, X is the intersection of BF and DE .

Let X_1 be the first intersection of the ray BF and ω . Then the homothety with centre B mapping ω_1 to ω sends S to X_1 . The tangent at X_1 to ω is hence parallel to the tangent at S to ω_1 , which is parallel to CA . In the same way, if X_2 is the first intersection of the ray ED and ω , then the tangent at X_2 to ω is also parallel to CA . This means $X_2 = X_1$ as both points lie on the same semicircle of ω determined by the diameter parallel to CA . Therefore, $X = X_2 = X_1$ lies on ω . $\square$

Here are the techniques and topics covered in this section.

- finding parallelograms ([problems 6.1, 6.2](#))
- constructing parallelograms ([problems 6.2, 6.3, 6.6](#))
- finding isosceles trapezoids ([problems 6.4, 6.5](#))
- constructing isosceles trapezoids ([problems 6.5, 6.6](#))
- quadrilaterals with incircles/excircles ([problems 6.7, 6.8, 6.9](#))
- pitot theorem ([problems 6.7, 6.9](#))

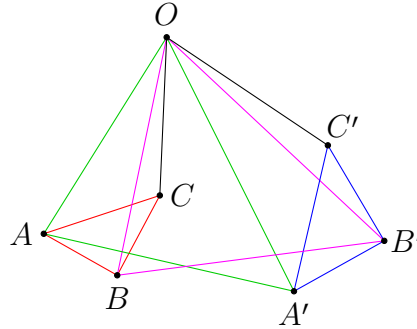
7 Spiral Similarity and Miquel Point

Spiral similarity is another kind of geometric transformations. It is commonly used because it is closely related to similar triangles. In particular, it can be used to derive some properties of the Miquel point of quadrilaterals. We are going to cover these two topics in this section.

7.1 Spiral Similarity

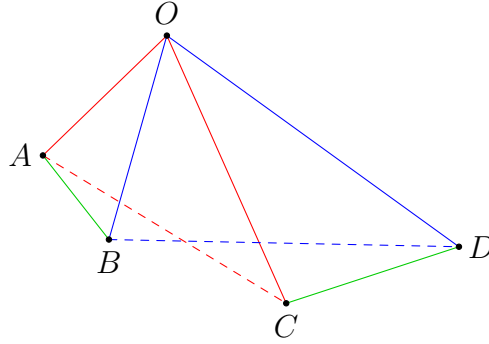
The transformation *spiral similarity*(旋轉相似) is the composition of a rotation and a homothety with the same centre. We first fix a centre O , an angle θ and a ratio $r > 0$. Every point A on the plane is first rotated about O by angle θ , and then dilated from O by ratio r .

Usually, we restrict the rotation angle θ to be nonzero, or otherwise, the resultant transformation is just a homothety (since there is no rotation). The condition that the ratio of homothety r being positive is not necessary, since a homothety with ratio -1 is the same as a rotation by 180° .



Spiral similarity preserves angles with the same orientation and preserves ratio of side lengths, and the image is similar to the original figure. This is because both rotation and homothety preserve these things. In particular, if A', B', C' are the images of A, B, C under a spiral similarity with centre O , then $\triangle A'B'C'$ and $\triangle ABC$ are directly similar. In addition, $\triangle OAB$ is directly similar to $\triangle OA'B'$, and $\triangle OAC$ is directly similar to $\triangle OA'C'$. This shows there are many pairs of similar triangles if a spiral similarity exists, suggesting the usefulness of this concept. The following provides an alternative way to formulate the results mentioned.

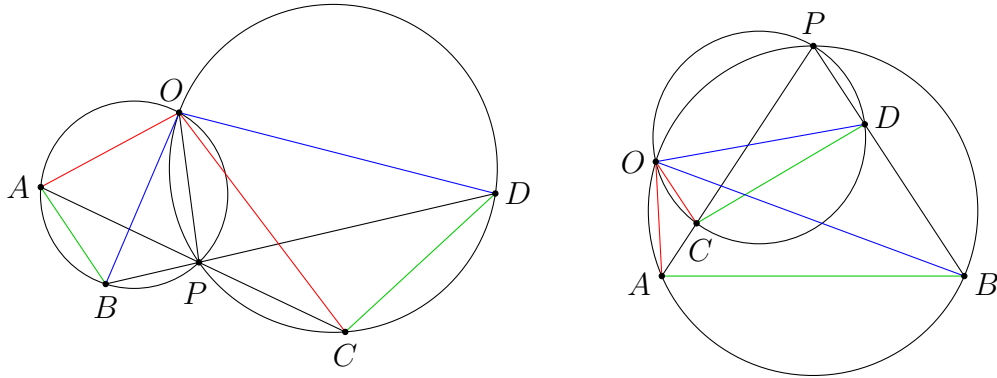
Proposition 7.1. Let O, A, B, C, D be distinct points on the plane. Then $\triangle OAB$ and $\triangle OCD$ are directly similar iff $\triangle OAC$ and $\triangle OBD$ are directly similar.



In other words, O is the centre of spiral similarity that sends A to C and B to D respectively iff O is the centre of spiral similarity that sends A to B and C to D respectively.

To find the centre of spiral similarity that sends two given points to two given images, we can use the following well-known result.

Proposition 7.2. Let A, B, C, D be distinct points on the plane such that AC and BD are not parallel, AB and CD are not parallel. Suppose the lines AC and BD intersect at P . Then the second intersection O of (PAB) and (PCD) is the unique centre of spiral similarity that sends A to C and B to D respectively.



Proof. We can always take A to C using a suitable rotation and homothety with centre O . It suffices to prove that the same transformation also takes B to D . In other words, we just need to prove $\triangle OAC$ and $\triangle OBD$ are directly similar.

Indeed, from the concyclic points, we have

$$\angle OAC = \angle OAP = \angle OBP = \angle OBD$$

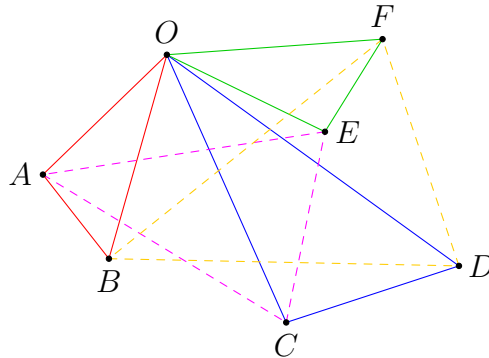
and

$$\angle OCA = \angle OCP = \angle ODP = \angle ODB.$$

Hence, $\triangle OAC$ and $\triangle OBD$ are directly similar as desired. To show that such a centre O is unique, we just need to observe that $\triangle OAC$ and $\triangle OBD$ being directly similar is equivalent to the centre of spiral similarity lying on both (PAB) and (PCD) . If this is the point P , then the transformation is just a homothety (when AB and CD are parallel), and hence is rejected. So O is the unique point satisfying the condition. $\square$

As a corollary of the proof, the fact that O, A, B, P (or O, C, D, P) are concyclic is another useful property of spiral similarity. **Proposition 7.2** includes the limiting case when, for example, P and B coincide (this occurs when A, B, C are collinear). In that case, the circumcircle of $\triangle PAB$ refers to the circle which passes through A and is tangent to BD at B . Thus, the angle $\angle OBP$ is just $\angle(OB, BD)$.

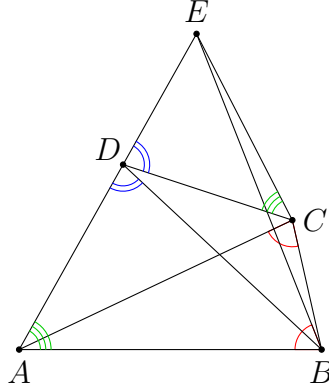
Next, suppose O is the centre of spiral similarity that sends AB to CD . Another spiral similarity with the same centre O and possibly a distinct ratio sends CD to EF . Then $\triangle ACE \sim \triangle BDF$. This can be easily proved via the known similarities from the transformations. The most useful corollary of this fact is that when A, C, E are collinear, then we immediately know that B, D, F are collinear. In general, to deduce similar results, one just needs to bear in mind that there are many groups of similarities in the figure. One can include more corresponding points of the similar figures to deduce more similar triangles.



Problem 7.1. Let D be a point on the side AE of $\triangle ABE$, and let C be a point inside $\angle BAE$. Suppose $\angle ADB = \angle CDE$, $\angle BAD = \angle ECD$ and $\angle ACB = \angle EBA$. Prove that B, C, E are collinear.

Analysis. When all equal angles are marked in the figure, it can be easily seen that $\triangle DAB$ and $\triangle DCE$ are directly similar. As they have a common point D , we can use **proposition 7.1** to deduce another pair of similar triangles. Then we can use the equal angles deduced from this to complete the proof easily. One simple way to show that B, C, E are collinear is to prove $\angle ECB = 180^\circ$.

Solution.



From the configuration, we have $\angle BAD = \angle ECD$ and $\angle ADB = \angle CDE$. Hence, $\triangle DAB$ and $\triangle DCE$ are directly similar. By [proposition 7.1](#), $\triangle DAC$ and $\triangle DBE$ are also directly similar. In particular, $\angle DCA = \angle DEB$. Hence,

$$\angle ECB = \angle ECD + \angle DCA + \angle ACB = \angle BAE + \angle AEB + \angle EBA = 0.$$

This shows B, C, E are collinear. □

Remarks. The use of directed angles is not necessary, since the assertion is only true under this configuration, i.e. when C lies inside $\angle BAE$. All angles can be written in the usual forms, so that we can deduce $\angle ECB = 180^\circ$ instead of 0° .

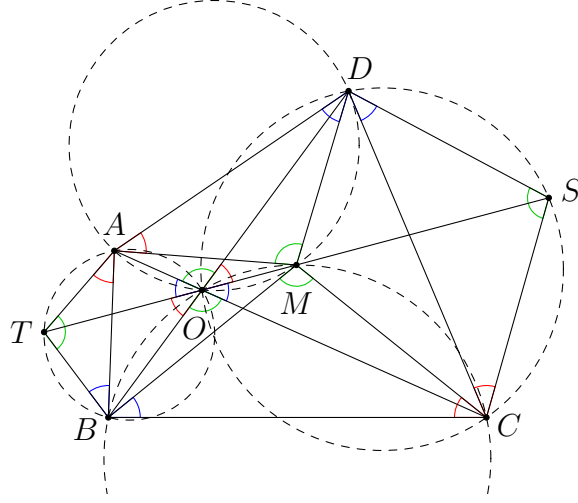
However, one should note that [proposition 7.1](#) can only be used when the pair of similar triangles has the same orientation. This means the configuration should be taken into account if directed angles are not used.

Problem 7.2. (CGMO 2006 Problem 2) Let O be the intersection of the diagonals of a convex quadrilateral $ABCD$. The circumcircles of $\triangle OAD$ and $\triangle OBC$ meet at O and M . Line OM meets the circumcircles of $\triangle OAB$ and $\triangle OCD$ at T and S respectively. Prove that M is the midpoint of ST .

Analysis. This problem can be solved without having any knowledge about spiral similarity. However, knowing such concepts may help us deduce more pairs of similar triangles easily. For example, in view of [proposition 7.2](#), we immediately observe that M is the centre of spiral similarity that takes BC to DA . There are many more spiral similarities in this figure using A, B, C, D as centres.

To observe similar triangles clearly from the concyclic points, we may try to mark the equal angles with the same colour. Drawing an accurate figure is also highly recommended.

Solution.



From the concyclic points, we have

$$\angle ABT = \angle AOT = \angle AOM = \angle ADM$$

and

$$\angle BTA = \angle BOA = \angle DOA = \angle DMA.$$

Hence, $\triangle ABT$ and $\triangle ADM$ are directly similar. By [proposition 7.1](#), $\triangle AMT$ and $\triangle ADB$ are also directly similar (one may prove this directly from the corresponding equal angles). We look for similar triangles containing the sides MT or MS in order to express these side lengths by the others. Indeed, $\triangle AMT \sim \triangle ADB$ implies $\frac{AM}{MT} = \frac{AD}{DB}$, i.e. $MT = AM \times \frac{DB}{AD}$.

However, the expression for MT found above is not suitable to compare with the one we are going to deduce for MS . The only method that works seems to be trigonometry, which is not the focus of this note. (Yet we should use any method we can think of in a competition!)

One useful trick is to seek for a more symmetric expression for MT . Indeed, the above deduction is only based on $(ATBO)$ and $(AOMD)$. Using similar arguments, we obtain another formula for MT by considering $(ATBO)$ and $(OBCM)$, namely, $MT = BM \times \frac{CA}{BC}$ (we just need to change the roles of A, B and D, C in the first formula). This is certainly not a symmetric formula. But we can obtain one by writing

$$MT = \sqrt{MT^2} = \sqrt{AM \times \frac{DB}{AD} \times BM \times \frac{CA}{BC}}.$$

Again by symmetry, we have

$$MS = \sqrt{DM \times \frac{AC}{DA} \times CM \times \frac{BD}{CB}}.$$

Clearly, $MS = MT$ once we have $AM \times BM = CM \times DM$. This is true since $\triangle MAD$ and $\triangle MCB$ are directly similar either by [proposition 7.2](#) or from the figure.

For the sake of completeness, we shall prove that S and T are distinct points (so that $MS = MT$ implies M is the midpoint of ST). Indeed, we first note that M cannot lie in $\angle AOD$ or $\angle BOC$. For example, if M lies in $\angle AOD$, then O lies inside $\triangle BCM$, and so it cannot lie on (BCM) . Hence, the line OM cuts $\angle AOB$ and $\angle COD$. For the same reason, S and O lie on different sides of CD , and T and O lie on different sides of AB . Then S and T cannot be the same point. $\square$

Remarks. Apart from the figure given above, it could happen that M lies between T and O . But such a configuration is essentially the same as this one due to symmetry. Using directed angles as shown is a possible way to handle this. But this does not help in deducing M should lie between S and T .

The trick used in this solution is useful. In short, if we get $XY = a$ where a is not a symmetric expression, we may try to obtain another formula $XY = b$ by changing the roles of the symmetric points (usually this can be done by changing labels). Then $XY = \sqrt{ab}$ would be a symmetric formula.

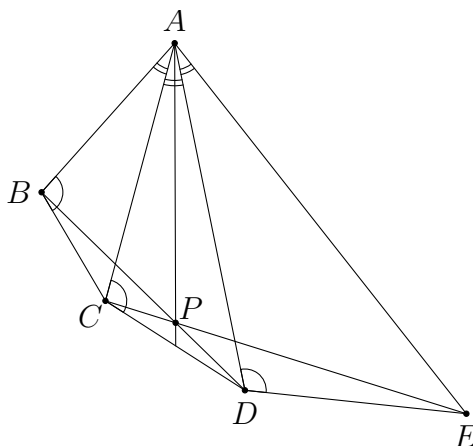
Problem 7.3. (IMO Shortlist 2006 G3) Let $ABCDE$ be a convex pentagon such that

$$\angle BAC = \angle CAD = \angle DAE \quad \text{and} \quad \angle ABC = \angle ACD = \angle ADE.$$

The diagonals BD and CE meet at P . Prove that the line AP bisects the side CD .

Analysis. Clearly, $\triangle ABC, \triangle ACD, \triangle ADE$ are directly similar. This suggests considering the spiral similarity mapping $\triangle ABC$ to $\triangle ADE$. This easily implies $ABCP$ and $AEDP$ are cyclic. Hence, AP is the radical axis of the two circles. In view of [proposition 5.1](#), we may try to prove that CD is a common tangent to these circles.

Solution.



From the equal angles, $\triangle ABC, \triangle ACD, \triangle ADE$ are directly similar. By [proposition 7.1](#), $\triangle ABD$ and $\triangle ACE$ are also directly similar. Then we can apply [proposition 7.2](#) to show that $ABCP$ and $AEDP$ are cyclic quadrilaterals.

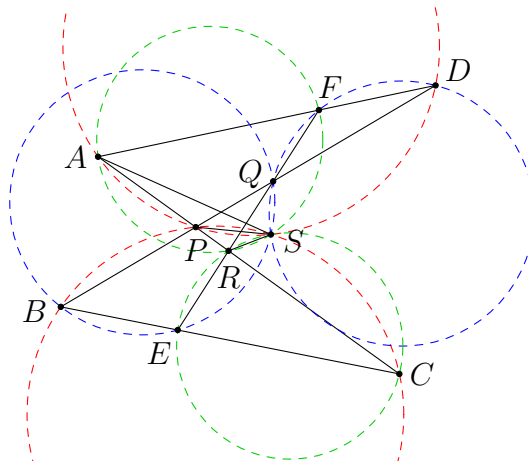
Note that $ABCD$ and $ACDE$ are directly similar. In particular, $\angle CBP = \angle DCE$ and $\angle PED = \angle BDC$. Therefore, CD is an external common tangent to $(ABCP)$ and $(AEDP)$ with C, D as the tangent points. By [proposition 5.1](#), the radical axis AP of the two circles bisects CD . $\square$

Problem 7.4. (IMO Shortlist 2005 G4/Problem 5) Let $ABCD$ be a fixed convex quadrilateral with $BC = DA$ and BC not parallel with DA . Let two variable points E and F lie on the sides BC and DA , respectively and satisfy $BE = DF$. The lines AC and BD meet at P , the lines BD and EF meet at Q , the lines EF and AC meet at R . Prove that the circumcircles of the triangles PQR , as E and F vary, have a common point other than P .

Analysis. Since $BC = DA$, the spiral similarity that takes B to D and C to A is indeed a rotation. Such a rotation also takes E to F since E and F cut BC and DA in the same ratio from $BE = DF$ and $EC = FA$. The centre of this rotation S can be found via [proposition 7.2](#).

To work out a solution, the best way is to determine the fixed point first. Indeed, after considering the rotation discussed above, it is natural to guess S is the fixed point. One reason is that such a fixed point must be highly symmetric w.r.t. the pairs B, D ; C, A and E, F . Otherwise, by changing the roles of these points, we would obtain more fixed points that lie on (PQR) , which cannot be true since the circles can have at most two common points (one of which is P) as the circles are clearly different from one another. Under a rotation, the only point which is highly symmetric is the centre. Such concepts about transformations provide the crucial observation to solve this IMO problem.

Solution.



Let (BCP) and (DAP) meet again at S . The fact that BC and AD are not parallel shows that S is distinct from P . By [proposition 7.2](#), S is the centre of rotation that takes B to D , C to A , and hence E to F . Again using [proposition 7.2](#), S lies on (BEQ) and (DFQ) , as well as (CER) and (AFR) . These concyclic points are sufficient to deduce P, Q, R, S are concyclic. In fact, this comes from

$$\angle SPQ = \angle SPD = \angle SAD = \angle SAF = \angle SRF = \angle SRQ.$$

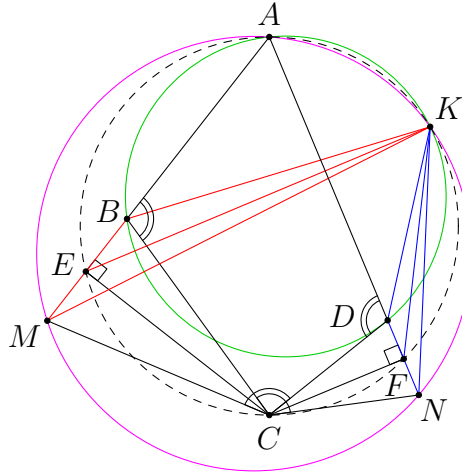
In other words, the point S , which is a fixed point distinct from P , always lies on (PQR) . $\square$

Problem 7.5. In a convex quadrilateral $ABCD$ with $\angle ABC = \angle ADC$, points M and N are chosen on the rays AB and AD respectively such that $\angle MCD = \angle NCB$. Let K be the second intersection of (AMN) and (ABD) . Prove that $AK \perp KC$.

Analysis. Firstly, the angle conditions allow us to deduce $\triangle CMB \sim \triangle CND$. Secondly, by [proposition 7.2](#), K is the centre of spiral similarity mapping BM to DN . These give us a number of similarities in this figure.

To show that K lies on the circle with diameter AC , we may form intersection points of this circle with AB and AD , and show that these points together with A and K are concyclic. The proof simply relies on the above similarities and [proposition 7.2](#).

[Solution.](#)



Since it is given that $\angle ABC = \angle ADC$, we have $\angle MBC = \angle NDC$. Also, since $\angle MCD = \angle NCB$, we have $\angle MCB = \angle NCD$. Thus, we get $\triangle CMB \sim \triangle CND$. Let E and F be the projections from C onto MB and ND respectively. Then E and F are the corresponding points in the above similarity. In particular, we have $BE : EM = DF : FN$.

By [proposition 7.2](#), K is the centre of spiral similarity mapping BM to DN . Since E and F cut the segments BM and DN in the same ratio, the same transformation

A, E, C, F, K are concyclic, and hence $\angle AKC = 90^\circ$. $\square$

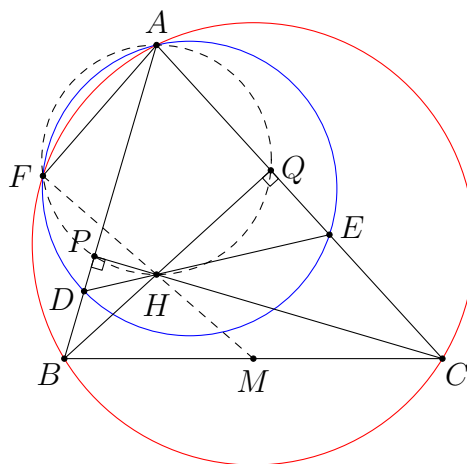
and it may not be easy for us to observe those facts without using spiral similarity.

 $\triangle ADE$.

that it coincides with the above definition.

in this spiral similarity which are related to this circle.

Solution.



Let P and Q be the feet of altitudes from C and B respectively, and let F be the second intersection of (ABC) and (ADE) . By [proposition 7.2](#), F is the centre of spiral similarity taking DB to EC . Since

$$\angle PHD = \frac{\pi}{2} - \angle HDA = \frac{\pi}{2} - \angle AEH = \angle EHQ = \angle DHB,$$

DH is an angle bisector of $\angle PHB$. (This also comes from the trivial but less well-known fact that the angle bisectors of the altitudes from B and C are parallel to the angle bisectors of $\angle BAC$. Note that there are two angle bisectors for each angle.)

Using the angle bisector theorem, we have $\frac{PD}{DB} = \frac{PH}{HB}$. Similarly, $\frac{QE}{EC} = \frac{QH}{HC}$. As B, C, Q, P are concyclic, these ratios are equal. Therefore,

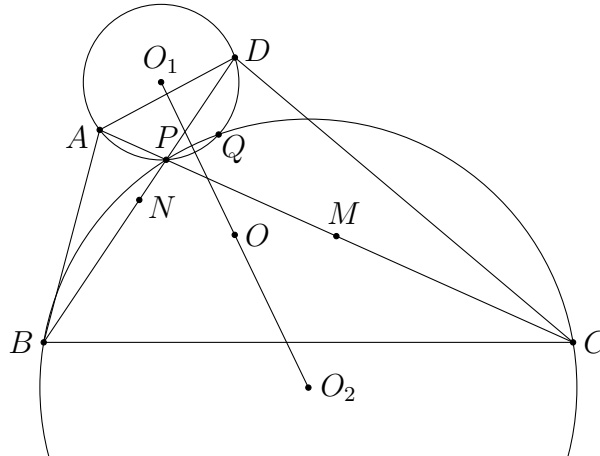
$$\frac{PD}{DB} = \frac{QE}{EC}.$$

Note that this relation holds even for directed lengths. As a result, the spiral similarity taking DB to EC also takes P to Q . Again by [proposition 7.2](#), A, F, P, Q are concyclic. Note that these points lie on the circle with diameter AH . In particular, $\angle HFA = 90^\circ$. By [problem 4.9](#), we find that H, M, F are collinear. The result follows readily. $\square$

Problem 7.7. Let P be the intersection of the diagonals AC and BD of a quadrilateral $ABCD$. Let O_1 and O_2 be the circumcentres of $\triangle PAD$ and $\triangle PBC$ respectively. Let M, N, O be the midpoints of AC, BD, O_1O_2 respectively. Prove that O is the circumcentre of $\triangle PMN$.

Analysis. We should think of spiral similarity because the problem involves two intersecting circles, and there are two lines passing through one of the intersection points. Indeed, this problem is trivialized by the use of spiral similarity. We shall see how powerful this tool is. The only idea is to find as many corresponding points in the similarity as possible.

Solution.



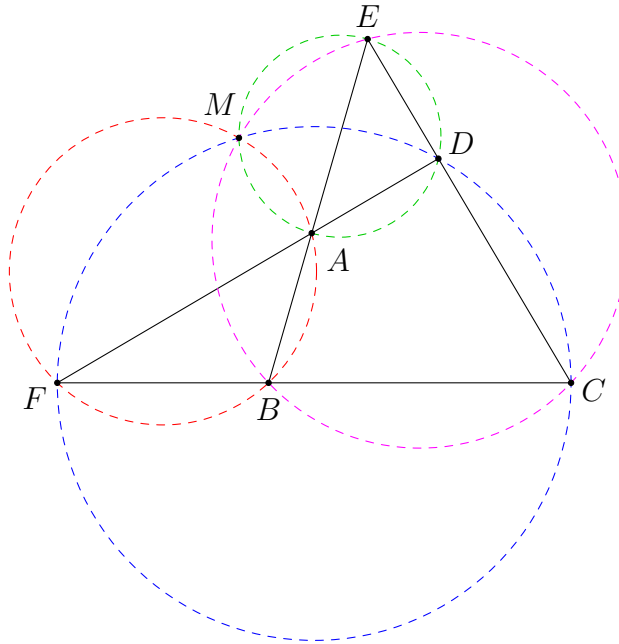
Let Q be the second intersection point of (PAD) and (PBC) . By [proposition 7.2](#), Q is the centre of spiral similarity that maps A to C and D to B . Thus, $\triangle QAD$ and $\triangle QCB$ are directly similar. Since O_1 and O_2 are the circumcentres of $\triangle QAD$ and $\triangle QCB$ respectively, $QADO_1$ and $QCBO_2$ are directly similar. It follows from [proposition 7.1](#) that $\triangle QAC$, $\triangle QDB$ and $\triangle QO_1O_2$ are directly similar. As M, N, O are the midpoints of the corresponding sides AC, DB, O_1O_2 , we also know that $QACM$, $QDBN$ and QO_1O_2O are directly similar. Finally, this in turn implies $QADO_1$ and $QMNO$ are directly similar. As O_1 is the circumcentre of $\triangle QAD$, O must be the circumcentre of $\triangle QMN$.

Finally, since P is the intersection point of AM and DN , it lies on (QMN) by [proposition 7.2](#). This implies O is the circumcentre of $\triangle PMN$. $\square$

Remarks. The relations of the points can be seen from the following table.

				circumcentre
	Q	A	D	O_1
	Q	C	B	O_2
midpoint	Q	M	N	O

7.2 Miquel Point



Let $ABCD$ be a quadrilateral such that AB meets CD at some point E , and AD meets BC at some point F . Let M be the centre of spiral similarity that maps AB to DC . By [proposition 7.2](#), M lies on both (ABF) and (CDF) . Recall that M is

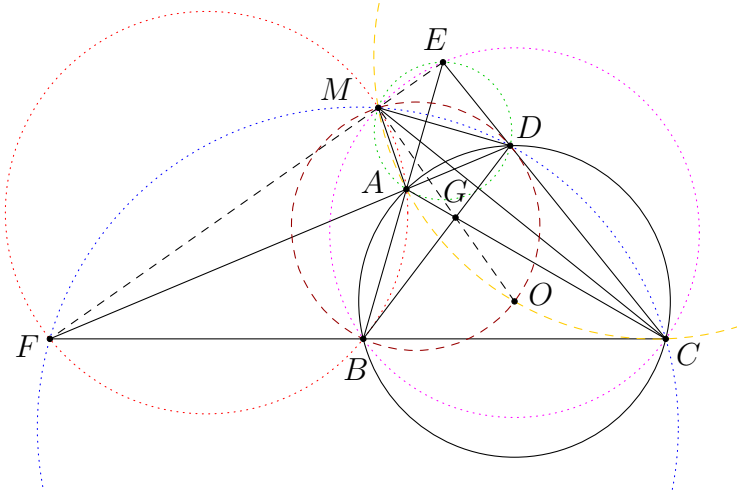
also the centre of spiral similarity that maps AD to BC . Therefore, M lies on both (ADE) and (BCE) . Since this point M is so special, we give a name to it, which is known as the *Miquel point* (密克點) of $ABCD$. This definition is related to the Miquel point of triangles defined earlier. Indeed, M is the Miquel point of the line FAD w.r.t. $\triangle EBC$, as well as the Miquel point of the line EAB w.r.t. $\triangle FCD$.

Note that M is also the centre of spiral similarity that maps FB to DE , as well as the centre of spiral similarity that maps FD to BE . A simple way to use the Miquel point is to define the Miquel point of some quadrilaterals in the figure. Then we would obtain a lot of concyclic points and similar triangles. This could probably simplify the problem.

Next, we introduce some more properties of the Miquel point of a cyclic quadrilateral.

Proposition 7.3. Let $ABCD$ be a cyclic quadrilateral such that its opposite sides are not parallel. Let AB meet CD at E , let AD meet BC at F , and let AC meet BD at G . Let O be the centre of $(ABCD)$. Let M be the Miquel point of $ABCD$. Then the following hold.

- (a) M lies on both (OAC) and (OBD)
- (b) MO bisects $\angle AMC$ and $\angle BMD$
- (c) M is the intersection point of EF and OG



Proof.

- (a) Since M, A, D, E are concyclic and M, F, C, D are concyclic, we have

$$\begin{aligned}\angle AMC &= \angle AMD + \angle DMC = \angle AED + \angle DFC = \angle(AB, CD) + \angle(AD, BC) \\ &= \angle(AB, BC) + \angle(AD, CD) = \angle ABC + \angle ADC = 2\angle ABC = \angle AOC.\end{aligned}$$

This implies M, A, O, C are concyclic. Similarly, M, B, O, D are concyclic.

(b) Since M, A, O, C are concyclic and $OA = OC$, the line MO bisects $\angle AMC$. Similarly, MO bisects $\angle BMD$.

(c) Firstly, as

$$\begin{aligned}\angle EMF &= \angle EMD + \angle DMA + \angle AMF = \angle EAD + \angle DEA + \angle ABF \\ &= \angle EAD + \angle DEA + \angle ABC = \angle EAD + \angle DEA + \angle ADC \\ &= \angle EAD + \angle DEA + \angle ADE = 0,\end{aligned}$$

the points E, M, F are collinear.

Secondly, using $\triangle MAB \sim \triangle MDC$ and $\triangle GAB \sim \triangle GDC$, we have

$$\frac{MA}{MD} = \frac{AB}{DC} = \frac{GB}{GC}.$$

Similarly, using $\triangle MAD \sim \triangle MBC$ and $\triangle GAD \sim \triangle GBC$, we have

$$\frac{MD}{MC} = \frac{AD}{BC} = \frac{GA}{GB}.$$

It follows that

$$\frac{MA}{MC} = \frac{MA}{MD} \times \frac{MD}{MC} = \frac{GB}{GC} \times \frac{GA}{GB} = \frac{GA}{GC}.$$

Thus, MG bisects $\angle AMC$. By part (b), M, G, O are collinear.

□

In view of the result of part (c) and Brokard's theorem (see **Projective Geometry and Inversion 3.1.1**), we know that $OM \perp EF$.

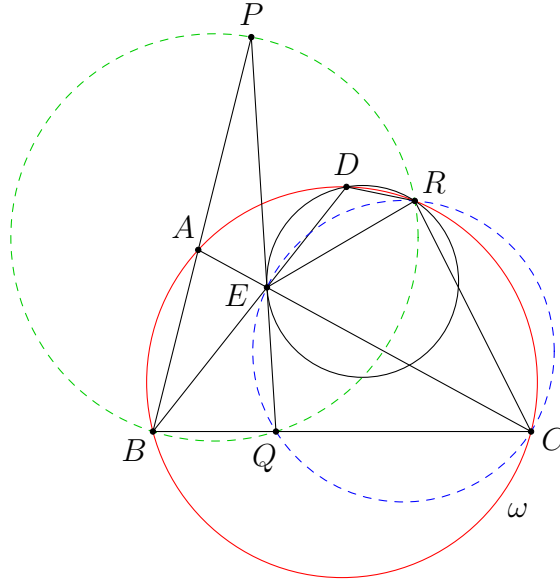
Problem 7.8. (Mongolia MO Final Round 2017 Grade 11 Problem 4) Let $ABCD$ be a cyclic quadrilateral with circumcircle ω , and E be the intersection of the diagonals AC and BD . A line passing through E intersects lines AB, BC at P, Q , respectively. Let R ($R \neq D$) be the intersection point of ω and a circle that passes through D, E and is tangent to the line PQ at E . Prove that B, P, Q, R are concyclic.

Analysis. Since R should lie on both (ABC) and (BPQ) , the assertion is the same as showing R is the Miquel point of $ABQE$. In view of this, there are more ways for us to solve the problem, such as proving R lies on (CEQ) or (PAE) . Although it is not necessary to use the Miquel point in solving this problem as everything is just angle chasing, this insight allows us to solve the problem quickly.

Solution. Since QE is tangent to (DER) , we have

$$\angle REQ = \angle RDE = \angle RDB = \angle RCB = \angle RCQ.$$

This implies E, Q, C, R are concyclic. As R lies on both (ABC) and (EQC) , it is the Miquel point of $ABQE$. Therefore, R also lies on (BPQ) as desired. □



Problem 7.9. (Morocco TST 2015) Let $ABA'B'$ be a convex quadrilateral, with $AA' \cap BB' = S$, and let T be the intersection of the circumcircles of $\triangle ABS$ and $\triangle A'B'S$. Let C and C' be points in lines AB and $A'B'$ respectively such that B is between A and C , and B' is between A' and C' . Let K and L be points in segments SB and SA respectively, such that K, B, C, T are concyclic and A', C', T, L are concyclic. Prove that C, C', K, L are collinear if and only if $\frac{CA}{BC} = \frac{C'A'}{C'B'}$.

Analysis. There are many circles in the figure, so it is natural to consider the Miquel points of some quadrilaterals. As all circles pass through T , the point T should probably be the Miquel point. Note that we can also define the Miquel point of a self-intersecting quadrilateral such as $ABB'A'$. Indeed, we immediately know that T is the Miquel point of $ABB'A'$, and it is the centre of spiral similarity mapping AB to $A'B'$.

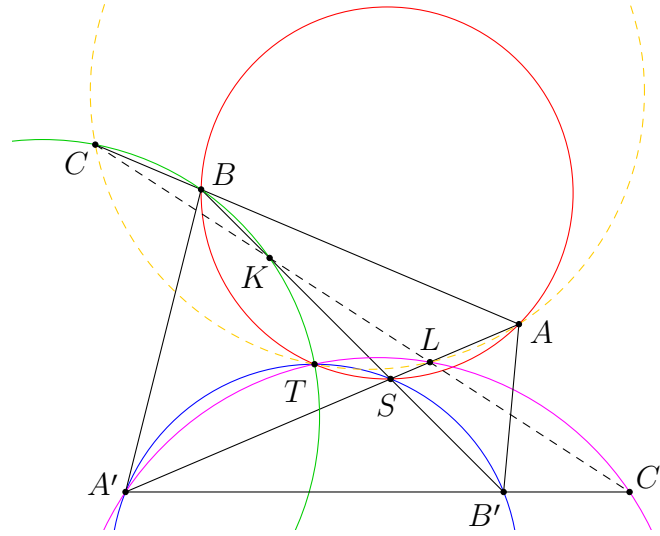
The equal ratio means C and C' cuts the line AB and the line $A'B'$ in the same ratio. Therefore, it is the same as saying that the above mentioned spiral similarity maps C to C' . Thus, there is a good chance that we can use Miquel points and spiral similarities to solve this problem. The only task is to spot the quadrilaterals that have T as their Miquel points.

Solution. Firstly, suppose C, K, L, C' are collinear. We have the following observations.

- As AA' meets BB' at S , and T lies on (ABS) and $(A'B'S)$, T is the Miquel point of $ABB'A'$. Therefore, T is the centre of spiral similarity mapping AB to $A'B'$.

- As BC meets LS at A , BS meets CL at K , and T lies on (ABS) and (BCK) , T is the Miquel point of $BCLS$. Therefore, T also lies on (ACL) .
- As AA' meets CC' at L , and T lies on both (ACL) and $(A'C'L)$, T is the Miquel point of $ACC'A'$. Therefore, T is the centre of spiral similarity mapping AC to $A'C'$.

Combining these, we know that T is the centre of spiral similarity mapping ABC to $A'B'C'$. Therefore, we have $\frac{AC}{BC} = \frac{A'C'}{B'C'}$.



Secondly, suppose $\frac{AC}{BC} = \frac{A'C'}{B'C'}$. As above, T is the centre of spiral similarity mapping AB to $A'B'$. From the configuration and the equal ratio, we know that the same spiral similarity maps C to C' . We can now argue the collinearity as follows.

- As T is the centre of spiral similarity mapping AC to $A'C'$, the intersection point $AA' \cap CC'$ must lie on $(A'C'T)$. Therefore, $AA' \cap CC' = L$. This implies C, L, C' are collinear.
- As T is the centre of spiral similarity mapping BC to $B'C'$, the intersection point $BB' \cap CC'$ must lie on (BCT) . Therefore, $BB' \cap CC' = K$. This implies C, K, C' are collinear.

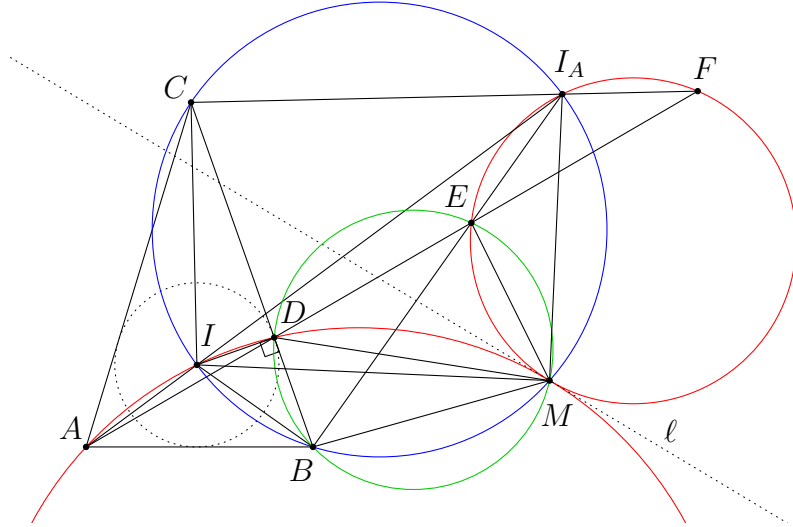
Combining these, we see that C, K, L, C' are collinear. $\square$

Problem 7.10. (IMO Shortlist 2020 G6) Let I and I_A be the incentre and the A -excentre of an acute-angled triangle ABC with $AB < AC$. Let the incircle meet BC at D . The line AD meets BI_A and CI_A at E and F , respectively. Prove that the circumcircles of triangles AID and I_AEF are tangent to each other.

Analysis. The hardest step in solving problems like this is usually to guess the position or properties of the tangential point. We should make various attempts to guess this and test one by one whether each guess is correct or not. This can be done by drawing different figures, assuming the results to see if there is any contradiction, etc. This time we guess that the tangential point is the Miquel point of a quadrilateral, because the figure consists of many circles. Indeed, apart from (AID) and (I_AEF) , we have another well-known group of concyclic points C, I, B, I_A . Also, in many problems involving the incentre and an excentre, the circumcircle of $\triangle ABC$ could be useful (for example, we know that the centre of $(CIBI_A)$ lies on (ABC)). In addition, the incircle and the A -excircle are two more circles that may be involved.

By drawing some of the circles mentioned above, it should not be hard to see that the Miquel point of $CDEI_A$ already lies on two of these circles. We would like to show that it is indeed the desired tangential point.

Solution.



Let M be the Miquel point of $CDEI_A$. Thus, M lies on (EFI_A) , (BCI_A) and (BDE) . Note that I also lies on (BCI_A) since $\angle IBI_A = \angle ICI_A = 90^\circ$. We shall show that M is the tangential point of (EFI_A) and (AID) .

Firstly, we prove that M lies on (AID) . Since A, I, I_A are collinear, we have

$$\angle ADM = \angle EDM = \angle EBM = \angle I_ABM = \angle I_AIM = \angle AIM.$$

This proves A, I, D, M are concyclic.

Secondly, we prove that (EFI_A) and (AID) share the same tangent at M . Let ℓ be a tangent at M to (ADI) . Then we have

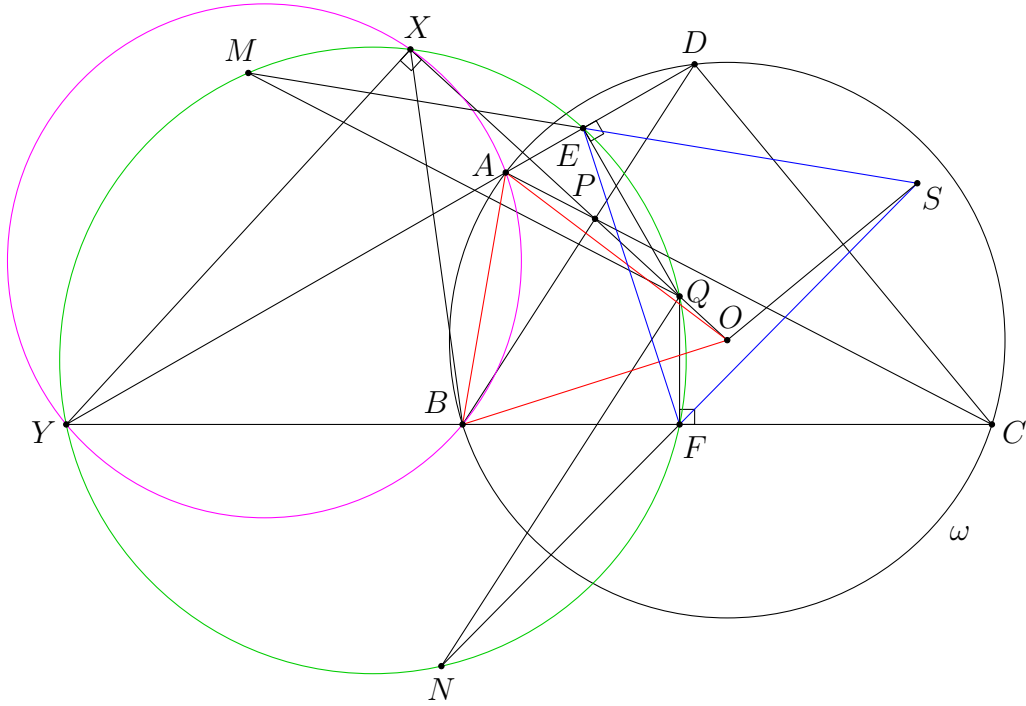
$$\angle(I_AM, \ell) = \angle I_AMI - \angle(\ell, IM) = \frac{\pi}{2} - \angle MDI = -\angle MDB = \angle BEM = \angle I_AEM.$$

This implies ℓ is tangent to (I_AEM) . Therefore, (EFI_A) and (AID) are tangent to each other at M . $\square$

Problem 7.11. (Iranian Geometric Olympiad 2022 Intermediate Problem 5) Let $ABCD$ be a quadrilateral inscribed in a circle ω with centre O . Let P be the intersection of two diagonals AC and BD . Let Q be a point lying on the segment OP . Let E and F be the orthogonal projections of Q on the lines AD and BC , respectively. The points M and N lie on the circumcircle of triangle QEF such that $QM \parallel AC$ and $QN \parallel BD$. Prove that the two lines ME and NF meet on the perpendicular bisector of segment CD .

Analysis. Let S be the intersection point of ME and NF . Initially, one may do some basic angle chasing to get more information about the point S . It is not hard to find that $\triangle SEF$ is an isoscles triangle with $SE = SF$, and that $\angle ESF = \angle AOB$. In other words, we have $\triangle SEF \sim \triangle OAB$. In view of this, it may be useful to consider the centre of spiral similarity that maps $\triangle SEF$ to $\triangle OAB$. In fact, this point is again just the Miquel point of $ABCD$. If we are aware of this, we can use some well-known properties of this Miquel point to finish the proof.

Solution.



Let S be the intersection point of ME and NF . The first step is to show that $\triangle SEF \sim \triangle OAB$. Note that

$$\angle SEF = \angle MEF = \angle MQF = \angle(AC, QF) = \frac{\pi}{2} + \angle ACB = \angle OAB.$$

By symmetry, we have $\angle SFE = \angle OBA$. Therefore, $\triangle SEF \sim \triangle OAB$.

Let X be the Miquel point of $ABCD$, and let AD meet BC at Y . Note that $\angle YEQ = \angle YFQ = 90^\circ$. Thus, Y lies on (EFQ) . By [proposition 7.3\(c\)](#) and Brokard's

theorem, we know that X lies on PQO and $\angle YXO = 90^\circ$. Therefore, X lies on (EFQ) as well.

Next, we prove that $\frac{AE}{ED} = \frac{BF}{FC}$. Let P_1, O_1 be the projections of P, O onto AD , and let P_2, O_2 be the projections of P, O onto BC . As $\triangle PAD \sim \triangle PBC$, we have $\frac{AP_1}{P_1D} = \frac{BP_2}{P_2C}$. Also, we have $\frac{AO_1}{O_1D} = 1 = \frac{BO_2}{O_2C}$ and $\frac{P_1E}{EO_1} = \frac{PQ}{QO} = \frac{P_2F}{FO_2}$. Combining these, we obtain $\frac{AE}{ED} = \frac{BF}{FC}$. (In short, the projections of each of P, Q, O onto AD and BC cut AD and BC in the same ratio.)

Now, since X is the centre of spiral similarity mapping AD to BC , and $\frac{AE}{ED} = \frac{BF}{FC}$, the spiral similarity also maps E to F . Thus, X is the centre of spiral similarity mapping AB to EF . As $\triangle OAB$ and $\triangle SEF$ are directly similar, the latter spiral similarity also maps O to S . Therefore, we deduce $\angle XOS = \angle XBF$. This angle relation is sufficient to show that S lies on the perpendicular bisector of CD . Indeed, as O lies on this perpendicular bisector, it suffices to show that $OS \perp CD$. We find that

$$\begin{aligned} \angle(CD, OS) &= \angle(CD, OX) + \angle XOS = \angle(CD, OX) + \angle XBF \\ &= \angle(CD, OX) + \angle(XB, BC) = \angle(CD, BC) + \angle(XB, OX) \\ &= \angle DCB + \angle BXO = \angle DAB + \left(\frac{\pi}{2} - \angle YXB\right) \\ &= \angle YAB + \frac{\pi}{2} - \angle YXB. \end{aligned}$$

This is equal to $\frac{\pi}{2}$ since X, Y, B, A are concyclic by the definition of the Miquel point. This completes the proof. $\square$

Here summarizes the techniques introduced in this section.

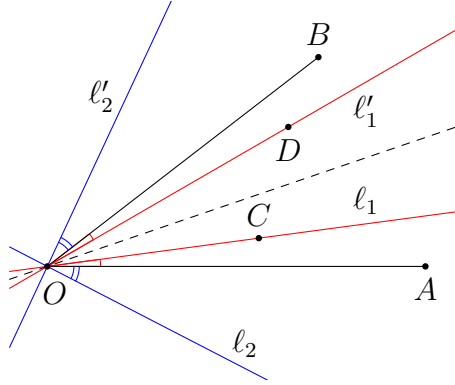
- using spiral similarity to find similar triangles ([problems 7.1, 7.2, 7.3, 7.7, 7.11](#))
- using spiral similarity to find concyclic points ([problems 7.3, 7.4, 7.5, 7.6](#))
- considering the centre of spiral similarity ([problems 7.4, 7.5, 7.6, 7.7, 7.11](#))
- using Miquel point to find concyclic points ([problems 7.8, 7.9, 7.10, 7.11](#))
- using Miquel point to find centre of spiral similarity ([problems 7.9, 7.11](#))
- considering Miquel point of quadrilateral ([problems 7.10, 7.11](#))

8 Isogonal Conjugate

Isogonal conjugate is a more advanced topic in plane geometry. It does not appear frequently in Olympiad, but in case it does, it can usually be used to solve more difficult problems. This section is designed to introduce the basic concepts and usages of this topic.

8.1 Isogonal Conjugate in Triangles

Definition 8.1. Let OA and OB be distinct lines and let ℓ be a line passing through O . The *isogonal line* (等角線) (or simply *isogonal*) of ℓ w.r.t. $\angle AOB$ is the line ℓ' obtained by reflecting ℓ in the angle bisector of $\angle AOB$.

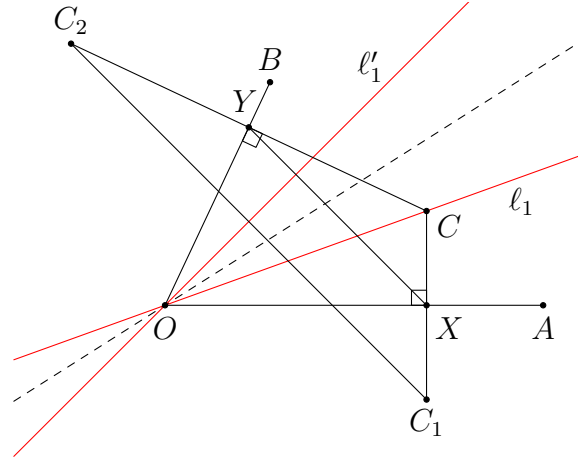


It does not matter which angle bisector of $\angle AOB$ we are using. Let ℓ'_1 be the isogonal line of ℓ_1 w.r.t. $\angle AOB$. For any point C on ℓ_1 and any point D on ℓ'_1 , we have $\angle AOC = \angle BOD$ and $\angle AOD = \angle BOC$. It is obvious that the isogonal of ℓ'_1 w.r.t. $\angle AOB$ is again ℓ_1 . If it is clear which angle we are talking about, we may simply say the isogonal of a line.

There are two important properties of isogonal lines.

Proposition 8.1. Let ℓ'_1 be the isogonal of ℓ_1 w.r.t. $\angle AOB$ and let C be a point on ℓ_1 distinct from O . Let X and Y be the projections of C onto OA and OB respectively, and let C_1 and C_2 be the reflections of C in OA and OB respectively. Then the following hold.

- (a) $XY \perp \ell'_1$
- (b) ℓ'_1 is the perpendicular bisector of C_1C_2



Proof.

- (a) The main observation is that O, X, C, Y are concyclic. Hence,

$$\begin{aligned}\angle(\ell'_1, XY) &= \angle(\ell'_1, OB) + \angle(OB, XY) = -\angle(\ell_1, OA) + \angle OYX \\ &= \angle XOC + \angle OCX = \frac{\pi}{2}.\end{aligned}$$

Note that $\angle(\ell'_1, OB) = -\angle(\ell_1, OA)$ as ℓ_1 and ℓ'_1 are isogonal lines. The assertion is also true even for the limiting cases when $O = X$ or $O = Y$, though some of the angles written above should be suitably defined.

- (b) Note that X and Y are the midpoints of CC_1 and CC_2 respectively. By the midpoint theorem, $XY \parallel C_1C_2$, so that $C_1C_2 \perp \ell'_1$ by part (a). Also, we have $OC_1 = OC = OC_2$, so that O lies on the perpendicular bisector of C_1C_2 . Hence, ℓ'_1 is the perpendicular bisector of C_1C_2 .

□

Next, the following result provides a simple way to generate new isogonal lines.

Theorem 8.1. (Isogonal line lemma) Let OC and OD be isogonal lines w.r.t. $\angle AOB$. Let AC meet BD at P , and let AD meet BC at Q . Then OP and OQ are isogonal lines w.r.t. $\angle AOB$.

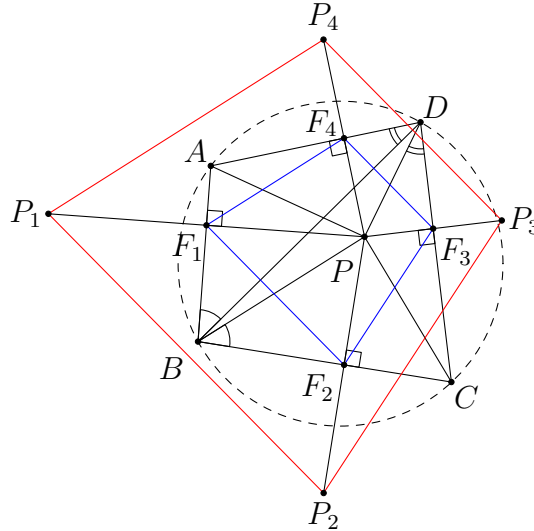
Proof. For simplicity, we stick to the configuration as shown. Let ℓ be the isogonal of OP w.r.t. $\angle AOB$. We are going to prove AD, BC, ℓ are concurrent. By Ceva's theorem, it suffices to prove

$$\frac{\sin \angle(AO, \ell)}{\sin \angle(BO, \ell)} \times \frac{\sin \angle OBC}{\sin \angle ABC} \times \frac{\sin \angle BAD}{\sin \angle OAD} = 1.$$

Analysis. Firstly, knowledge about isogonals helps us determine the position of P . Suppose $ABCD$ is given. We can take any point E on BD (or simply take $E = D$) and reflect E in AB, BC to get E_1, E_2 . According to [proposition 8.1\(b\)](#), BP is the perpendicular bisector of E_1E_2 , as BP is the isogonal of BE w.r.t. $\angle CBA$. Similarly, we can determine the line PD so that the intersection of these lines is P . Note that the condition guarantees the two lines have a unique intersection.

Next, we may try to see in the other way round. We can reflect P in the four sides and consider its projections onto the four sidelines. We can use the properties to deduce some useful results. Also, PA and PC are the diameters of some circles which pass through some of the projection points. These may be useful to relate different things together. For example, we can use the sine law to express the lengths of these diameters as the lengths of other chords.

Solution.



Let P_1, P_2, P_3, P_4 be the reflections of P in AB, BC, CD, DA respectively, and let F_i be the midpoint of PP_i for each $i = 1, 2, 3, 4$. As BP is the isogonal of BD w.r.t. $\angle CBA$, BD is the perpendicular bisector of P_1P_2 by [proposition 8.1\(b\)](#). Similarly, BD is the perpendicular bisector of P_3P_4 . This shows $P_1P_2P_3P_4$ is an isosceles trapezoid with $P_1P_4 = P_2P_3$.

Consider the homothety with centre P and ratio $\frac{1}{2}$. It is then obvious that $F_1F_2F_3F_4$ is an isosceles trapezoid with $F_1F_4 = F_2F_3$. Note that F_1, F_4 lie on the circle with diameter PA . Hence, $PA = \frac{F_1F_4}{\sin A}$, and similarly, $PC = \frac{F_2F_3}{\sin C}$. Therefore, $PA = PC$ iff $\sin A = \sin C$, which is equivalent to A, B, C, D being concyclic as the case $A = C$ yields the contradiction BD bisects $\angle CBA, \angle ADC$ (since P would lie on BD in that case). $\square$

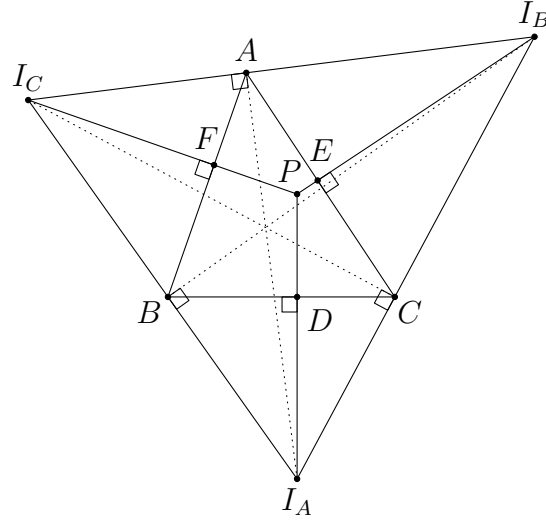
Problem 8.2. (IMO Shortlist 2003 G3) Let ABC be a triangle and let P a point in its interior. Denote by D, E, F the feet of the perpendiculars from P to the lines BC, CA, AB , respectively. Suppose that

$$AP^2 + PD^2 = BP^2 + PE^2 = CP^2 + PF^2.$$

Denote by I_A, I_B, I_C the excentres of the triangle ABC . Prove that P is the circumcentre of the triangle $I_AI_BI_C$.

Analysis. We consider one of the equations $AP^2 + PD^2 = BP^2 + PE^2$. One may try to use Pythagoras' theorem to simplify it. Indeed, this simply means $AE = BD$, and similarly, $BF = CE$ and $CD = AF$. These imply D, E, F are the contact points of the excircles. It is then not hard to figure out a proof. Properties of isogonals will be used to simplify the proof. This is certainly not necessary.

Solution.



By Pythagoras' theorem, $AP^2 + PD^2 = BP^2 + PE^2$ becomes $AE^2 = BD^2$, and hence $AE = BD$. Similarly, we get $BF = CE$ and $CD = AF$. From these, $AE = BD = s - c$, $BF = CE = s - a$ and $CD = AF = s - b$. Therefore, D is the contact point of the A -excircle and the side BC . In particular, D is the projection of I_A onto BC , so that $PI_A \perp BC$, etc.

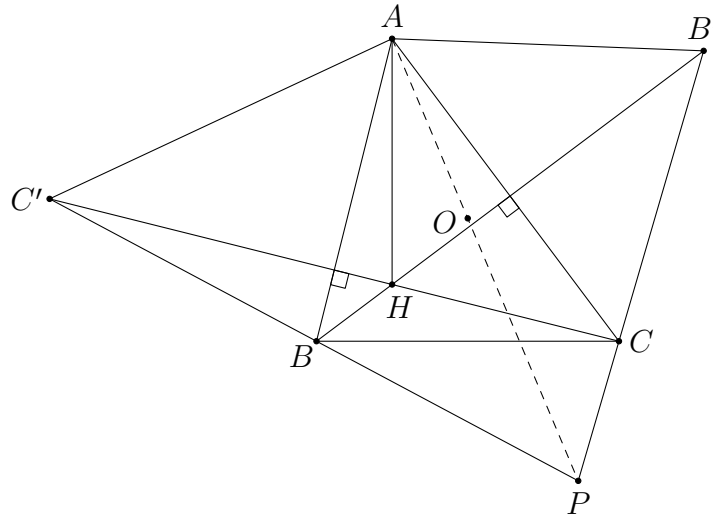
Note that $\triangle ABC$ is the orthic triangle of $\triangle I_AI_BI_C$ (that means A, B, C are the feet of altitudes). Since B and C are the projections of the orthocentre of $\triangle I_AI_BI_C$ onto I_AI_C and I_AI_B , and $BC \perp I_AP$, the line I_AP is the isogonal of the I_A -altitude w.r.t $\angle I_BI_AI_C$ by [proposition 8.1\(a\)](#). Therefore, the circumcentre of $\triangle I_AI_BI_C$ lies on I_AP (see remark). Similarly, the circumcentre lies on both I_BP and I_CP . Thus, it is simply the point P . $\square$

Remarks. Let H and O be the orthocentre and the circumcentre of $\triangle ABC$ respectively. Then AH and AO are isogonals w.r.t. $\angle BAC$ since $\angle BAH = \angle OAC$.

Problem 8.3. Given $\triangle ABC$, let B' be the reflection of B in the line AC , and let C' be the reflection of C in the line AB . Let O be the circumcentre of $\triangle ABC$. Prove that $C'B, B'C, AO$ are concurrent.

Analysis. Let P be the intersection point of $C'B$ and $B'C$. We have to prove that A, O, P are collinear. Equivalently, we have to show that AP and the A -altitude are isogonals w.r.t. $\angle BAC$. This approach seems to be on the right track, because BB' and CC' also meet at the orthocentre of $\triangle ABC$. This problem demonstrates the use of the isogonal line lemma.

Solution.



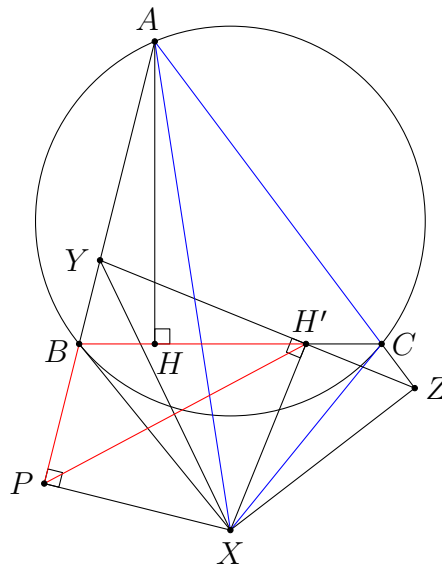
Let P be the intersection point of $C'B$ and $B'C$, and let H be the orthocentre of $\triangle ABC$. Since $\angle C'AB = \angle BAC = \angle B'AC$, AB' and AC' are isogonals w.r.t. $\angle BAC$. Now, as $BB' \perp AC$ and $CC' \perp AB$, BB' and CC' meet at H . Also, as $C'B$ and $B'C$ meet at P , we know that AH and AP are isogonals w.r.t. $\angle BAC$ by the isogonal line lemma. Recall that AH and AO are isogonals. Therefore, A, O, P are collinear. Equivalently, $C'B, B'C, AO$ are concurrent. $\square$

Problem 8.4. (Iran TST 2015 Test 3 Problem 6) AH is the altitude of triangle ABC and H' is the reflection of H through the midpoint of BC . If the tangent lines to the circumcircle of $\triangle ABC$ at B and C intersect each other at X , and the perpendicular line to XH' at H' intersects AB and AC at Y and Z respectively, prove that $\angle ZXC = \angle YXB$.

Analysis. The final assertion is to show that XB and XZ are isogonals w.r.t. $\angle CXY$ (or vice versa). Using the isogonal line lemma, it is the same as showing XA and XH' are isogonals w.r.t. $\angle CXY$. Although this may not necessarily simplify the problem, it gives us another possible direction to tackle the problem, which is always better than nothing. In this new version of the problem, we do not need the point Z . Thus, the problem is simplified in some sense.

Next, we mainly need to see how to use the right angle condition. We choose to define the projection of X onto the line AB , because this can generate a group of concyclic points using the right angles. With some angle chasing, we see that it remains to prove a pair of similar triangles. This transforms a hard problem to a standard problem about proving similar triangles.

Solution.



WLOG assume $AB \leq AC$. Let P be the projection of X onto the line AB . We are going to prove that $\triangle PBH' \sim \triangle XCA$. Firstly, we have

$$\angle PBH' = 180^\circ - B = \angle XCA$$

as XC is tangent to (ABC) . Note that $\triangle XPB \sim \triangle AHC$ as both are right-angled triangles and $\angle PBX = \angle HCA$ (as XB is tangent to (ABC)). Therefore, we have $\frac{XB}{AC} = \frac{PB}{HC}$. It follows that

$$\frac{PB}{XC} = \frac{PB}{XB} = \frac{HC}{AC} = \frac{BH'}{CA}.$$

Hence, $\triangle PBH' \sim \triangle XCA$. Now, as P, X, H', Y are concyclic, we have

$$\angle H'XY = \angle H'PY = \angle H'PB = \angle AXC.$$

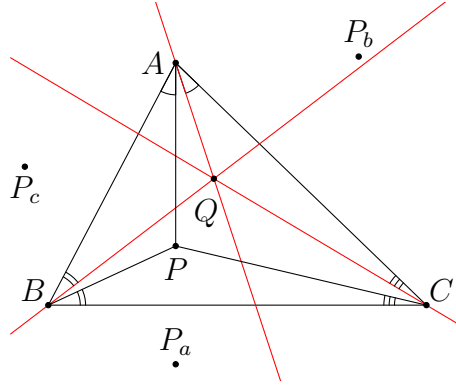
This means XA and XH' are isogonals w.r.t. $\angle CXY$. As AY meets $H'C$ at B , and AC meets $H'Y$ at Z , we know that XB and XZ are isogonals w.r.t. $\angle CXY$. This means exactly $\angle ZXC = \angle YXB$. $\square$

Remarks. Here is an easier way to memorize the isogonal line lemma. Suppose we want to prove that OC and OD are isogonals w.r.t. $\angle AOB$. By defining $E = AC \cap BD$ and $F = AD \cap BC$, it is the same as proving OE and OF are isogonals w.r.t. $\angle AOB$. Also, note that it is quite flexible in choosing A and B , since they can be replaced by any other point on the lines OA and OB .

Suppose $\triangle ABC$ and a point P not lying on the lines BC, CA, AB are given. Let P_a, P_b, P_c be the reflections of P in BC, CA, AB respectively. By [proposition 8.1\(b\)](#), the isogonal lines of AP w.r.t. $\angle BAC$, that of BP w.r.t. $\angle CBA$, that of CP w.r.t. $\angle ACB$ are the perpendicular bisectors ℓ_a, ℓ_b, ℓ_c of P_bP_c, P_cP_a, P_aP_b respectively. By [proposition 4.1\(a\)](#), P_a, P_b, P_c are collinear iff P lies on (ABC) . In that case, ℓ_a, ℓ_b, ℓ_c are parallel to the Steiner line of P .

If we further assume P does not lie on (ABC) , then ℓ_a, ℓ_b, ℓ_c intersect at the circumcentre of $\triangle P_aP_bP_c$. This motivates the following definition.

Definition 8.2. Suppose $\triangle ABC$ is given. Let P be a point not lying on the lines BC, CA, AB and (ABC) . Then the isogonal lines of AP w.r.t. $\angle BAC$, that of BP w.r.t. $\angle CBA$, that of CP w.r.t. $\angle ACB$ are concurrent at a point Q , called the *isogonal conjugate* (等角共軛) of P w.r.t. $\triangle ABC$.



It is not hard to see that Q is distinct from A, B, C . Therefore, it is meaningful to talk about the isogonals of AQ, BQ, CQ w.r.t. the three angles respectively. Clearly, these are simply the lines AP, BP, CP , and hence they are concurrent at P . In other words, the isogonal conjugate of Q is again P . Therefore, we sometimes say that P and Q is an isogonal conjugate pair w.r.t. $\triangle ABC$. Usually, we need not specify the triangle as it is clear from the context. Note that we restrict P to be a point not lying on BC, CA, AB , since otherwise, the ‘isogonal conjugate’ would be one of A, B, C , whose isogonal conjugate cannot be defined (as the line AA is not well-defined, etc.).

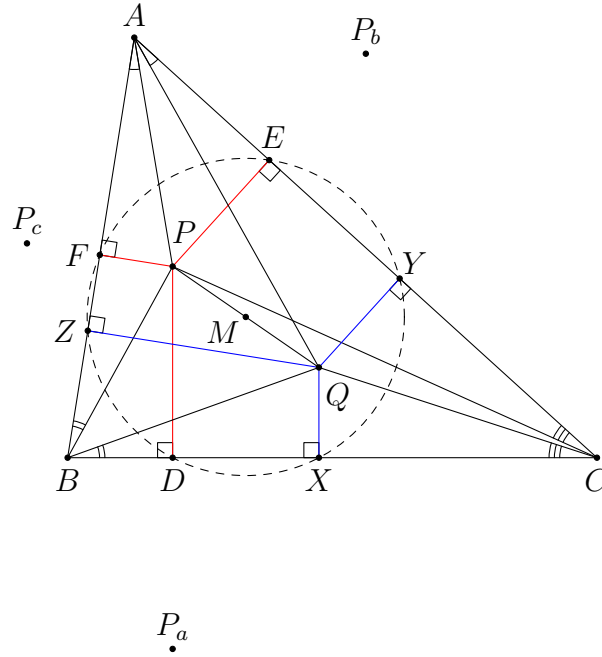
A straightforward way to use this concept is the definition. For example, if P and Q are two points inside $\triangle ABC$ such that $\angle PAB = \angle QAC$ and $\angle PBC = \angle QBA$, then we immediately know that $\angle PCB = \angle QCA$ since P and Q are isogonal conjugates.

The isogonal conjugates of the incentre and excentres of a triangle are the incentre and excentres themselves, as angle bisectors are fixed under reflections in them. A less trivial example is that the orthocentre and the circumcentre form an isogonal conjugate pair (because AH and AO are isogonals w.r.t. $\angle BAC$, etc.).

The following proposition gives some important properties about isogonal conjugate pairs.

Proposition 8.2. Let P and Q be an isogonal conjugate pair w.r.t. $\triangle ABC$. Let P_a, P_b, P_c be the reflections of P in BC, CA, AB respectively. Let D, E, F be the projections of P onto BC, CA, AB respectively, and let X, Y, Z be the projections of Q onto BC, CA, AB respectively. Also, let M be the midpoint of PQ . Then the following hold.

- (a) the circumcentre of $\triangle P_a P_b P_c$ is Q
- (b) D, E, F, X, Y, Z lie on the same circle with centre M
- (c) $PD \times QX = PE \times QY = PF \times QZ$



Proof.

- (a) This part follows easily from the construction of Q as discussed above.
- (b) Note that $\triangle DEF$ is called the *pedal triangle*(垂足三角形) of P w.r.t. $\triangle ABC$, while (DEF) is called the *pedal circle*(垂足圓). This part states that the pedal circles of P and Q coincide. In fact, D, E, F are the midpoints of PP_a, PP_b, PP_c respectively. By considering the homothety with centre P and ratio $\frac{1}{2}$, $\triangle P_a P_b P_c$ is mapped to $\triangle DEF$. The circumcentre Q is mapped to M , which is the centre of (DEF) . Similarly, M is also the centre of (XYZ) .

Next, as $PD \perp DX$, $QX \perp DX$ and M is the midpoint of PQ , we know that M lies on the perpendicular bisector of DX . Thus, $MD = MX$. Therefore, the pedal circles of P and Q have the same centre and radius. This shows the two circles coincide as desired.

(c) It is not hard to see that $PFBD$ and $QXBZ$ are oppositely similar from the angles. Therefore, $PD \times QX = PF \times QZ$, etc.

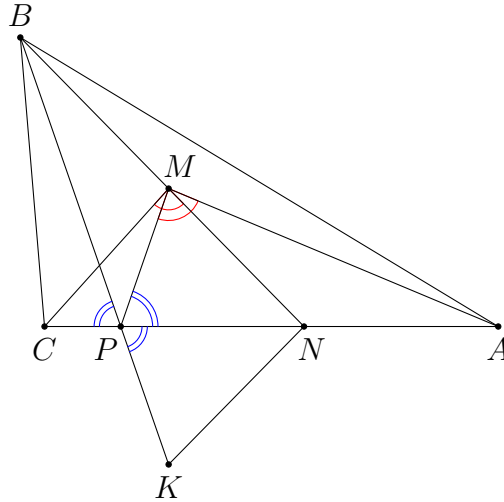
□

As a simple application, note that the orthocentre H and circumcentre O of $\triangle ABC$ form an isogonal conjugate pair. Their common pedal circle is the nine-point circle with centre N which is the midpoint of H and O .

Problem 8.5. (International Zhautykov Olympiad 2015 Problem 2) Inside the triangle ABC a point M is given. The line BM meets the side AC at N . The point K is symmetrical to M with respect to AC . The line BK meets AC at P . If $\angle AMP = \angle CMN$, prove that $\angle ABP = \angle CBN$.

Analysis. Both the condition and the assertion are about some isogonal lines. This suggests using isogonal conjugates to solve the problem. The only task is to spot a pair of isogonal conjugate in the figure. Note that we need not always use $\triangle ABC$ as the reference triangle.

Solution.



Since $\angle AMP = \angle CMN$, MA and MC are isogonals w.r.t. $\angle PMN$, which means they are isogonals w.r.t. $\angle BPM$. Also, we have

$$\angle BPC = \angle KPN = \angle MPN.$$

From the configuration, this implies PC and PA are isogonals w.r.t. $\angle BPM$. Therefore, A and C are isogonal conjugates w.r.t. $\triangle BPM$. This yields $\angle CBP = \angle ABM$, and hence $\angle ABP = \angle CBN$. □

Problem 8.6. (IMO Shortlist 2011 G6) Let ABC be a triangle with $AB = AC$, and let D be the midpoint of AC . The angle bisector of $\angle BAC$ intersects the circle through D, B , and C in a point E inside the triangle ABC . The line BD intersects the circle through A, E , and B in two points B and F . The lines AF and BE meet at a point I , and the lines CI and BD meet at a point K . Show that I is the incentre of triangle KAB .

Analysis. To begin with, we probably try to do some angle chasing using the concyclic points. It is not hard to deduce $\angle FBI = \angle IBA$, but the other two angle bisectors cannot be easily proved. Next, one may deduce that DE is the perpendicular bisector of AF , and hence $\angle CFA = 90^\circ$. These facts are all useful, but it seems that they cannot lead to a proof directly.

When we get stuck in a problem, we may try to use different ways to consider the problem, even if such ways seem to be unrelated to it. For example, in this problem, we may consider the radical axes of some circles. It should be remarked that considering the circle with diameter AC is useful as well, because of the above right angle. Another approach is to use isogonal conjugates. This is quite natural since isogonal simply means equal angles, and is thus related to angle bisectors. Also, isogonal conjugates sometimes occur naturally in some configurations, just like similar triangles and concyclic points. So thinking in this direction is not as odd as one may think.

The first step in this solution is to consider the reflection I' of I in AE , the perpendicular bisector of BC . Since we are going to prove $\angle BAI = \angle IAK$, it suffices to show AK and AI' are isogonals w.r.t. $\angle FAC$. Indeed, it can be shown that K and I' are isogonal conjugates w.r.t. $\triangle AFC$. More interestingly, F and I' are also isogonal conjugates w.r.t. $\triangle ABC$. To notice and prove all these, one may need better observations, more experiences, and, probably, more luck!

Solution. Firstly, as B, C, D, E are concyclic and AE is the axis of symmetry of the isosceles $\triangle ABC$, we have

$$\angle FBI = \angle DBE = \angle DCE = \angle ACE = \angle EBA = \angle IBA.$$

This establishes one angle bisector. Hence, it suffices to prove $\angle BAI = \angle IAK$. By some angle chasing, we get

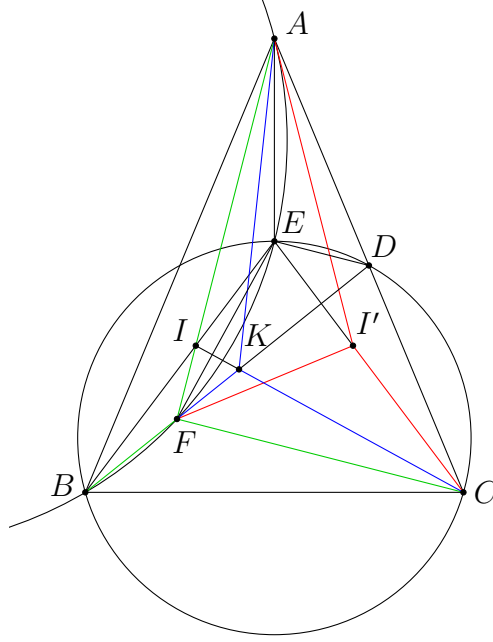
$$\angle ADE = \angle CDE = \angle CBE = \angle ECB = \angle EDB = \angle EDF.$$

Together with $\angle EAD = \angle BAE = \angle BFE = \angle DFE$ and $DE = DE$, we obtain $\triangle AED \cong \triangle FED$ (oppositely). In particular, $FD = AD = CD$, and so $\angle CFA = 90^\circ$, and DE is the perpendicular bisector of AF (so that $DE \parallel CF$).

Let I' be the reflection of I in AE . We claim that F and I' are isogonal conjugates w.r.t. $\triangle ABC$. Clearly, $\angle BAF = \angle I'AC$. We also have

$$\angle I'CB = \angle ECB = \angle EDB = \angle EDF = \angle ADE = \angle ACF.$$

These prove the claim.



Next, we shall prove that K and I' are isogonal conjugates w.r.t. $\triangle AFC$. This is a bit more difficult. Note that our final goal is to prove $\angle FAK = \angle I'AC$, so we shall not consider this pair of angles at this stage.

Firstly, as F and I' are isogonal conjugates w.r.t. $\triangle ABC$,

$$\angle ICB = \angle CBI' = \angle FBA = 2\angle EBA = 2\angle ACE = \angle ACI' + \angle FCB,$$

so that $\angle KCF = \angle ACI'$. Secondly, we observe that

$$\angle EFI = \angle EFA = \angle EBA = \angle FBE,$$

and hence EF is a tangent to (BFI) . Therefore,

$$EF^2 = EI \cdot EB = EI' \cdot EC.$$

This in turn gives $\angle I'FE = \angle ECF$, and hence

$$\begin{aligned} \angle I'FA &= \angle I'FE + \angle EFA = \angle ECF + \angle EBA = \angle ECF + \angle ACE \\ &= \angle ACF = \angle DCF = \angle CFD = \angle CFK. \end{aligned}$$

This shows K and I' are isogonal conjugates w.r.t. $\triangle AFC$, so that

$$\angle IAK = \angle FAK = \angle I'AC = \angle BAI.$$

As a result, I is the incentre of $\triangle ABK$. □

We introduce two more results before we work on another problem. Recall that a common way to use isogonal conjugates is to deduce a pair of equal angles from

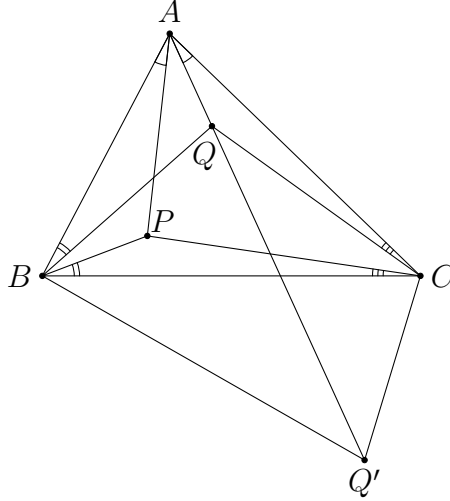
two pairs of equal angles. However, sometimes we could only find one pair of equal angles, and fail to prove the other two pairs (which are symmetric in many cases). The following proposition provides an alternative method to prove that two points are isogonal conjugates.

Proposition 8.3. Given $\triangle ABC$, let P and Q be points on the plane not lying on the lines BC, CA, AB and (ABC) . Suppose

$$\angle BAP = -\angle CAQ \quad \text{and} \quad \angle BQC = \angle ABP + \angle PCA.$$

Then one of the following holds.

- P and Q form an isogonal conjugate pair w.r.t. $\triangle ABC$
- $\triangle ABP$ and $\triangle AQC$ are directly similar



Proof. Firstly, if P and Q are isogonal conjugates, then $\angle BAP = -\angle CAQ$ and

$$\angle BQC = -\angle CBQ - \angle QCB = \angle ABP + \angle PCA.$$

Suppose there is another point Q' instead of the isogonal conjugate of P satisfying both conditions. In particular,

$$\angle BQ'C = \angle ABP + \angle PCA = \angle BQC.$$

As we have assumed $Q' \neq Q$, this shows B, C, Q, Q' are concyclic. Clearly, there can be at most one such point Q' (as the first condition fixes the line AQ' and this line only meets (BQC) at two points, one of which is Q). Then

$$\angle PBA = \angle CBQ = \angle CQ'Q = \angle CQ'A$$

and

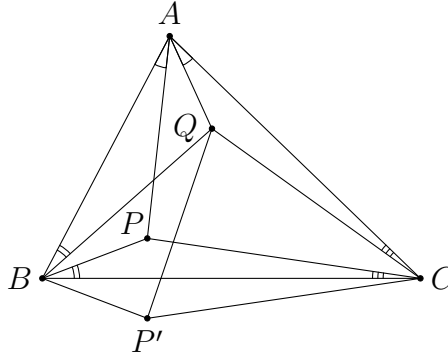
$$\angle BAP = \angle Q'AC,$$

which imply $\triangle ABP$ and $\triangle AQ'C$ are directly similar. □

The motivation of the conditions in this proposition is that both can be deduced easily if P and Q are isogonal conjugates. However, it turns out that they are not sufficient conditions. There is another possibility for the two conditions to hold. But this is not that bad because the second possibility is a standard configuration of spiral similarity. Thus, both results have important geometric meanings. Usually one of the cases can be easily rejected using the configuration (depending on which result we want).

The condition $\angle BQC = \angle ABP + \angle PCA$ may also be written in the usual angle notation. Yet we need to be careful about the configuration when it is used.

Proposition 8.4. Let P and Q be an isogonal conjugate pair w.r.t. $\triangle ABC$. Let P' be the reflection of P in BC . Then A and P' is an isogonal conjugate pair w.r.t. $\triangle QBC$.



Proof. By definition, we have $\angle PBC = \angle ABQ$ and $\angle PCB = \angle ACQ$. One may suspect P and A are isogonal conjugates w.r.t. $\triangle QBC$. But this is not true since a negative sign is missing from each equation. This motivates the reflection we consider in this proposition. Indeed, we have $-\angle P'BC = \angle ABQ$ and $-\angle P'CB = \angle ACQ$, which leads to the desired result. $\square$

In particular, [proposition 8.4](#) yields $\angle BQP' = -\angle CQA$.

Problem 8.7. (IMO Shortlist 2014 G5/Problem 3) Convex quadrilateral $ABCD$ has $\angle ABC = \angle CDA = 90^\circ$. Point H is the foot of the perpendicular from A to BD . Points S and T lie on sides AB and AD , respectively, such that H lies inside triangle SCT and

$$\angle CHS - \angle CSB = 90^\circ, \quad \angle THC - \angle DTC = 90^\circ.$$

Prove that line BD is tangent to the circumcircle of triangle TSH .

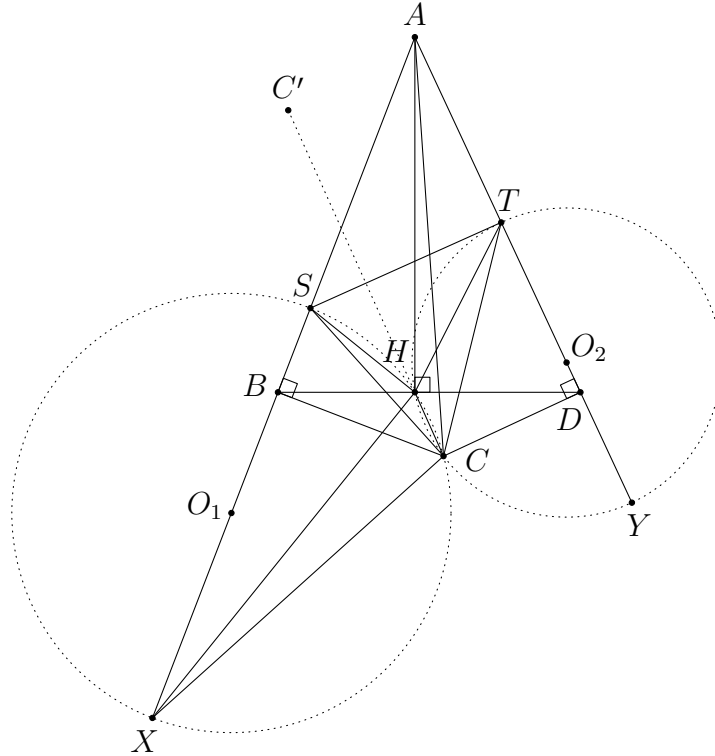
Analysis. Firstly, we try to see how to locate points S and T . From the condition $\angle CHS = \angle CSB + 90^\circ$, it is natural to divide $\angle CHS$ into two parts, one of which is

90° . If we take X on the line AB such that $\angle SHX = 90^\circ$, then $\angle XHC = \angle XSC$, which implies X, C, H, S are concyclic, lying on the circle with diameter XS . This provides one possible way to draw point S (and similarly point T).

It is obvious that AC and AH are isogonals w.r.t. $\angle BAD$. This suggests a possibility of solving the problem in this direction. We may try to guess whether points C and H would be isogonal conjugates w.r.t. a certain triangle. Certainly it cannot be $\triangle ABD$ (as H lies on BD). The next simple guess is $\triangle AST$, which is indeed the case. To prove this, note that we get one pair of equal angles $\angle SAC = \angle HAT$. The other two pairs are symmetric to each other so there is not much hope to prove this by definition (or otherwise the concept of isogonal conjugate is not needed since we can obtain all pairs of equal angles directly). Hence, we may try to apply [proposition 8.3](#).

Once this is established, we can focus our attention on $\triangle TSH$ (for example, by using [proposition 8.4](#)). The equal angles we have obtained up to this point should be sufficient to yield the final result.

[Solution.](#)



Let X be the point on the ray AB for which $\angle SHX = 90^\circ$. By the first condition, we have $\angle XHC = \angle XSC$, so that X, C, H, S are concyclic. The circle on which these points lie has SX as a diameter since $\angle SHX = 90^\circ$. Therefore, if we define O_1 to be the intersection of the perpendicular bisector of CH and the line AB , then the

circle with centre O_1 passing through C and H would meet AB at S and X , where S lies between A and O_1 . We can define Y and O_2 analogous to X and O_1 on the line AD to locate point T .

We first prove that C and H are isogonal conjugates w.r.t. $\triangle AST$ using [proposition 8.3](#). Firstly, by [proposition 8.1\(a\)](#), AC and AH are isogonals w.r.t. $\angle BAD$. Hence, we have $\angle SAC = \angle HAT$. Next, we need to prove $\angle SCT = \angle ASH + \angle HTA$. This follows from

$$\begin{aligned}\angle SCT &= \angle SCH + \angle HCT = \angle SXH + \angle HYT = \left(\frac{\pi}{2} + \angle XSH\right) + \left(\frac{\pi}{2} + \angle HTY\right) \\ &= \angle ASH + \angle HTA.\end{aligned}$$

In addition, we have to reject the case for which $\triangle ASC$ and $\triangle AHT$ are directly similar. This relies on the configuration (that S and T lie on the segments AB and AD), and so directed angles should not be used. Indeed,

$$\angle ASC > \angle ABC = 90^\circ = \angle AHD > \angle AHT,$$

so that the two angles cannot be equal. By [proposition 8.3](#), C and H are isogonal conjugates w.r.t. $\triangle AST$. In view of [proposition 8.4](#), if we define C' to be the reflection of C in ST , then A and C' are isogonal conjugates w.r.t. $\triangle TSH$. Now we are much closer to our goal as this provides some information of this triangle. That is the reason why we try to use this proposition.

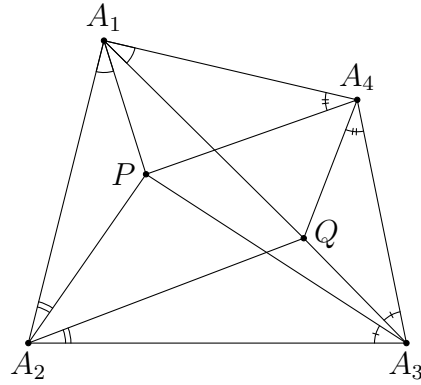
Recall that the isogonal of AH w.r.t. $\angle THS$ should be the altitude from H (of $\triangle TSH$), as AH should pass through the centre of (TSH) . Therefore, we should prove that $C'H \perp ST$. Since it is obvious that $CC' \perp ST$, we may simply prove $CH \perp ST$. This can be done by simple angle chasing. For example, using the isogonal conjugates and conditions,

$$\begin{aligned}\angle(CH, ST) &= \angle(CH, HS) + \angle(HS, ST) = \angle CHS + \angle HST \\ &= \left(\frac{\pi}{2} + \angle CHX\right) + \angle ASC = \frac{\pi}{2} + \angle CSX + \angle XSC = \frac{\pi}{2}.\end{aligned}$$

This yields $CH \perp ST$, and hence C, H, C' are collinear with $C'H \perp ST$. Then AH passes through the centre of (TSH) , so that BD is tangent to (TSH) (using the fact that $AH \perp BD$). $\square$

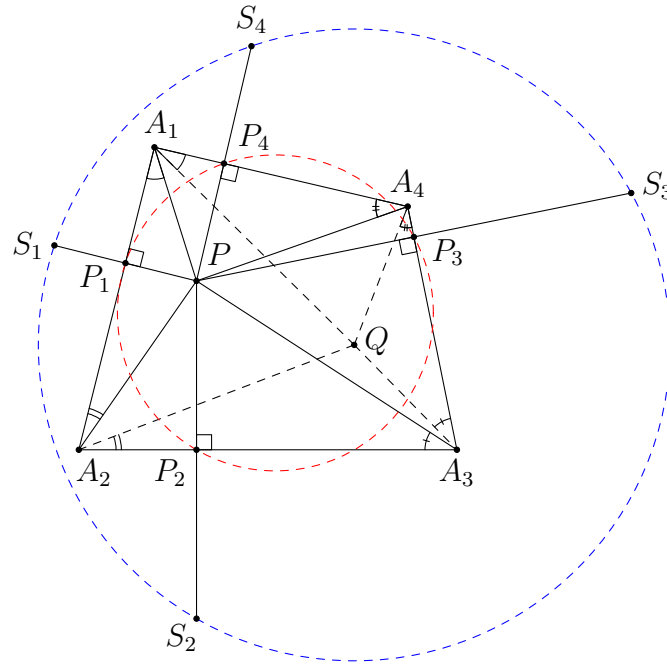
8.2 Isogonal Conjugate in Polygons

We can define the isogonal conjugate of a point P w.r.t. a polygon analogously. However, it is not guaranteed that such an isogonal conjugate always exists. Indeed, let $A_1A_2\cdots A_n$ be a convex polygon. Given any point P not lying on the sidelines of the polygon, we can construct the isogonal of A_iP w.r.t. $\angle A_i$ for $i = 1, 2, \dots, n$. If these n lines are concurrent at a point Q , then Q is called the isogonal conjugate of P w.r.t. the polygon.



The following gives a criterion for the isogonal conjugate to exist.

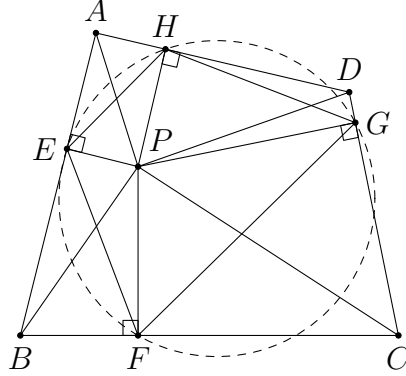
Proposition 8.5. Let $A_1A_2\cdots A_n$ be a convex polygon. Let P be a point inside the polygon, and let P_i be the projection of P onto A_iA_{i+1} for each $i = 1, 2, \dots, n$ (where $A_{n+1} = A_1$). Then the isogonal conjugate of P w.r.t. $A_1A_2\cdots A_n$ exists iff $P_1, P_2, \dots, P_n$ are concyclic.



Proof. Let S_i be the reflection of P in A_iA_{i+1} for each $i = 1, 2, \dots, n$. By [proposition 8.1\(b\)](#), the isogonal of A_iP w.r.t. $\angle A_i$ is the perpendicular bisector of S_iS_{i-1} (where $S_0 = S_n$). Thus, these n isogonals are concurrent iff these perpendicular bisectors are concurrent at the circumcentre of $S_1S_2\cdots S_n$, or equivalently, $S_1S_2\cdots S_n$ is cyclic. Note that $P_1P_2\cdots P_n$ is similar to $S_1S_2\cdots S_n$ by considering the homothety at P and ratio $\frac{1}{2}$. This proves the equivalence. $\square$

The most commonly used result is the case $n = 4$. In that case, we have a much simpler condition as given below.

Proposition 8.6. Let P be a point inside a convex quadrilateral $ABCD$. Then the isogonal conjugate of P w.r.t. $ABCD$ exists iff $\angle APB + \angle CPD = 180^\circ$.



Proof. This is merely angle chasing. Let E, F, G, H be the projections of P onto AB, BC, CD, DA respectively. Note that A, E, P, H are concyclic, etc. Thus, we have

$$\begin{aligned} \angle EHG + \angle EFG &= (180^\circ - \angle AHE - \angle DHG) + (180^\circ - \angle BFE - \angle CFG) \\ &= 360^\circ - \angle APE - \angle DPG - \angle BPE - \angle CPG \\ &= 360^\circ - (\angle APE + \angle BPE) - (\angle DPG + \angle CPG) \\ &= 360^\circ - \angle APB - \angle CPD. \end{aligned}$$

Therefore, E, F, G, H are concyclic iff $360^\circ - \angle APB - \angle CPD = 180^\circ$, which is the same as $\angle APB + \angle CPD = 180^\circ$. By [proposition 8.5](#), the result follows immediately. $\square$

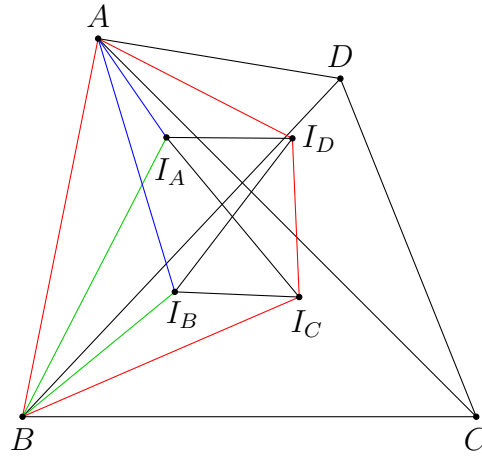
Problem 8.8. (All-Russian MO 2017 Grade 11 Problem 8) Given a convex quadrilateral $ABCD$. We denote I_A, I_B, I_C and I_D centres of $\omega_A, \omega_B, \omega_C$ and ω_D inscribed in the triangles DAB, ABC, BCD and CDA , respectively. It turns out that $\angle BI_A A + \angle I_C I_A I_D = 180^\circ$. Prove that $\angle BI_B A + \angle I_C I_B I_D = 180^\circ$.

Analysis. By [proposition 8.6](#), we know that I_A has an isogonal conjugate in $ABI_C I_D$, and we need to prove I_B has an isogonal conjugate in $ABI_C I_D$. It is not hard to guess that I_A and I_B are indeed isogonal conjugates of each other. This gives a simple proof to this problem.

Solution. By [proposition 8.6](#), we know that I_A has an isogonal conjugate w.r.t. $ABI_C I_D$. Since

$$\angle I_A A I_D = \angle I_A A D - \angle I_D A D = \frac{1}{2} \angle BAD - \frac{1}{2} \angle CAD = \frac{1}{2} \angle BAC = \angle B A I_B$$

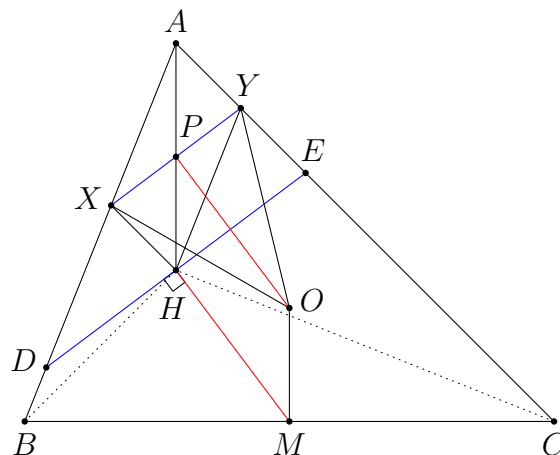
and $\angle I_B B I_C = \angle A B I_A$ by symmetry, the isogonal conjugate of I_A is just I_B . Again by [proposition 8.6](#), we have $\angle BI_B A + \angle I_C I_B I_D = 180^\circ$. $\square$



Problem 8.9. Let H be the orthocentre of an acute $\triangle ABC$. Let M be the midpoint of BC . Suppose the line through H perpendicular to HM meets AB and AC at D and E respectively. Prove that $HD = HE$.

Analysis. To use the perpendicular condition, one approach is to look for some lines parallel to HM . Let P be the midpoint of AH and let O be the circumcentre of $\triangle ABC$. Recall that $PHMO$ is a parallelogram. Therefore, $HM \parallel PO$. On the other hand, we may also think of the parallelogram trick since there are many midpoints in this problem. For example, we can construct a parallelogram $AXHY$ where X and Y lie on AB and AC respectively. Then P is also the midpoint of XY , and we are going to show that X and Y are the midpoints of AD and AE respectively (afterwards we can use a homothety at A to map XPY to DHE). We shall see how to use isogonal conjugates to prove this.

Solution.



Let P be the midpoint of AH and let O be the circumcentre of $\triangle ABC$. Also, let X and Y be points on AB and AC respectively such that $AXHY$ is a parallelogram.

Then P is the midpoint of XY . Since

$$\angle BHC + \angle XHY = (180^\circ - A) + A = 180^\circ,$$

H has an isogonal conjugate w.r.t. $BCYX$ by [proposition 8.6](#). As $\angle CBO = \angle HBA$ and $\angle BCO = \angle HCA$, this isogonal conjugate is simply O . Thus, we have

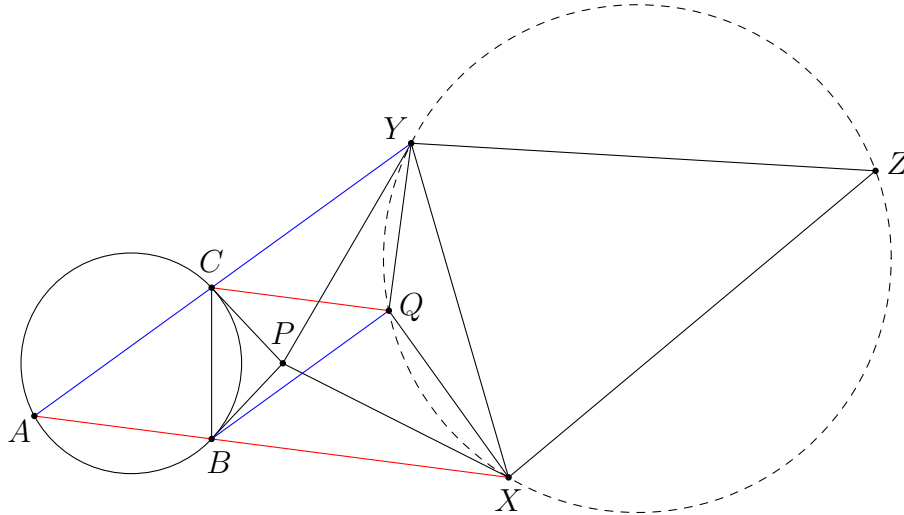
$$\angle OXY = \angle BXH = A = \angle HYC = \angle XYO.$$

This shows $\triangle OXY$ is isosceles, and thus $OP \perp XY$. Recall that $PHMO$ is a parallelogram. Therefore, we have $HM \parallel PO$. It follows that $XY \parallel DE$. Finally, as $PX = PY$, we have $HD = HE$ by the intercept theorem (or homothety). $\square$

Problem 8.10. (Czech-Polish-Slovak Match 2020 Problem 6) Let ABC be an acute triangle. Let P be a point such that PB and PC are tangent to circumcircle of triangle ABC . Let X and Y be variable points on AB and AC , respectively, such that $\angle XPY = 2\angle BAC$ and P lies in the interior of triangle AXY . Let Z be the reflection of A across XY . Prove that the circumcircle of triangle XYZ passes through a fixed point.

Analysis. With some angle chasing, one observes that $\angle BPC + \angle XPY = 180^\circ$. This suggests considering the isogonal conjugate of P w.r.t. $CBXY$ to do some more angle chasing. It turns out that this point is exactly the fixed point we need.

Solution.



Since

$$\angle BPC + \angle XPY = 180^\circ - \angle PBC - \angle PCB + 2\angle BAC = 180^\circ,$$

P has an isogonal conjugate w.r.t. $CBXY$ by [proposition 8.6](#). Let Q be this isogonal conjugate. As $\angle QBX = \angle PBC = \angle BAC$, we have $BQ \parallel AC$. Similarly, $CQ \parallel AB$.

This shows $QCAB$ is a parallelogram, and hence Q does not depend on X and Y . Again by [proposition 8.6](#), we have $\angle BQC + \angle XQY = 180^\circ$. This implies

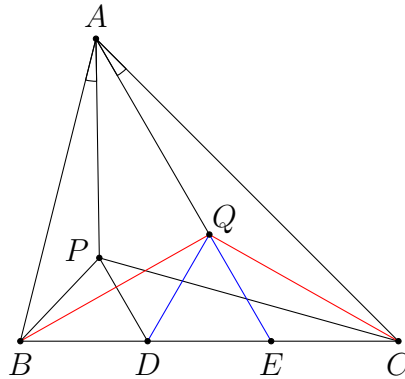
$$\angle XQY = 180^\circ - \angle BQC = 180^\circ - \angle BAC = 180^\circ - \angle XZY.$$

As Q and Z lie on different sides of XY , Q always lies on (XYZ) , and we are done. $\square$

Problem 8.11. (ELMO 2012 Problem 5) Let ABC be an acute triangle with $AB < AC$, and let D and E be points on side BC such that $BD = CE$ and D lies between B and E . Suppose there exists a point P inside ABC such that $PD \parallel AE$ and $\angle PAB = \angle EAC$. Prove that $\angle PBA = \angle PCA$.

Analysis. It is clear that the problem is related to isogonals. Thus, we may consider the isogonal conjugate of P w.r.t. $\triangle ABC$. Assuming the final result, this point should be the intersection of AE and the perpendicular bisector of BC . Therefore, we may define this point in this alternative way, and prove that it is the isogonal conjugate of P . In the solution, we demonstrate the use of [proposition 8.6](#) on a degenerate quadrilateral.

Solution.



Let Q be the intersection of AE and the perpendicular bisector of BC . Note that Q also lies on the perpendicular bisector of DE , and $\triangle QBD \cong \triangle QCE$. Therefore, we have

$$\angle BQD + \angle AQC = \angle CQE + \angle AQC = 180^\circ.$$

By [proposition 8.6](#), Q has an isogonal conjugate w.r.t. the degenerate quadrilateral $ABDC$. As $\angle BAP = \angle QAC$ and $\angle BDP = \angle DEQ = \angle QDC$, this isogonal conjugate is exactly P . This implies $\angle DCP = \angle QCA$. Together with $\angle BAP = \angle QAC$, we see that P and Q form an isogonal conjugate pair w.r.t. $\triangle ABC$ as well. Therefore, we obtain

$$\angle ABP = \angle QBC = \angle QCB = \angle ACP$$

as desired. $\square$

Here is the summary of the things used in the problems in this section.

- properties of isogonal lines ([problems 8.1, 8.2, 8.7](#))
- isogonal line lemma ([problems 8.3, 8.4](#))
- isogonal conjugates in triangles ([problems 8.5, 8.6, 8.7, 8.11](#))
- isogonal conjugates in quadrilaterals ([problems 8.8, 8.9, 8.10, 8.11](#))

9 More on Triangle Geometry

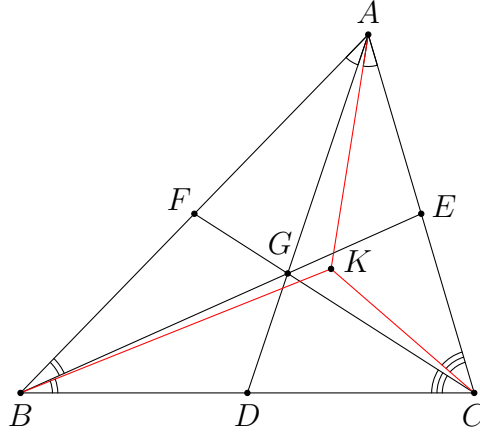
In this section, we investigate more about triangle geometry. We shall introduce a few more triangle centres other than those revised in section 4.

9.1 Symmedians

Symmedians are defined using isogonal lines. They are closely related to medians and tangents to the circumcircle.

Definition 9.1. Let D be the midpoint of the side BC of $\triangle ABC$. The isogonal line of the median AD w.r.t. $\angle BAC$ is called the A -*symmedian* (類似中線) of $\triangle ABC$.

One can define the symmedians from B and C in the same way. The three symmedians are concurrent at the isogonal conjugate of the centroid G , called the *symmedian point* (類似重心) or *Lemoine point* (萊莫恩點) of $\triangle ABC$, usually denoted by K .



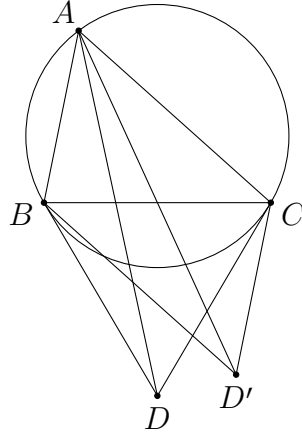
The following provides two common configurations involving the symmedians.

Proposition 9.1. In $\triangle ABC$, let D be the intersection point of the tangents at B and C to (ABC) . Then AD is the A -symmedian.

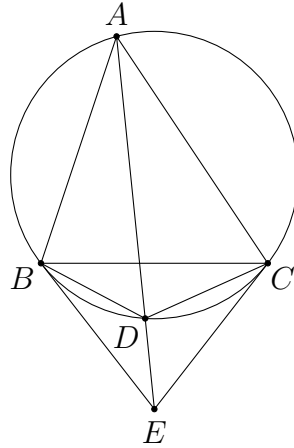
Proof. Let D' be the isogonal conjugate of D w.r.t. $\triangle ABC$. Then

$$\angle D'BC = \angle ABD = \angle ACB,$$

which implies $AC \parallel BD'$. Similarly, $AB \parallel CD'$. This shows $ABD'C$ is a parallelogram. In particular, AD' bisects BC , and hence AD' coincides with the median from A . By definition, its isogonal AD is the A -symmedian. $\square$



Proposition 9.2. Given $\triangle ABC$, let D be a point on $\widehat{BC}$ of (ABC) not containing A . Then AD is the A -symmedian of $\triangle ABC$ iff $\frac{DB}{DC} = \frac{AB}{AC}$.



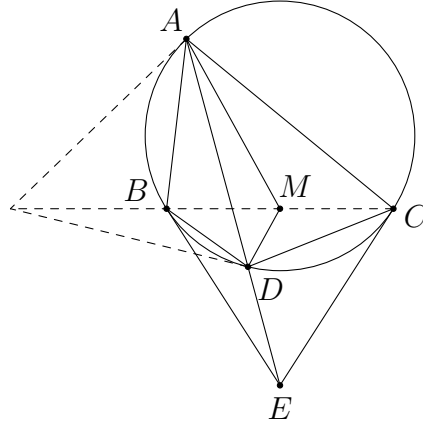
Proof. Suppose AD is the A -symmedian. Let E be the intersection point of the tangents at B and C . Then A, D, E are collinear by [proposition 9.1](#). This gives $\triangle EBD \sim \triangle EAB$, and hence $\frac{BD}{AB} = \frac{EB}{EA}$. Similarly, we have $\frac{CD}{AC} = \frac{EC}{EA}$. It follows that

$$\frac{BD}{AB} = \frac{EB}{EA} = \frac{EC}{EA} = \frac{CD}{AC}.$$

This proves the ‘only if’ part. The ‘if’ part also holds since the A -symmedian has a unique intersection point with $\widehat{BC}$ not containing A , and there is only one point D on this arc such that $\frac{DB}{DC} = \frac{AB}{AC}$. $\square$

A cyclic quadrilateral $ABDC$ satisfying $\frac{DB}{DC} = \frac{AB}{AC}$ is called a *harmonic quadrilateral* (調和四邊形). Note that this condition means the products of the lengths of the

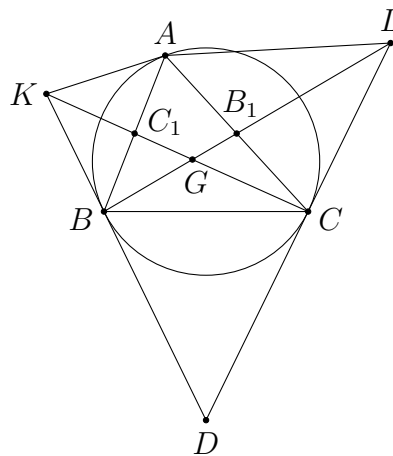
opposite sides are equal. In particular, the condition is symmetric in all vertices. For example, we immediately know from [proposition 9.2](#) that AD is the D -symmedian of $\triangle BCD$, and BC is the B -symmedian of $\triangle ABD$ as well as the C -symmedian of $\triangle ACD$. Thus, by [proposition 9.1](#), the tangents at A and D are concurrent with BC . In addition, let M be the midpoint of BC . Then we can easily use angles to prove that $\triangle AMC \sim \triangle ABD \sim \triangle CMD$.



Problem 9.1. (Sharygin Geometry Olympiad 2018 Final Grade 9 Problem 7) Let B_1, C_1 be the midpoints of sides AC, AB of a triangle ABC respectively. The tangents to the circumcircle at B and C meet the rays CC_1, BB_1 at points K and L respectively. Prove that $\angle BAK = \angle CAL$.

Analysis. Since the problem involves the tangents at the vertices to the circumcircle, and we are going to prove AK and AL are isogonals w.r.t. $\angle BAC$, it is likely that symmedians can be used. Indeed, the result follows immediately from the isogonal line lemma.

Solution.



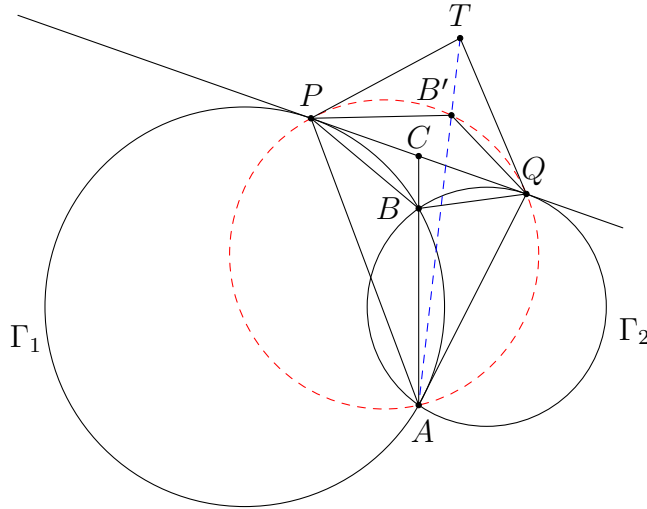
Let BK and CL meet at D , and let $G = BL \cap CK$ be the centroid of $\triangle ABC$. By [proposition 9.1](#), AD is the A -symmedian of $\triangle ABC$. Therefore, AD and AG are isogonals w.r.t. $\angle BAC$. Now, as $BD \cap CG = K$ and $CD \cap BG = L$, AK and AL are isogonals w.r.t. $\angle BAC$ by the isogonal line lemma. This means exactly $\angle BAK = \angle CAL$. $\square$

Problem 9.2. Two circles Γ_1 and Γ_2 intersect at distinct points A and B . A common tangent to Γ_1 and Γ_2 touches the circles at P and Q respectively. Let B' be the reflection of B in PQ , and let T be the intersection of the tangents at P and Q to (APQ) . Prove that A, B', T are collinear.

Analysis. Using our new knowledge, we know that AT is the A -symmedian of $\triangle APQ$. Also, using the radical axis, we know that AB bisects PQ . Equivalently, AB is the A -median of $\triangle APQ$. Thus, these things can probably be put together to yield a proof.

It could happen that B lies inside $\triangle APQ$ or A lies inside $\triangle BPQ$. When we think of a proof, we mainly focus on one of the cases. By using directed angles, the same solution works in both cases.

Solution.



Since PQ is a common tangent to Γ_1 and Γ_2 , we have

$$\angle PB'Q = \angle QBP = \angle BQP + \angle QPB = \angle BAQ + \angle PAB = \angle PAQ.$$

This implies A, B', P, Q are concyclic. Then we have

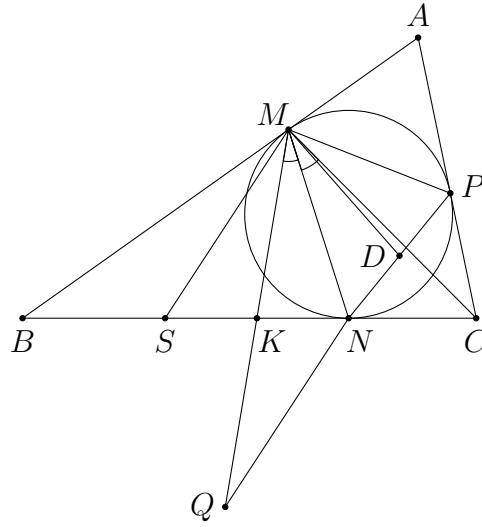
$$\angle QAB' = \angle QPB' = \angle BPQ = \angle BAP.$$

This means AB' and AB are isogonals w.r.t. $\angle QAP$. By [proposition 5.1](#), AB bisects PQ . Thus, AB is the A -median of $\triangle APQ$, and hence AB' is the A -symmedian of $\triangle APQ$. By [proposition 9.1](#), AT is also the A -symmedian of $\triangle APQ$. Therefore, A, B', T are collinear. $\square$

Problem 9.3. (Bulgaria Regional MO 2021 Grade 9) The incircle of $\triangle ABC$ touches the sides AB, BC and CA at the points M, N and P , respectively. The point K on BC is such that MN bisects $\angle CMK$, and the point Q is such that K is the midpoint of MQ . Prove that $\angle KNQ = \angle PNM$.

Analysis. Note that MC is the M -symmedian of $\triangle MNP$. In view of the condition $\angle KMN = \angle NMC$, we may consider the midpoint D of NP to generate another angle which is equal to these. Another idea is to construct a parallelogram $MSQN$ to handle the midpoint condition.

Solution.



Let D be the midpoint of NP . As MC is the M -symmedian of $\triangle MNP$ by [proposition 9.1](#), we have

$$\angle DMP = \angle NMC = \angle KMN.$$

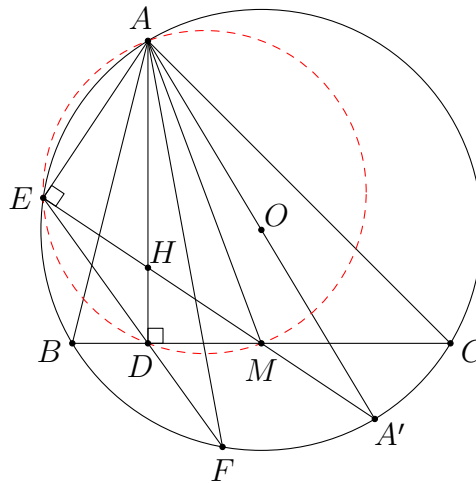
Together with $\angle MPD = \angle MNK$, we deduce $\triangle MDP \sim \triangle MKN$. Let S be the reflection of N in K . As D is the midpoint of NP and K is the midpoint of SN , we have $MNDP \sim MSKN$. Also, note that $MSQN$ is a parallelogram. Therefore, we have

$$\angle KNQ = \angle KSM = \angle DNM = \angle PNM.$$

This completes the proof. □

Problem 9.4. (APMO 2012 Problem 4) Let ABC be an acute triangle. Denote by D the foot of the perpendicular line drawn from the point A to the side BC , by M the midpoint of BC , and by H the orthocentre of ABC . Let E be the point of intersection of the circumcircle Γ of the triangle ABC and the half line MH , and F be the point of intersection (other than E) of the line ED and the circle Γ . Prove that $\frac{BF}{CF} = \frac{AB}{AC}$ must hold.

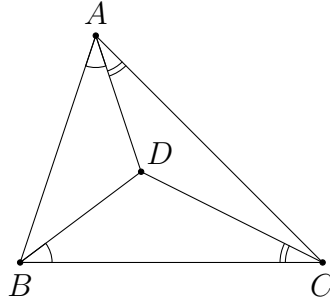
Solution.


$$\angle CAF = \angle DAB + \angle MED = \angle DAB + \angle MAD = \angle MAB.$$

This shows AF is the A -symmedian of $\triangle ABC$. By [proposition 9.2](#), the result follows. $\square$

In this subsection, we introduce a pair of isogonal conjugates that have some useful properties.

$$\angle DAB = \angle DBC \quad \text{and} \quad \angle DAC = \angle DCB.$$



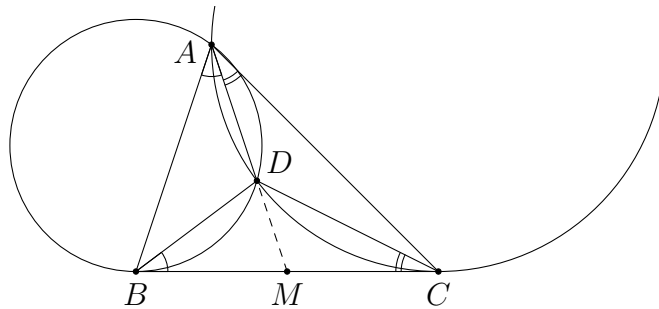
Equivalently, D is the point such that both (ABD) and (ACD) are tangent to BC . The following gives the properties of the humpty point.

Proposition 9.3. In $\triangle ABC$, let D be the A -humpty point, let H be the orthocentre of $\triangle ABC$, and let E be the point such that $ABEC$ is a harmonic quadrilateral. Then the following hold.

- (a) D lies on the A -median of $\triangle ABC$
- (b) $\frac{DB}{DC} = \frac{AB}{AC}$
- (c) D lies on (BCH)
- (d) D lies on the circle with diameter AH
- (e) $\triangle BCD \cong \triangle BCE$

Proof.

- (a) This follows from [proposition 5.1](#) since AD is the radical axis of (ABD) and (ACD) , and BC is an external common tangent of these circles.



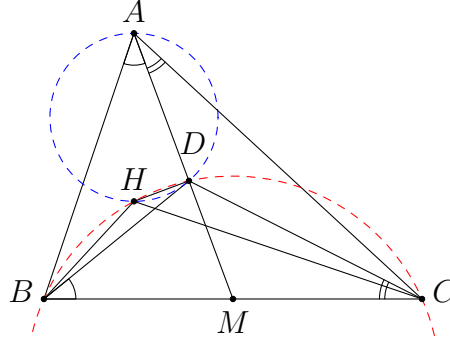
- (b) Let M be the midpoint of BC . Then $\triangle MBD \sim \triangle MAB$ and $\triangle MCD \sim \triangle MAC$. Thus, we have

$$\frac{BD}{AB} = \frac{MB}{MA} = \frac{MC}{MA} = \frac{CD}{AC}.$$

The result follows readily.

(c) The points B, C, D, H are concyclic since

$$\angle BDC = \angle DBC + \angle BCD = \angle DAB + \angle CAD = \angle CAB = \angle BHC.$$

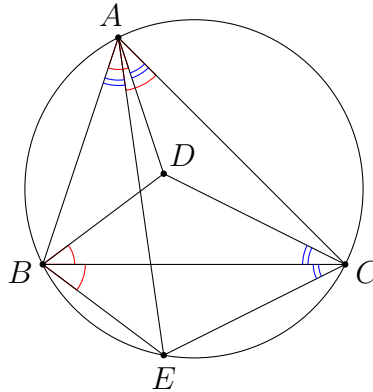


(d) Note that A is the orthocentre of $\triangle BCH$. Thus, AM is the line joining the midpoint of BC and the orthocentre of $\triangle BCH$. By [problem 4.9](#), this line passes through the second intersection point of (BCH) and the circle with diameter AH . This intersection point is exactly D by parts (a) and (c). This proves D lies on the circle with diameter AH . As a corollary, we have $\angle ADH = 90^\circ$.

(e) By [proposition 9.2](#), AE is the A -symmedian of $\triangle ABC$. Therefore, we have

$$\angle EBC = \angle EAC = \angle BAD = \angle CBD,$$

and similarly $\angle ECB = \angle BCD$. Together with the common side BC , we obtain $\triangle DBC \cong \triangle ECB$ (oppositely). As a corollary, D and E are symmetric about the line BC .



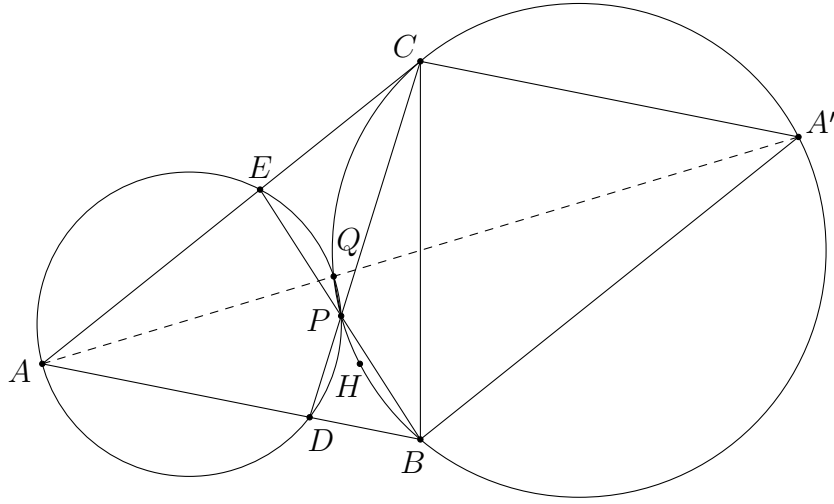
□

To use humpty points in solving problems, one way is to observe some humpty points in the figure and use their properties given in the above proposition. Therefore, one should be familiar with all the above properties.

Problem 9.5. Let D and E be points on the sides AB and AC of $\triangle ABC$ respectively. Let P be the intersection point of BE and CD , and let H be the orthocentre of $\triangle ABC$. Suppose A, D, P, E are concyclic, and let Q be the second intersection point of $(ADPE)$ and (PBC) . Prove that $\angle AQH = 90^\circ$.

Analysis. It is not hard to show that H lies on (PBC) . Therefore, Q should be the point lying on (HBC) such that $\angle AQH = 90^\circ$, which means Q should be the A -humpty point of $\triangle ABC$. One possible approach is to show that Q lies on the A -median. We may consider the point A' such that $ABA'C$ is a parallelogram. Note that A' lies on (BCH) , and AA' is the A -median. Thus, different parts of the problem can be related together.

Solution.



Firstly, as

$$\angle BPC = \angle EPD = \angle EAD = \angle CAB = \angle BHC,$$

the points B, C, P, H are concyclic. Secondly, let A' be the point such that $ABA'C$ is a parallelogram. Then A' also lies on (BCH) since $\angle BA'C = \angle CAB = \angle BHC$. In addition, we have

$$\angle AQP = \angle ADP = \angle A'CP = \angle A'QP.$$

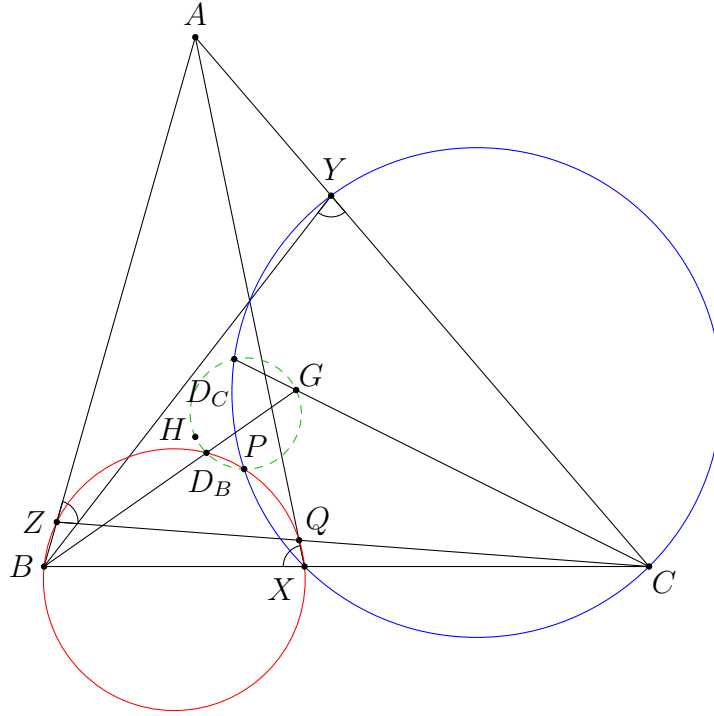
This implies A, Q, A' are collinear. As AA' bisects BC , it is the A -median of $\triangle ABC$. Now, we have found that Q is the intersection point of the A -median and (BCH) (other than the point A'). By [proposition 9.3\(a\)](#) and [\(c\)](#), the point Q must be the A -humpty point of $\triangle ABC$. Thus, by [proposition 9.3\(d\)](#), we have $\angle AQH = 90^\circ$. $\square$

Remarks. The result of this problem is also a useful property of the humpty point. In other words, the humpty point always lies on $(ADPE)$. We shall demonstrate the use of this property in the next problem.

Problem 9.6. (Brazil MO 2015 Problem 6) Let $\triangle ABC$ be a scalene triangle and X, Y and Z be points on the lines BC, AC and AB , respectively, such that $\angle AXB = \angle BYC = \angle CZA$. The circumcircles of $\triangle BXZ$ and $\triangle CXY$ intersect at P . Prove that P is on the circle which diameter has ends in the orthocentre H and in the centroid G of $\triangle ABC$.

Analysis. The circle with diameter GH passes through all the humpty points of $\triangle ABC$. This is the main reason why we would think of humpty points. Also, the intersection point Q of AX and CZ lies on (BXZ) . Thus, we know from [problem 9.5](#) that the B -humpty point lies on (BXZ) . These allow us to work out a proof easily.

Solution.



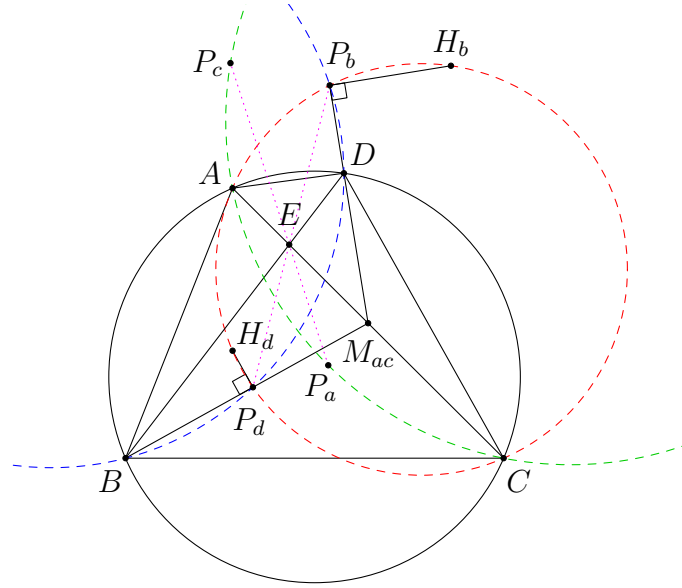
Let Q be the intersection point of AX and CZ . As $\angle AXB = \angle CZA$, the points B, X, Q, Z are concyclic. By [problem 9.5](#), the B -humpty point D_B of $\triangle ABC$ lies on (BXZ) . Similarly, the C -humpty point D_C of $\triangle ABC$ lies on (CXY) . Observe that P is the Miquel point of $\triangle XD_C D_B$ w.r.t. $\triangle GBC$ as it lies on both (BXD_B) and (CXD_C) . Thus, P also lies on $(GD_B D_C)$.

Next, by [proposition 9.3\(a\)](#) and [\(d\)](#), we know that D_B lies on BG and $\angle HD_B G = 90^\circ$. Therefore, D_B lies on the circle with diameter GH . Similarly, D_C lies on the same circle. It follows that $(GD_B D_C)$ is the circle with diameter GH . This proves P lies on this circle. $\square$

Problem 9.7. (Sharygin Geometry Olympiad 2023 Correspondence Problem 21)
 Let $ABCD$ be a cyclic quadrilateral; M_{ac} be the midpoint of AC ; H_d, H_b be the orthocentres of $\triangle ABC, \triangle ADC$ respectively; P_d, P_b be the projections of H_d and H_b to BM_{ac} and DM_{ac} respectively. Define similarly P_a, P_c for the diagonal BD . Prove that P_a, P_b, P_c, P_d are concyclic.

Analysis. We immediately know that P_d is the B -humpty point of $\triangle ABC$, etc. The main idea in the solution is to find out many groups of concyclic points. Knowing the properties of humpty points allow us to find out such groups quickly.

Solution.



By [proposition 9.3\(a\)](#) and [\(d\)](#), we know that P_d is the B -humpty point of $\triangle ABC$. Therefore, by [proposition 9.3\(c\)](#), the points A, H_d, P_d, C are concyclic. Similarly, A, H_b, P_b, C are concyclic. Also, A, H_d, C, H_b are concyclic as

$$\angle AH_dC = \angle CBA = \angle CDA = \angle AH_bC.$$

Thus, A, H_d, P_d, C, H_b, P_b lie on the same circle.

Next, recall that AC is the external common tangent of (BAP_d) and (BCP_d) . Therefore, we have

$$M_{ac}B \times M_{ac}P_d = M_{ac}A^2.$$

By symmetry, we have

$$M_{ac}D \times M_{ac}P_b = M_{ac}A^2 = M_{ac}B \times M_{ac}P_d.$$

This implies B, P_d, D, P_b are concyclic. By symmetry, A, P_a, C, P_c are concyclic.

Now, by considering the radical centre of $(ABCD)$, (AP_dCP_b) , (BP_dDP_b) , we find that AC, BD, P_bP_d are concurrent at a point E . Similarly, P_aP_c passes through E ,

and hence

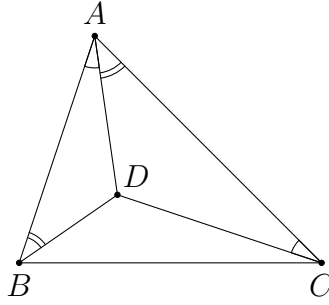
$$EP_b \times EP_d = EB \times ED = EA \times EC = EP_a \times EP_c.$$

This proves P_a, P_b, P_c, P_d are concyclic. $\square$

Next, we introduce another triangle centre with similar properties.

Definition 9.3. Given $\triangle ABC$, the *A-dumpty point* of $\triangle ABC$ is defined as the point D such that

$$\angle DAB = \angle DCA \quad \text{and} \quad \angle DAC = \angle DBA.$$



Equivalently, D is the point such that (ABD) is tangent to AC and (ACD) is tangent to AB . Also, D is the centre of spiral similarity that maps BA to AC . The following gives the properties of the dumpty point.

Proposition 9.4. In $\triangle ABC$, let D be the A -dumpty point, let O be the circumcentre of $\triangle ABC$, and let E be the point such that $ABEC$ is a harmonic quadrilateral. Then the following hold.

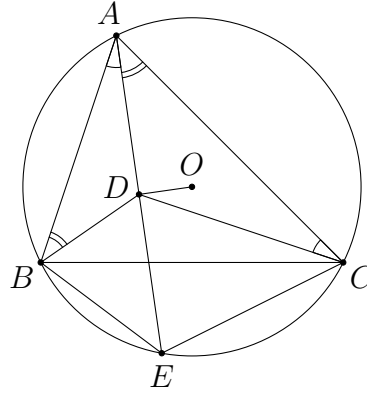
- (a) D is the midpoint of AE
- (b) D lies on (BCO)
- (c) D lies on the circle with diameter AO
- (d) D is the isogonal conjugate of the A -humpty point

Proof.

- (a) Let D' be the midpoint of AE . By the properties of harmonic quadrilaterals, we know that $\triangle ABD' \sim \triangle CBE \sim \triangle CAD'$ (directly). Therefore, D' is the centre of spiral similarity that maps BA to AC . This proves $D' = D$ as desired. As a corollary, the A -dumpty point lies on the A -symmedian.

- (b) The points B, C, D, O are concyclic since

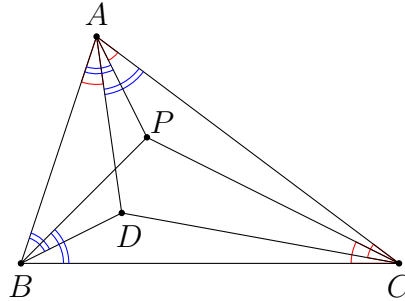
$$\begin{aligned} \angle BDC &= \angle DBA + \angle BAC + \angle ACD = \angle DAC + \angle BAC + \angle BAD \\ &= 2\angle BAC = \angle BOC. \end{aligned}$$



- (c) By part (a), D is the midpoint of a chord AE of the circumcircle. Therefore, we have $OD \perp AE$. The result then follows.
- (d) Let P be the A -humpty point of $\triangle ABC$. By [proposition 9.3\(a\)](#) and [proposition 9.4\(a\)](#), we know that P lies on the A -median, while D lies on the A -symmedian. Therefore, AD and AP are isogonals w.r.t. $\angle BAC$. Also, we have

$$\angle PBC = \angle PAB = \angle CAD = \angle ABD.$$

This implies BD and BP are isogonals w.r.t. $\angle ABC$. Thus, D and P are isogonal conjugates w.r.t. $\triangle ABC$.

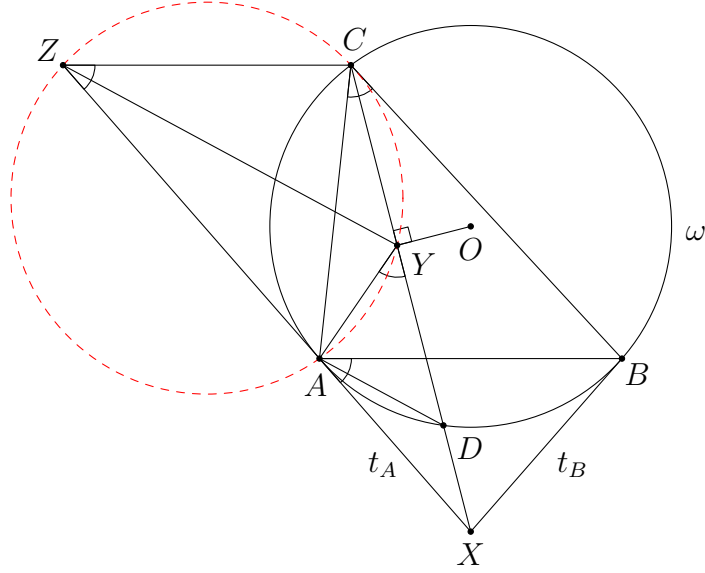


□

Problem 9.8. (Balkan MO 2022 Problem 1) Let ABC be an acute triangle such that $CA \neq CB$ with circumcircle ω and circumcentre O . Let t_A and t_B be the tangents to ω at A and B respectively, which meet at X . Let Y be the foot of the perpendicular from O onto the line segment CX . The line through C parallel to line AB meets t_A at Z . Prove that the line YZ passes through the midpoint of the line segment AC .

Analysis. Observe that Y is the C -dumpty point of $\triangle ABC$. Thus, Y is the midpoint of CD where D is the second intersection point of CX and ω . Therefore, the problem is the same as proving $ZY \parallel AD$. This can be done by angle chasing and using some properties of the symmedian CD .

Solution.



Let D be the second intersection point of CX and ω . By [proposition 9.1](#), CDX is the C -symmedian of $\triangle ABC$. By [proposition 9.4\(a\)](#) and (c), we know that Y is the C -dumpty point of $\triangle ABC$, and Y is the midpoint of CD .

Since $CADB$ is a harmonic quadrilateral, we have $\triangle ABC \sim \triangle ADY$ (directly). It follows that

$$\angle AYD = \angle ACB = \angle XAB = \angle AZC.$$

Therefore, A, Y, C, Z are concyclic. This implies

$$\angle XAD = \angle ACD = \angle ACY = \angle AZY,$$

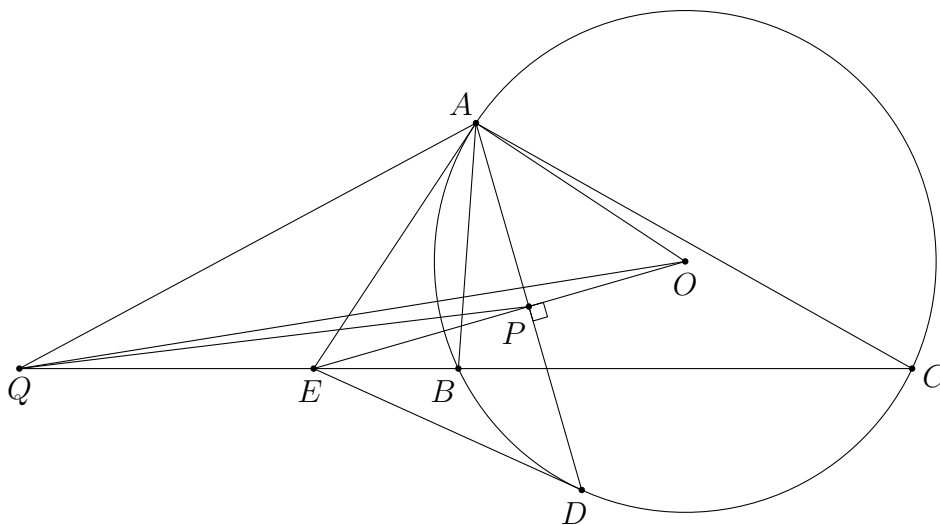
and hence $AD \parallel ZY$. Finally, as Y is the midpoint of CD , YZ bisects AC by the intercept theorem. $\square$

Problem 9.9. (ELMO Shortlist 2012 G7) Let $\triangle ABC$ be an acute triangle with circumcentre O such that $AB < AC$, let Q be the intersection of the external bisector of $\angle A$ with BC , and let P be a point in the interior of $\triangle ABC$ such that $\triangle BPA$ is similar to $\triangle APC$. Show that $\angle QPA + \angle OQB = 90^\circ$.

Analysis. Clearly, P is the A -dumpty point of $\triangle ABC$. To rewrite the assertion in a nicer way, we extend the line OP to a point E on the line BC , so that it remains to show $\angle QPE = \angle OQB$. Equivalently, we have to prove that EQ is tangent to (POQ) .

Another important observation is that AE is tangent to (ABC) . This again follows easily from some properties of dumpty points and symmedians.

Solution.



Let D be the second intersection point of AP and (ABC) , and let E be the intersection point of the lines BC and OP . By definition, P is the A -dumpty point of $\triangle ABC$. By [proposition 9.4\(c\)](#), we have $\angle OPA = 90^\circ$. Therefore,

$$\begin{aligned} & \angle QPA + \angle OQB = 90^\circ \\ \Leftrightarrow & \angle OQB = \angle QPE \\ \Leftrightarrow & \triangle EQO \sim \triangle EPQ. \end{aligned}$$

Next, by [proposition 9.4\(a\)](#), P is the midpoint of AD . Therefore, the tangents at A and D to (ABC) and the line OP are concurrent. Note that the tangents at A and D and the line BC are also concurrent as $ABDC$ is harmonic. Thus, all these lines meet at E . In particular, EA is tangent to (ABC) . This implies $\angle EAO = 90^\circ = \angle EPA$, and hence

$$EA^2 = EP \times EO.$$

Comparing with the final goal, we should prove $EA = EQ$. Indeed, as AQ is the external angle bisector of $\angle BAC$, we have

$$\angle QAE = \angle QAB - \angle EAB = \left(90^\circ - \frac{A}{2}\right) - C$$

and

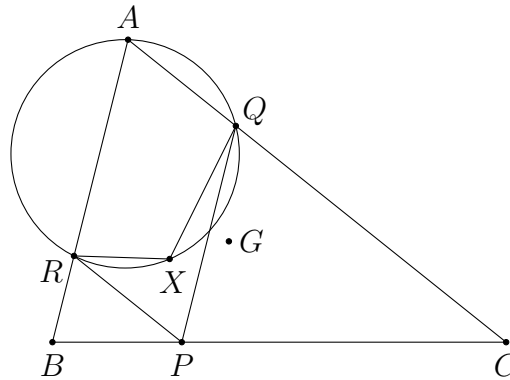
$$\angle AQE = 180^\circ - \angle QAC - \angle ACQ = \left(90^\circ - \frac{A}{2}\right) - C = \angle QAE.$$

This proves $EA = EQ$, and hence $EQ^2 = EA^2 = EP \times EO$. Finally, this implies $\triangle EQO \sim \triangle EPQ$ as desired. $\square$

Problem 9.10. (USA TST 2008 Problem 7) Let ABC be a triangle with G as its centroid. Let P be a variable point on segment BC . Points Q and R lie on sides AC and AB respectively, such that $PQ \parallel AB$ and $PR \parallel AC$. Prove that, as P varies along segment BC , the circumcircle of triangle AQR passes through a fixed point X such that $\angle BAG = \angle CAX$.

Analysis. Since the fixed point X should lie on the A -symmedian, we may first think of a few special points lying on this symmedian and see if those points could be the desired point X or not. These may include the symmedian point and the A -dumpty point. Indeed, we shall prove that X is exactly the A -dumpty point.

Solution.



Let X be the A -dumpty point of $\triangle ABC$. By [proposition 9.4\(a\)](#), X lies on the A -symmedian of $\triangle ABC$. Therefore, $\angle BAG = \angle CAX$. It suffices to prove X lies on (AQR) .

Recall that X is the centre of spiral similarity mapping BA to AC . As

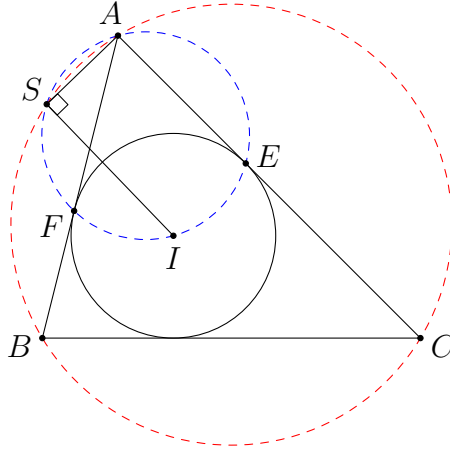
$$\frac{BR}{RA} = \frac{BP}{PC} = \frac{AQ}{QC},$$

the spiral similarity maps R to Q . Thus, we have $\triangle XBR \sim \triangle XAQ$. It follows that $\angle XRB = \angle XQA$, and hence A, R, X, Q are concyclic. So we are done. $\square$

9.3 Sharky-Devil Point

In this subsection, we introduce one more special point which is related to the circumcircle and the incircle.

Definition 9.4. Given $\triangle ABC$, let E, F be the contact points of the incircle and the sides AC, AB respectively. The A -*Sharky-Devil point* of $\triangle ABC$ is defined as the Miquel point of $BCEF$.



Let S be the A -Sharky-Devil point of $\triangle ABC$. This means S is the centre of spiral similarity mapping BF to CE , and the centre of spiral similarity mapping FE to BC . Thus, we immediately know the following properties of the Sharky-Devil point.

- S lies on (ABC)
- S lies on (AEF) , i.e. the circle with diameter AI where I is the incentre of $\triangle ABC$
- $\angle ASI = 90^\circ$

Indeed, many properties of the Sharky-Devil point can be deduced from the spiral similarities. Two more useful results are given below.

Proposition 9.5. In $\triangle ABC$, let I be the incentre and let D, E, F be the contact points of the incircle and the sides BC, CA, AB respectively. Let S be the A -Sharky-Devil point, let M be the midpoint of $\widehat{BC}$ of the circumcircle not containing A , and let P be the intersection point of IS and EF . Then the following hold.

- S, D, M are collinear
- $DP \perp EF$

Proof.

- We shall prove that $\triangle MID \sim \triangle MSI$. Recall that S is the centre of spiral similarity mapping FE to BC . Since I is the midpoint of $\widehat{FE}$ on (SFE) and M is the midpoint of $\widehat{BC}$ on (SBC) , the spiral similarity maps I to M . Thus, we have $SFIE \sim SBMC$, and hence

$$\frac{SI}{SM} = \frac{FI}{BM} = \frac{ID}{IM}.$$

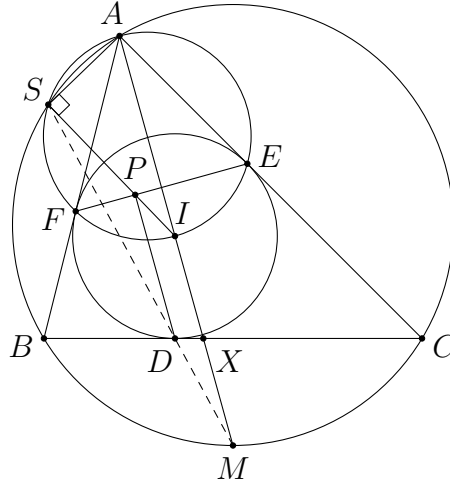
Also, we have

$$\angle ISM = \angle ASM - 90^\circ = \angle ABM - 90^\circ = B + \frac{A}{2} - 90^\circ$$

and

$$\angle DIM = \angle AXC - 90^\circ = B + \frac{A}{2} - 90^\circ = \angle ISM$$

where X is the intersection point of BC and AM . This proves $\triangle MID \sim \triangle MSI$, and hence S, D, M are collinear.



- (b) Recall that $SFIE \sim SBMC$. As P is the intersection point of SI and FE , while D is the intersection point of SM and BC , we have $SFIEP \sim SBMCD$. In particular, we have

$$\frac{SP}{PI} = \frac{SD}{DM}.$$

This implies $PD \parallel IM$ by the intercept theorem, and hence $PD \perp EF$.

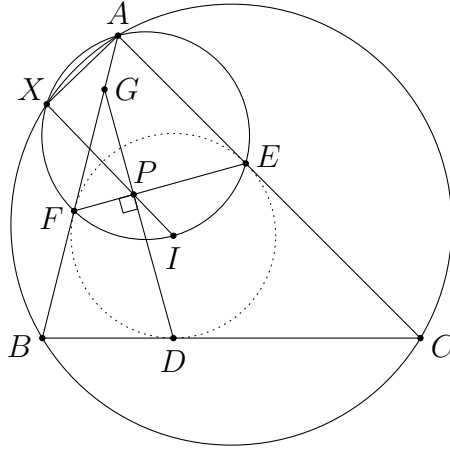
□

We give two examples below to demonstrate the use of the Sharky-Devil points.

Problem 9.11. (Francophone Mathematical Olympiad 2020 Senior Problem 1) Let ABC be an acute triangle with $AC > AB$. Let DEF be the intouch triangle with $D \in BC$, $E \in AC$, $F \in AB$. Let G be the intersection of the perpendicular from D to EF with AB , and $X = (ABC) \cap (AEF)$. Prove that B, D, G and X are concyclic.

Analysis. Note that X is the A -Sharky-Devil point. The problem is trivialized using the properties of the Sharky-Devil point. One mainly needs to recall the spiral similarities with centre X .

Solution. Let P be the intersection point of EF and DG . By proposition 9.5(b), P lies on IX . Also, by the proof of that result, we know that X is the centre of spiral similarity mapping FPE to BDC . Thus, X is the Miquel point of $BDPF$. As BF meets DP at G , the points B, D, G, X are concyclic.



Remarks. This problem is not at all simple if one does not know those facts about the Sharky-Devil points and spiral similarities. But now we basically need to do nothing in the solution. This shows the importance of learning a lot of facts in geometry.

Problem 9.12. (MODS Mathematical Olympiad 2020 Advanced Problem 2) Let I be the incentre of a triangle ABC , and let D, E , and F be the intersections of the incircle of $\triangle ABC$ with BC, CA , and AB respectively. Suppose that the circumcircles of $\triangle ABC$ and $\triangle AID$ intersect again at $P \neq A$, and that AP intersects EF at Q . Show that DQ is perpendicular to EF .

Analysis. Among the interior angles of the cyclic quadrilateral $AIDP$, we can compute $\angle AID = 180^\circ - \frac{|B - C|}{2}$. This implies $\angle APD = \frac{|B - C|}{2}$. In view of this angle, it is natural to consider the midpoint M of $\widehat{BC}$ containing A on (ABC) , since $\angle ABM = \frac{|B - C|}{2}$. In other words, M should lie on PD .

On the other hand, we are going to prove that Q is the intersection of EF and SI , where S is the A -Sharky-Devil point of $\triangle ABC$. Note that the spiral similarity mapping FE to BC also maps A to M . This connects different parts together, and suggests us using this spiral similarity to work out a proof.

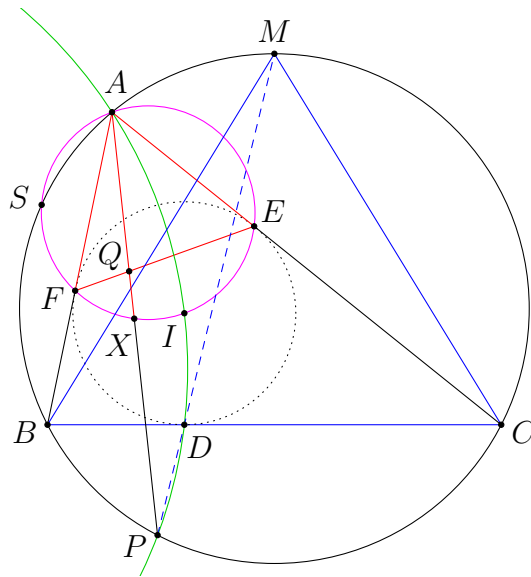
Solution. WLOG assume $AB < AC$. Let M be the midpoint of $\widehat{BC}$ containing A on (ABC) . Let $\angle ABM = \angle ACM = \theta$. Then

$$B - \theta = \angle CBM = \angle BCM = C + \theta,$$

which implies $\theta = \frac{B - C}{2}$. Also, we have

$$\begin{aligned} \angle APD &= 180^\circ - \angle AID = 180^\circ - \angle AIB - \angle BID \\ &= 180^\circ - \left(90^\circ + \frac{C}{2}\right) - \left(90^\circ - \frac{B}{2}\right) = \frac{B - C}{2}. \end{aligned}$$

This shows $\angle APM = \angle APD$, and hence P, D, M are collinear.



Let S be the A -Sharky-Devil point of $\triangle ABC$, and let AP meet (AEF) again at X . Recall that S lies on (AEF) . As BF meets PX at A , and S lies on both (AFX) and (ABP) , S is the centre of spiral similarity mapping FX to BP by [proposition 7.2](#). Recall that this spiral similarity also maps E to C . In addition, since A is the midpoint of $\widehat{FE}$ on (SFE) and M is the midpoint of $\widehat{BC}$ on (SBC) , the spiral similarity maps A to M . Thus, it also maps $Q = AX \cap FE$ to $D = MP \cap BC$. We know from the proof of [proposition 9.5\(b\)](#) that Q must be the intersection point of FE and SI , and that $DQ \perp EF$ as desired. $\square$

In this section, we introduce several useful objects and points related to a triangle.

- symmedian (problems 9.1, 9.2, 9.3, 9.4, 9.8, 9.10)
- harmonic quadrilateral (problems 9.4, 9.9)
- humpty point (problems 9.5, 9.6, 9.7)
- dumpty point (problems 9.8, 9.9, 9.10)
- Sharky-Devil point (problems 9.11, 9.12)

10 Reconstruction of Figures

Lastly, we introduce a general technique in solving geometry problems. Reconstruction of figures refers to the trick of constructing the figure in the problem in an alternative way compared to the problem statement, so that the problem is rephrased to another equivalent problem. The main purpose is to simplify the problem until we are able to solve the whole problem completely. During the process, we may need to add extra points or use some results we know. But how should we reconstruct the figure? Here lists out some possible directions that we may consider.

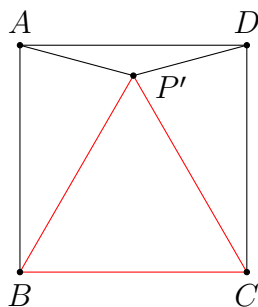
- Reduce the number of points.
- Construct the same point in an alternative way, and show that this agrees with the original construction.
- Reverse the order of constructing the points.
- Remove strange conditions such as equations.
- Avoid proving concurrency since we do not have many tools to prove this.

The second point refers to the method of false position. We have already seen one application of this method in [problem 2.1](#). We now investigate more about this method.

Problem 10.1. Let $ABCD$ be a square. Let P be an interior point such that $\angle PAD = \angle PDA = 15^\circ$. Prove that $\triangle PBC$ is an equilateral triangle.

Analysis. We first start with an extremely simple example. Although it is simple, it is impossible for one to prove the result directly by angle chasing or length chasing without adding extra points. However, it is easy to prove that $\angle PAD = \angle PDA = 15^\circ$ if we assume $\triangle PBC$ is an equilateral triangle. This is not equivalent to the original problem, but we can use a little trick to solve the original problem in this direction.

Solution.



Let P' be the point in $ABCD$ such that $\triangle P'BC$ is an equilateral triangle. As $P'B = BC = AB$, we have

$$\angle BAP' = \frac{180^\circ - \angle ABP'}{2} = \frac{180^\circ - (90^\circ - 60^\circ)}{2} = 75^\circ,$$

and hence

$$\angle P'AD = 90^\circ - 75^\circ = 15^\circ.$$

By symmetry, $\angle P'DA = 15^\circ$. Since there is a unique point P inside $ABCD$ satisfying $\angle PAD = \angle PDA = 15^\circ$, we must have $P' = P$. Thus, $\triangle PBC$ is an equilateral triangle. $\square$

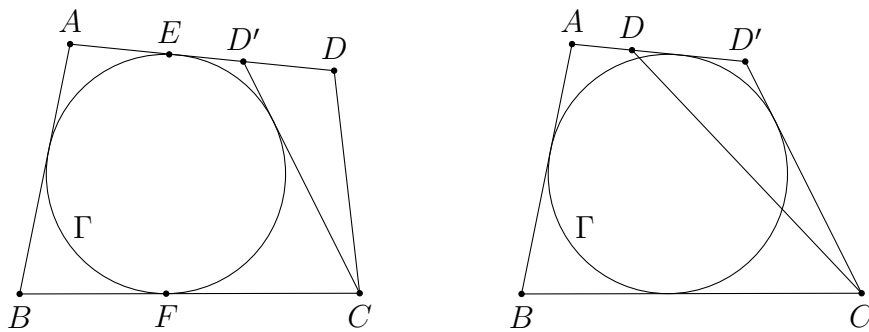
Remarks. When using the method of false position, we have to ensure that the points we redefine are uniquely determined. For example, if we do not know P or P' lie inside $ABCD$ in this problem, we cannot conclude $P' = P$ after proving $\angle P'AD = \angle P'DA = 15^\circ$ as there are two points satisfying this condition (one lying outside the square). This rarely happens if there is a unique configuration.

From this example, we see the advantage of the method of false position. In some sense, it helps us swap a condition in the question with the conclusion. Sometimes it is easier for us to work in the converse direction. For example, sometimes it is easier to use a concyclic condition rather than proving the concyclicity since there are more angle relations we can tell if we assume some points are concyclic.

Problem 10.2. Let $ABCD$ be a convex quadrilateral such that $AB + CD = AD + CB$. Prove that $ABCD$ is a circumscribed quadrilateral.

Analysis. This is the converse of the Pitot theorem. We do not have many ways to show that a quadrilateral has an incircle. Therefore, we use the method of false position and first construct a circumscribed quadrilateral. This remains to compare the lengths of some segments, which is an easier task. There are more configuration issues in this problem, so we have to be careful.

Solution.



We first construct a circle Γ tangent to the segment AB and the rays AD and BC . This circle is uniquely determined since its centre lies on the angle bisectors of $\angle A$ and $\angle B$. Let E and F be the contact points of Γ with the lines AD and BC . Since

$$AD + CB = AB + CD > AB = AE + BF,$$

we may assume F lies on the segment BC . Thus, there is a unique point D' on the line AD such that Γ is the incircle of $ABCD'$. Therefore, we have $AB + CD' = AD' + CB$ by the Pitot theorem.

- If D' lies on the segment AD , then

$$CD - CD' = (AB + CD) - (AB + CD') = (AD + CB) - (AD' + CB) = DD',$$

contradicting the triangle inequality in $\triangle CDD'$.

- If D' lies on the extension of AD , then

$$CD' - CD = (AB + CD') - (AB + CD) = (AD' + CB) - (AD + CB) = DD',$$

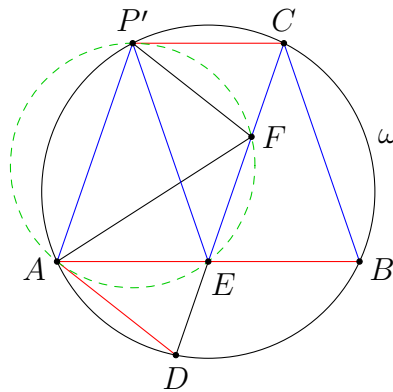
contradicting the triangle inequality in $\triangle CDD'$.

Thus, we must have $D' = D$, which means $ABCD$ is a circumscribed quadrilateral. $\square$

Problem 10.3. (Sharygin Geometry Olympiad 2022 Final Grade 9 Problem 5) Chords AB and CD of a circle ω meet at point E in such a way that $AD = AE = EB$. Let F be a point of segment CE such that $ED = CF$. The bisector of angle AFC meets an arc DAC at point P . Prove that A, E, F , and P are concyclic.

Analysis. The angle bisector condition is hard to use. Therefore, we may first assume P is the second intersection point of ω and (AEF) , and prove that PF bisects $\angle AFC$. Since $ED = CF$, we note that there is a symmetry between $\triangle ADE$ and $\triangle PCF$ about the perpendicular bisector of CD . This observation allows us to define P in a more convenient way, which gives us more information.

Solution.



Let P' be the point on ω such that $AP' \parallel DC$. Then $ADCP'$ is an isosceles trapezoid. As $DE = CF$, $\triangle ADE$ and $\triangle CP'F$ are symmetric about the perpendicular bisector of CD . Thus, $AEP'F$ is an isosceles trapezoid. In particular, it is cyclic. This suggests P' should be the point P in the problem.

Note that $AE = EB = AD = P'C$ and $\angle AED = \angle ADE = \angle P'CE$. This implies $AB \parallel P'C$. Thus, both $AEC P'$ and $EBC P'$ are parallelograms, while $ABCP'$ is an isosceles trapezoid. It follows that $P'A = CB = P'E$, and so

$$\angle CFP' = \angle EAP' = \angle P'EA = \angle P'FA.$$

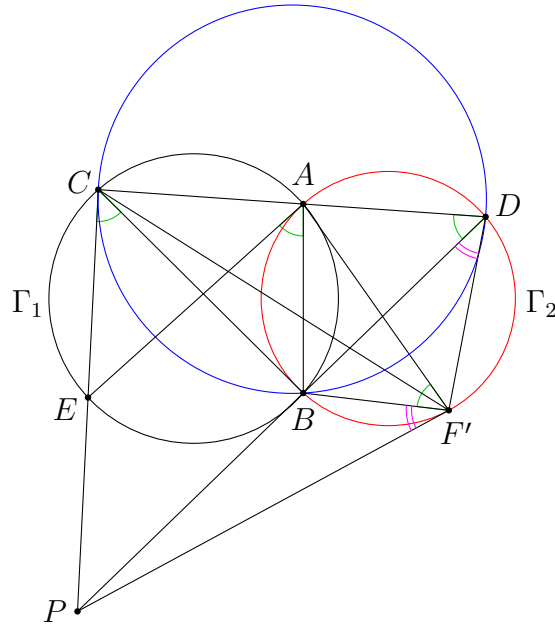
This proves $P'F$ bisects $\angle CFA$. Therefore, $P' = P$, and hence A, E, F, P are concyclic. $\square$

Problem 10.4. (Iranian Geometric Olympiad 2021 Advanced Problem 2) Two circles Γ_1 and Γ_2 meet at two distinct points A and B . A line passing through A meets Γ_1 and Γ_2 again at C and D respectively, such that A lies between C and D . The tangent at A to Γ_2 meets Γ_1 again at E . Let F be a point on Γ_2 such that F and A lie on different sides of BD , and $2\angle AFC = \angle ABC$. Prove that the tangent at F to Γ_2 , and lines BD and CE are concurrent.

Analysis. In fact, Γ_1 should not pass through the centre of Γ_2 , since otherwise the three lines are parallel instead of concurrent.

The main barrier in this problem is the condition $2\angle AFC = \angle ABC$. There are different ways to interpret this condition geometrically. Our approach is to redefine the point F and attempt to prove this angle relation instead. This makes the construction more natural, and the whole proof mainly relies on angle chasing. Also, we need not prove the concurrency, which is usually harder as discussed.

Solution.



We first assume Γ_1 does not pass through the centre of Γ_2 as discussed. We claim that BD and CE are not parallel. Indeed, if $BD \parallel CE$, then $\angle ECA + \angle ADB = 0$. It follows that

$$\angle ADB = \angle ACE = \angle ACB + \angle BCE = \angle ACB + \angle BAE = \angle ACB + \angle BDA.$$

This implies $2\angle ADB = \angle ACB$, and hence the centre of Γ_2 lies on Γ_1 , contradiction.

Now, let P be the intersection point of BD and CE , and let F' be the contact point of the tangent from P to Γ_2 such that A and F' lie on different sides of BD . We are going to prove that $2\angle AF'C = \angle ABC$.

Firstly, since $\angle ECB = \angle EAB = \angle ADB = \angle CDB$, the line CP is tangent to (BCD) . Therefore, by the power of points, we have

$$PC = \sqrt{PB \times PD} = PF'.$$

This means $\triangle PCF'$ is isosceles. The remaining task is purely angle chasing. We check that

$$\begin{aligned} \angle ABC &= \angle(AB, BF') + \angle(BF', F'P) + \angle(F'P, CF') + \angle(CF', CP) + \angle(CP, BC) \\ &= \angle ADF' + \angle BDF' + \angle PF'C + \angle F'CP + \angle CDB \\ &= \angle ADF' + (\angle BDF' + \angle CDB) + (\angle PF'C + \angle F'CP) \\ &= 2\angle ADF' + 2\angle PF'C \\ &= 2\angle AF'P + 2\angle PF'C \\ &= 2\angle AF'C. \end{aligned}$$

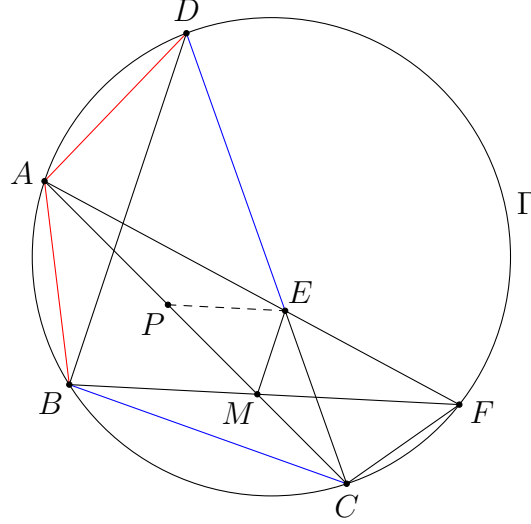
From the configuration, F' must lie outside Γ_1 , and hence $\angle AF'C < \angle ABC$. Therefore, we must have $\angle ABC = 2\angle AF'C$. Note that there is a unique point F on Γ_2 such that F and A lie on different sides of BD , and $2\angle AFC = \angle ABC$ (because the angle relation and the configuration determine a circular arc, which intersects Γ_2 at A and another point). Therefore, we have proved $F' = F$. The assertion follows readily. $\square$

Remarks. Interestingly, this problem is a reconstruction of IMO Shortlist 2020 G8. The two problems look totally different because a lot of intermediate steps are needed. (And now I forget how to transform one problem to another!) This is one possible way to make up some new geometry problems.

Next, we go through several examples to see how to reconstruct the figures so as to simplify the problem in various ways. One should learn the ideas involved rather than the ways to solve these particular problems.

Problem 10.5. (CHKMO 2020 Problem 3) Let $ABCD$ be a cyclic quadrilateral inscribed in a circle Γ such that $AB = AD$. Let E be a point on the segment CD such that $BC = DE$. The line AE intersects Γ again at F . The chords AC and BF

meet at M . Let P be the symmetric point of C about M . Prove that PE and BF are parallel.



Analysis. A simple idea to reconstruct the figure is to remove some points by rephrasing its property in terms of the other points.

Firstly, in view of the midpoint and the parallel result, we may use the midpoint theorem. We see that it is the same as proving BF bisects CE . This interpretation of the problem is simpler because we do not need to define the points P and M . We may need to define the intersection point of BF and CE , but still one point is removed from the figure.

Secondly, the point A does not play too much role in the problem. The only condition related to A is $AB = AD$. This can be rephrased as an angle relation not involving A , such as EF bisects $\angle DFB$.

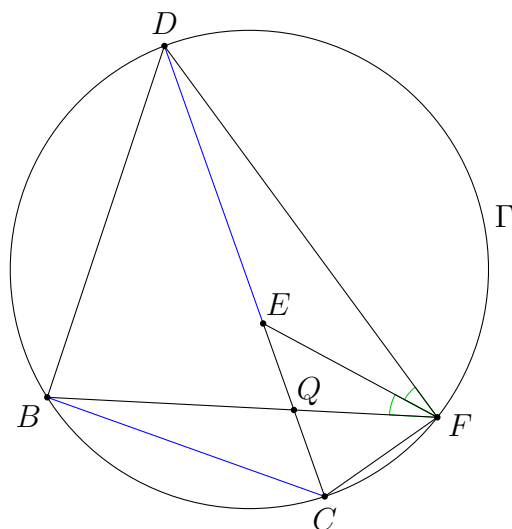
Now, the figure is reduced to an arbitrary cyclic quadrilateral $BCFD$ with two points on the diagonal CD (including the intersection of CD and BF). This new problem looks much simpler, and can be solved immediately. For example, to use the angle bisector condition, we may think of the angle bisector theorem. This is on the right track because two of the lengths involved in the ratios obtained are DE and EQ , which are related to the condition and the final result.

Solution. Since $AB = AD$, we have $\angle DFE = \angle EFB$. Let Q be the intersection point of CD and BF . By the angle bisector theorem, we have

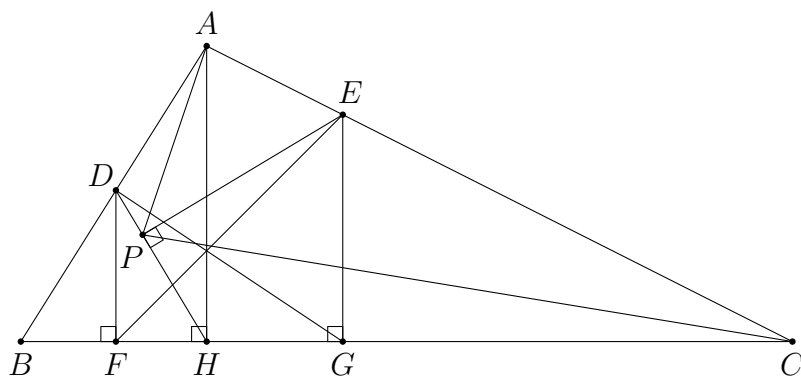
$$\frac{ED}{EQ} = \frac{FD}{FQ} = \frac{BC}{CQ}.$$

The last equality holds since $\triangle DFQ \sim \triangle BCQ$.

Now, as $BC = DE$, this implies $CQ = EQ$. By the midpoint theorem, we know that $QM \parallel EP$, which means $PE \parallel BF$. $\square$



Problem 10.6. (IMO HK TST 2014 Test 2 Problem 2) Let ABC be an acute triangle and H be the perpendicular foot at the side BC from A . Let D and E be points on the segments AB and AC , F and G be the perpendicular feet at BC from D and E respectively. Let P be the perpendicular foot at the segment DH from E . Suppose that DG, EF, AH are concurrent. Show that $\angle APE = \angle CPE$.



Analysis. In the above figure, $\triangle ABC$ is obtuse instead. This is just to make the figure nicer, which does not affect the arguments below.

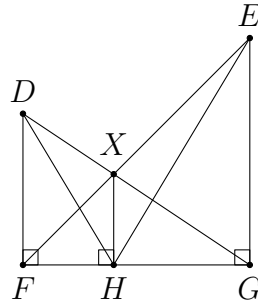
The setting is complicated because there are many unrelated things in the figure. We should study the relations of the points clearly to see whether some points can be removed, and whether some conditions can be rephrased to a better meaning. First of all, when we draw the figure, it is not ideal to start with $\triangle ABC$. This is because if D and E are randomly put on AB and AC , it is almost hopeless that DG, EF, AH are concurrent. Rather than that, we may first start with the points D, E, F, G . This allows us to fix the intersection point $DG \cap EF$, and hence H is uniquely determined. Next, we still have one more flexibility. For example, by choosing any point A on the altitude, points B and C are uniquely determined.

Do we really need to locate the point B ? No! This is because none of the conditions or the final assertion is related to B . This means the point B can be removed if we reconstruct the figure in the above way. This is often the case when we reconstruct a figure in a suitable way. This means a point X may be the first point defined in the original problem, but after a reconstruction, this point may no longer be needed.

Next, can we remove further points from the figure? It looks like F is not too useful. The only property of F is the concurrency of the three lines including EF . We can rewrite this concurrency to another condition. For example, it is indeed well-known that the concurrency is equivalent to $\triangle DHF \sim \triangle EHG$, or simply $\angle AHD = \angle AHE$. Using this equivalent condition, it is surprising that we do not need all of D, F, G in the figure.

Now, there are two perpendicular conditions remained and there are two angle bisectors (one to be proved). The problem looks nicer because these conditions seem to be related. Actually, those who are familiar with projective geometry (in particular, harmonic division) should now find that the whole problem is trivialized.

Solution.



We first show that $\angle AHD = \angle AHE$. Let X be the intersection point of DG and EF . Using similar triangles, we have

$$\frac{FH}{GH} = \frac{DX}{GX} = \frac{DF}{EG}.$$

Together with the included angle $\angle DFH = \angle EGH$, we obtain $\triangle DFH \sim \triangle EGH$. This implies $\angle DHF = \angle EHG$, and hence AH bisects $\angle DHE$.

Let Q be the intersection point of AC and PH . As $AH \perp CH$, AH and CH are the angle bisectors of $\angle EQH$. By the angle bisector theorem, we have

$$\frac{AE}{AQ} = \frac{HE}{HQ} = \frac{CE}{CQ}.$$

In exactly the same way, if we define A' to be the point on the ray CE such that PE bisects $\angle CPA'$, then PE and PQ are the angle bisectors of $\angle CPA'$. This implies

$$\frac{A'E}{CE} = \frac{A'P}{CP} = \frac{A'Q}{CQ}.$$



- If C, E, Q lie in this order, then the ratio in (1) is less than 1. Since both A and A' lie on the ray EQ , we must have $A = A'$ from (1).
- If E, C, Q lie in this order, then both A and A' do not lie on the segment EQ . Therefore, we still have $A = A'$ from (1).

In any case, we have $A = A'$. This proves $\angle APE = \angle CPE$.

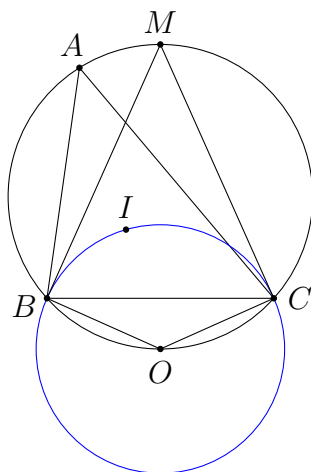
Analysis. This time it does not seem there is any redundant point in the figure. The points I, H, M are certainly necessary. To define I and M , we need $\triangle ABC$. To define H , we need $\triangle DEF$. There is not much room for us to redefine the points if we start with $\triangle ABC$.

If we want a reconstruction to yield a big change in the difficulty of the problem, we need more courage to think of a totally different way to define the points. Usually we can start with some points which are needed in the final assertion. For example, this time we start with $\triangle IBC$. Recall that the circumcentre of $\triangle IBC$ is the midpoint of $\widehat{BC}$. Although this point is not M , it is indeed the antipode of M on (ABC) . This allows us to conclude that MB and MC are tangent to (IBC) , and hence IM is indeed the I -symmedian of $\triangle IBC$. In view of this, we do not need M anymore, as it suffices to prove H lies on this symmedian.

How to define H in a simpler way? The points D and E are necessary because of the major condition $DB = BC = CE$. From $BD = BC$ and BI bisects $\angle CBD$, we see that D is the reflection of C in BI . Using this interpretation, we do not need point A as well. We can also ignore the point F by using $DH \perp BE$ and $EH \perp CD$ to define H . As a result, we have simplified the problem so that only 6 points remain. All kinds of tools such as trigonometry and coordinate geometry can be used to solve this problem easily.

Let us focus on pure geometry. To use $DH \perp BE$ and $EH \perp CD$, it may be easier to use $DH \parallel IC$ and $EH \parallel IB$. Since there are two pairs of parallel lines, we may construct a parallelogram based on this. Although this generates two extra points, we should find this useful as parallelograms are closely related to midpoints, which are further related to symmedians.

Solution.



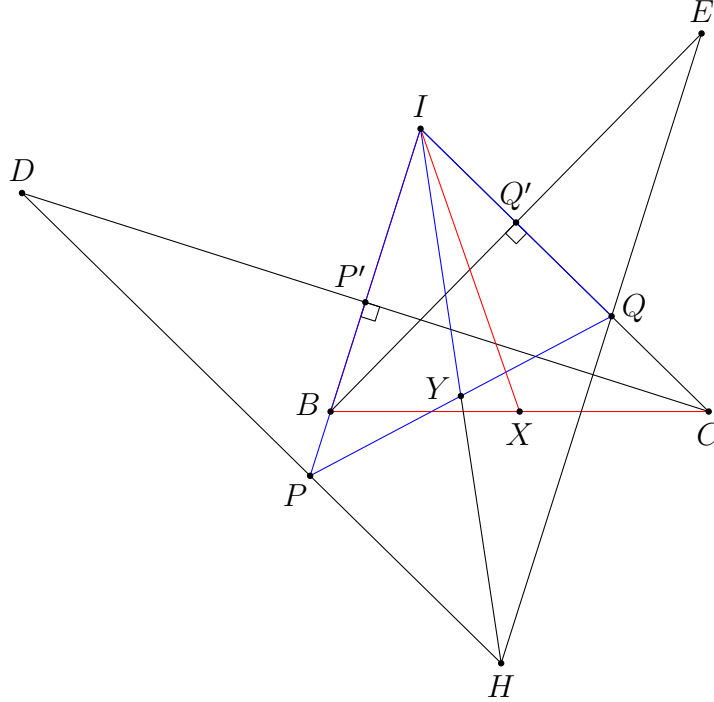
Recall that the circumcentre O of $\triangle IBC$ is the midpoint of $\widehat{BC}$ not containing A on (ABC) . Thus, MO is a diameter of (ABC) . This implies $\angle MBO = \angle MCO = 90^\circ$,

and hence MB and MC are tangent to (IBC) . By [proposition 9.1](#), IM is the I -symmedian of $\triangle IBC$. It remains to show that H lies on this symmedian.

Since $BD = BC$ and BI bisects $\angle CBD$, we see that D is the reflection of C in BI . Similarly, E is the reflection of B in CI . Also, as $DH \perp BE$ and $IC \perp BE$, we have $DH \parallel IC$. Similarly, $EH \parallel IB$. We can rephrase the problem as follows.

- Given $\triangle IBC$, let D and E be the reflections of C and B in BI and CI respectively. Let H be the point such that $DH \parallel IC$ and $EH \parallel IB$. Prove that IH is the I -symmedian of $\triangle IBC$.

Note that we do not even need $\triangle IBC$ to be obtuse (which must be the case in the original problem). In other words, we may actually draw a (wrong) figure where $\triangle IBC$ is acute to work out a proof. Sometimes a problem could become simpler if the triangle changes from obtuse to acute! We do this below to emphasize that we should always be flexible.



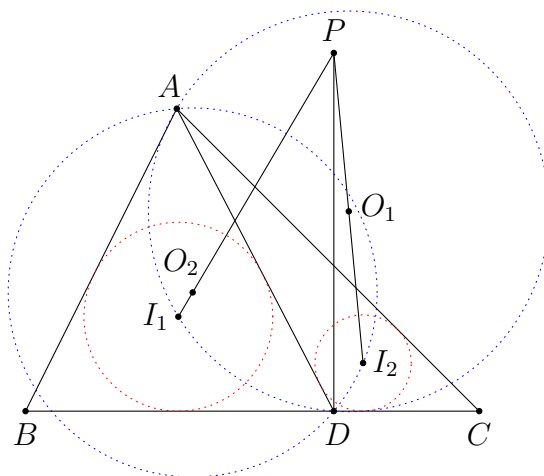
Let IB meet DH at P , and let IC meet EH at Q . Then $IPHQ$ is a parallelogram, and so the midpoint Y of PQ lies on IH . It remains to prove IY is the I -symmedian of $\triangle IBC$.

Let P' and Q' be the feet of altitudes from C and B of $\triangle IBC$ respectively. Then they are the midpoints of CD and BE respectively. Since C and D are symmetric about IP , and $IC \parallel DP$, it is obvious that $CIDP$ is a rhombus. In particular, P' is the midpoint of IP . Similarly, Q' is the midpoint of IQ . It follows that

$$\frac{IP}{IQ} = \frac{IP'}{IQ'} = \frac{IC}{IB}.$$

The last equality holds since B, C, Q', P' are concyclic. This implies $\triangle IBC \sim \triangle IQP$ (oppositely). Thus, if we let X be the midpoint of BC , then $IBC X \sim IQPY$, and hence $\angle BIX = -\angle QIY$. This shows IY and IX are isogonals w.r.t. $\angle BIC$. This proves the result as IX is the median. $\square$.

Problem 10.8. (CWMI 2017 Problem 3) In a triangle ABC , let D be a point on BC . Let I_1 and I_2 be the incentres of triangles ABD and ACD , respectively. Let O_1 and O_2 be the circumcentres of triangles AI_1D and AI_2D , respectively. Let lines I_1O_2 and I_2O_1 meet at P . Show that $PD \perp BC$.



Analysis. We should recall the basic result that O_1 is the midpoint of $\widehat{AD}$ on (ABD) , etc. Also, using the angle bisectors, it is not hard to deduce $I_1D \perp I_2D$. This is the crucial observation in the following solution.

Again, we try to reconstruct the figure by not starting with $\triangle ABC$. Indeed, we start with the points I_1, D, I_2 because of the right angle. (The motivation was that I attempted to solve this problem using coordinate geometry.) Next, O_1 and O_2 are points on the perpendicular bisectors of DI_1 and DI_2 respectively. But can they be arbitrary points? We just need to see whether the other points can always be defined or not. Point P is still the intersection of I_1O_2 and I_2O_1 . Point A is uniquely determined because (AI_1D) and (AI_2D) are fixed. Then the incircles of $\triangle ABD$ and $\triangle ACD$ are also fixed, which means B and C can be uniquely determined. The condition $\angle I_1DI_2 = 90^\circ$ guarantees that B, D, C must be collinear, so everything goes well.

Under this reconstruction of the figure, we can remove all of points A, B, C ! (Strictly speaking, we still need the line BC to prove $PD \perp BC$. But it is easy to use angles to give a direction to the line BC , so the exact positions of B and C are unimportant.) Again, we only have 6 points left with some simple conditions (right angle and perpendicular bisectors). Now, just a few minutes is sufficient to solve this seemingly hard problem (maybe using coordinate geometry).

In the following solution, we shall reconstruct the figure many times in addition to the above one. The main purpose is not to solve this particular problem, but to demonstrate how a problem can be investigated in different ways. At the end, you will see an amazing proof to this problem using conics!

Solution. Firstly, we have

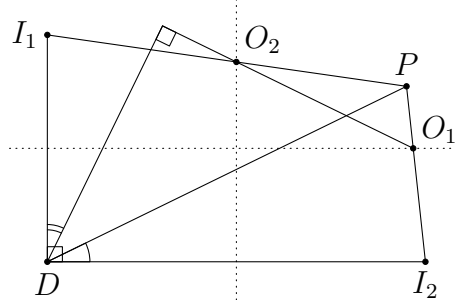
$$\angle I_1DI_2 = \angle ADI_1 + \angle ADI_2 = \frac{1}{2}\angle ADB + \frac{1}{2}\angle ADC = \frac{1}{2} \times 180^\circ = 90^\circ.$$

Note that

$$\begin{aligned} PD &\perp BC \\ \Leftrightarrow \quad \angle PDB &= \angle I_2DI_1 \\ \Leftrightarrow \quad \angle I_2DP &= \angle I_1DB \\ \Leftrightarrow \quad \angle I_2DP &= \angle ADI_1. \end{aligned}$$

Noting that $AD \perp O_1O_2$, it suffices to prove the following.

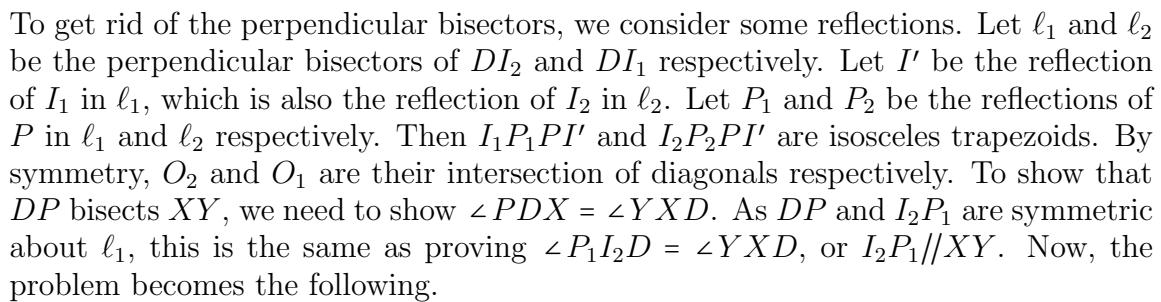
- Let PI_1DI_2 be a quadrilateral such that $\angle I_1DI_2 = 90^\circ$. Let O_1 be the intersection of I_2P and the perpendicular bisector of DI_1 . Let O_2 be the intersection of I_1P and the perpendicular bisector of DI_2 . Prove that the perpendicular line from D to O_1O_2 is the isogonal of DP w.r.t. $\angle I_1DI_2$.



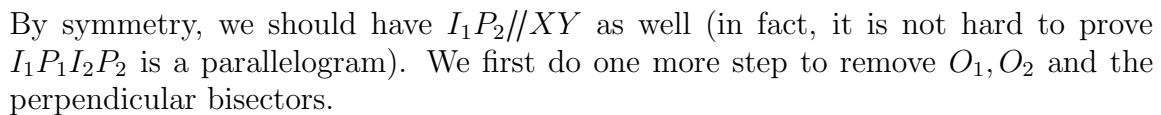
We try to rephrase the problem again. Let O_1O_2 meet the lines DI_2 and DI_1 at X and Y respectively. Since $\angle XDY = 90^\circ$, the isogonal of the D -altitude of $\triangle DXY$ w.r.t. $\angle XDY$ is the line passing through D and the circumcentre of $\triangle DXY$. As the circumcentre is the midpoint of XY , it suffices to show DP bisects XY . We rephrase the problem as follows.

- Let $\triangle YDX$ be a right-angled triangle with $\angle XDY = 90^\circ$. Let I_1 and I_2 be points on the sides DY and $respectively. Let O_1 be the intersection of XY and the perpendicular bisector of DI_1 . Let O_2 be the intersection of XY and the perpendicular bisector of DI_2 . Let P be the intersection point of I_1O_2 and I_2O_1 . Prove that DP bisects XY .$

In this formulation, we only need to prove a collinearity. Ceva's and/or Menelaus' theorem can be used to solve it.

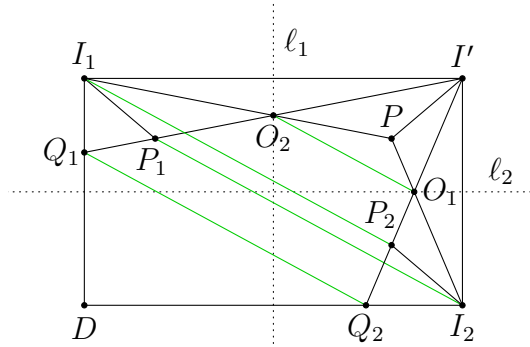


- In this formulation, we do not need points X and Y .

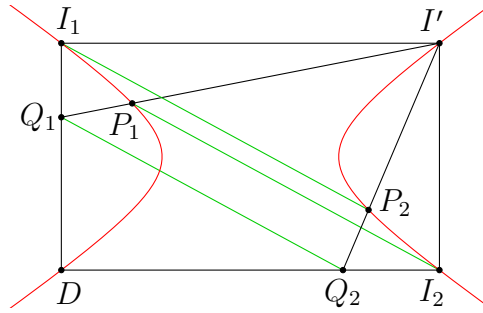


Consider the homothety with centre I' and ratio 2. Let Q_1 and Q_2 be the images of O_2 and O_1 respectively. Note that Q_1 lies on DI_1 and Q_2 lies on DI_2 . We must have $Q_2Q_1//O_1O_2$. Therefore, we further transform the problem to the following.

- 176



We now give a proof to this final version of the problem. One fact that we need is the following: every conic passing through the vertices I_1, D, I_2, I' of the rectangle must be symmetric about the centre of this rectangle (which can be easily proved using coordinate geometry by setting the centre to be the origin). Thus, the unique conic passing through I_1, D, I_2, I', P_1 must pass through P_2 by symmetry. Finally, applying Pascal's theorem to $I'P_2I_1DI_2P_1$, we see that $I'P_2 \cap DI_2 = Q_2$, $P_2I_1 \cap I_2P_1$ and $I_1D \cap P_1I' = Q_1$ are collinear. But since $P_2I_1P_1I_2$ is a parallelogram, this means $P_2I_1 // I_2P_1 // Q_2Q_1$ as desired. $\square$



Remarks. We have reconstructed the figure for a few times and obtained four more versions of the same problem above. They look quite different from each other, especially when compared to the original contest problem. Although the above solution is mainly for fun, it shows us how powerful this idea could be, since it gives us more possibilities to explore the problem. To use this idea, we need to know a lot of geometric facts. Also, we should be creative and think of various ways to reconstruct the points.

We mainly introduce two key ideas in reconstructing the figures. They are only some general ideas, but the way of using them greatly depends on the problems.

- false position ([problems 10.1, 10.2, 10.3, 10.4](#))
- reducing the number of points ([problems 10.5, 10.6, 10.7, 10.8](#))

Links

Theorems

- [Central angle theorem](#)
- [Fagnano's problem](#)
- [Intercept theorem 1](#)
- [Intercept theorem 2](#)
- [Iran lemma 1](#)
- [Iran lemma 2](#)
- [Isogonal line lemma](#)
- [Midpoint theorem](#)
- [Monge's theorem](#)
- [Pitot theorem](#)
- [Power of a point theorem](#)
- [Reim's theorem](#)
- [Thébault's theorem](#)
- [Y. Sawayama's lemma](#)

Propositions

- [1.1](#): Basic facts about directed angles
- [1.2](#): Basic facts about directed lengths
- [2.1](#): Parallel lines and similar triangles
- [2.2](#): Concyclic points and similar triangles
- [4.1](#): Steiner line and anti-Steiner point
- [4.2](#): Homothety between incircle and excircle
- [5.1](#): Radical axis bisects external common tangent
- [6.1](#): Properties of circumscribed quadrilaterals
- [7.1](#): Basic facts about spiral similarity
- [7.2](#): Centre of spiral similarity
- [7.3](#): Properties of the Miquel point of cyclic quadrilateral
- [8.1](#): Basic facts about isogonals
- [8.2](#): Basic facts about isogonal conjugates

- 8.3: Test for isogonal conjugates
- 8.4: Generating new isogonal conjugates
- 8.5: Existence of isogonal conjugates in polygons
- 8.6: Existence of isogonal conjugates in quadrilaterals
- 9.1: Symmedian and tangents
- 9.2: Symmedian and harmonic quadrilateral
- 9.3: Properties of humpty points
- 9.4: Properties of dumpty points
- 9.5: Properties of Sharky-Devil points

Problems

- 1.1: Miquel point of triangle
- 1.2: Directed angles and tangent
- 1.3: Directed angles and concyclic points
- 1.4: Generalization of Pythagoras' theorem
- 2.1: Intercept theorem
- 2.2: Angle chasing and finding concyclic points
- 2.3: Corresponding points of similar triangles
- 2.4: IMO HK TST 2003 Test 1 Problem 2
- 2.5: CGMO 2003 Problem 3
- 2.6: CWMO 2005 Problem 2
- 3.1: Translation
- 3.2: Rotation and square
- 3.3: 1st Fermat point
- 3.4: Fagnano's problem
- 3.5: IMO Prelim 2012 Problem 16
- 3.6: Homothety 1
- 3.7: Homothety 2
- 3.8: IMO Shortlist 2007 G3
- 3.9: IMO Shortlist 2021 G4
- 4.1: Centroid and the 2 : 1 ratio
- 4.2: Dutch TST 2021 Test 3 Problem 2

- 4.3: Iran lemma 1
- 4.4: Midpoint theorem and circles
- 4.5: Argentina TST 2011 Problem 3
- 4.6: EGMO 2012 Problem 1
- 4.7: CWMO 2002 Problem 5
- 4.8: Iran MO 2nd Round 2013 Problem 3
- 4.9: Orthocentre and midpoint
- 4.10: CHKMO 2013 Problem 3
- 4.11: Iran MO 2nd Round 2015 Problem 4
- 4.12: IMO Shortlist 1995 G2
- 4.13: IMO Shortlist 2000 G3
- 4.14: Belarus TST 2019 Test 2 Problem 2
- 4.15: Nine-point circle and homothety
- 4.16: IMO 2011 Problem 6
- 4.17: Iran lemma 2
- 4.18: CWMO 2004 Problem 2
- 4.19: IMO Shortlist 2005 G1
- 4.20: CGMO 2016 Problem 7
- 5.1: IMO HK TST 2011 Test 1 Problem 1
- 5.2: IMO HK TST 2011 Test 2 Problem 1
- 5.3: Power and concyclic points
- 5.4: Power and tangents
- 5.5: CGMO 2022 Problem 5
- 5.6: Power and tangential circles
- 5.7: Y. Sawayama's lemma and Thébault's theorem
- 5.8: IMO 2009 Problem 2
- 5.9: Radical axis and concyclic points
- 5.10: IMO 2000 Problem 1
- 5.11: Radical axis and perpendicular lines
- 5.12: CHKMO 2006 Problem 3
- 5.13: Radical axis and point circle
- 5.14: IMO HK TST 2021 Test 2 Problem 3

- 5.15: IMO 2015 Problem 3
- 6.1: IMO 2002 Problem 2
- 6.2: Balkan MO Shortlist 2017 G4
- 6.3: IMO Shortlist 2010 G5
- 6.4: IMO Shortlist 2013 G2
- 6.5: Thailand Online MO 2021 Problem 7
- 6.6: All-Russian MO 2019 Grade 11 Problem 6
- 6.7: Pitot theorem
- 6.8: Bicentric quadrilateral
- 6.9: IMO 2008 Problem 6
- 7.1: Spiral similarity
- 7.2: CGMO 2006 Problem 2
- 7.3: IMO Shortlist 2006 G3
- 7.4: IMO 2005 Problem 5
- 7.5: Centre of spiral similarity
- 7.6: IMO Shortlist 2005 G5
- 7.7: Corresponding points of spiral similarities
- 7.8: Mongolia MO Final Round 2017 Grade 11 Problem 4
- 7.9: Morocco TST 2015
- 7.10: IMO Shortlist 2020 G6
- 7.11: Iranian Geometric Olympiad 2022 Intermediate Problem 5
- 8.1: IMO 2004 Problem 5
- 8.2: IMO Shortlist 2003 G3
- 8.3: Isogonal line lemma
- 8.4: Iran TST 2015 Test 3 Problem 6
- 8.5: International Zhautykov Olympiad 2015 Problem 2
- 8.6: IMO Shortlist 2011 G6
- 8.7: IMO 2014 Problem 3
- 8.8: All-Russian MO 2017 Grade 11 Problem 8
- 8.9: Isogonal conjugate in quadrilateral
- 8.10: Czech-Polish-Slovak Match 2020 Problem 6
- 8.11: ELMO 2012 Problem 5

- [9.1](#): Sharygin Geometry Olympiad 2018 Final Grade 9 Problem 7
- [9.2](#): Symmedian and tangents
- [9.3](#): Bulgaria Regional MO 2021 Grade 9
- [9.4](#): APMO 2012 Problem 4
- [9.5](#): Humpty point and concyclic points
- [9.6](#): Brazil MO 2015 Problem 6
- [9.7](#): Sharygin Geometry Olympiad 2023 Correspondence Problem 21
- [9.8](#): Balkan MO 2022 Problem 1
- [9.9](#): ELMO Shortlist 2012 G7
- [9.10](#): USA TST 2008 Problem 7
- [9.11](#): Francophone Mathematical Olympiad 2020 Senior Problem 1
- [9.12](#): MODS Mathematical Olympiad 2020 Advanced Problem 2
- [10.1](#): False position and square
- [10.2](#): Converse of the Pitot theorem
- [10.3](#): Sharygin Geometry Olympiad 2022 Final Grade 9 Problem 5
- [10.4](#): Iranian Geometric Olympiad 2021 Advanced Problem 2
- [10.5](#): CHKMO 2020 Problem 3
- [10.6](#): IMO HK TST 2014 Test 2 Problem 2
- [10.7](#): Reconstruction of figures
- [10.8](#): CWMI 2017 Problem 3

Terminologies

- [1st Fermat point](#)
- [anti-Steiner point](#)
- [antiparallel](#)
- [bicentric quadrilateral](#)
- [centroid \$G\$](#)
- [circumcentre \$O\$](#)
- [circumcircle](#)
- [circumradius \$R\$](#)
- [circumscribed quadrilateral](#)
- [dilation](#)

- directed angle $\sphericalangle ABC$, $\sphericalangle (AB, CD)$
- directed length
- directly similar
- dumpty point
- Euler line
- excentre E_a, E_b, E_c
- excircle
- harmonic quadrilateral
- homothety
- humpty point
- incentre I
- incircle
- inradius r
- isogonal
- isogonal conjugate
- isogonal line
- isometry
- isotomic conjugate
- Lemoine point K
- Miquel point (triangle)
- Miquel point (quadrilateral)
- nine-point centre N
- nine-point circle
- oppositely similar
- orthocentre H
- pedal circle
- pedal triangle
- power
- radical axis
- radical centre
- reflection
- rigid motion

- rigid transformation
- rotation
- semiperimeter s
- Sharky-Devil point
- spiral similarity
- Steiner line
- symmedian
- symmedian point K
- transformation
- translation

References

- Properties of the Sharky-Devil point (by aops29)
(Available from: <https://artofproblemsolving.com/community/c946900h1911664>)
- Euclidean Geometry in Mathematical Olympiads (by Evan Chen)
- Anti-Steiner Points with respect to a Triangle (by Darij Grinberg)
- Isogonal Conjugation with respect to a Triangle (by Darij Grinberg)
- A Special Point on the Median (by Anant Mudgal, Gunmay Handa)
- On Two Special Points in Triangle (by Kapil Pause)
- Isogonal Line Lemma (by utkarshgupta)
(Available from: <https://artofproblemsolving.com/community/c2771h1181729>)
- Advanced Lemmas in Geometry (by Fedir Yudin)
- Three Lemmas in Geometry (by Yufei Zhao)

「水因地而制行，兵因敵而制勝。故兵無常勢，水無常形。能因敵變化而取勝者，謂之神。」

《孫子兵法—實虛篇》